

FOREWORD

This wiring diagram manual has been prepared to provide information on the electrical system of the 1992 TOYOTA CELICA.

Applicable models: AT180 Series
 ST184, 185 Series

For service specifications and repair procedures of the above models other than those listed in this manual, refer to the following manuals:

Manual Name	Pub. No.
• 1992 CELICA Repair Manual Volume 1 Volume 2 • 1992 Model New Car Features	RM251U1 RM251U2 NCF079U

All information in this manual is based on the latest product information at the time of publication. However, specifications and procedures are subject to change without notice.

TOYOTA MOTOR CORPORATION

NOTICE

Servicing vehicles with an SRS AIRBAG (referred to as the airbag in the remainder of this manual) installed.

When handling airbag components (removal, installation or inspection, etc.), always follow the directions given in the repair manuals listed above to prevent the occurrence of accidents and airbag malfunction.

INTRODUCTION

This manual consists of the following 11 sections:

No.	Section	Description
A	INDEX	Index of the contents of this manual.
	INTRODUCTION	Brief explanation of each section.
B	HOW TO USE THIS MANUAL	Instructions on how to use this manual.
C	TROUBLE-SHOOTING	Describes the basic inspection procedures for electrical circuits.
D	ABBREVIATIONS	Defines the abbreviations used in this manual.
E	GLOSSARY OF TERMS AND SYMBOLS	Defines the symbols and functions of major parts.
F	RELAY LOCATIONS	Shows position of the Electronic Control Unit, Relays, Relay Block, etc. This section is closely related to the system circuit.
G	ELECTRICAL WIRING ROUTING	Describes position of Parts Connectors, Splice points, Ground points, etc. This section is closely related to the system circuit.
H	POWER SOURCE (Current Flow Chart)	Describes power distribution from the power supply to various electrical loads.
I	INDEX	Index of the system circuits.
	SYSTEM CIRCUITS	Electrical circuits of each system are shown from the power supply through ground points. Wiring connections and their positions are shown and classified by code according to the connection method. (Refer to the section, "How to use this manual"). The "System Outline" and "Service Hints" useful for troubleshooting are also contained in this section.
J	GROUND POINTS	Shows ground positions of all parts described in this manual.
K	OVERALL WIRING DIAGRAM	Provides circuit diagrams showing the circuit connections.

This manual provides information on the electrical circuits in on vehicles by dividing them into a circuit for each system.

The actual wiring of each system circuit is shown from the point where the power source is received from the battery as far as each ground point. (All circuit diagrams are shown with the switches in the OFF position.)

When troubleshooting any problem, first understand the operation of the circuit where the problem was detected (see System Circuit section), the power source supplying power to that circuit (see Power Source section), and the ground points (see Ground Points section). See the System Outline to understand the circuit operation.

When the circuit operation is understood, begin troubleshooting of the problem circuit to isolate the cause. Use Relay Location and Electrical Wire Routing sections to find each part, junction block and wiring harness connectors, wiring harness and wiring harness connectors, splice points, and ground points of each system circuit. Internal wiring for each junction block is also provided for better understanding of connection within a junction block.

Wiring related to each system is indicated in each system circuit by arrows (from __ , to __). When overall connections are required, see the Overall Wiring Diagram at the end of this manual.

4

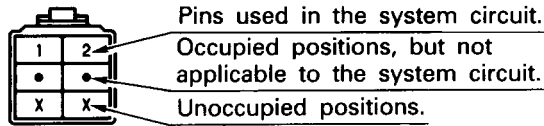


- (A) : System Title
- (B) : Indicates a Relay Block. No shading is used and only the Relay Block No. is shown to distinguish it from the J/B.

Example:  Indicates Relay Block No. 1.

- (C) : Indicates the connector to be connected to a part (the numeral indicates the pin No.)

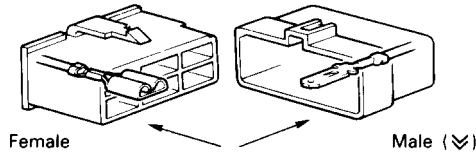
Explanation of pin use.



The pins shown are only for the highest grade, or only include those in the specification.

- (D) : Connector Color
Connectors not indicated are milky white in color.
- (E) : () is used to indicate different wiring and connector, etc. when the vehicle model, engine type, or specification is different.
- (F) : Indicates related system.
- (G) : Indicates the wiring harness and wiring harness connector. The wiring harness with male terminal is shown with arrows (⇨).

Outside numerals are pin numbers.

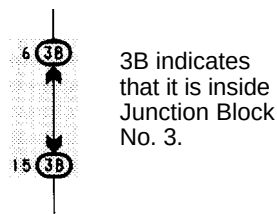


The first letter of the code for each wiring harness and wiring harness connector(s) indicates the component's location, e.g., "E" for the Engine Compartment, "I" for the Instrument Panel and Surrounding area, and "B" for the Body and Surrounding area.

When more than one code has the first and second letters in common, followed by numbers (e.g., IH1, IH2), this indicates the same type of wiring harness and wiring harness connector.

- (H) : Represents a part (all parts are shown in sky blue). The code is the same as the code used in parts position.
- (I) : Junction Block (The number in the circle is the J/B No. and the connector code is shown beside it). Junction Blocks are shaded to clearly separate them from other parts (different junction blocks are shaded differently for further clarification).

Example:

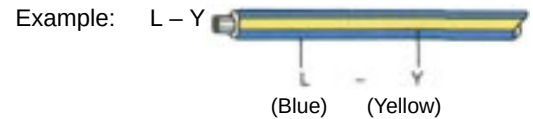


- (J) : Indicates the wiring color.

Wire colors are indicated by an alphabetical code.

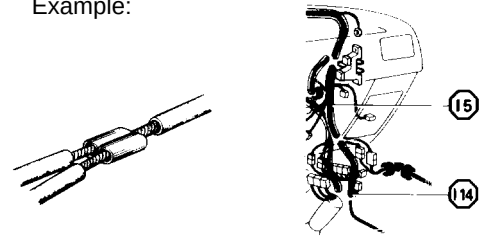
B = Black L = Blue R = Red
BR = Brown LG = Light Green V = Violet
G = Green O = Orange W = White
GR = Gray P = Pink Y = Yellow

The first letter indicates the basic wire color and the second letter indicates the color of the stripe.



- (K) : Indicates a wiring Splice Point (Codes are "E" for the Engine Room, "I" for the Instrument Panel, and "B" for the Body).

Example:



The Location of Splice Point I 5 is indicated by the shaded section.

- (L) : Page No.

- (M) : Indicates a shielded cable.



- (N) : Indicates a ground point.

The first letter of the code for each ground point(s) indicates the component's location, e.g., "E" for the Engine Compartment, "I" for the Instrument Panel and Surrounding area, and "B" for the Body and Surrounding area.

- (O) : Indicates the pin number of the connector.

The numbering system is different for female and male connectors.

Example: Numbered in order from upper left to lower right Numbered in order from upper right to lower left



- (P) : When 2 parts both use one connector in common, the parts connector name used in the wire routing section is shown in square brackets [].

HOW TO USE THIS MANUAL



SYSTEM OUTLINE

WITH THE IGNITION SW TURNED ON, THE CURRENT FLOWS TO TERMINAL 3 OF THE POWER WINDOW MASTER SW, TERMINAL 2 OF THE POWER WINDOW CONTROL RELAY AND TERMINAL 8 OF THE POWER WINDOW SW THROUGH THE DOOR FUSE.

1. DRIVER'S WINDOW "MANUAL UP" OPERATION BY MASTER SW

HOLDING MANUAL SW (DRIVER'S) ON "UP" POSITION LOCATED IN POWER WINDOW MASTER SW, THE CURRENT FLOWS TO TERMINAL 5 OF THE POWER WINDOW CONTROL RELAY THROUGH TERMINAL 3 OF THE MASTER SW → TERMINAL 2 TO OPERATE A POWER WINDOW CONTROL RELAY. THUS THE CURRENT INSIDE THE RELAY FLOWS FROM TERMINAL 2 OF THE RELAY → TERMINAL 1 → TERMINAL 2 OF THE POWER WINDOW MOTOR → TERMINAL 1 → TERMINAL 4 OF THE RELAY → TERMINAL 3 → TO GROUND. THE MOTOR TURNS TO ASCENT THE WINDOW. RELEASING THIS SW, THE ROTATION OF MOTOR IS STOPPED AND THE WINDOWS CAN STOP AT WILL POINT.

(FOR THE "MANUAL DOWN" OPERATION, CURRENT FLOWS IN THE REVERSE DIRECTION BECAUSE THE TERMINALS WHERE IT FLOWS ARE CHANGED).

2. DRIVER'S WINDOW "AUTO DOWN" OPERATION BY MASTER SW

ONCE THE "AUTO DOWN" BUTTON OF THE MASTER SW IS PUSHED, THE CURRENT FLOWS TERMINAL 9 OF THE POWER WINDOW CONTROL RELAY THROUGH TERMINAL 3 OF THE MASTER SW → TERMINALS 8 AND 9 TO OPERATE THE RELAY. THUS THE CURRENT INSIDE THE POWER WINDOW CONTROL RELAY FLOWS FROM TERMINAL 2 OF THE RELAY → TERMINAL 4 → TERMINAL 1 OF THE POWER WINDOW MOTOR → TERMINAL 2 → TERMINAL 1 OF THE RELAY → TERMINAL 3 → TO GROUND. THE MOTOR CONTINUES THE ROTATION ENABLING TO DESCENT THE WINDOW.

THE WINDOW DESCENDS TO THE END POSITION. THE CURRENT WILL BE CUT OFF TO RELEASE THE AUTO DOWN FUNCTION BASED ON THE INCREASING CURRENT BETWEEN TERMINAL 2 OF THE RELAY AND TERMINAL 1 IN RELAY.

3. DRIVER'S WINDOW AUTO DOWN RELEASE OPERATION BY MASTER SW

HOLDING THE MANUAL SW (DRIVER'S) ON "UP" POSITION IN OPERATING AUTO DOWN. THE CURRENT FROM TERMINAL 3 OF THE MASTER SW PASSING TERMINAL 2 FLOWS TERMINAL 5 OF THE RELAY AND RELEASES THE AUTO DOWN FUNCTION IN THE POWER WINDOW CONTROL RELAY. RELEASING THE HAND FROM SW, WINDOW STOPS AND CONTINUING ON TOUCHING SW, THE FUNCTION SWITCHES TO MANUAL UP OPERATION.

4. PASSENGER'S WINDOW UP OPERATION (MASTER SW) AND WINDOW LOCK SW OPERATION

HOLDING PASSENGER'S WINDOW SW (MASTER SW) ON "UP", THE CURRENT FLOWS FROM TERMINAL 3 OF THE MASTER SW PASSING TERMINAL 6 TO TERMINAL 3 OF THE POWER WINDOW SW (PASSENGER'S) → TERMINAL 4 → TERMINAL 2 OF THE MOTOR → TERMINAL 1 → TERMINAL 9 OF THE POWER WINDOW SW → TERMINAL 7 → TERMINAL 1 OF THE MASTER SW → TERMINAL 4 TO GROUND. THE MOTOR RUNS TO ASCENT THE WINDOW. RELEASING THIS SW, THE ROTATION OF MOTOR IS STOPPED AND WINDOW CAN STOP AT WILL PLACE.

SWITCHING THE WINDOW LOCK SW IN "LOCK" POSITION, THE CIRCUIT IS OPENED AND STOPPED THE MOTOR ROTATION.

(FOR THE DOWN OPERATION, CURRENT FLOWS IN THE REVERSE DIRECTION BECAUSE THE TERMINALS WHERE IT FLOWS ARE CHANGED).



SERVICE HINTS

P2 POWER WINDOW CONTROL RELAY

3-GROUND: ALWAYS CONTINUITY

2-GROUND: APPROX. 12 VOLTS WITH IGNITION SW AT ON POSITION

5-GROUND: APPROX. 12 VOLTS WITH IGNITION SW AT ON POSITION AND MASTER SW AT UP POSITION

8-GROUND: APPROX. 12 VOLTS WITH IGNITION SW AT ON POSITION AND MASTER SW AT AUTO DOWN POSITION

9-GROUND: APPROX. 12 VOLTS WITH IGNITION SW AT ON POSITION AND MASTER SW AT DOWN OR AUTO DOWN POSITION

P 4 POWER WINDOW MASTER SW

4-GROUND: ALWAYS CONTINUITY

3-GROUND: APPROX. 12 VOLTS WITH IGNITION SW AT ON POSITION

WINDOW LOCK SW

OPEN WITH WINDOW LOCK SW AT LOCK POSITION



: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
P2	21	P4	21	P6	21
P3	21	P5	21		21



: RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCK (RELAY BLOCK LOCATION)
1	16	R/B NO. 1 (INSTRUMENT PANEL LEFT SIDE)



: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
3B	14	J/B NO. 3 AND COWL WIRE (INSTRUMENT PANEL LEFT SIDE)



: CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
ID1	26	FRONT DOOR RH WIRE AND COWL WIRE (RIGHT KICK PANEL)
IH1	26	FRONT DOOR LH WIRE AND COWL WIRE (LEFT KICK PANEL)



: GROUND POINTS

CODE	SEE PAGE	GROUND POINT LOCATION
IC	24	COWL LEFT



: SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESSES WITH SPLICE POINTS
I5	24	COWL WIRE

Ⓚ : Explains the system outline.

Ⓡ : Indicates values or explains the function for reference during troubleshooting.

Ⓢ : Indicates the reference page showing the position on the vehicle of the parts in the system circuit.

Example: Part "P4" (Power Window Master SW) is on page 21 of the manual.

* The letter in the code is from the first letter of the part, and the number indicates its order in parts starting with that letter.

Example: P 4
└───┬───┬───
 └───┬─── Part is 4th in order
 └─── Power Window Master SW

Ⓣ : Indicates the reference page showing the position on the vehicle of Relay Block Connectors in the system circuit.

Example: Connector "1" is described on page 16 of this manual and is installed on the left side of the instrument panel.

Ⓤ : Indicates the reference page showing the position on the vehicle of J/B and Wire Harness in the system circuit.

Example: Connector "3B" connects the Cowl Wire and J/B No. 3. It is described on page 14 of this manual, and is installed on the instrument panel left side.

Ⓥ : Indicates the reference page describing the wiring harness and wiring harness connector (the female wiring harness is shown first, followed by the male wiring harness).

Example: Connector "ID1" connects the front door RH wire (female) and cowl wire (male). It is described on page 26 of this manual, and is installed on the right side kick panel.

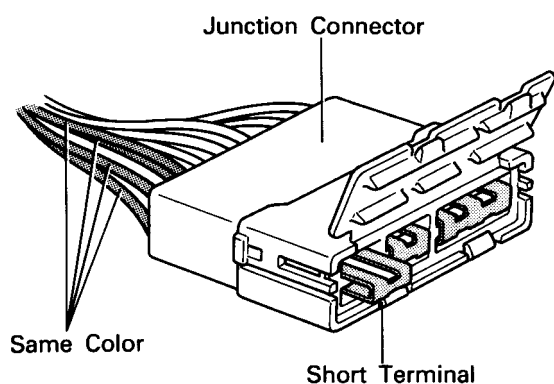
Ⓦ : Indicates the reference page showing the position of the ground points on the vehicle.

Example: Ground point "IC" is described on page 24 of this manual and is installed on the cowl left side.

Ⓧ : Indicates the reference page showing the position of the splice points on the vehicle.

Example: Splice point "I 5" is on the Cowl Wire Harness and is described on page 24 of this manual.

HINT:



Junction connector (code: J1, J2, J3, J4, J5, J6) in this manual include a short terminal which is connected to a number of wire harnesses. Always perform inspection with the short terminal installed. (When installing the wire harnesses, the harnesses can be connected to any position within the short terminal grouping. Accordingly, in other vehicles, the same position in the short terminal may be connected to a wire harness from a different part.)

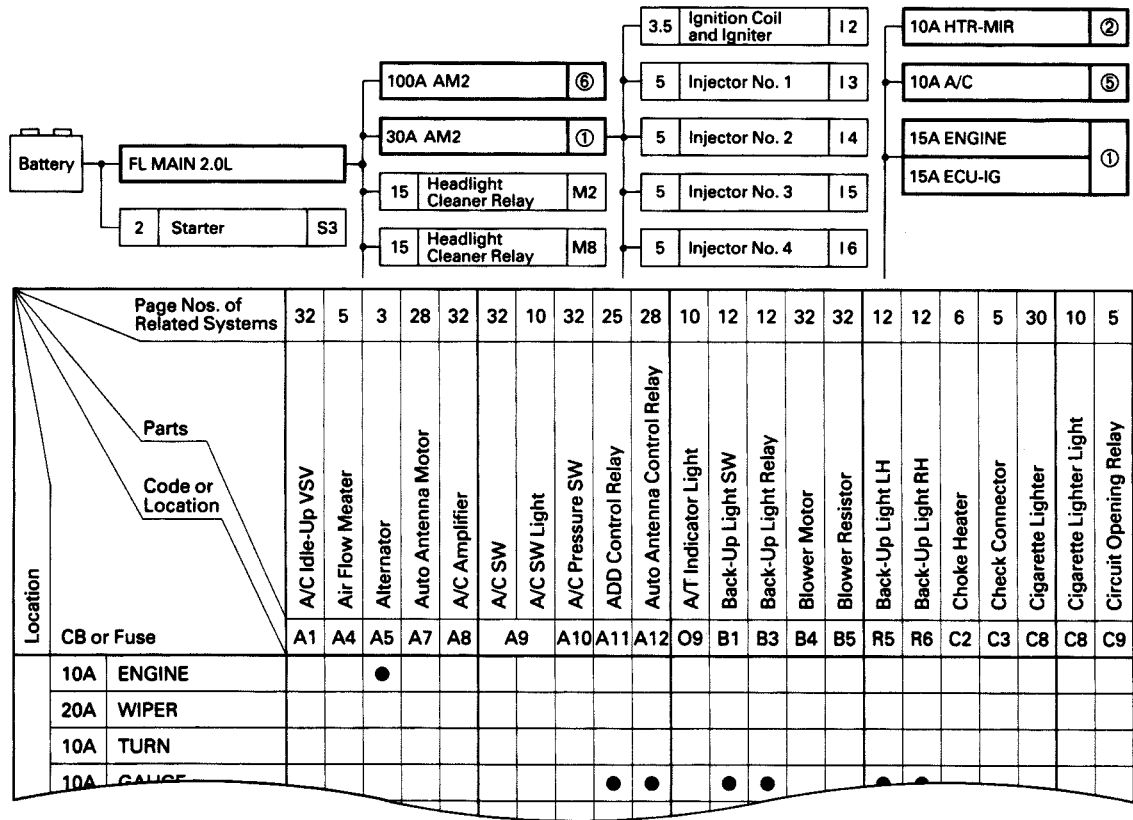
Wire harness sharing the same short terminal grouping have the same color.

HOW TO USE THIS MANUAL

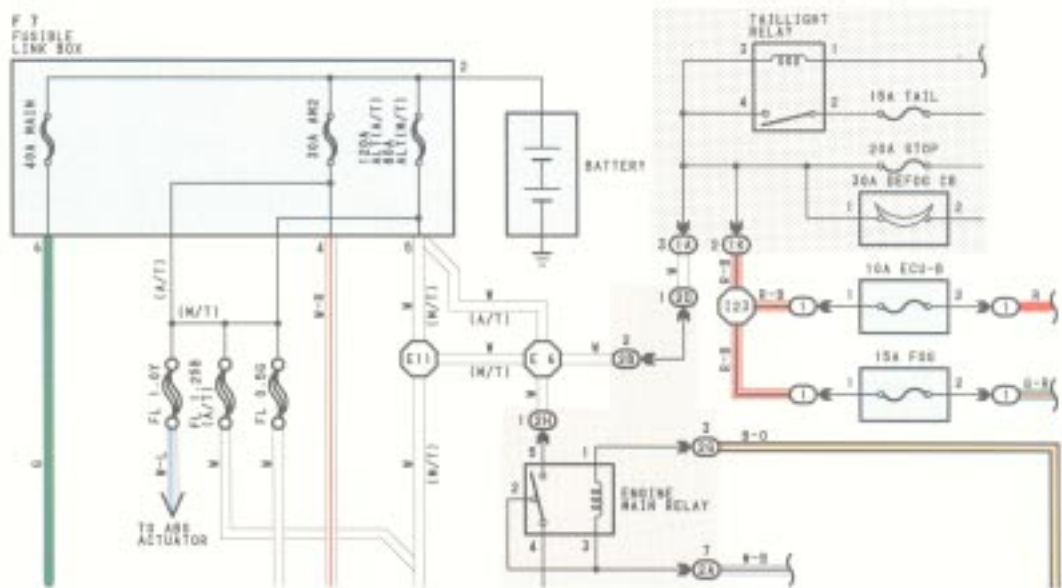
The “Current Flow Chart” section, describes which parts each power source (fuses, fusible links, and circuit breakers) transmits current to. In the Power Source circuit diagram, the conditions when battery power is supplied to each system are explained. Since all System Circuit diagrams start from the power source, the power source system must be fully understood.

POWER SOURCE (Current Flow Chart)

The chart below shows the route by which current flows from the battery to each electrical source (Fusible Link, Circuit Breaker, Fuse, etc.) and other parts.
The next page and following pages shown the parts to which each electrical source outputs current.



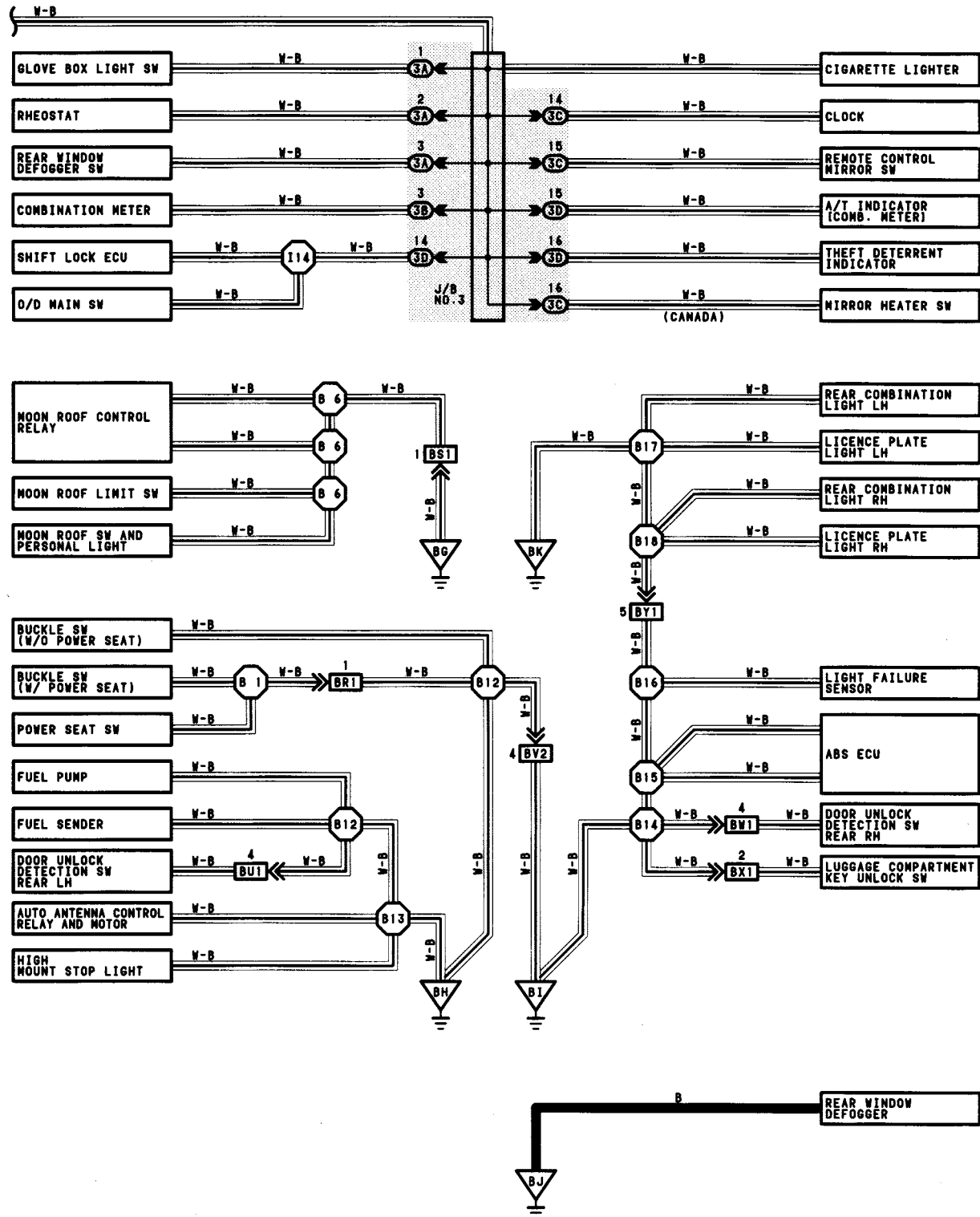
POWER SOURCE



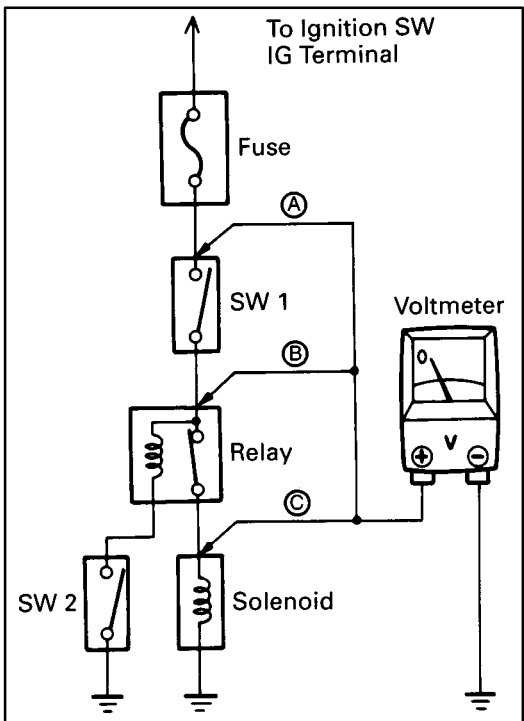
* The system shown here is an EXAMPLE ONLY. It is different to the actual circuit shown in the SYSTEM CIRCUITS SECTION.

The ground points circuit diagram shows the connections from all major parts to the respective ground points. When troubleshooting a faulty ground point, checking the system circuits which use a common ground may help you identify the problem ground quickly. The relationship between ground points ( ,  , and  shown below) can also be checked this way.

GROUND POINTS



TROUBLESHOOTING

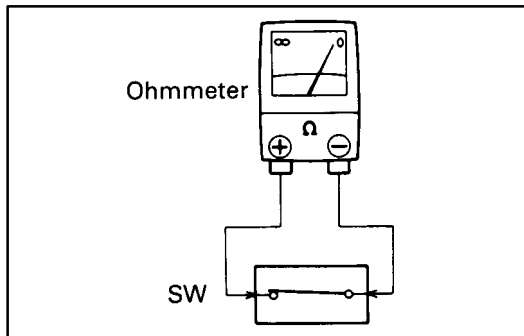


VOLTAGE CHECK

- (a) Establish conditions in which voltage is present at the check point.

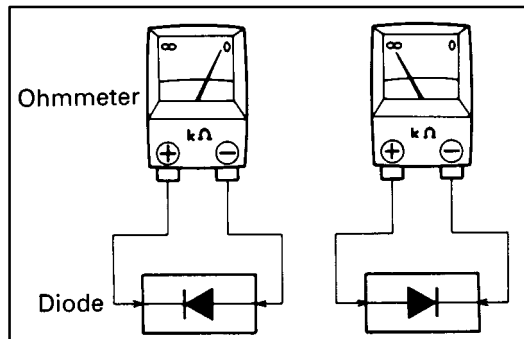
Example:

- Ⓐ – Ignition SW on
 - Ⓑ – Ignition SW and SW 1 on
 - Ⓒ – Ignition SW, SW 1 and Relay on (SW 2 off)
- (b) Using a voltmeter, connect the negative lead to a good ground point or negative battery terminal, and the positive lead to the connector or component terminal. This check can be done with a test a light instead of a voltmeter.



CONTINUITY AND RESISTANCE CHECK

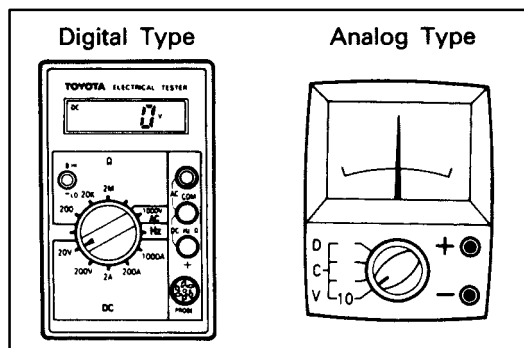
- (a) Disconnect the battery terminal or wire so there is no voltage between the check points.
- (b) Contact the two leads of an ohmmeter to each of the check points.



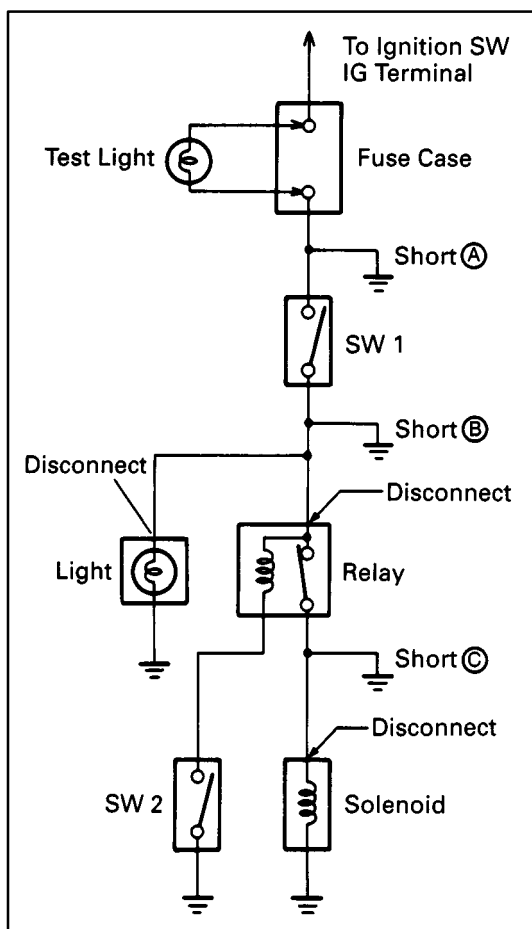
If the circuit has diodes, reverse the two leads and check again.

When contacting the negative lead to the diode positive side and the positive lead to the negative side, there should be continuity.

When contacting the two leads in reverse, there should be no continuity.



- (c) Use a volt/ohmmeter with high impedance (10 kΩ/V minimum) for troubleshooting of the electrical circuit.



FINDING A SHORT CIRCUIT

- Remove the blown fuse and disconnect all loads of the fuse.
- Connect a test light in place of the fuse.
- Establish conditions in which the test light comes on.

Example:

- Ignition SW on
 - Ignition SW and SW 1 on
 - Ignition SW, SW 1 and Relay on (Connect the Relay) and SW 2 off (or Disconnect SW 2)
- Disconnect and reconnect the connectors while watching the test light.
The short lies between the connector where the test light stays lit and the connector where the light goes out.
 - Find the exact location of the short by lightly shaking the problem wire along the body.

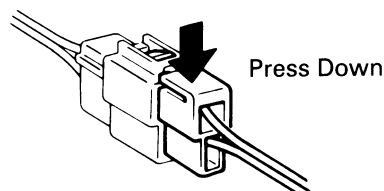
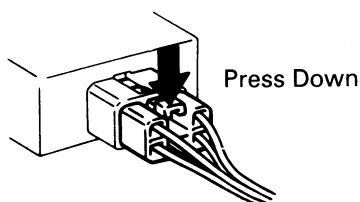
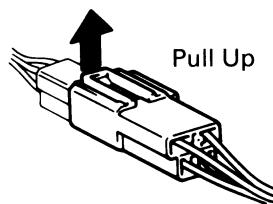
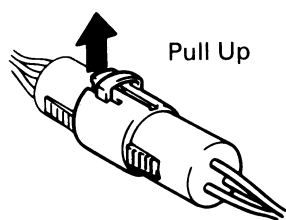
CAUTION:

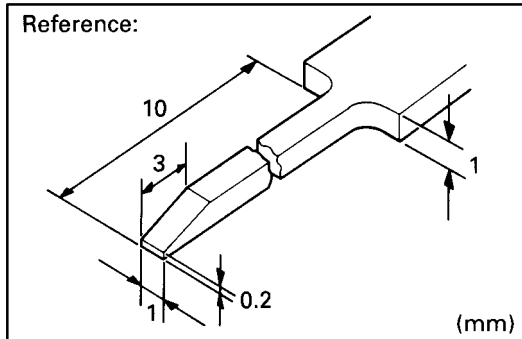
Do not open the cover or the case of the ECU unless absolutely necessary. (If the IC terminals are touched, the IC may be destroyed by static electricity.)

DISCONNECTION OF MALE AND FEMALE CONNECTORS

To pull apart the connectors, pull on the connector itself, not the wire harness.

HINT: Check to see what kind of connector you are disconnecting before pulling apart.





HOW TO REPLACE TERMINAL (with terminal retainer or secondary locking device)

1. PREPARE THE SPECIAL TOOL

HINT: To remove the terminal from the connector, please construct and use the special tool or like object shown on the left.

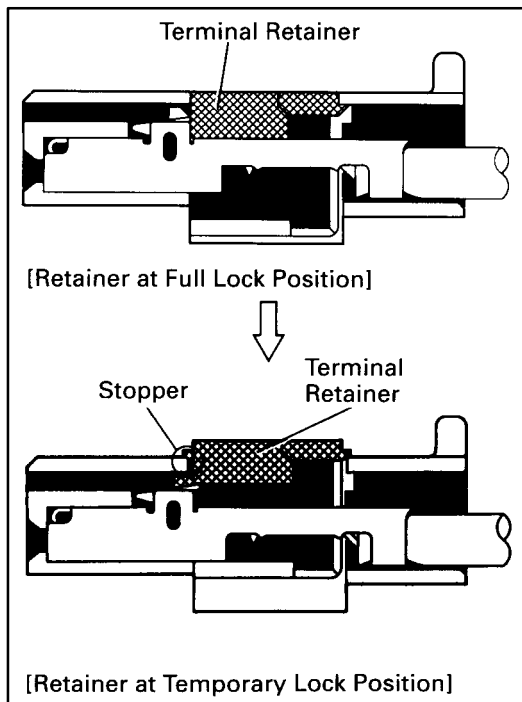
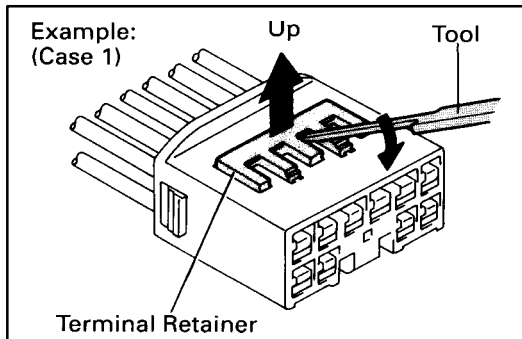
2. DISCONNECT CONNECTOR

3. DISENGAGE THE SECONDARY LOCKING DEVICE OR TERMINAL RETAINER.

- (a) Locking device must be disengaged before the terminal locking clip can be released and the terminal removed from the connector.
- (b) Use a special tool or the terminal pick to unlock the secondary locking device or terminal retainer.

NOTICE:

Do not remove the terminal retainer from connector body.

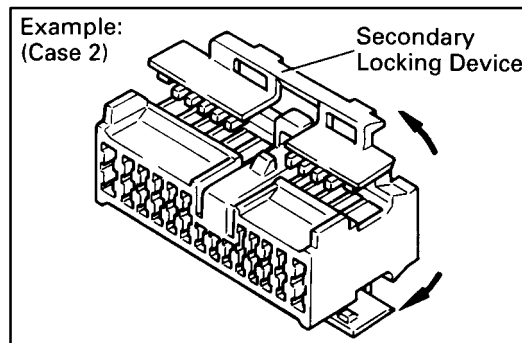


Ⓐ For Non-Waterproof Type Connector

HINT: The needle insertion position varies according to the connector's shape (number of terminals etc.), so check the position before inserting it.

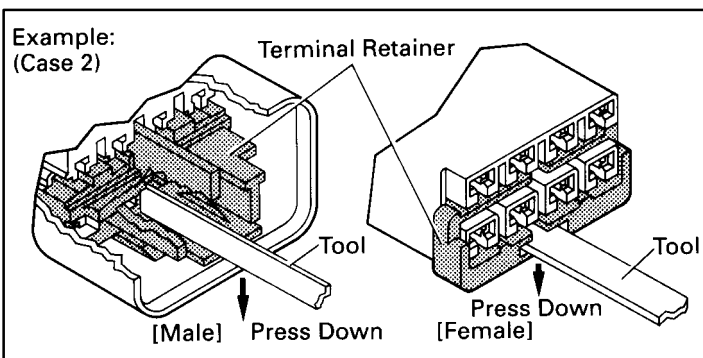
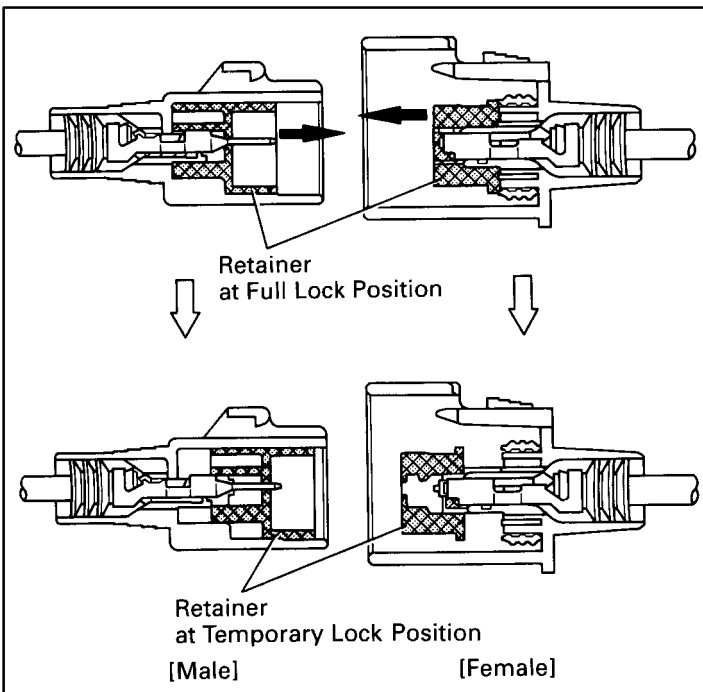
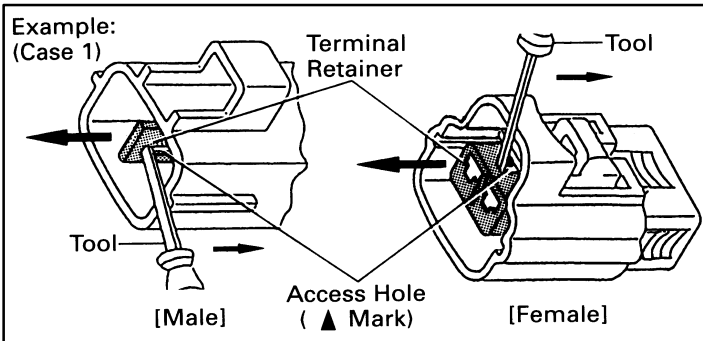
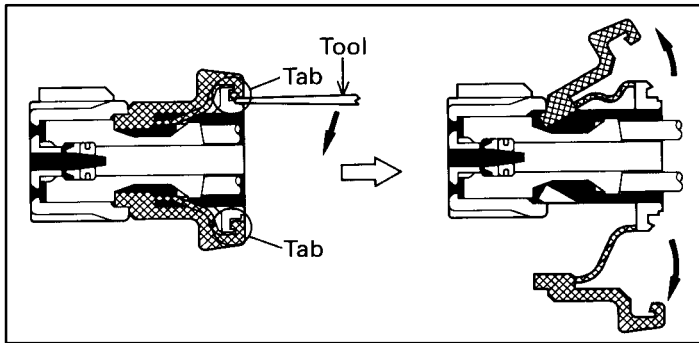
“Case 1”

Raise the terminal retainer up to the temporary lock position.



“Case 2”

Open the secondary locking device.



② For Waterproof Type Connector

HINT: Terminal retainer color is different according to connector body.

Example:

Terminal Retainer: Connector Body

Black or White : Gray

Black or White : Dark Gray

Gray or White : Black

“Case 1”

Type where terminal retainer is pulled up to the temporary lock position (Pull Type).

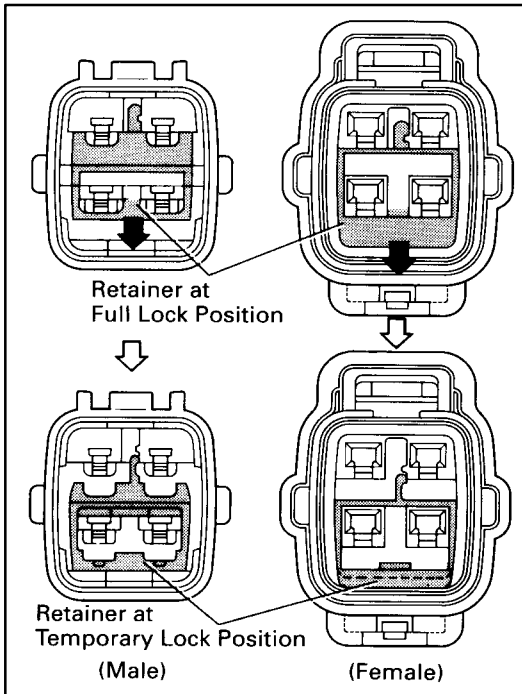
Insert the special tool into the terminal retainer access hole (▲ Mark) and pull the terminal retainer up to the temporary lock position.

HINT: The needle insertion position varies according to the connector's shape (Number of terminals, etc.), so check the position before inserting it.

“Case 2”

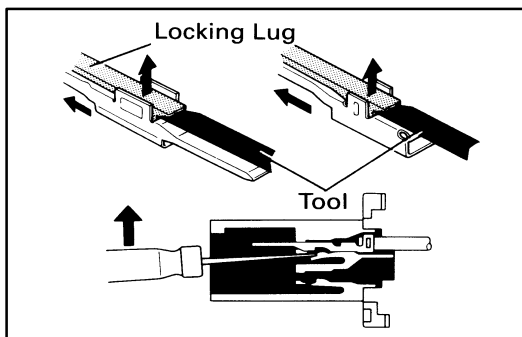
Type which cannot be pulled as far as Power Lock.

TROUBLESHOOTING

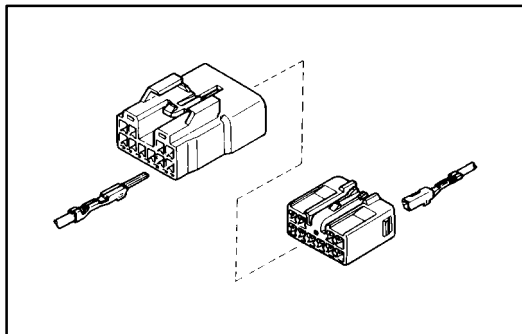


Insert the tool straight into the access hole of terminal retainer as shown.

Push the terminal retainer down to the temporary lock position.



- (c) Release the locking lug from terminal and pull the terminal out from rear.

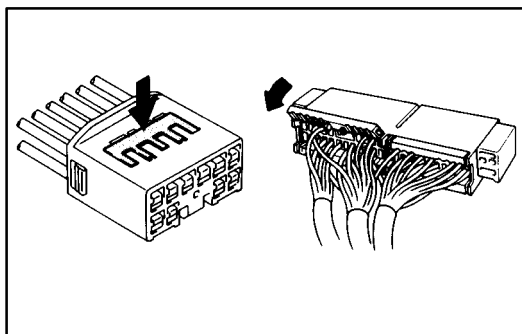


4. INSTALL TERMINAL TO CONNECTOR

- (a) Insert the terminal.

HINT:

1. Make sure the terminal is positioned correctly.
2. Insert the terminal until the locking lug locks firmly.
3. Insert the terminal with terminal retainer in the temporary lock position.



- (b) Push the secondary locking device or terminal retainer in to the full lock position.

5. CONNECT CONNECTOR

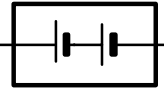
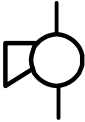
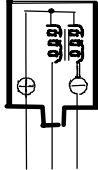
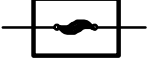
ABBREVIATIONS

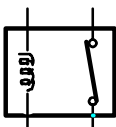
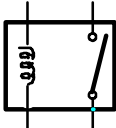
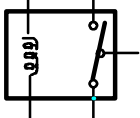
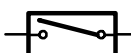
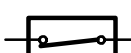
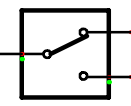
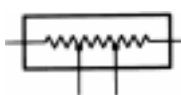
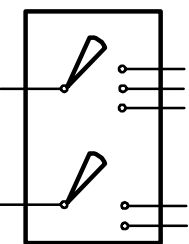
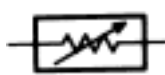
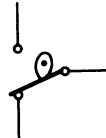
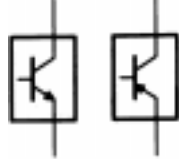
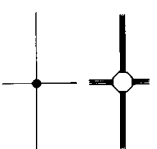
The following abbreviations are used in this manual.

A/C	= Air Conditioner
ABS	= Anti-Lock Brake System
A/T	= Automatic Transmission
C/P	= Coupe Type
COMB.	= Combination
ECT	= Electronic Controlled Transmission
ECU	= Electronic Control Unit
EFI	= Electronic Fuel Injection
EGR	= Exhaust Gas Recirculation
EX.	= Except
FL	= Fusible Link
I/A	= Integrated Ignition Assembly
ISC	= Idle Speed Control
J/B	= Junction Block
L/B	= Liftback Type
LH	= Left-Hand
M/T	= Manual Transmission
O/D	= Overdrive
R/B	= Relay Block
RH	= Right-Hand
SRS	= Supplemental Restraint System
SW	= Switch
TEMP.	= Temperature
VSV	= Vacuum Switching Valve
W/	= With
W/O	= Without
2WD	= Two Wheel Drive
4WD	= Four Wheel Drive

- * The titles given inside the components are the names of the terminals (terminal codes) and are not treated as being abbreviations.

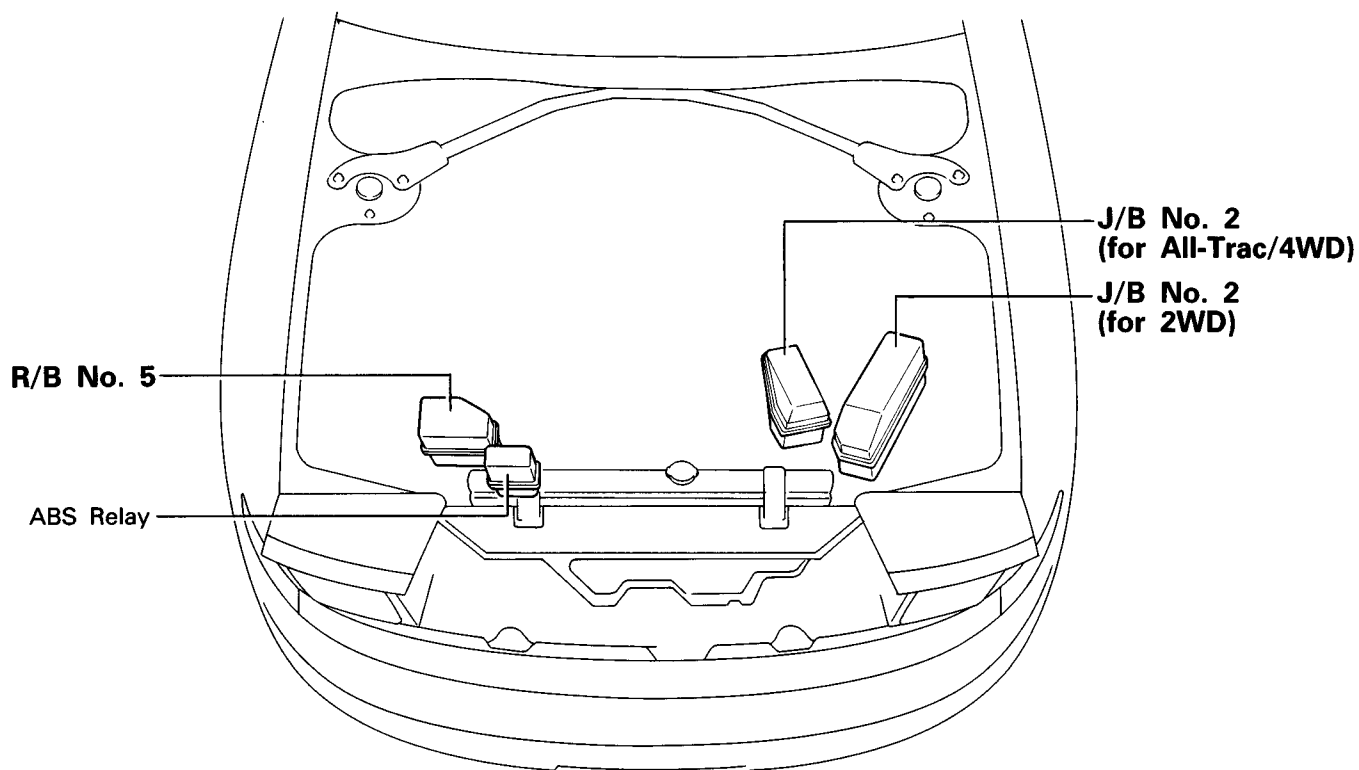
GLOSSARY OF TERMS AND SYMBOLS

 BATTERY Stores chemical energy and converts it into electrical energy. Provides DC current for the auto's various electrical circuits.	HEADLIGHTS Current flow causes a headlight filament to heat up and emit light. A headlight may have either a single (1) filament or a double (2) filament. 1. SINGLE FILAMENT  2. DOUBLE FILAMENT 
 CAPACITOR (Condenser) A small holding unit for temporary storage of electrical voltage.	
 CIGARETTE LIGHTER An electric resistance heating element.	HORN An electric device which sounds a loud audible signal. 
 CIRCUIT BREAKER Basically a reusable fuse, a circuit breaker will heat and open if too much current flows through it. Some units automatically reset when cool, others must be manually reset.	IGNITION COIL Converts low-voltage DC current into high-voltage ignition current for firing the spark plugs. 
 DIODE A semiconductor which allows current flow in only one direction.	
 DIODE, ZENER A diode which allows current flow in one direction but blocks reverse flow only up to a specific voltage. Above that potential, it passes the excess voltage. This acts as a simple voltage regulator.	LIGHT Current flow through a filament causes the filament to heat up and emit light. 
 DISTRIBUTOR, IIA Channels high-voltage current from the ignition coil to the individual spark plugs.	LED (LIGHT EMITTING DIODE) Upon current flow, these diodes emit light without producing the heat of a comparable light. 
 FUSE A thin metal strip which burns through when too much current flows through it, thereby stopping current flow and protecting a circuit from damage.  FUSIBLE LINK A heavy-gauge wire placed in high amperage circuits which burns through on overloads, thereby protecting the circuit. The numbers indicate the cross-section surface area of the wires.  (for Medium Current Fuse) (for High Current Fuse or Fusible Link.)	METER, ANALOG Current flow activates a magnetic coil which causes a needle to move, thereby providing a relative display against a background calibration. 
	METER, DIGITAL Current flow activates one or many LED's, LCD's, or fluorescent displays, which provide a relative or digital display. 
 GROUND The point at which wiring attaches to the Body, thereby providing a return path for an electrical circuit; without a ground, current cannot flow.	MOTOR A power unit which converts electrical energy into mechanical energy, especially rotary motion. 

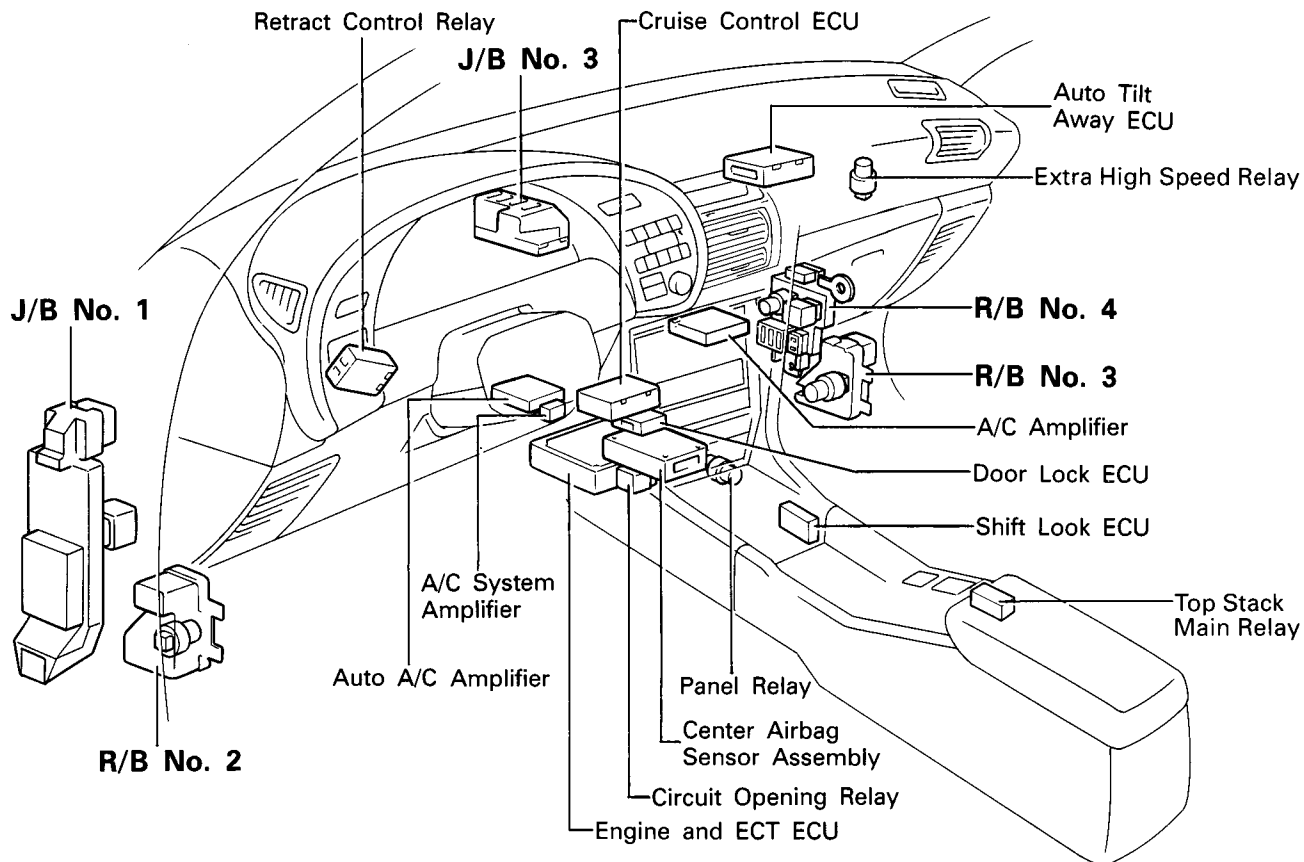
RELAY  1. NORMALLY CLOSED  2. NORMALLY OPEN Basically, an electrically operated switch which may be normally closed (1) or open (2). Current flow through a small coil creates a magnetic field which either opens or closes an attached switch.	SPEAKER  An electromechanical device which creates sound waves from current flow.
RELAY, DOUBLE THROW  A relay which passes current through one set of contacts or the other.	SWITCH, MANUAL  1. NORMALLY OPEN  2. NORMALLY CLOSED Opens and closes circuits, thereby stopping (1) or allowing (2) current flow.
RESISTOR  An electrical component with a fixed resistance, placed in a circuit to reduce voltage to a specific value.	SWITCH, DOUBLE THROW  A switch which continuously passes current through one set of contacts or the other.
RESISTOR, TAPPED  A resistor which supplies two or more different non adjustable resistance values.	SWITCH, IGNITION  A key operated switch with several positions which allows various circuits, particularly the primary ignition circuit, to become operational.
RESISTOR, VARIABLE or RHEOSTAT  A controllable resistor with a variable rate of resistance. Also called a potentiometer or rheostat.	SWITCH, WIPER PARK  Automatically returns wipers to the stop position when the wiper switch is turned off.
SENSOR (Thermistor)  A resistor which varies its resistance with temperature.	TRANSISTOR  A solidstate device typically used as an electronic relay; stops or passes current depending on the voltage applied at "base."
SENSOR, ANALOG SPEED  Uses magnetic impulses to open and close a switch to create a signal for activation of other components.	WIRES (1) NOT CONNECTED  (2) SPliced  Wires are always drawn as straight lines on wiring diagrams. Crossed wires (1) without a black dot at the junction are not joined; crossed wires (2) with a black dot or octagonal (O) mark at the junction are spliced (joined) connections.
SHORT PIN  Used to provide an unbroken connection within a junction block.	
SOLENOID  An electromagnetic coil which forms a magnetic field when current flows, to move a plunger, etc.	

RELAY LOCATIONS

[Engine Compartment]

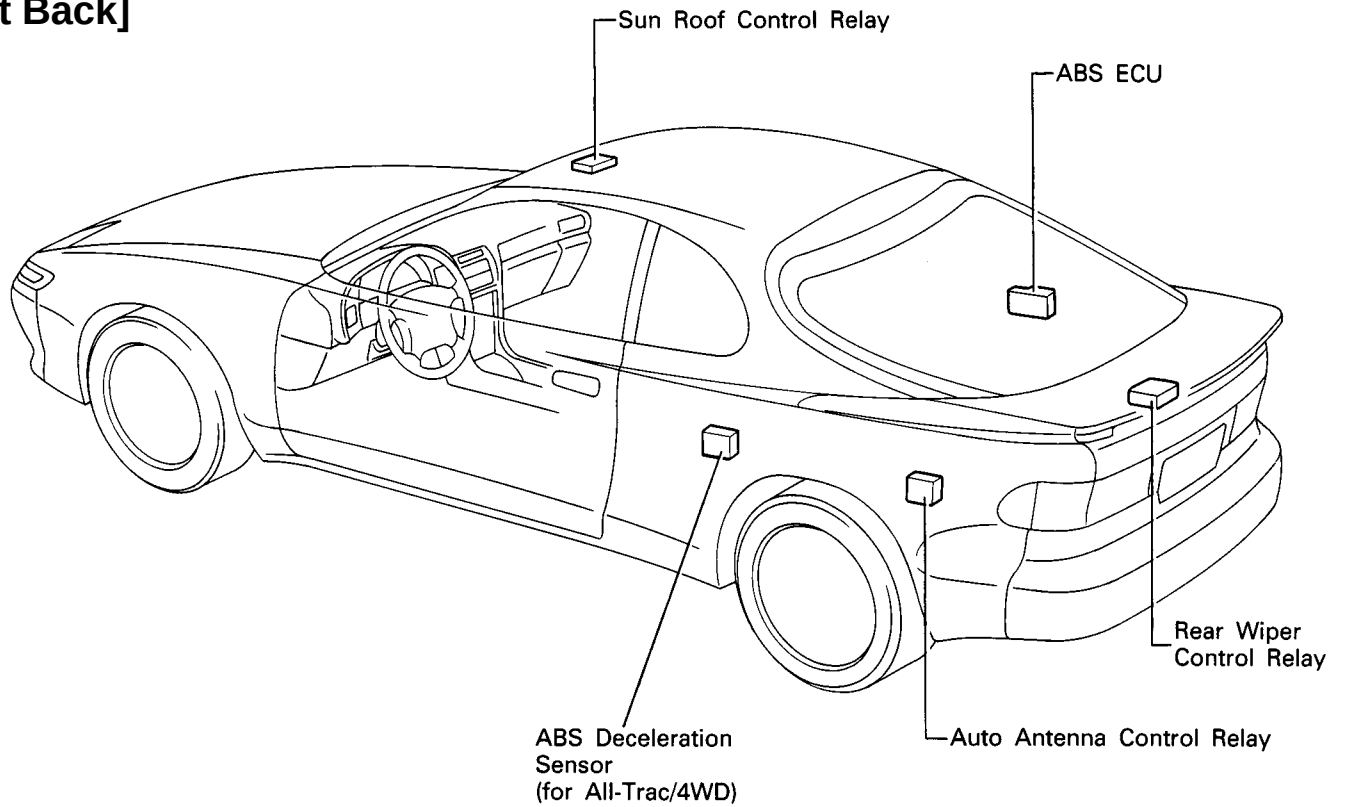


[Instrument Panel]



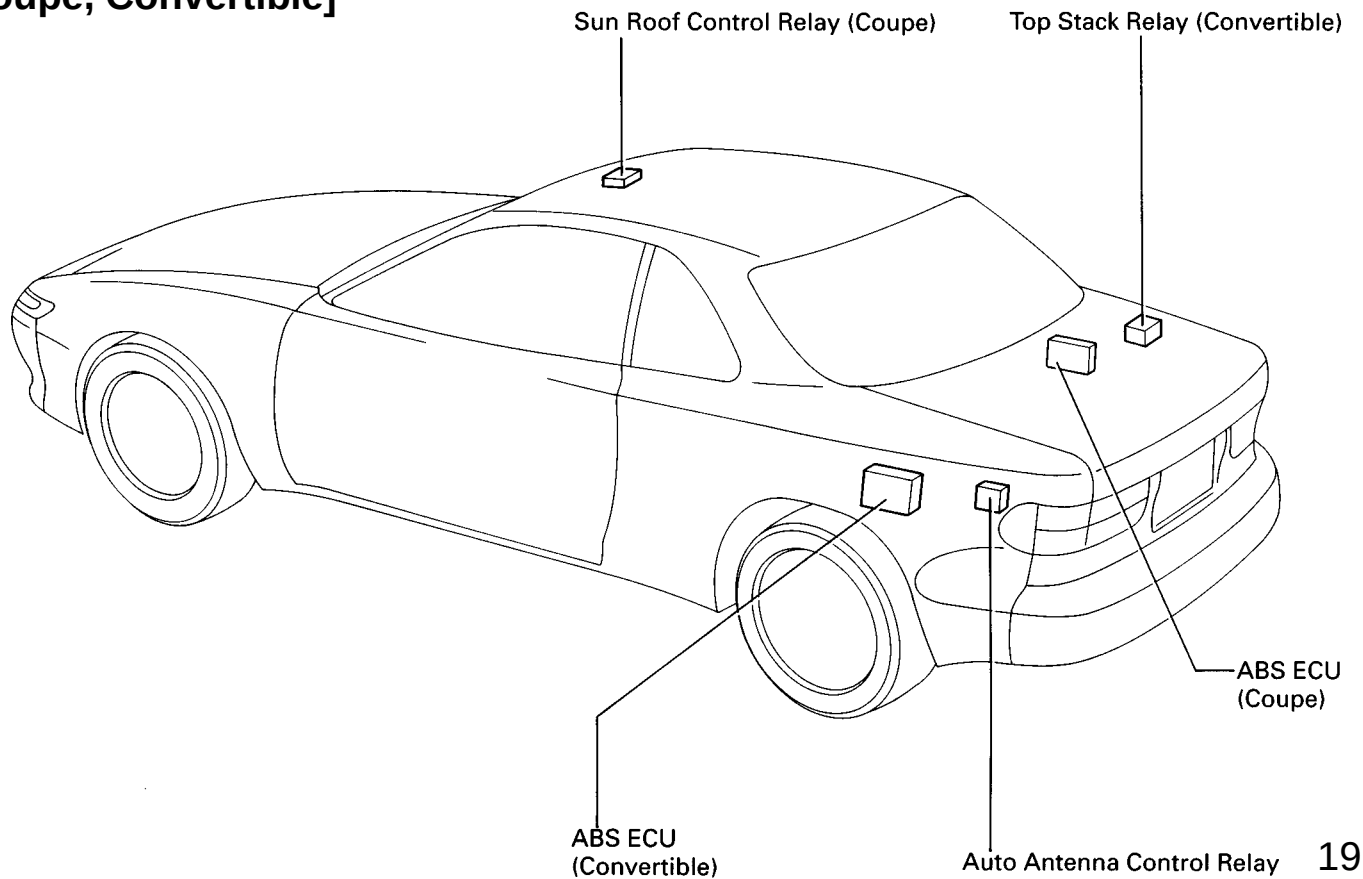
[Body]

[Lift Back]



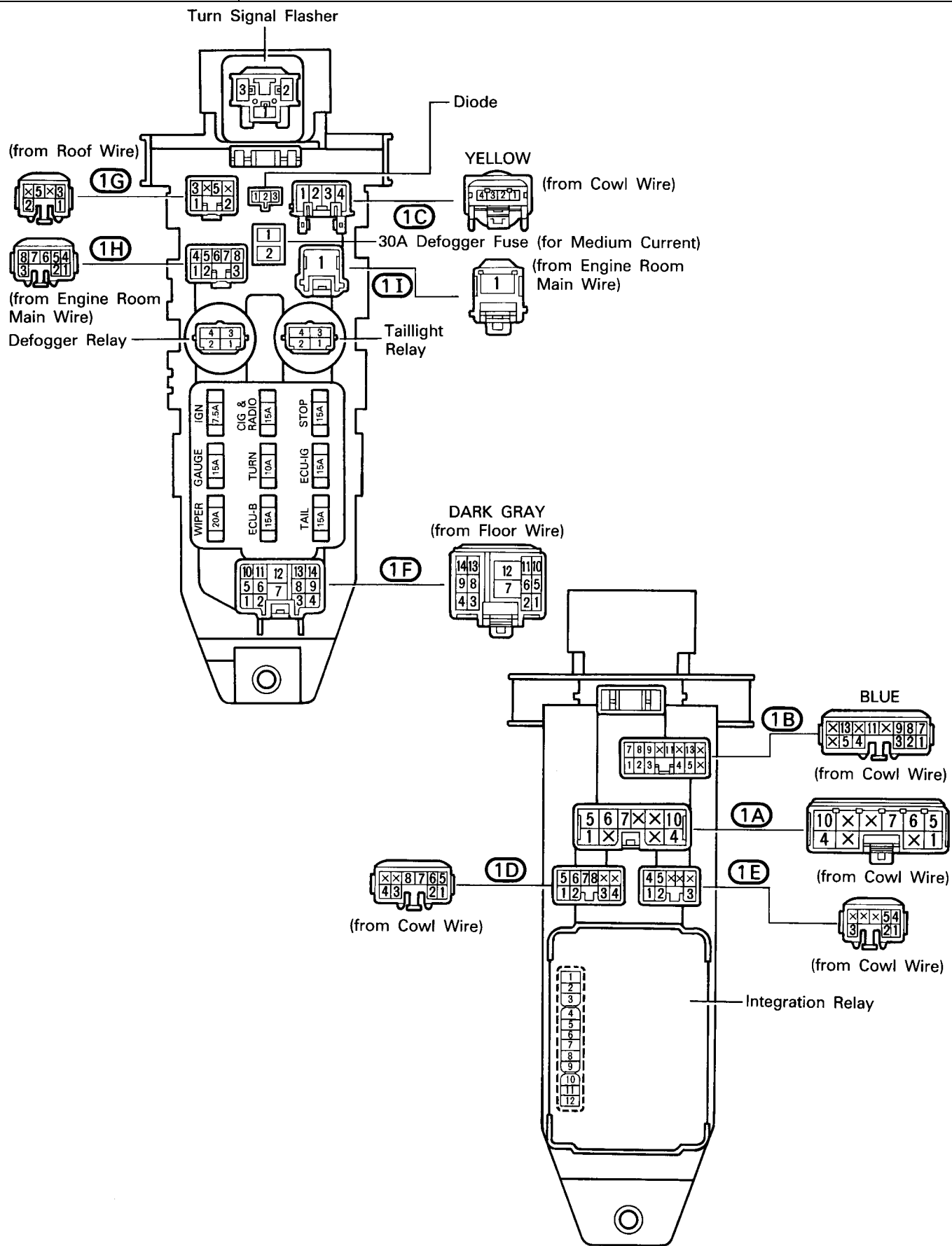
[Body]

[Coupe, Convertible]

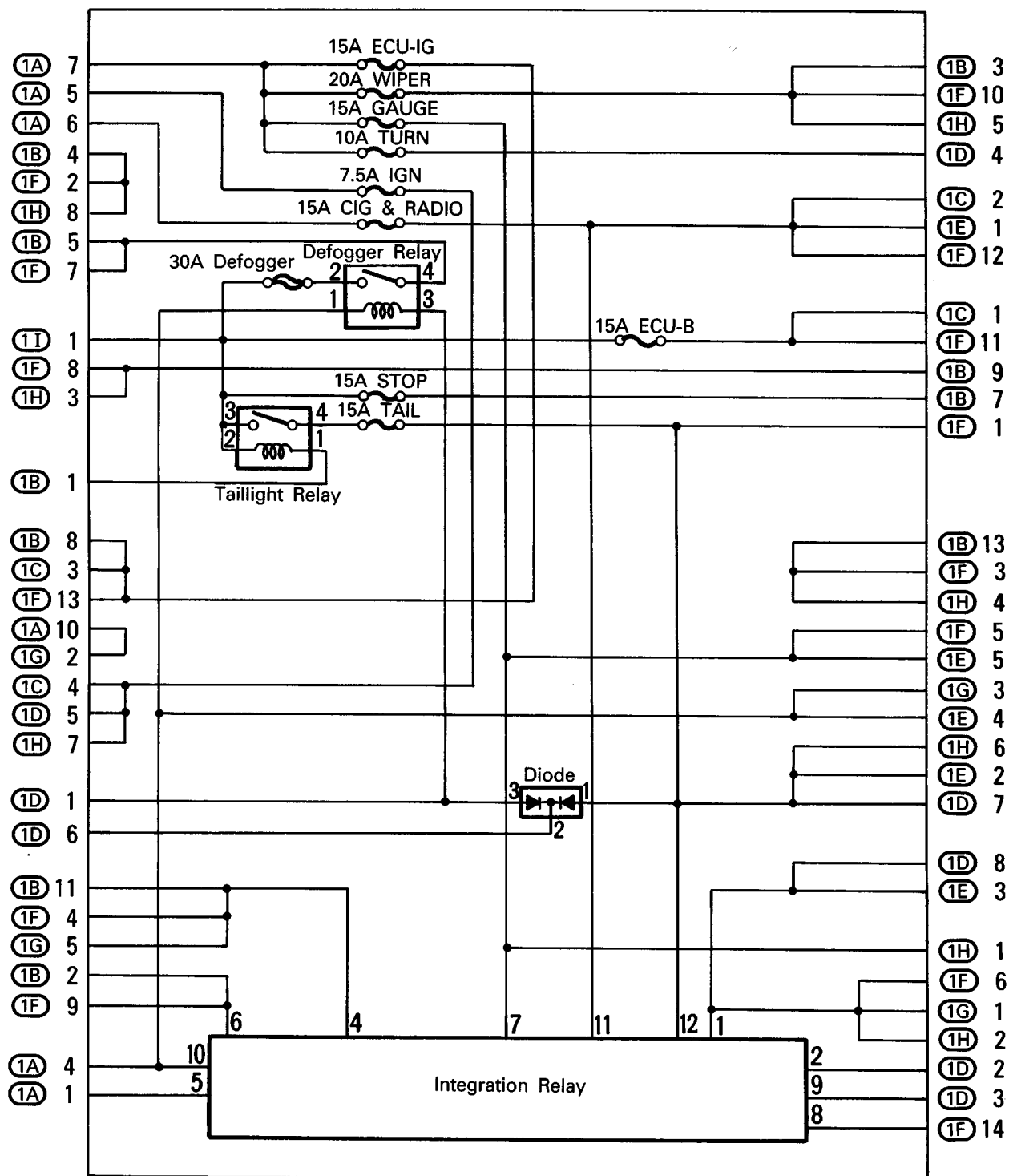


RELAY LOCATIONS

○ : J/B No. 1 Left Kick Panel (See Page 18)



[J/B No. 1 Inner Circuit]



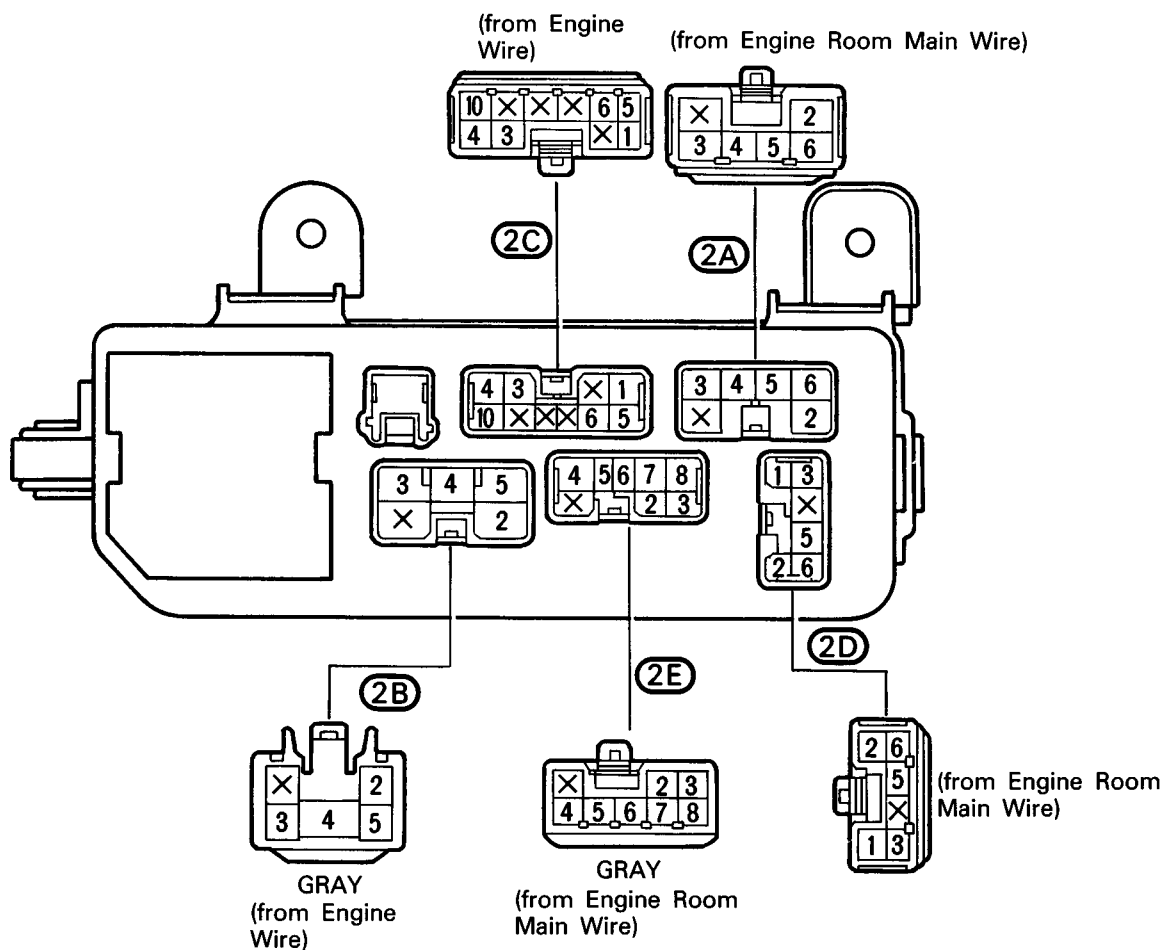
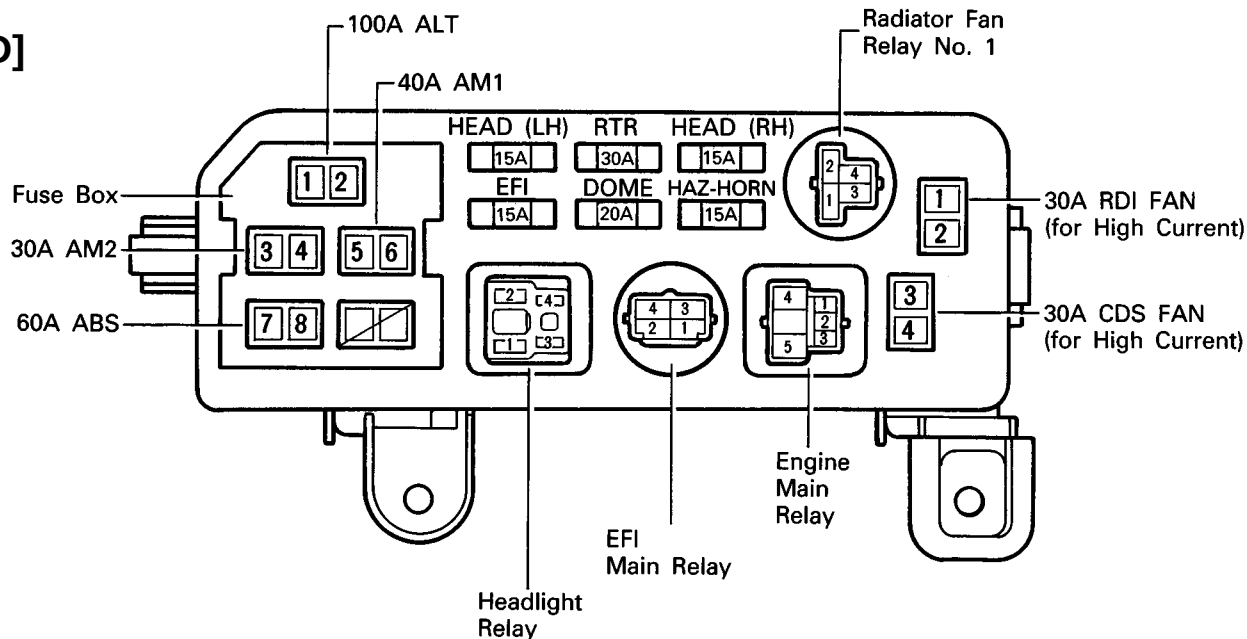
RELAY LOCATIONS

○ : J/B No. 2

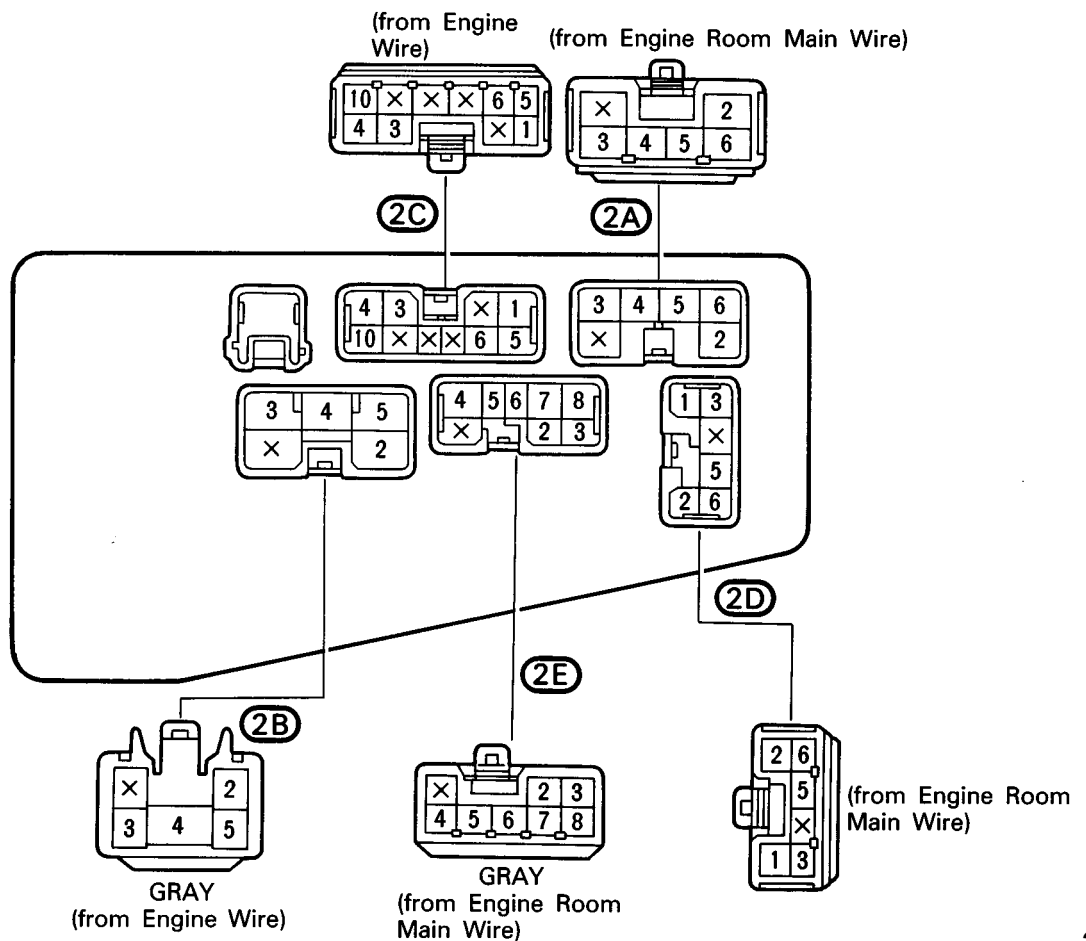
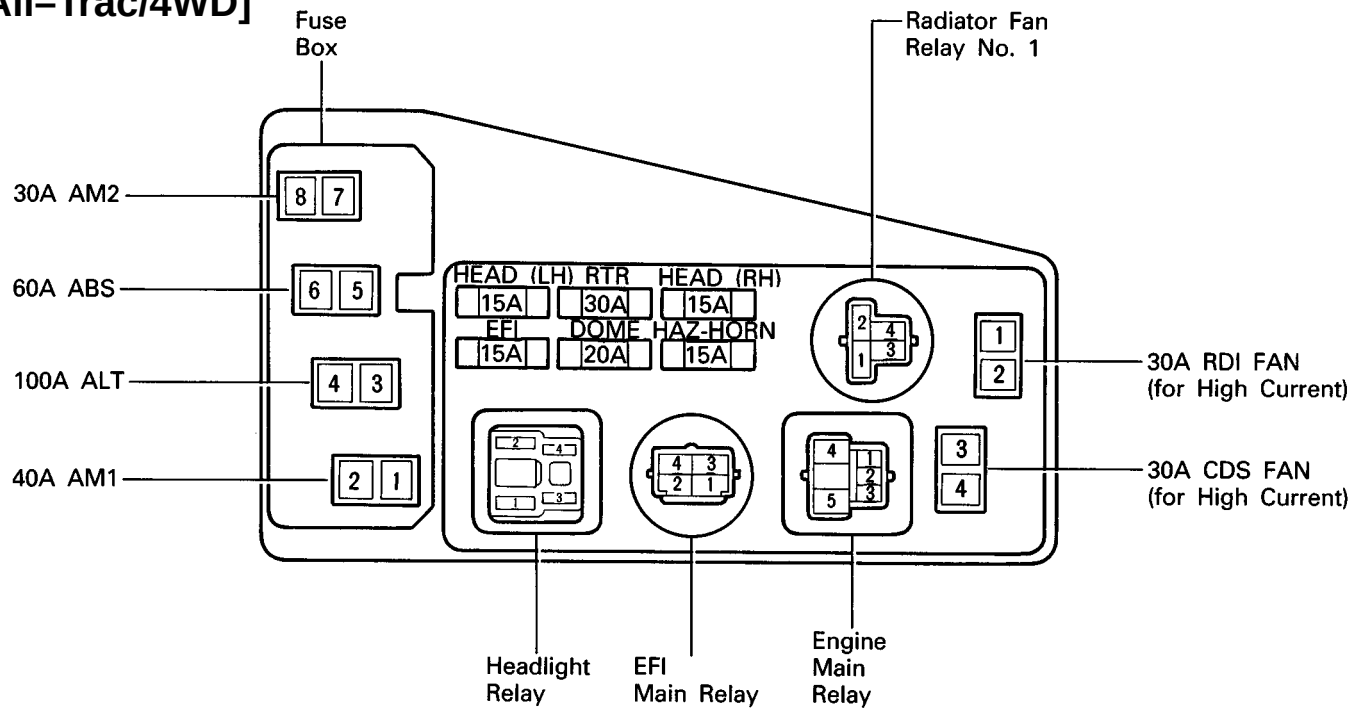
Near the Battery

(See Page 18)

[2WD]

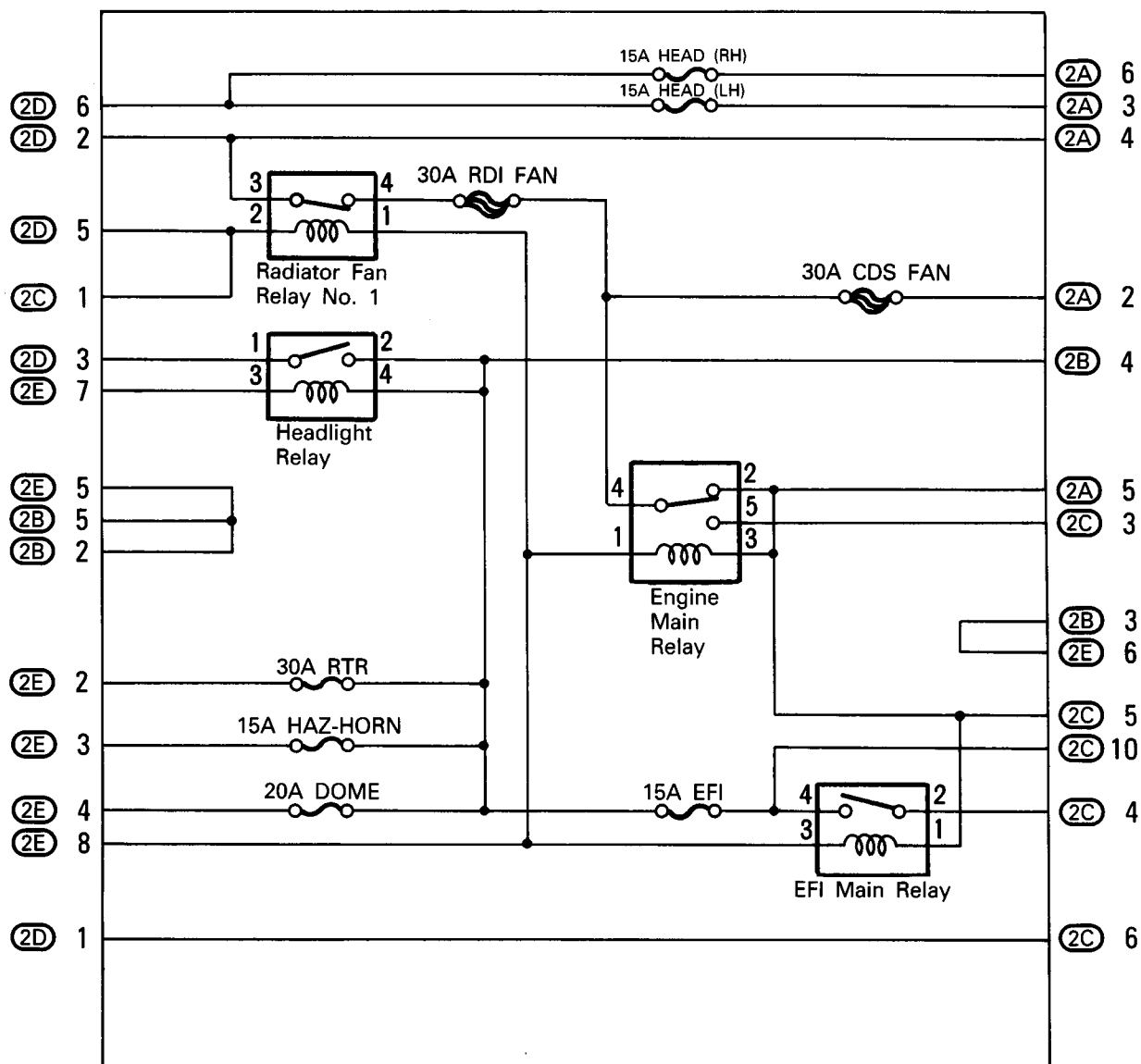


[All-Trac/4WD]



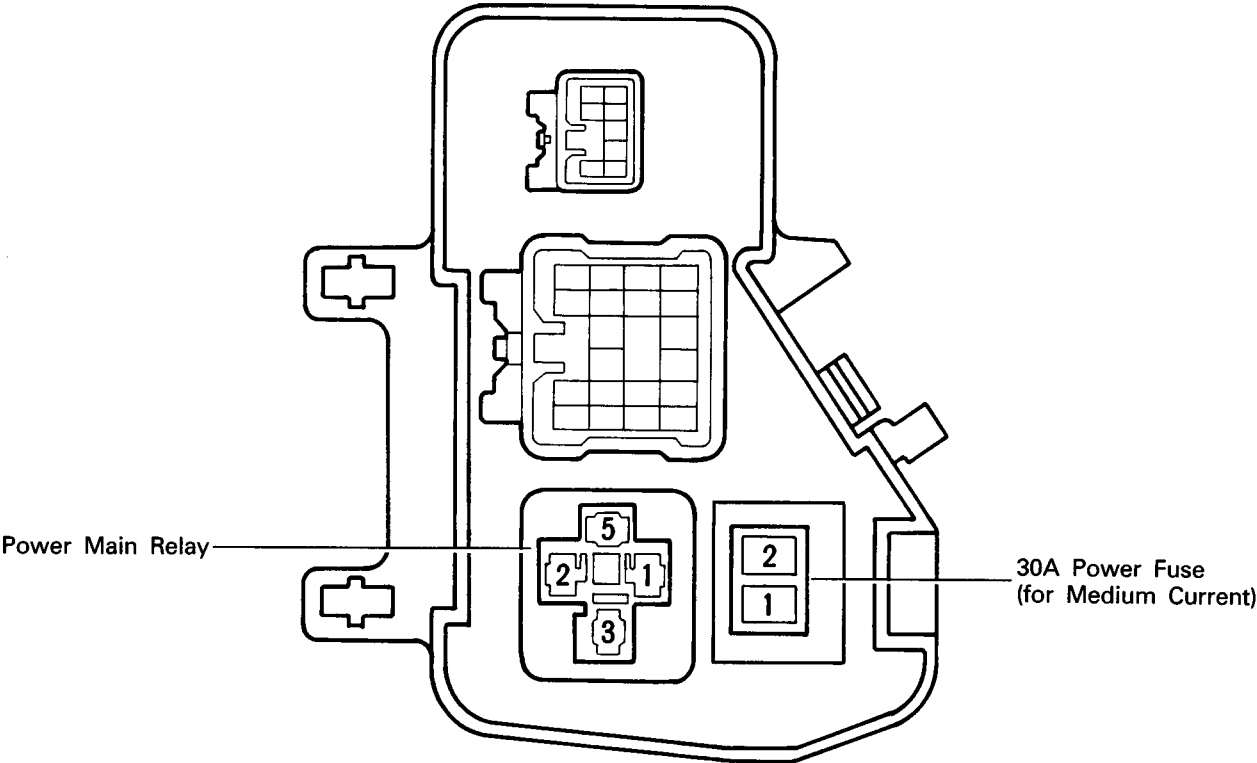
RELAY LOCATIONS

[J/B No. 2 Inner Circuit]

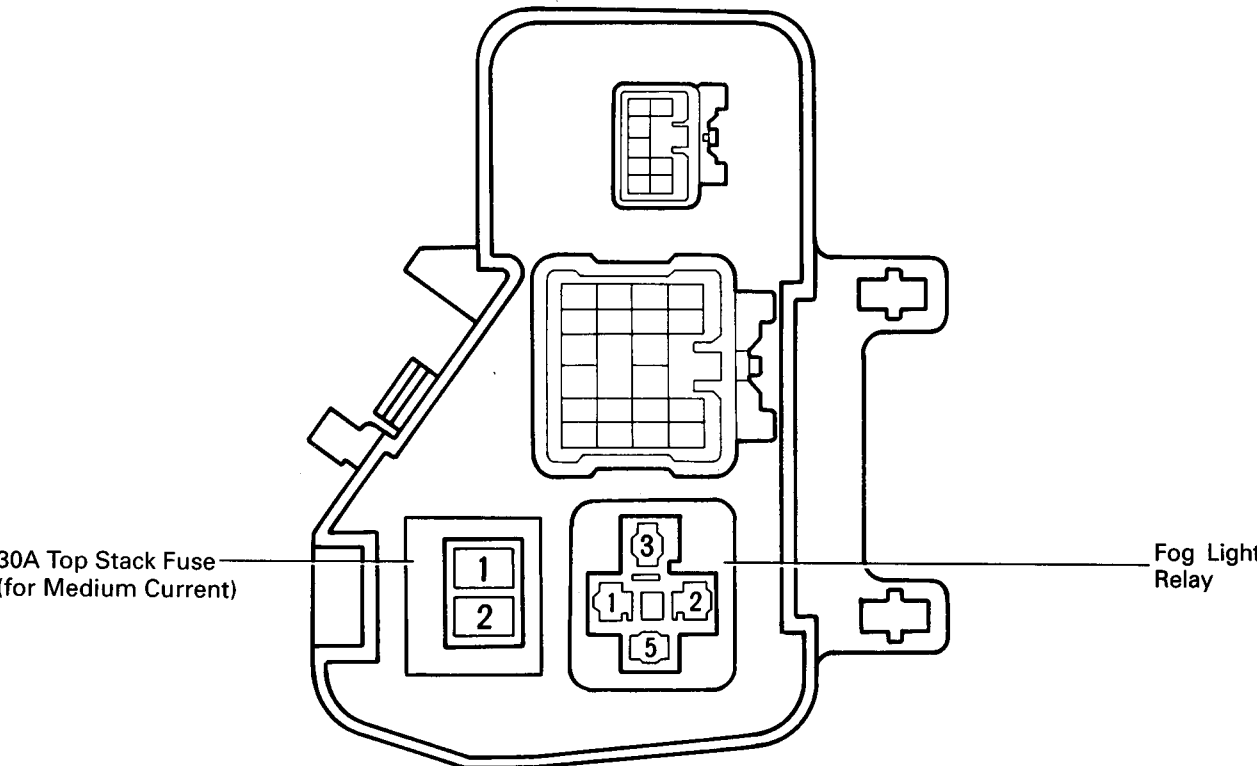


RELAY LOCATIONS

② : R/B No. 2	Left Kick Panel	(See Page 18)
---------------	-----------------	---------------

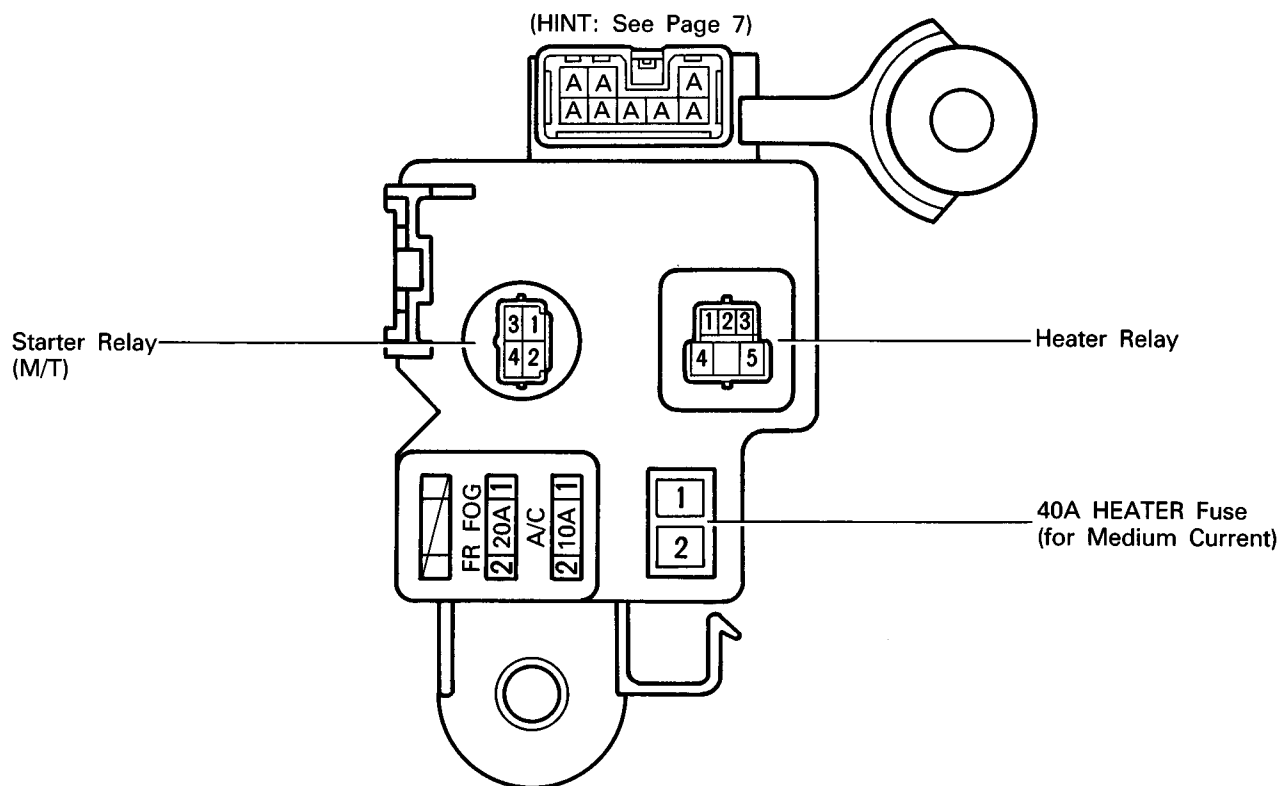


③ : R/B No. 3	Right Kick Panel	(See Page 18)
---------------	------------------	---------------



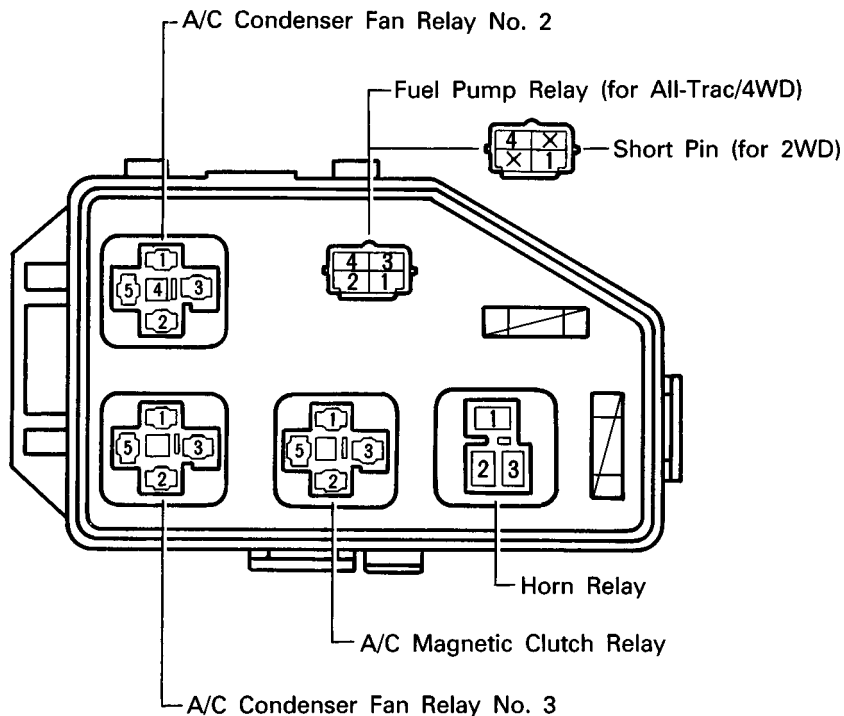
④ : R/B No. 4

Right Kick Panel (See Page 18)



⑤ : R/B No. 5

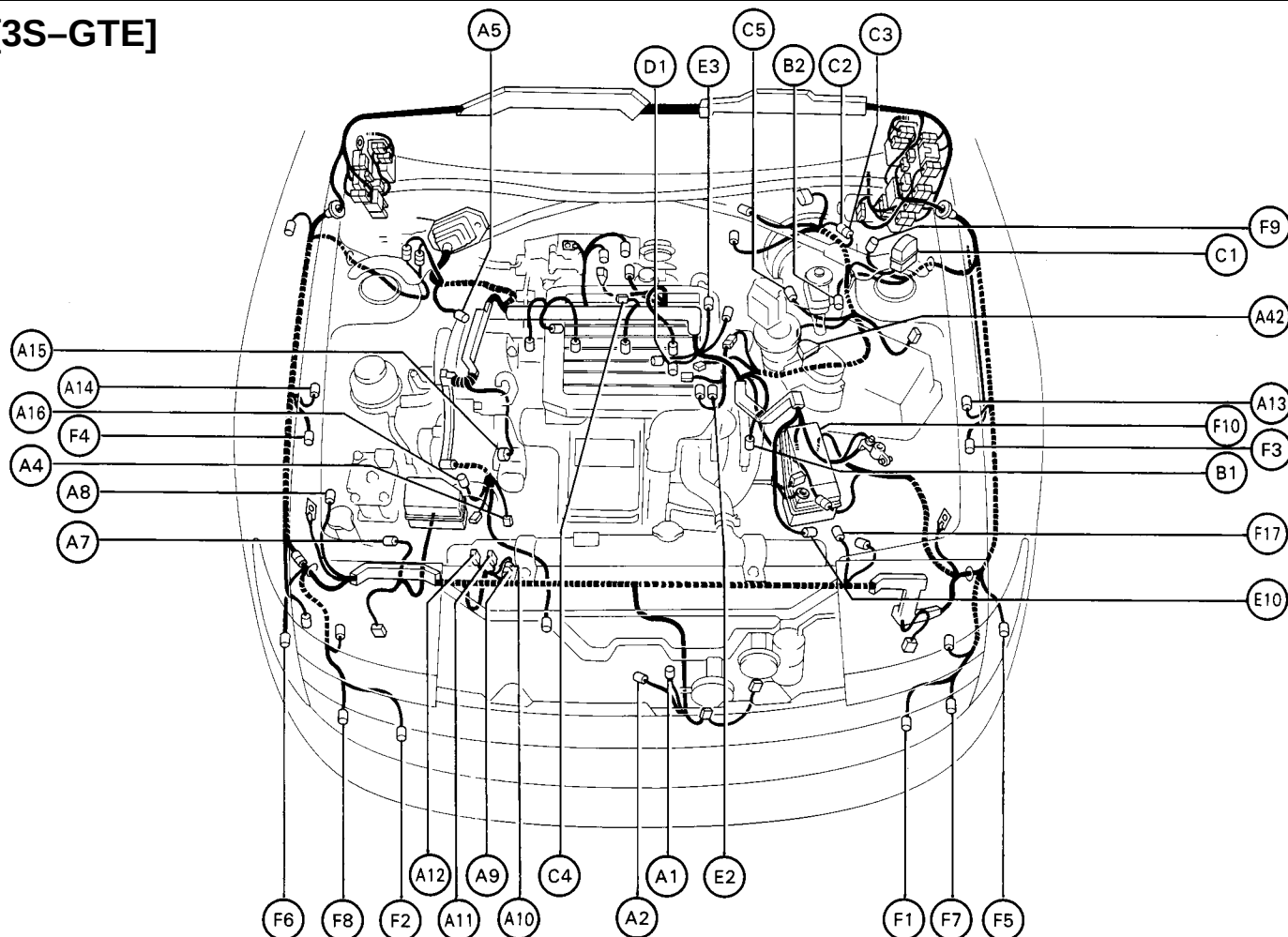
Engine Compartment Front Right (See Page 18)



ELECTRICAL WIRING ROUTING

Position of Parts in Engine Compartment

[3S-GTE]



- A 1 A/C Ambient Temp. Sensor
- A 2 A/C Condenser Fan Motor
- A 4 A/C Magnetic Clutch and Compressor Sensor
- A 5 A/C Pressure SW or Short Pin
- A 7 ABS Actuator
- A 8 ABS Actuator
- A 9 ABS Check Connector
- A 10 ABS Check Connector
- A 11 ABS Relay
- A 12 ABS Relay
- A 13 ABS Speed Sensor Front LH
- A 14 ABS Speed Sensor Front RH
- A 15 Alternator
- A 16 Alternator
- A 42 Air Flow Meter

- B 1 Back-Up Light SW (M/T)
- B 2 Brake Fluid Level SW

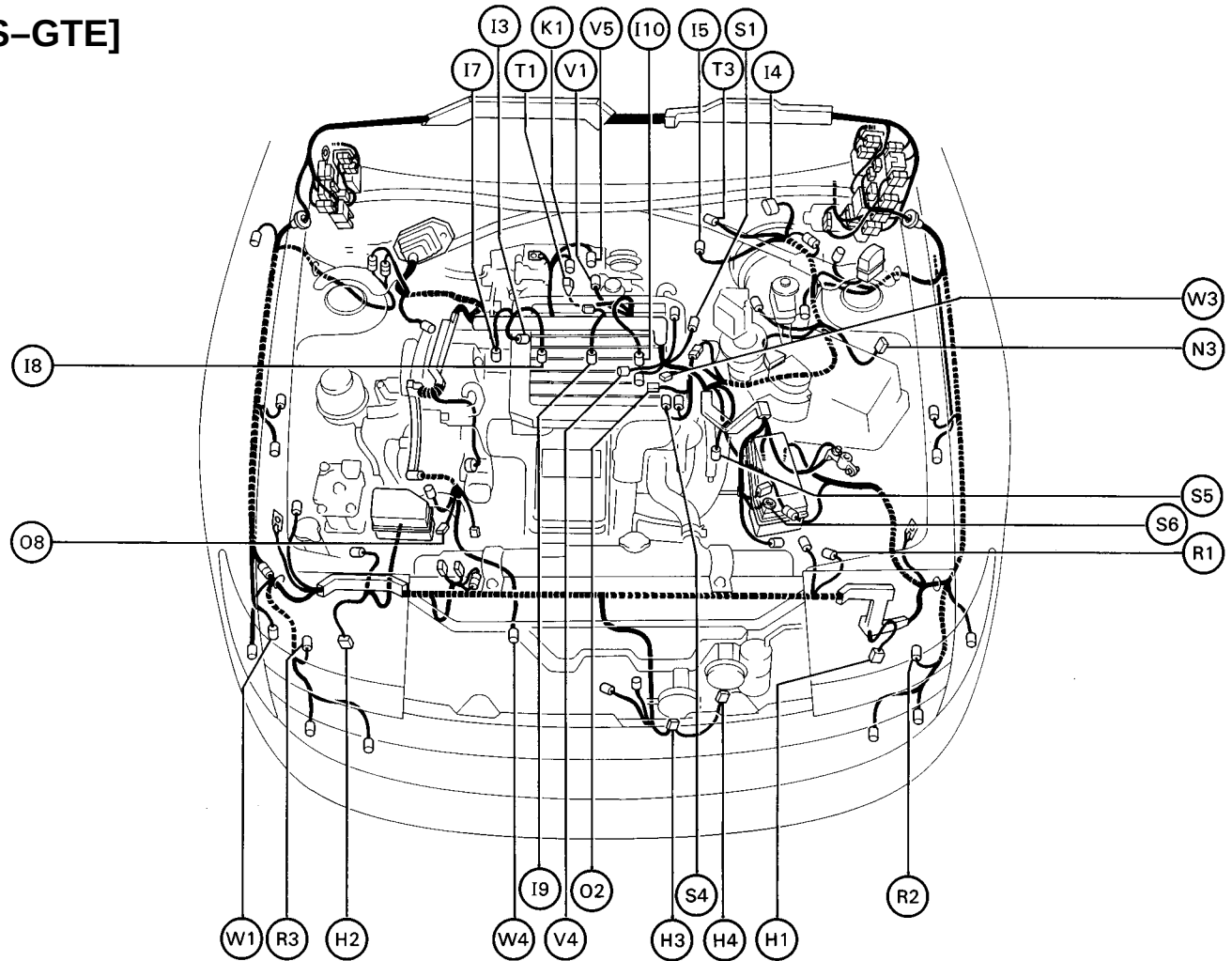
- C 1 Check Connector
- C 2 Check Connector (for Fan Check)
- C 3 Check Connector (for Fan Check)
- C 4 Cold Start Injector
- C 5 Cruise Control Actuator

- D 1 Distributor
- E 2 EFI Water Temp. Sensor
- E 3 EGR Gas Temp. Sensor (for California) or Short Pin (for Ex. California)
- E 10 EFI Resistor

- F 1 Fog Light LH
- F 2 Fog Light RH
- F 3 Front Airbag Sensor LH
- F 4 Front Airbag Sensor RH
- F 5 Front Side Marker Light LH
- F 6 Front Side Marker Light RH
- F 7 Front Turn Signal Light and Clearance Light LH
- F 8 Front Turn Signal Light and Clearance Light RH
- F 9 Front Wiper Motor
- F 10 Fuse Box (All-Trac/4WD)
- F 17 Fuel Pump Resistor

Position of Parts in Engine Compartment

[3S-GTE]



H 1 Headlight LH
H 2 Headlight RH
H 3 Horn LH
H 4 Horn RH

i 3 ISC Valve
I 4 Igniter
I 5 Ignition Coil
I 7 Injector No. 1
I 8 Injector No. 2
I 9 Injector No. 3
I 10 Injector No. 4

K 1 Knock Sensor

N 3 Noise Filter (for Ignition System)

O 2 Oil Pressure SW
O 8 Oxygen Sensor

R 1 Radiator Fan Motor
R 2 Retract Motor LH
R 3 Retract Motor RH

S 1 Speed Sensor (for Cruise Control System)
S 4 Start Injector Time SW
S 5 Starter
S 6 Starter

T 1 Throttle Position Sensor
T 3 Turbo Charging Pressure Sensor

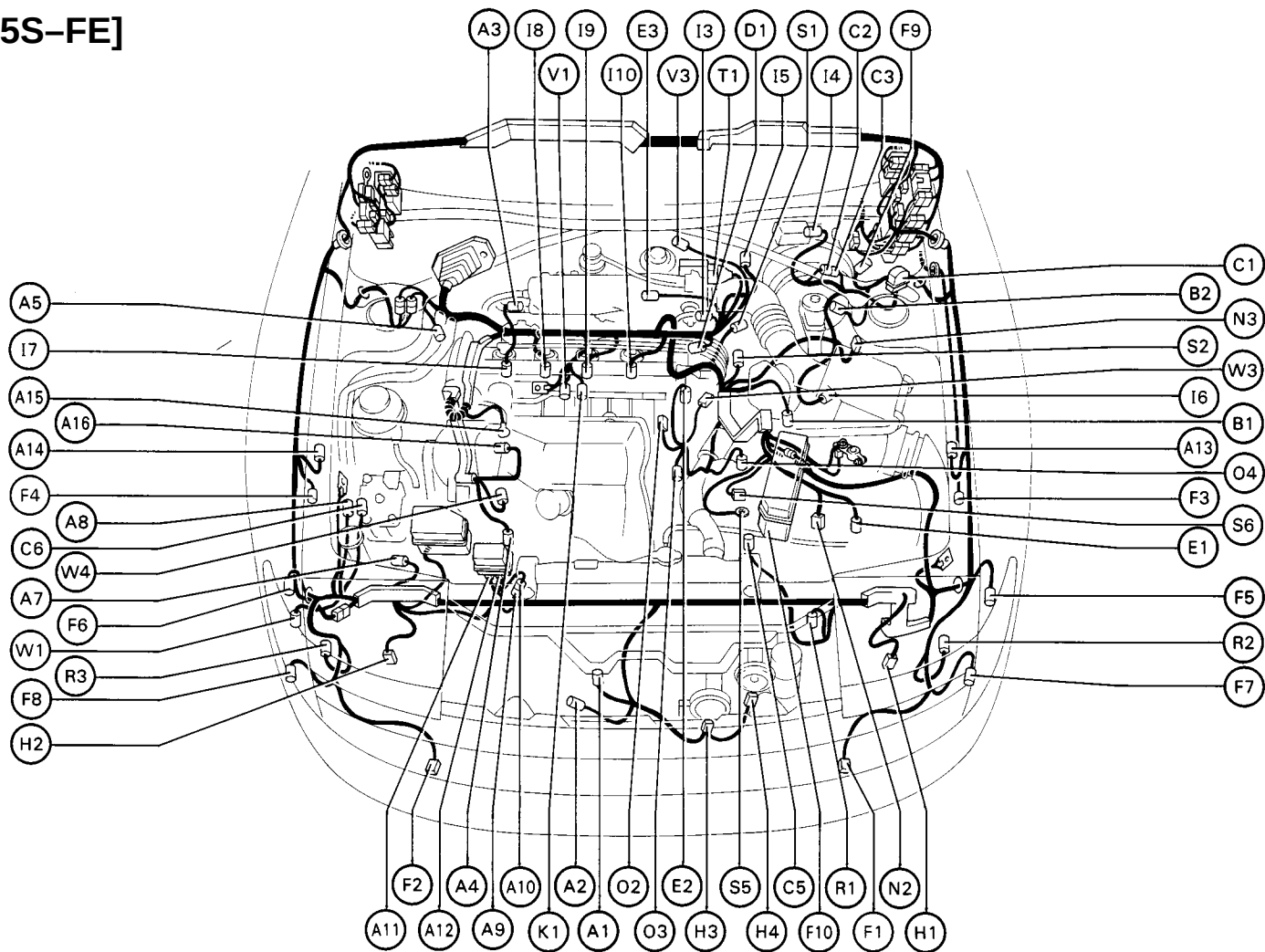
V 1 VSV (for EGR System)
V 4 VSV (for Turbo Charging Pressure)
V 5 VSV (for T-VIS)

W 1 Washer Motor
W 3 Water Temp. Sender
W 4 Water Temp. SW (for Fans Control)

ELECTRICAL WIRING ROUTING

Position of Parts in Engine Compartment

[5S-FE]



- A 1 A/C Ambient Temp. Sensor
- A 2 A/C Condenser Fan Motor
- A 3 A/C Idle-Up VSV
- A 4 A/C Magnetic Clutch (4A-FE) or A/C Magnetic Clutch Clutch and Compressor Sensor (5S-FE)
- A 5 A/C Pressure SW or Short Pin
- A 6 A/C Water Temp. SW (5S-FE)
- A 7 ABS Actuator (5S-FE)
- A 8 ABS Actuator (5S-FE)
- A 9 ABS Check Connector (5S-FE)
- A 10 ABS Check Connector (5S-FE)
- A 11 ABS Relay (5S-FE)
- A 12 ABS Relay (5S-FE)
- A 13 ABS Speed Sensor Front LH (5S-FE)
- A 14 ABS Speed Sensor Front RH (5S-FE)
- A 15 Alternator
- A 16 Alternator

- B 1 Back-Up Light SW (M/T)
- B 2 Brake Fluid Level SW

- C 1 Check Connector
- C 2 Check Connector (for Fan Check)
- C 3 Check Connector (for Fan Check)
- C 5 Cruise Control Actuator (w/ ABS of 5S-FE)
- C 6 Cruise Control Actuator (w/o ABS)

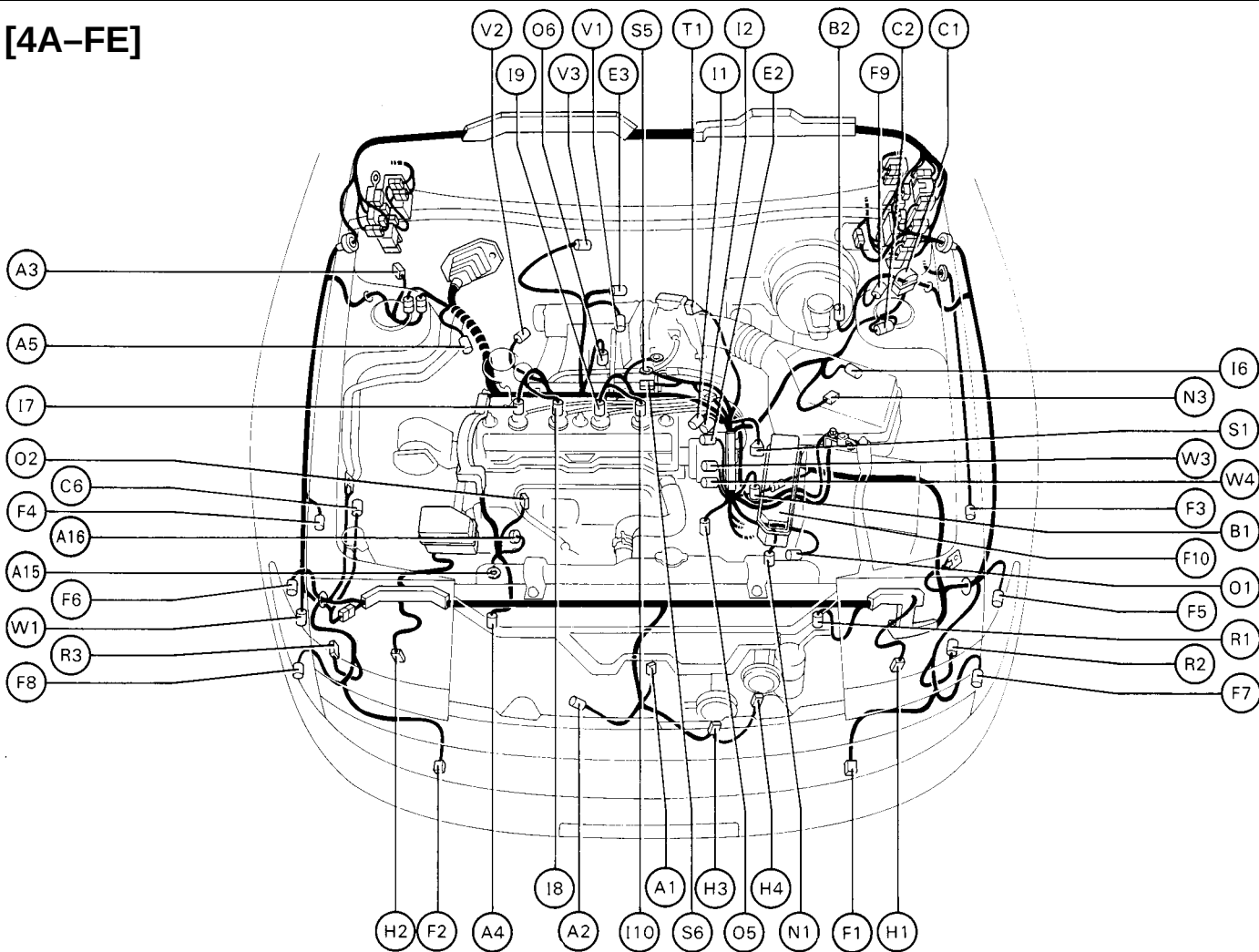
- D 1 Distributor (5S-FE)
- E 1 ECT Solenoid (5S-FE)
- E 2 EFI Water Temp. Sensor
- E 3 EGR Gas Temp. Sensor (for California)

- F 1 Fog Light LH
- F 2 Fog Light RH
- F 3 Front Airbag Sensor LH
- F 4 Front Airbag Sensor RH
- F 5 Front Side Marker Light LH
- F 6 Front Side Marker Light RH
- F 7 Front Turn Signal Light and Clearance Light LH
- F 8 Front Turn Signal Light and Clearance Light RH
- F 9 Front Wiper Motor
- F 10 Fuse Box (2WD)

- H 1 Headlight LH
- H 2 Headlight RH
- H 3 Horn LH
- H 4 Horn RH

Position of Parts in Engine Compartment	
---	--

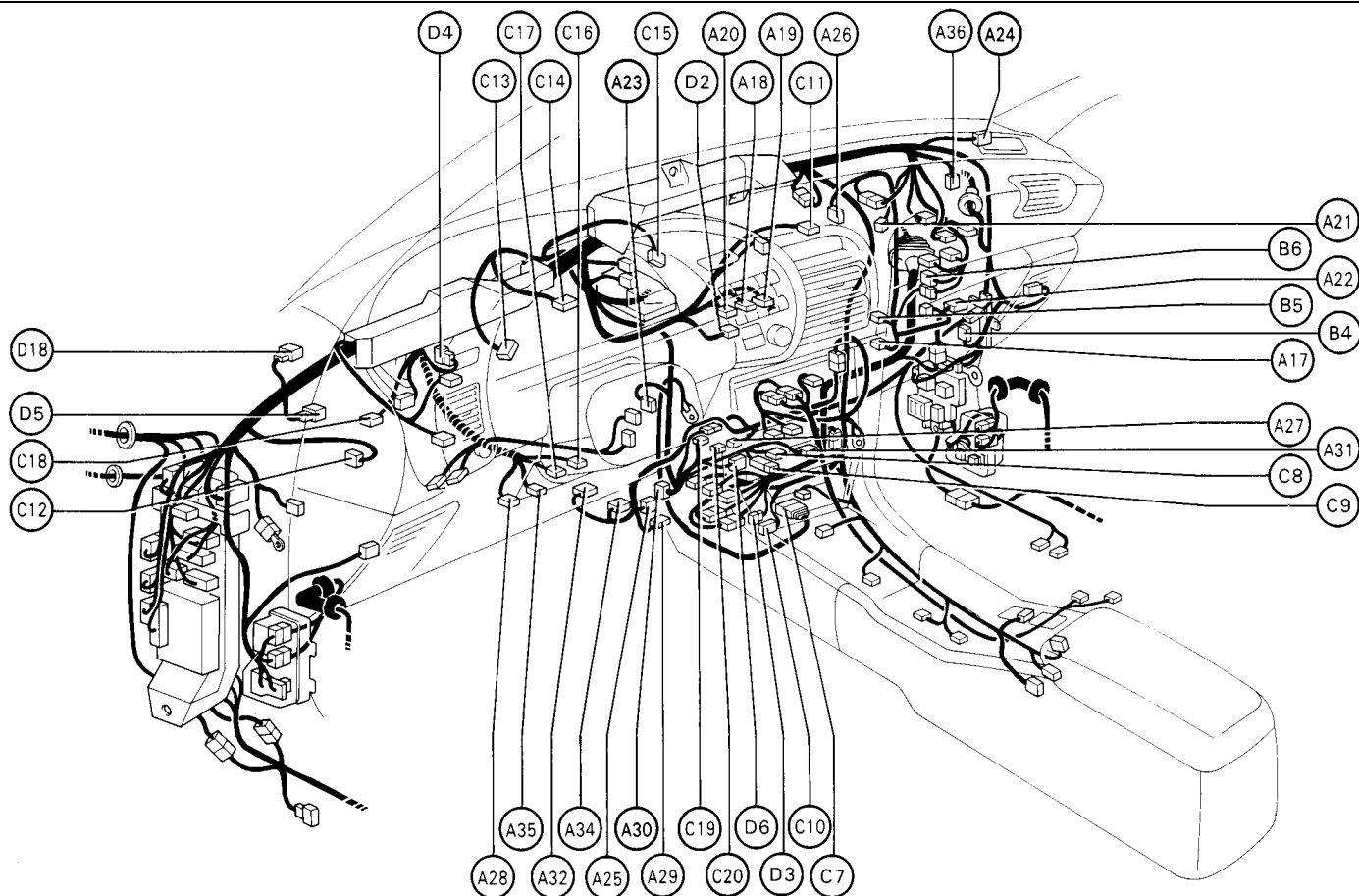
[4A-FE]



- | | | | | | |
|---|----|--|---|---|--|
| I | 1 | IIA (4A-FE) | R | 1 | Radiator Fan Motor |
| I | 2 | IIA (4A-FE) | R | 2 | Retract Motor LH |
| I | 3 | ISC Valve (5S-FE) | R | 3 | Retract Motor RH |
| I | 4 | Igniter (5S-FE) | | | |
| I | 5 | Ignition Coil (5S-FE) | S | 1 | Speed Sensor (for Cruise Control System) |
| I | 6 | In-Air Temp. Sensor | S | 2 | Speed Sensor (for ECT of 5S-FE) |
| I | 7 | Injector No.1 | S | 5 | Starter |
| I | 8 | Injector No.2 | S | 6 | Starter |
| I | 9 | Injector No.3 | | | |
| I | 10 | Injector No.4 | T | 1 | Throttle Position Sensor |
| K | 1 | Knock Sensor | V | 1 | VSV (for EGR System) |
| | | | V | 2 | VSV (for Electrical Idle-Up System of 4A-FE) |
| N | 1 | Neutral Start SW and Back-Up Light SW (A/T w/o ECT) | V | 3 | Vacuum Sensor (for EFI System) |
| N | 2 | Neutral Start SW, Back-Up Light SW and A/T Indicator (A/T w/ ECT of 5S-FE) | W | 1 | Washer Motor |
| N | 3 | Noise Filter (for Ignition System) | W | 3 | Water Temp. Sender |
| | | | W | 4 | Water Temp. SW (for Fans Control) |
| O | 1 | O/D Solenoid | | | |
| O | 2 | Oil Pressure SW | | | |
| O | 3 | Oxygen Sensor (Main, 5S-FE) | | | |
| O | 4 | Oxygen Sensor (Sub, 5S-FE) | | | |
| O | 5 | Oxygen Sensor (for California of 4A-FE) | | | |
| O | 6 | Oxygen Sensor (for Ex. California of 4A-FE) | | | |

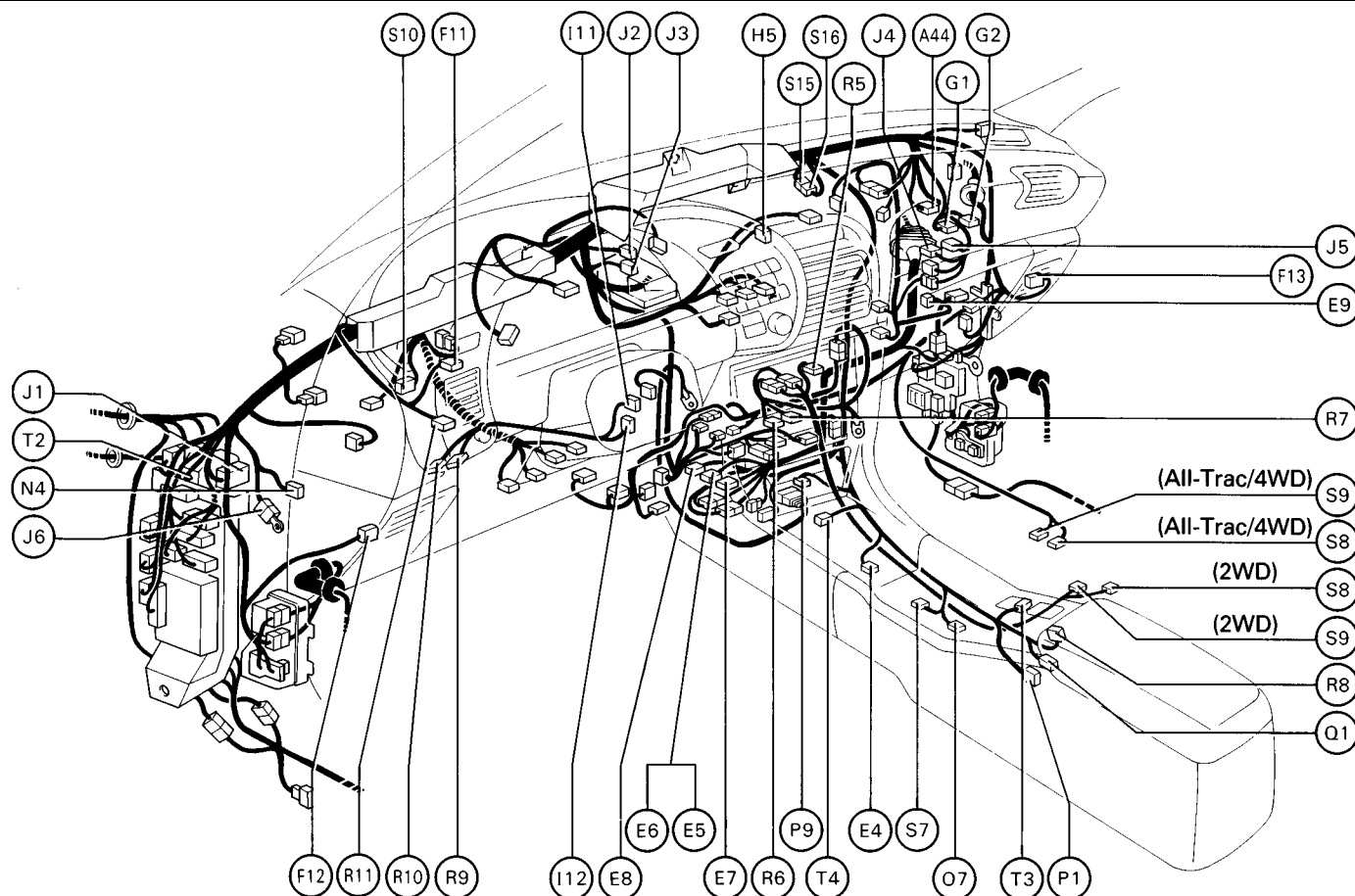
ELECTRICAL WIRING ROUTING

Position of Parts in Instrument Panel



- | | | | |
|------|-----------------------------------|------|--|
| A 17 | A/C Amplifier | C 7 | Center Airbag Sensor Assembly |
| A 18 | A/C Control Assembly | C 8 | Cigarette Lighter |
| A 19 | A/C Control Assembly | C 9 | Cigarette Lighter Illumination |
| A 20 | A/C Control Assembly | C 10 | Circuit Opening Relay |
| A 21 | A/C Evaporator Temp. Sensor | C 11 | Clock |
| A 22 | A/C Power Transistor | C 12 | Clutch Start SW |
| A 23 | A/C Room Temp. Sensor | C 13 | Combination Meter |
| A 24 | A/C Solar Sensor | C 14 | Combination Meter |
| A 25 | A/C System Amplifier | C 15 | Combination Meter |
| A 26 | A/C Thermistor | C 16 | Combination SW |
| A 27 | A/C Water Temp. Sensor | C 17 | Combination SW |
| A 28 | Airbag Squib | C 18 | Cruise Control Clutch SW |
| A 29 | Air Mix Control Servo Motor | C 19 | Cruise Control ECU |
| A 30 | Air Vent Mode Control Servo Motor | C 20 | Cruise Control ECU |
| A 31 | Ashtray Illumination | | |
| A 32 | Auto A/C Amplifier | D 2 | Defogger SW |
| A 33 | Auto A/C Amplifier | D 3 | Diode (for O/D System) |
| A 34 | Auto A/C Amplifier | D 4 | Diode (for Cruise Control System) |
| A 35 | Auto Tilt Away Actuator | D 5 | Diode (for Key Off) |
| A 36 | Auto Tilt Away ECU | D 6 | Door Lock ECU |
| A 44 | Air Inlet Control Servo Motor | D 18 | Diode (for Daytime Running Light System) |
| B 3 | Blower Control Relay | | |
| B 4 | Blower Motor | | |
| B 5 | Blower Resistor | | |
| B 6 | Blower Resistor | | |

Position of Parts in Instrument Panel

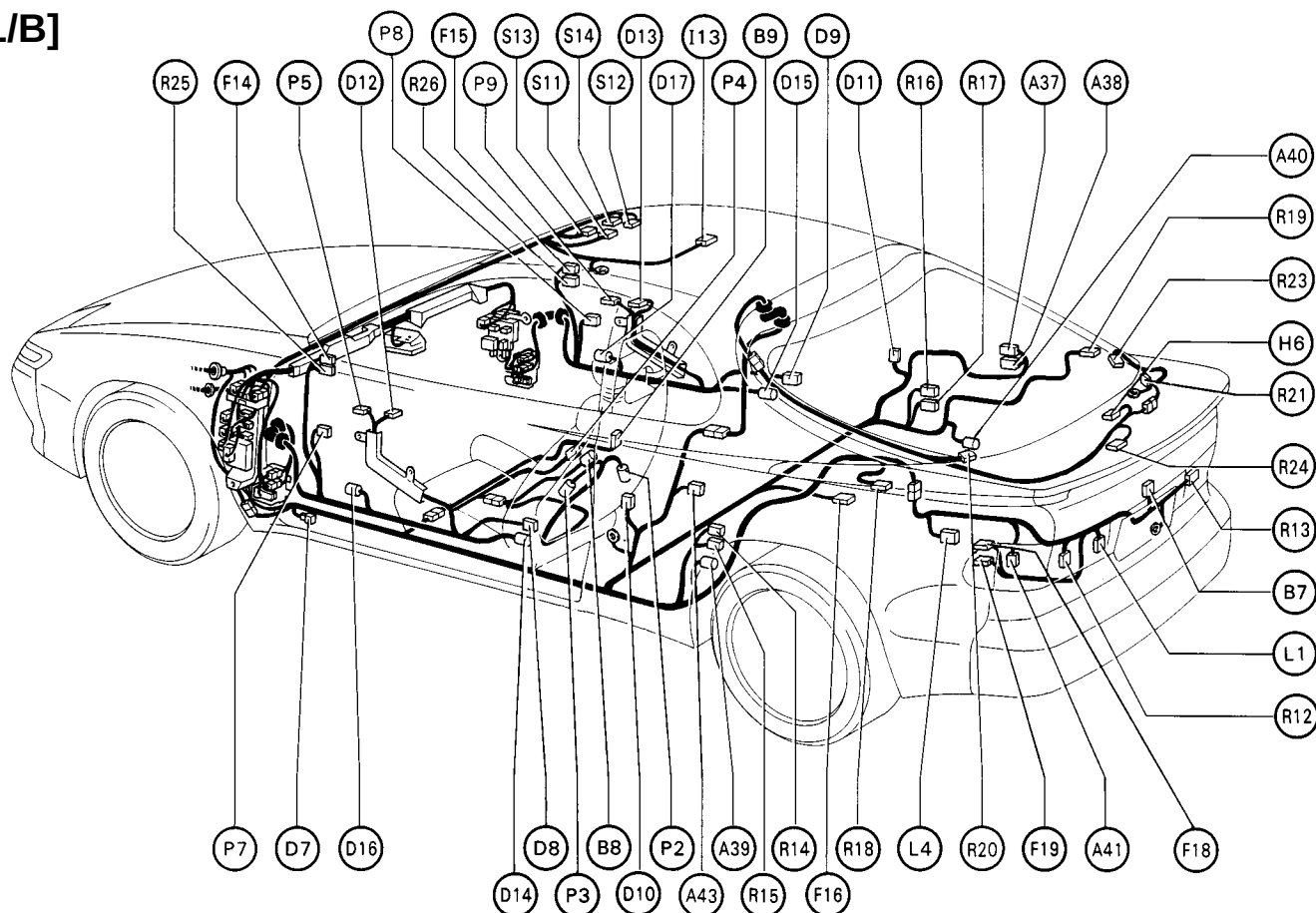


- | | |
|--|--|
| <p>E 4 ECT Pattern Select SW</p> <p>E 5 Engine ECU (4A-FE)</p> <p>E 6 Engine and ECT ECU (5S-FE, 3S-GTE)</p> <p>E 7 Engine and ECT ECU (5S-FE, 3S-GTE)</p> <p>E 8 Engine and ECT ECU</p> <p>E 9 Extra High Speed Relay</p>
<p>F 11 Fog Light SW</p> <p>F 12 Front Speaker LH</p> <p>F 13 Front Speaker RH</p>
<p>G 1 Glove Box Light</p> <p>G 2 Glove Box Light SW</p>
<p>H 5 Hazard SW</p>
<p>I 11 Ignition Key Cylinder Light</p> <p>I 12 Ignition SW and Unlock Warning SW</p>
<p>J 1 Junction Connector</p> <p>J 2 Junction Connector</p> <p>J 3 Junction Connector</p> <p>J 4 Junction Connector</p> <p>J 5 Junction Connector</p> <p>J 6 Junction Connector (for Earth)</p> | <p>K 2 Key Inter Lock Solenoid</p>
<p>N 4 Noise Filter (for Defogger System)</p>
<p>O 7 O/D Main SW and A/T Indicator (Shift Lever)</p>
<p>P 1 Parking Brake SW</p> <p>P 9 Panel Relay (for Daytime Running Light System)</p>
<p>Q 1 Quarter Power Window SW</p>
<p>R 4 RECIRC/FRESH Control Servo Motor</p> <p>R 5 Radio and Player (w/ CD Player)</p> <p>R 6 Radio and Player (w/o CD Player)</p> <p>R 7 Radio and Player (w/o CD Player)</p> <p>R 8 Remote Control Mirror SW</p> <p>R 9 Retract Control Relay</p> <p>R 10 Retract Control Relay (for CANADA)</p> <p>R 11 Rheostat</p>
<p>S 7 Shift Lock ECU</p> <p>S 8 Stereo Component Amplifier (w/CD Player)</p> <p>S 9 Stereo Component Amplifier (w/CD Player)</p> <p>S 10 Stop Light SW (w/o Cruise Control System) or Stop Light and Cruise Control Stop SW</p>
<p>S 15 Short Pin (for Daytime Running Light System)</p> <p>S 16 Short Pin (for Daytime Running Light System)</p>
<p>T 2 Turn Signal Flasher</p> <p>T 3 Top Stack Main Relay</p> <p>T 4 Top Stack SW</p> |
|--|--|

ELECTRICAL WIRING ROUTING

Position of Parts in Body

[L/B]



- A 37 ABS ECU (L/B)
- A 38 ABS ECU (L/B)
- A 39 ABS Speed Sensor Rear LH (L/B)
- A 40 ABS Speed Sensor Rear RH (L/B)
- A 41 Auto Antenna Control Relay and Motor
- A 43 ABS Deceleration Sensor (All-Trac/4WD)

- B 7 Back Door Courtesy SW (L/B)
- B 8 Buckle SW (w/ Power Seat)
- B 9 Buckle SW (w/o Power Seat)

- D 7 Diode
- D 8 Door Courtesy Light LH
- D 9 Door Courtesy Light RH
- D 10 Door Courtesy SW LH
- D 11 Door Courtesy SW RH

- D 12 Door Lock Control SW LH
- D 13 Door Lock Control SW RH
- D 14 Door Lock Motor, Door Unlock Detection SW LH and Key Lock and Unlock SW LH
- D 15 Door Lock Motor, Door Unlock Detection SW RH and Key Lock and Unlock SW RH
- D 16 Door Speaker LH
- D 17 Door Speaker RH

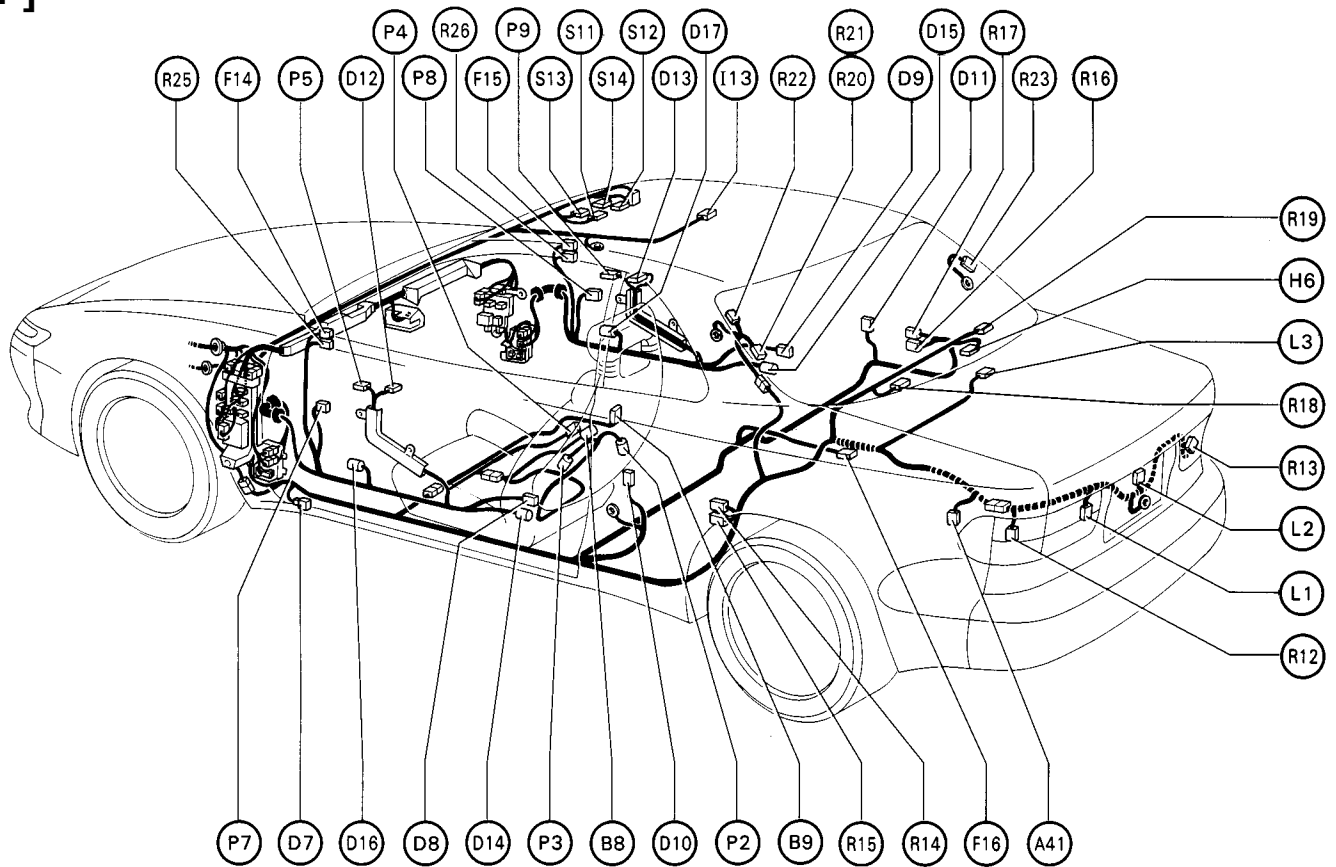
- F 14 Front Tweeter (Speaker) LH
- F 15 Front Tweeter (Speaker) RH
- F 16 Fuel Pump and Fuel Sender (2WD)
- F 18 Fuel Pump (All-Trac/4WD)
- F 19 Fuel Sender (All-Trac/4WD)

- H 6 High Mount Stop Light

- I 13 Interior Light

Position of Parts in Body

[C/P]



L 1 License Plate Light
 L 2 Luggage Compartment Door Courtesy SW (C/P)
 L 3 Luggage Compartment Light (C/P)
 L 4 Luggage Compartment Light (L/B)

P 2 Power Seat Motor (for Lumbar Support)
 P 3 Power Seat Motor (for Side Support)
 P 4 Power Seat SW
 P 5 Power Window Master SW
 P 7 Power Window Motor LH
 P 8 Power Window Motor RH
 P 9 Power Window SW RH

R 12 Rear Combination Light LH
 R 13 Rear Combination Light RH
 R 14 Rear Speaker LH
 R 15 Rear Speaker LH
 R 16 Rear Speaker RH
 R 17 Rear Speaker RH

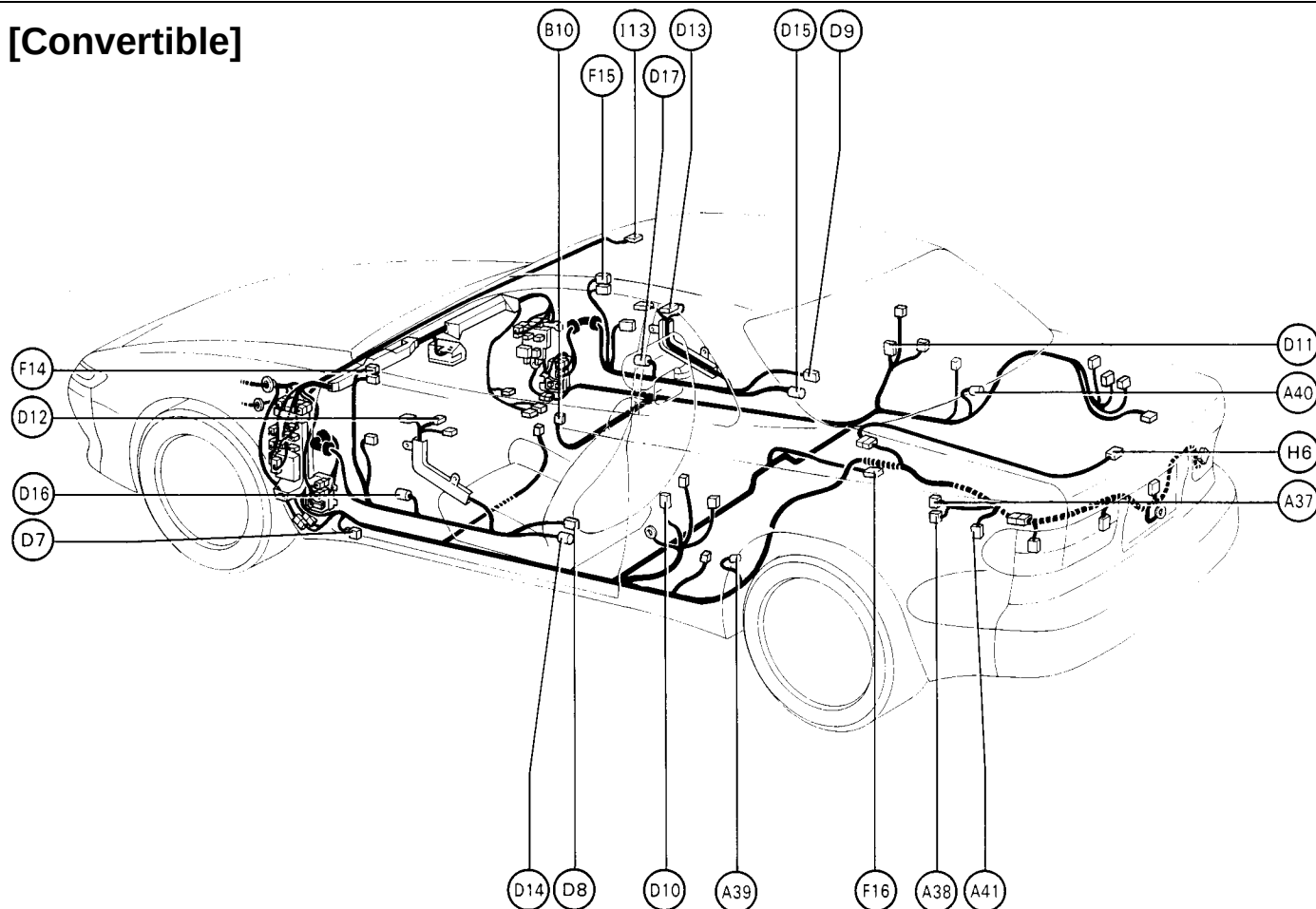
R 18 Rear Tweeter (Speaker) LH
 R 19 Rear Tweeter (Speaker) RH
 R 20 Rear Window Defogger
 (+, w/o Diversity Antenna)
 R 21 Rear Window Defogger
 (+, w/ Diversity Antenna)
 R 22 Rear Window Defogger
 (-, C/P, w/ Diversity Antenna)
 R 23 Rear Window Defogger
 (-, w/o Diversity Antenna)
 R 24 Rear Wiper Motor and Relay (L/B)
 R 25 Remote Control Mirror LH
 R 26 Remote Control Mirror RH

S 11 Sun Roof Control Relay
 S 12 Sun Roof Control SW and Personal Light
 S 13 Sun Roof Limit SW
 S 14 Sun Roof Motor

ELECTRICAL WIRING ROUTING

Position of Parts in Body

[Convertible]



- A 37 ABS ECU
- A 38 ABS ECU
- A 39 ABS Speed Sensor Rear LH
- A 40 ABS Speed Sensor Rear RH
- A 41 Auto Antenna Control Relay and Motor

- B 9 Buckle SW LH (w/o Power Seat)
- B 10 Buckle SW RH (w/o Power Seat)

- D 7 Diode
- D 8 Door Courtesy Light LH
- D 9 Door Courtesy Light RH
- D 10 Door Courtesy SW LH
- D 11 Door Courtesy SW RH

- D 12 Door Lock Control SW LH
- D 13 Door Lock Control SW RH
- D 14 Door Lock Motor, Door Unlock Detection SW LH and Key Lock and Unlock SW LH
- D 15 Door Lock Motor, Door Unlock Detection SW RH and Key Lock and Unlock SW RH
- D 16 Door Speaker LH
- D 17 Door Speaker RH

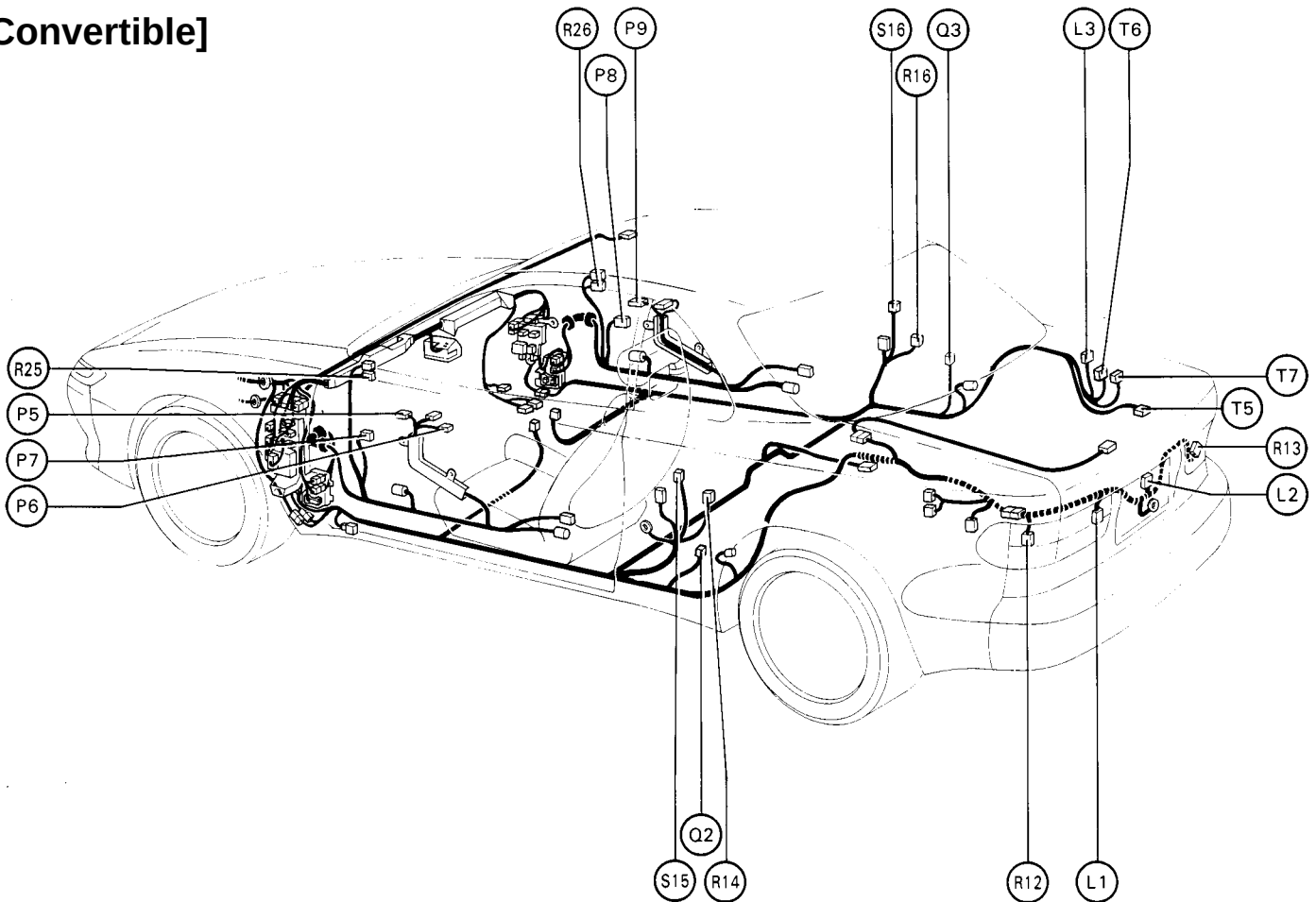
- F 14 Front Tweeter (Speaker) LH
- F 15 Front Tweeter (Speaker) RH
- F 16 Fuel Pump and Fuel Sender

- H 6 High Mount Stop Light

- I 13 Interior Light

Position of Parts in Body

[Convertible]



- L 1 Licence Plate Light
- L 2 Luggage Compartment Door Courtesy SW
- L 3 Luggage Compartment Light

- P 5 Power Window Master SW
- P 6 Power Window Master SW
- P 7 Power Window Motor LH
- P 8 Power Window Motor RH
- P 9 Power Window SW RH

- Q 2 Quarter Power Window Motor LH
- Q 3 Quarter Power Window Motor RH

- R 12 Rear Combination Light LH
- R 13 Rear Combination Light RH
- R 14 Rear Speaker LH
- R 16 Rear Speaker RH
- R 25 Remote Control Mirror LH
- R 26 Remote Control Mirror RH

- S 15 Seat Belt Solenoid (Driver's)
- S 16 Seat Belt Solenoid (Passenger's)

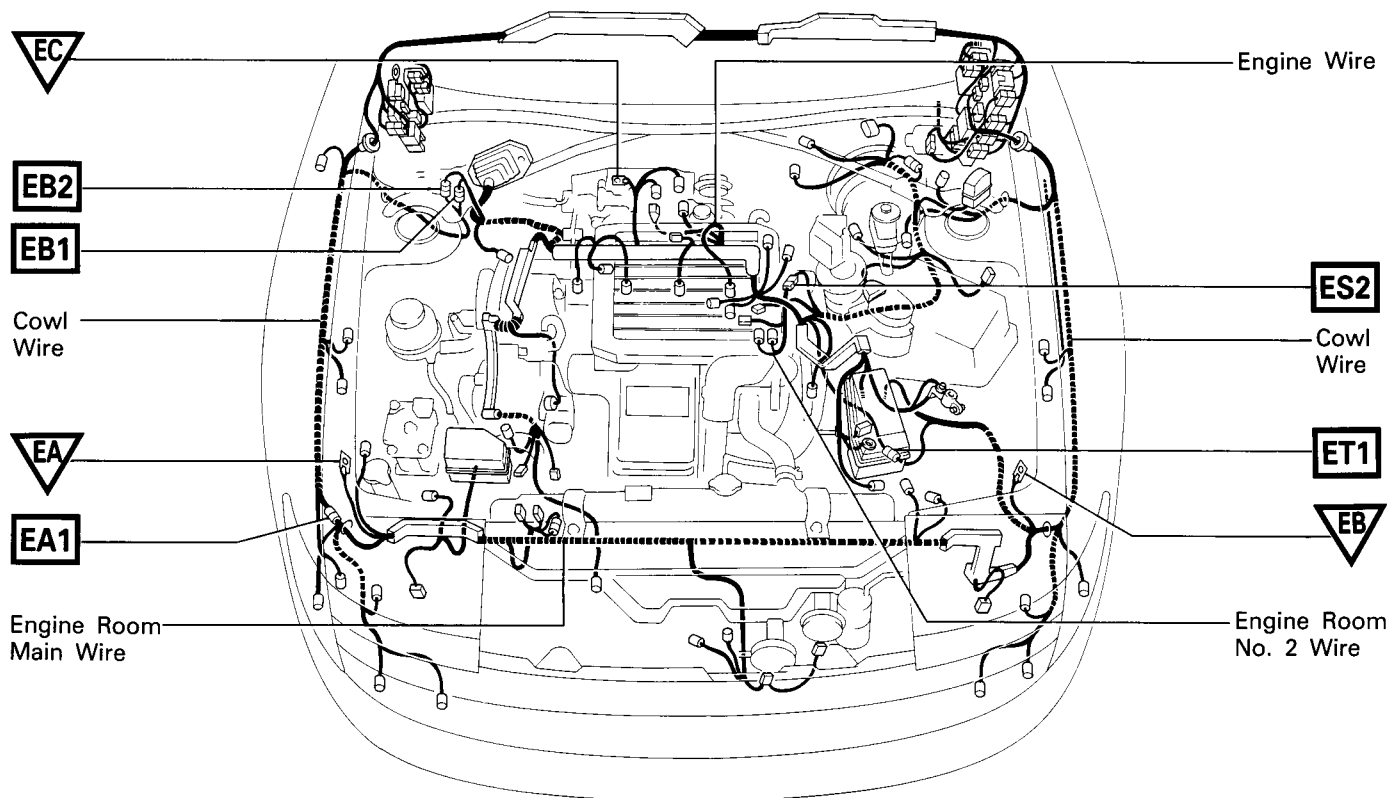
- T 5 Top Stack Motor
- T 6 Top Stack Relay
- T 7 Top Stack Relay

ELECTRICAL WIRING ROUTING

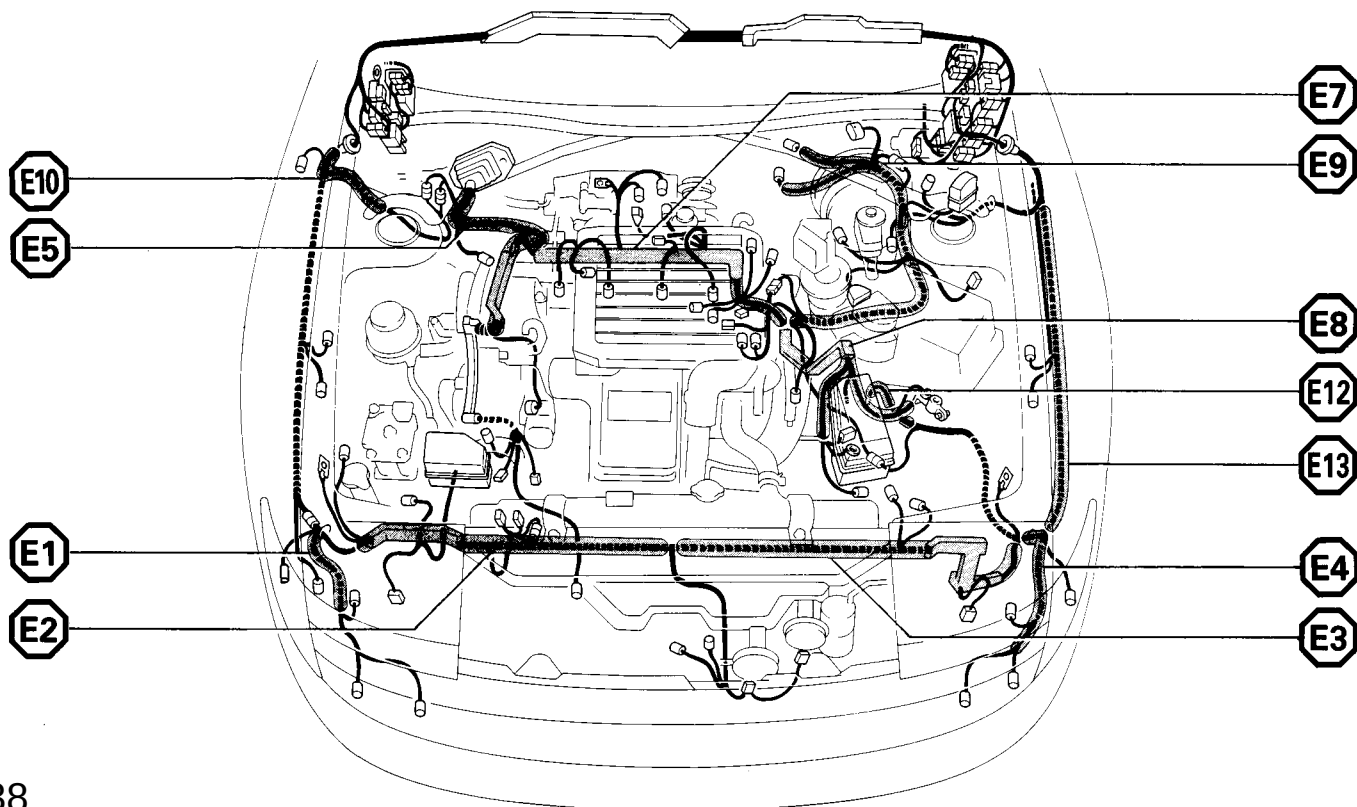
□ : Location of Connector Joining Wire Harness and Wire Harness

▽ : Location of Ground Points

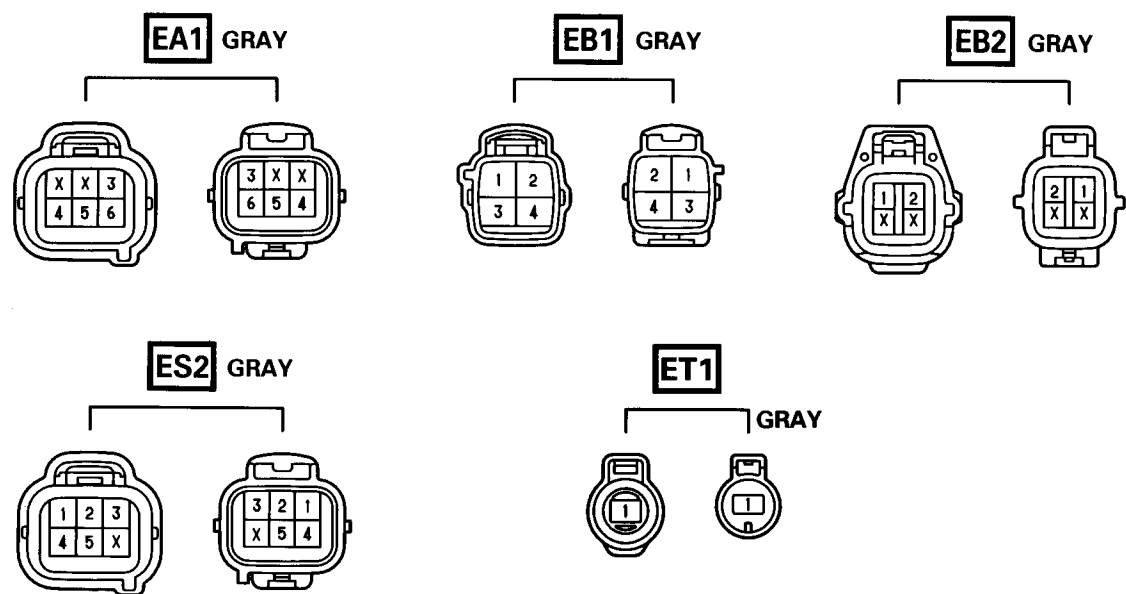
[3S-GTE]



○ : Location of Splice Points



Connector Joining Wire Harness and Wire Harness



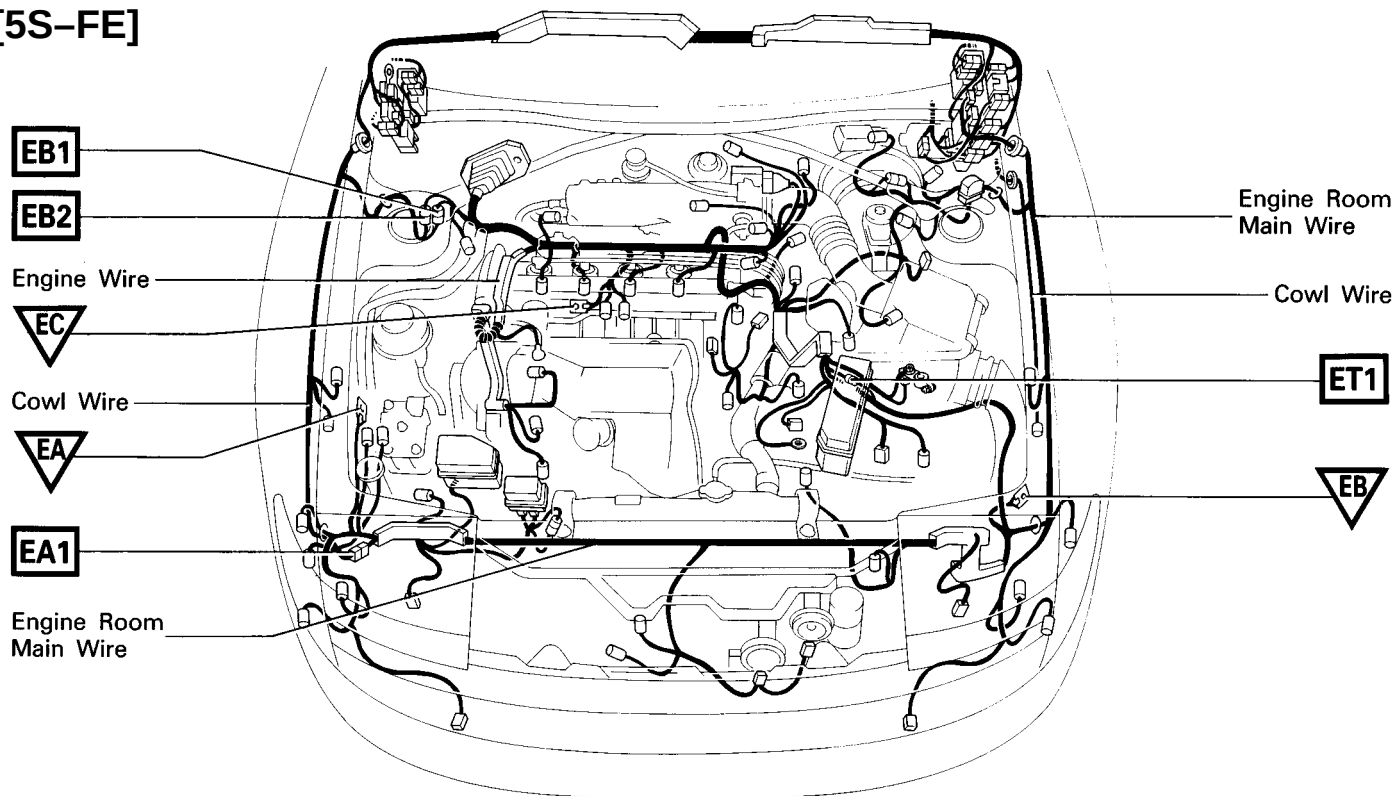
CODE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EA1	COWL WIRE AND ENGINE ROOM MAIN WIRE (FRONT SIDE OF RIGHT FRONT FENDER)
EB1	ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER)
EB2	
ES2	ENGINE WIRE AND ENGINE ROOM NO. 2 WIRE (NEAR THE DISTRIBUTOR)
ET1	ENGINE ROOM MAIN WIRE AND ENGINE WIRE (NEAR THE J/B NO. 2)

ELECTRICAL WIRING ROUTING

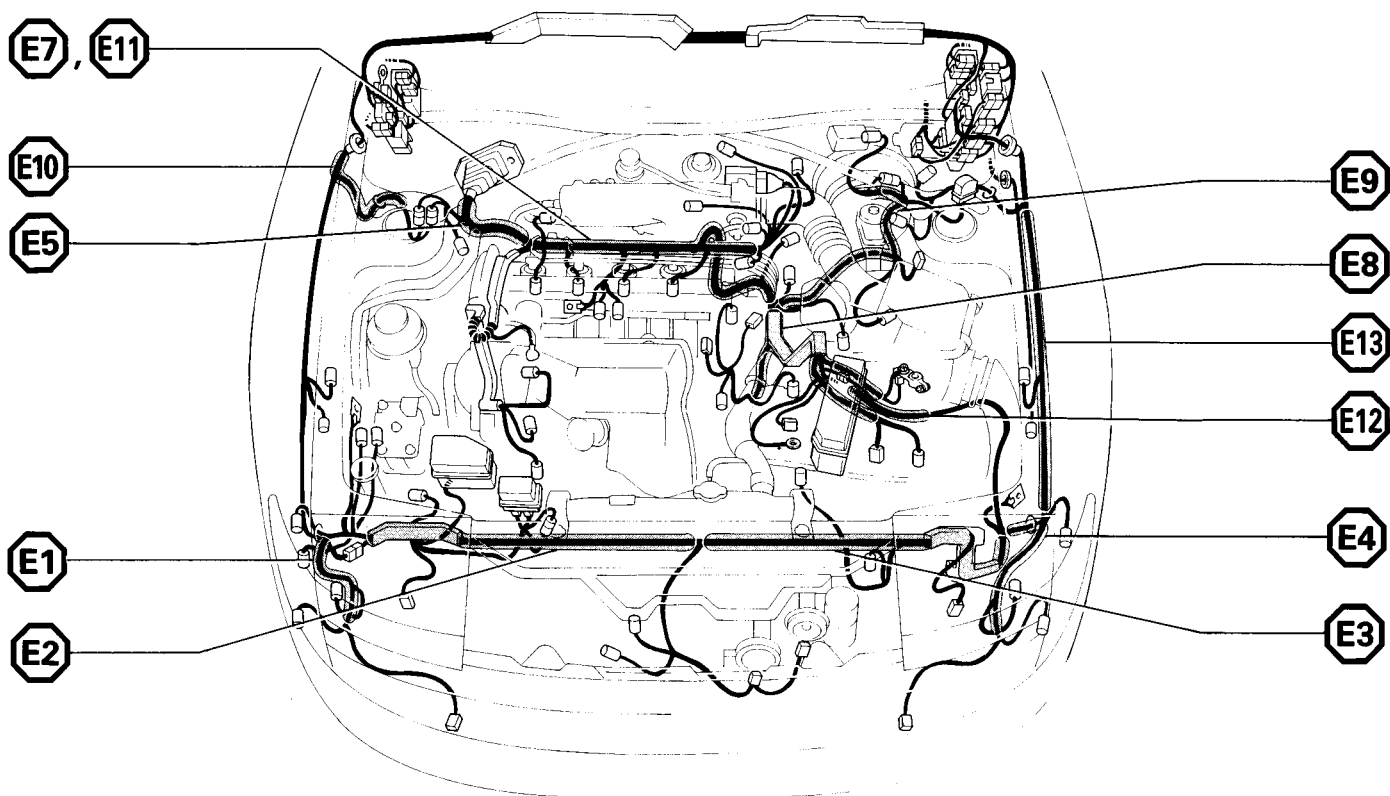
□ : Location of Connector Joining Wire Harness and Wire Harness

▽ : Location of Ground Points

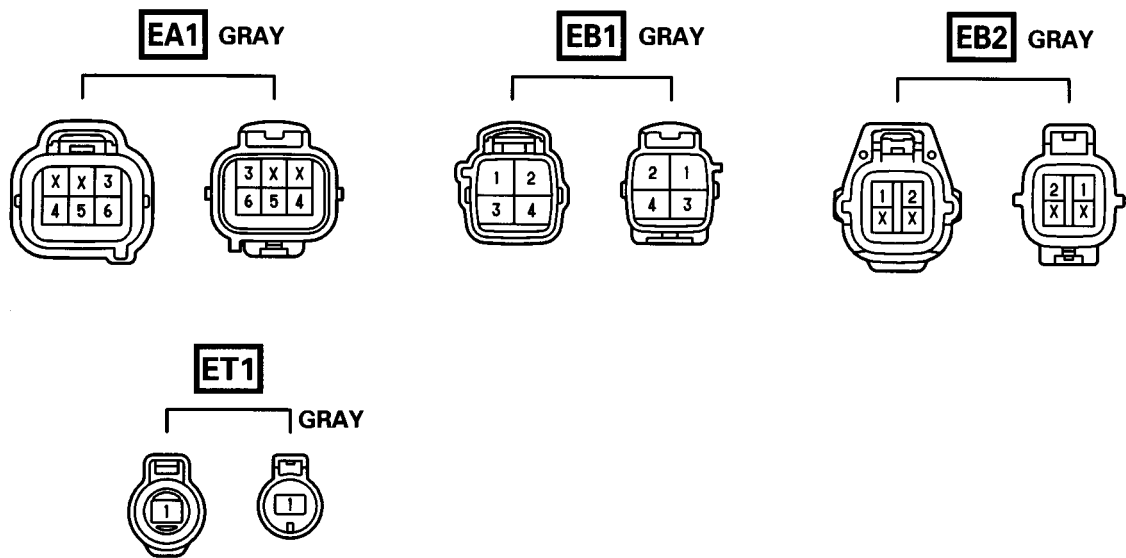
[5S-FE]



○ : Location of Splice Points



Connector Joining Wire Harness and Wire Harness



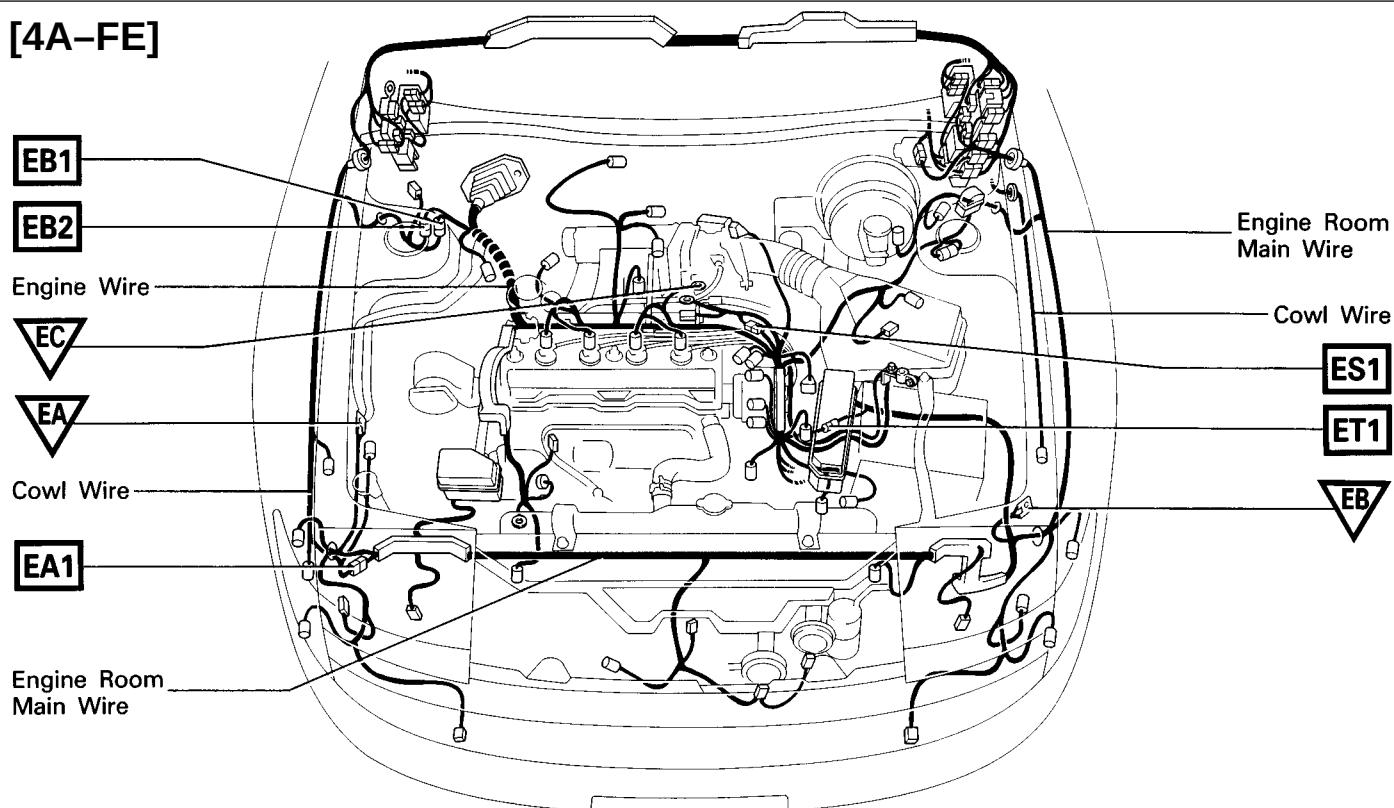
CODE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EA1	COWL WIRE AND ENGINE ROOM MAIN WIRE (FRONT SIDE OF RIGHT FRONT FENDER)
EB1	ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER)
EB2	
ET1	ENGINE ROOM MAIN WIRE AND ENGINE WIRE (NEAR THE J/B NO. 2)

ELECTRICAL WIRING ROUTING

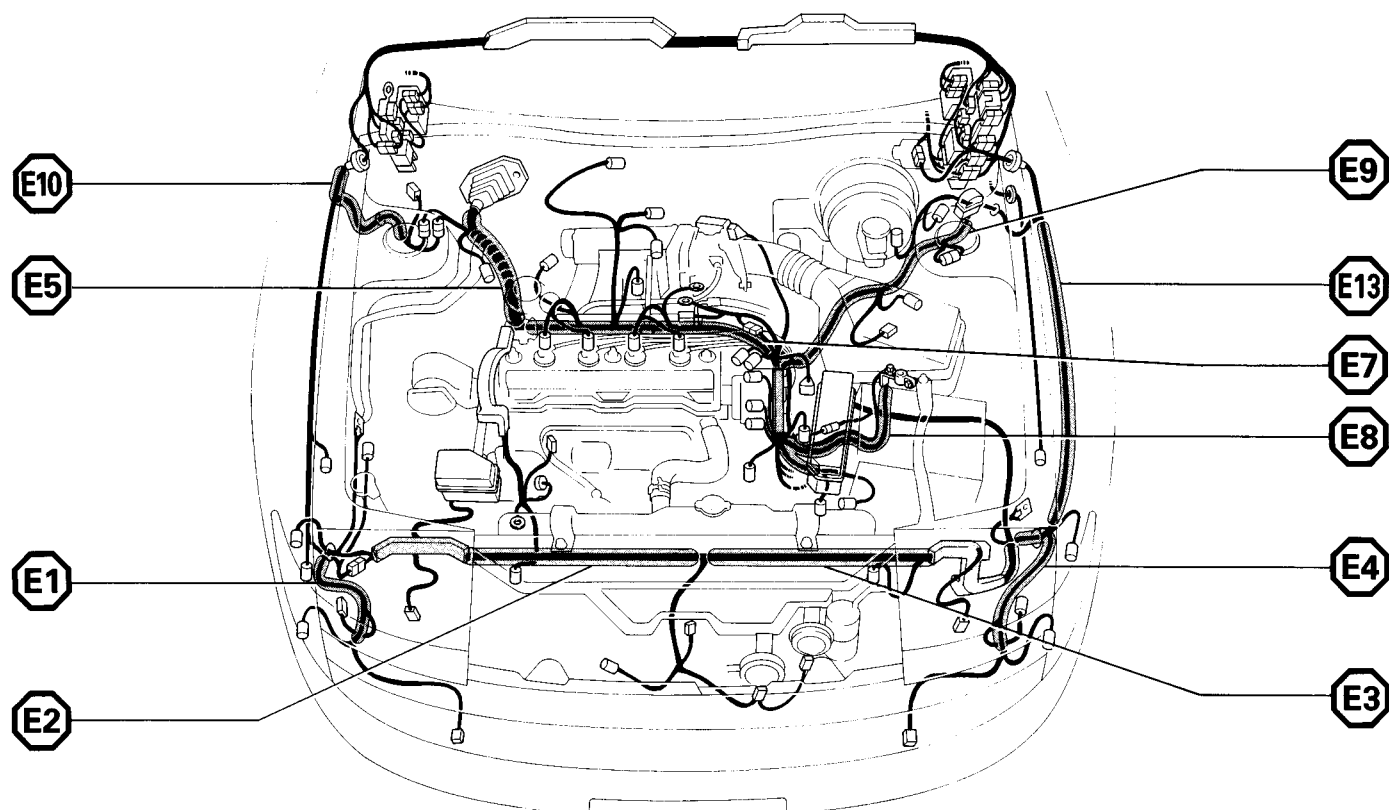
□ : Location of Connector Joining Wire Harness and Wire Harness

▽ : Location of Ground Points

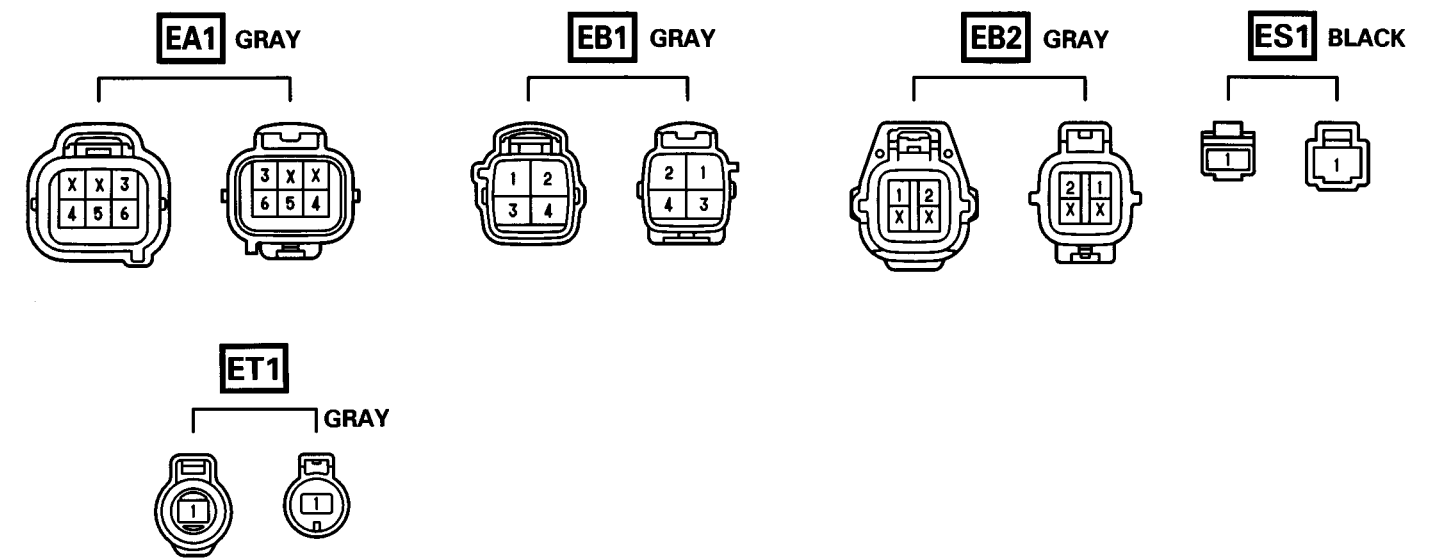
[4A-FE]



○ : Location of Splice Points



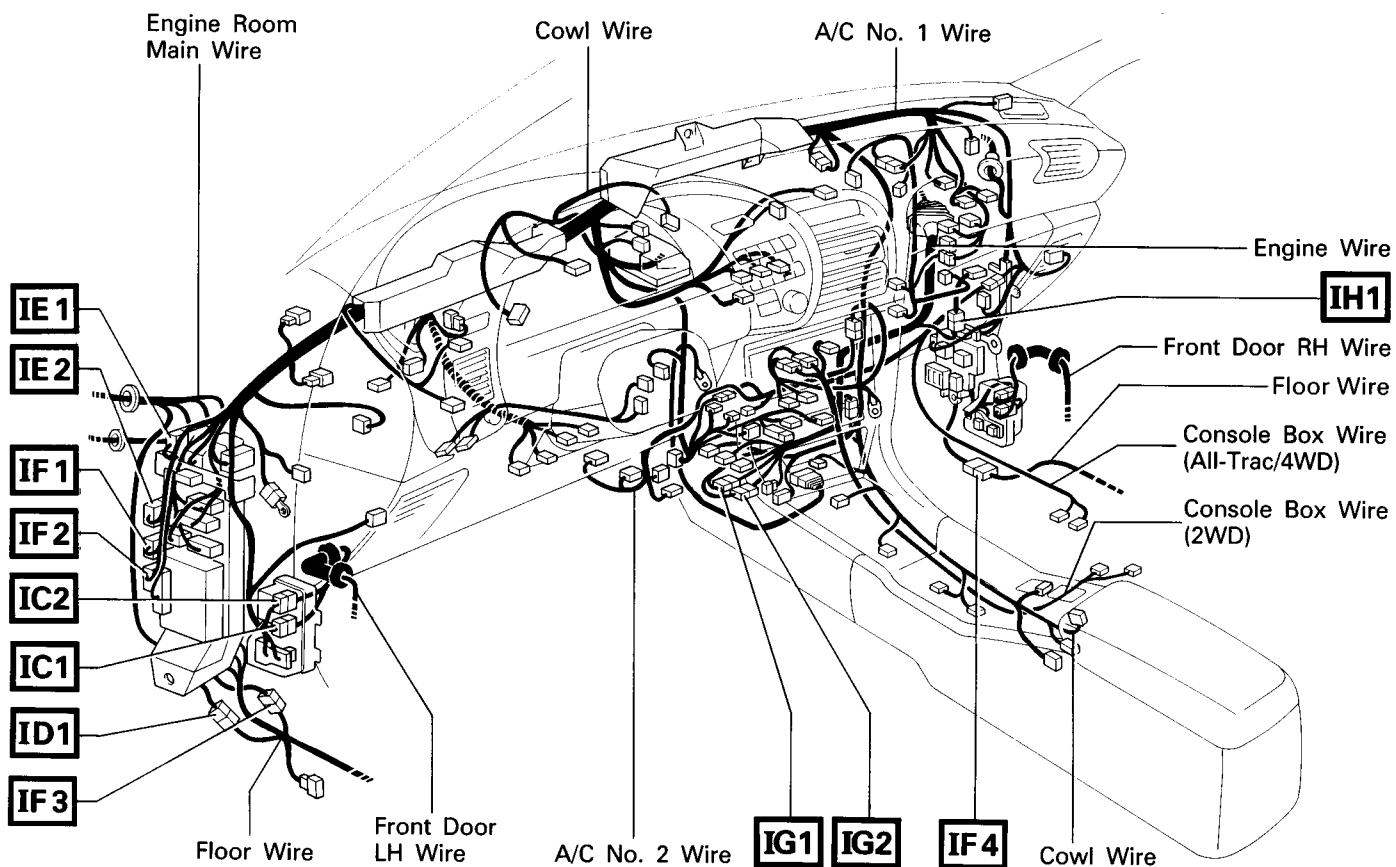
Connector Joining Wire Harness and Wire Harness



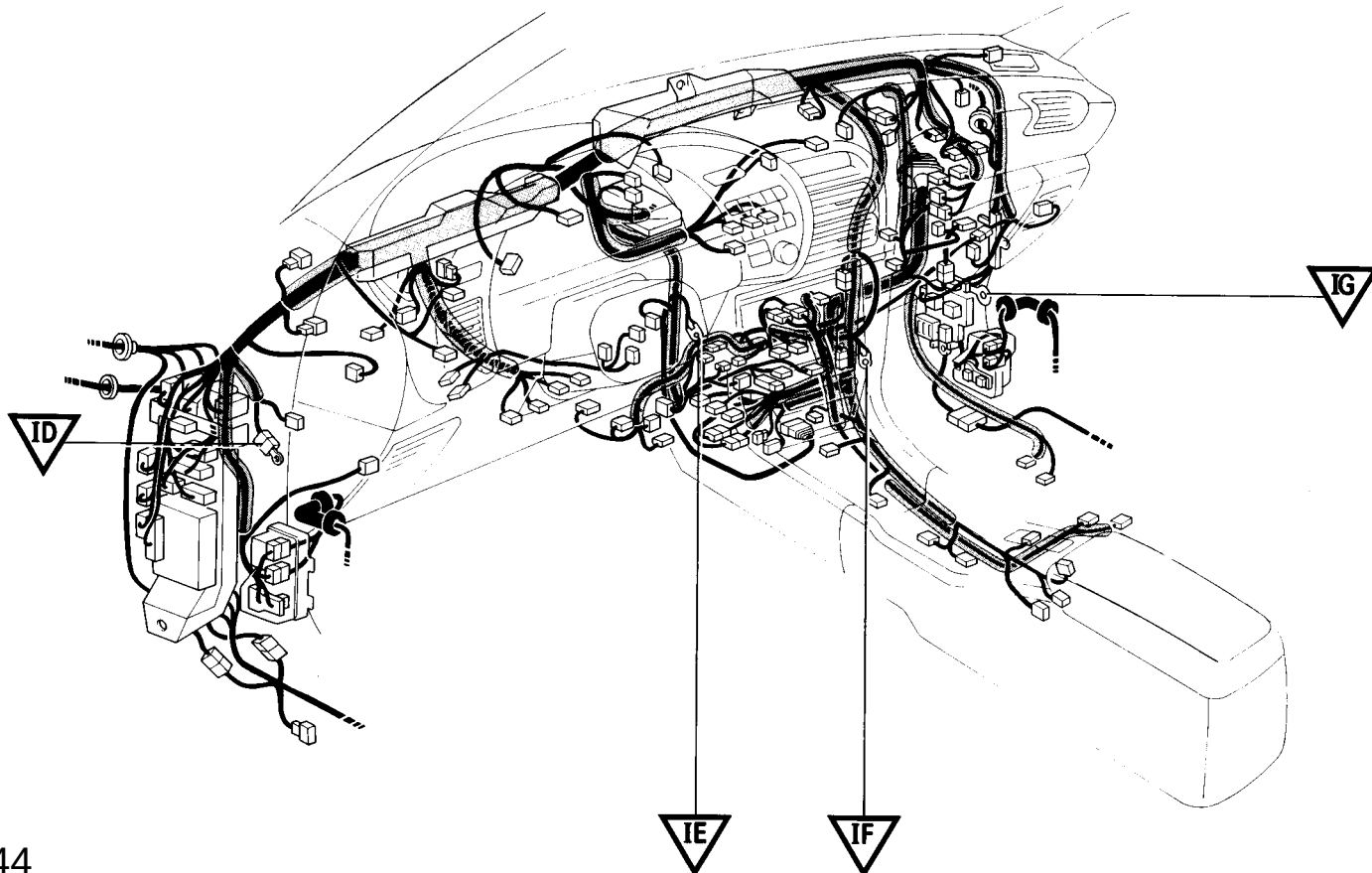
CODE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EA1	COWL WIRE AND ENGINE ROOM MAIN WIRE (FRONT SIDE OF RIGHT FRONT FENDER)
EB1	ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER)
EB2	
ES1	ENGINE WIRE AND ENGINE ROOM NO. 2 WIRE (NEAR THE STARTER)
ET1	ENGINE ROOM MAIN WIRE AND ENGINE WIRE (NEAR THE J/B NO. 2)

ELECTRICAL WIRING ROUTING

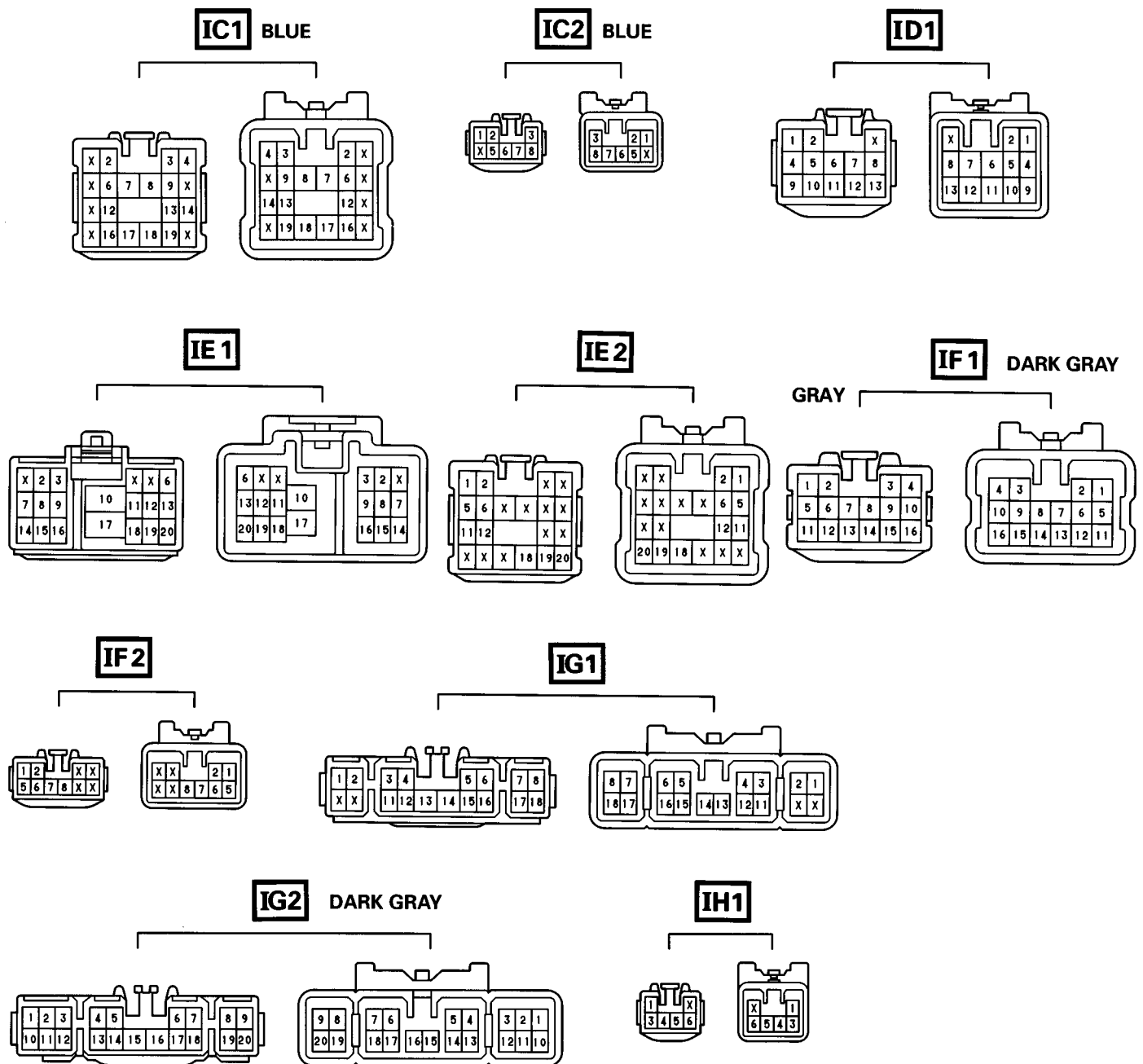
□ : Location of Connector Joining Wire Harness and Wire Harness



▽ : Location of Ground Points



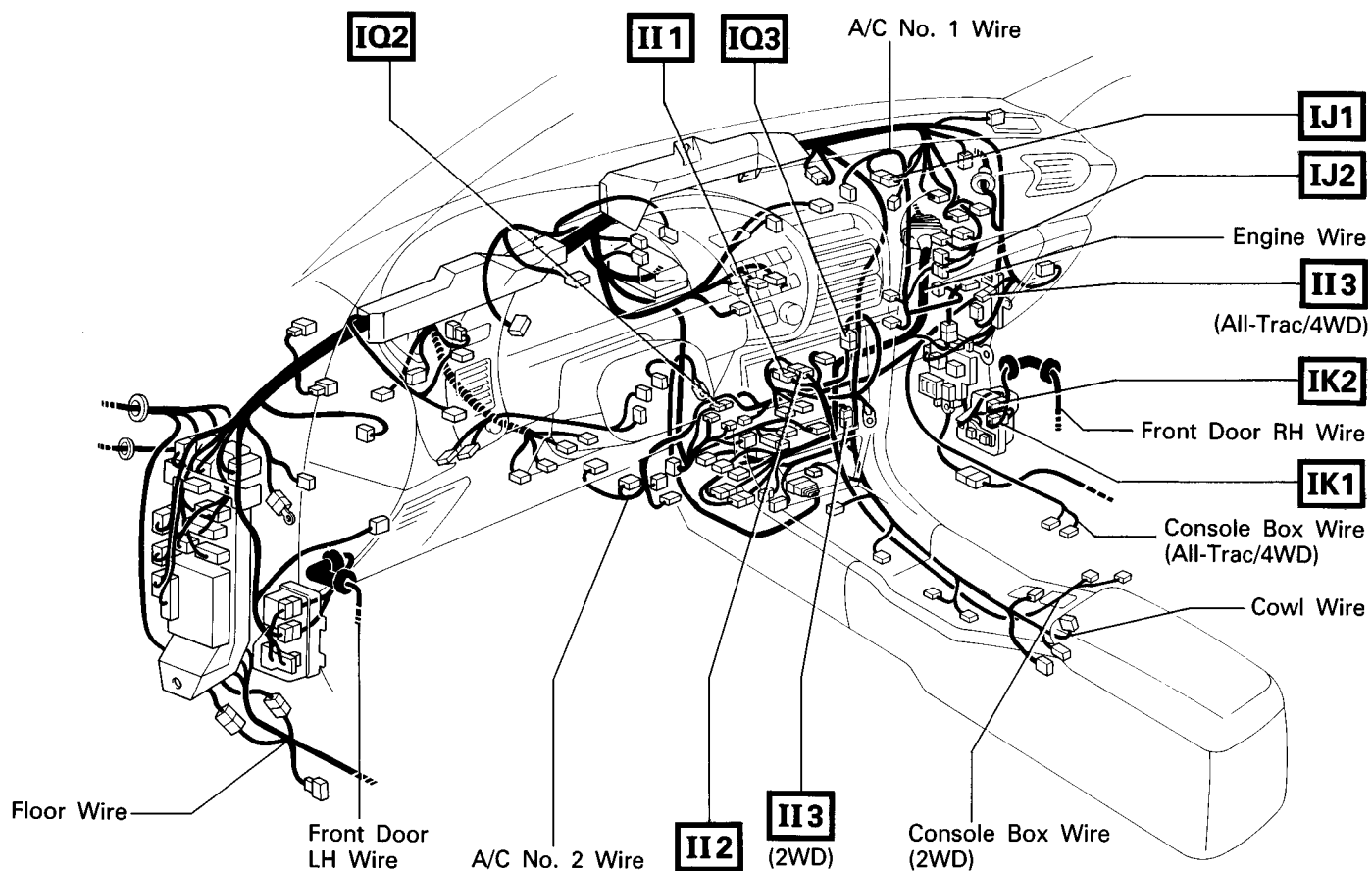
Connector Joining Wire Harness and Wire Harness



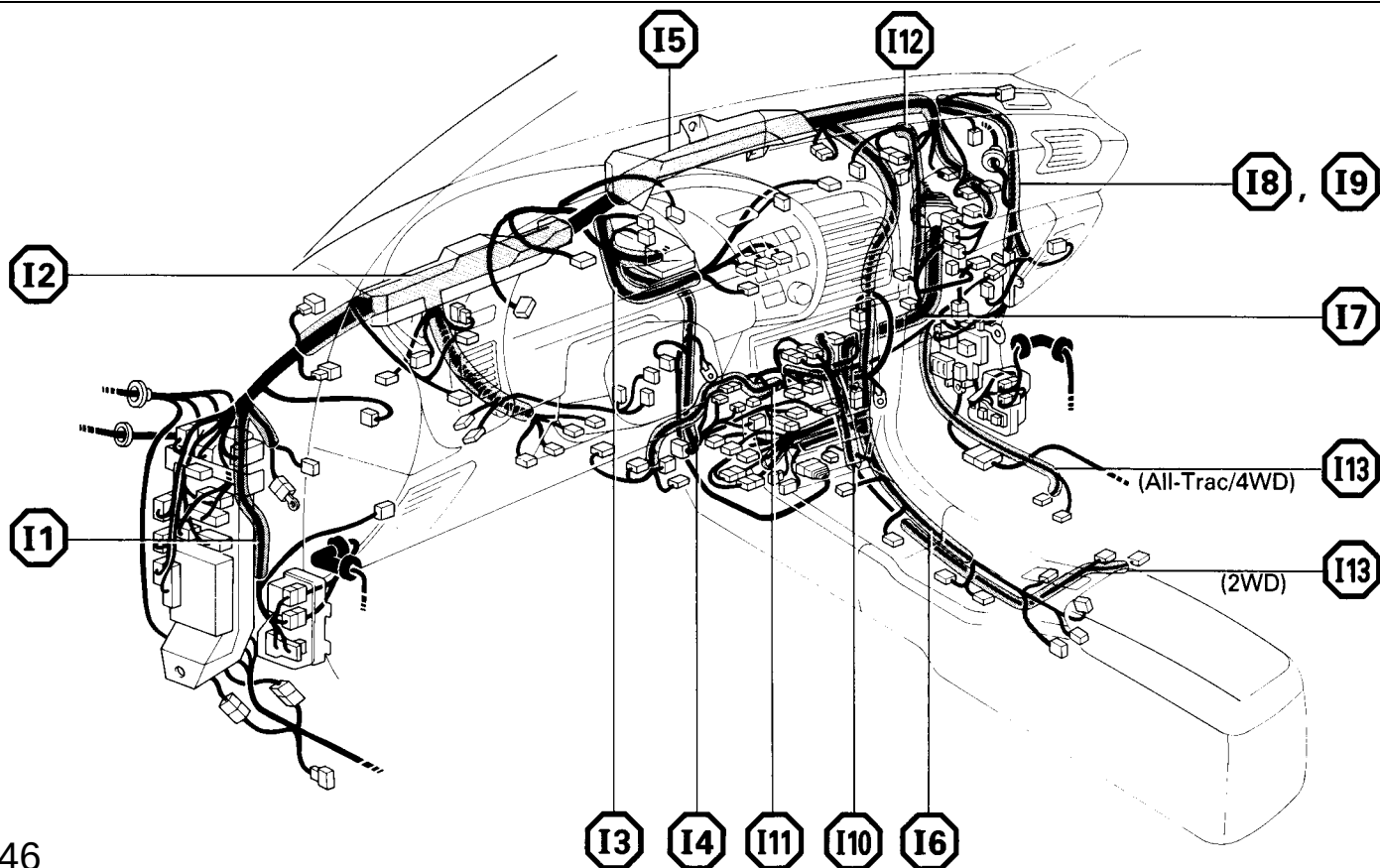
CODE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IC1	FRONT DOOR LH WIRE AND COWL WIRE (LEFT KICK PANEL)
IC2	FRONT DOOR LH WIRE AND COWL WIRE (LEFT KICK PANEL)
ID1	ENGINE ROOM MAIN WIRE AND FLOOR WIRE (LEFT KICK PANEL)
IE1	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)
IE2	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)
IF1	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
IF2	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
IG1	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)
IG2	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)
IH1	ENGINE WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)

ELECTRICAL WIRING ROUTING

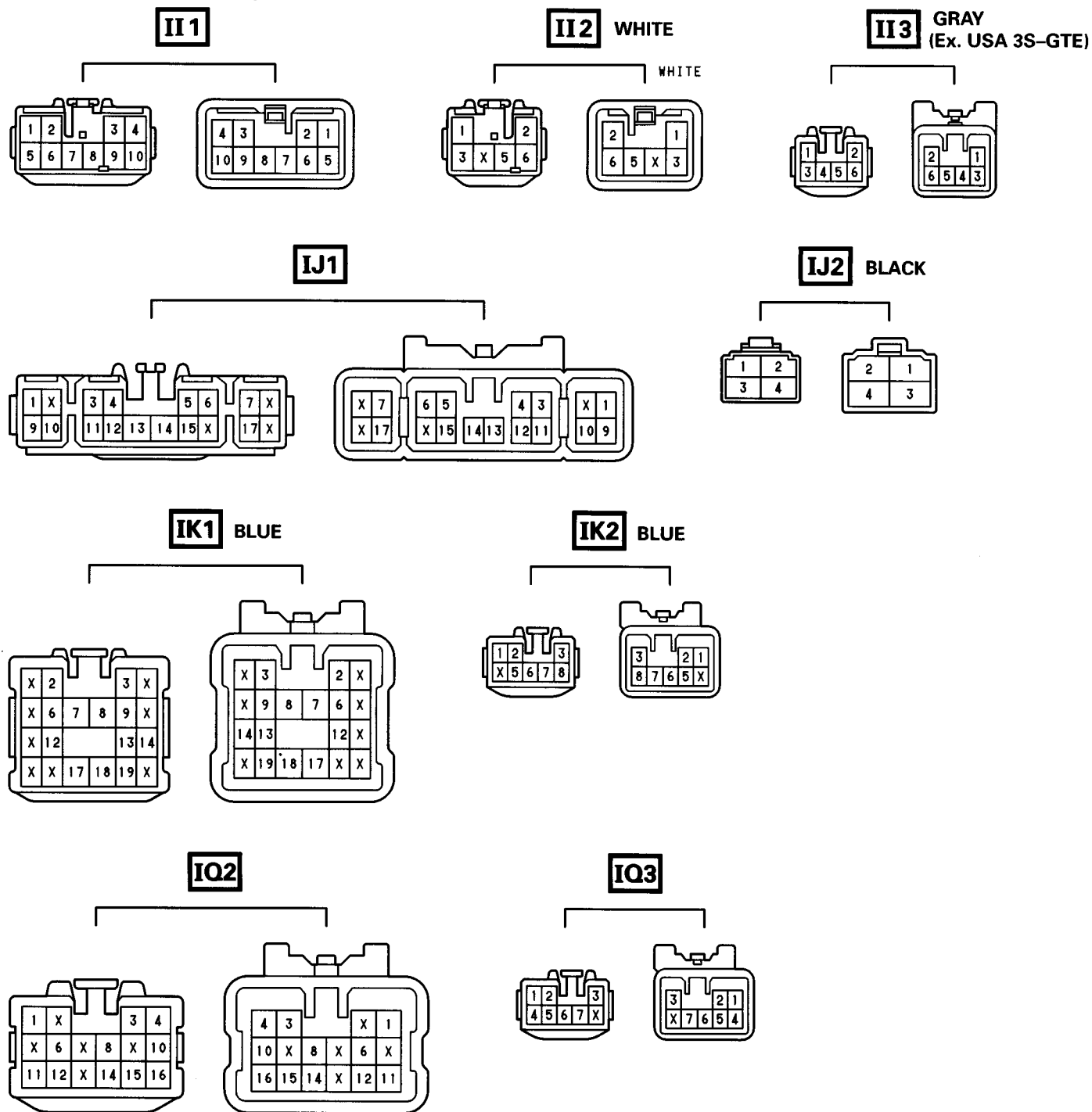
□ : Location of Connector Joining Wire Harness and Wire Harness



○ : Location of Splice Points



Connector Joining Wire Harness and Wire Harness



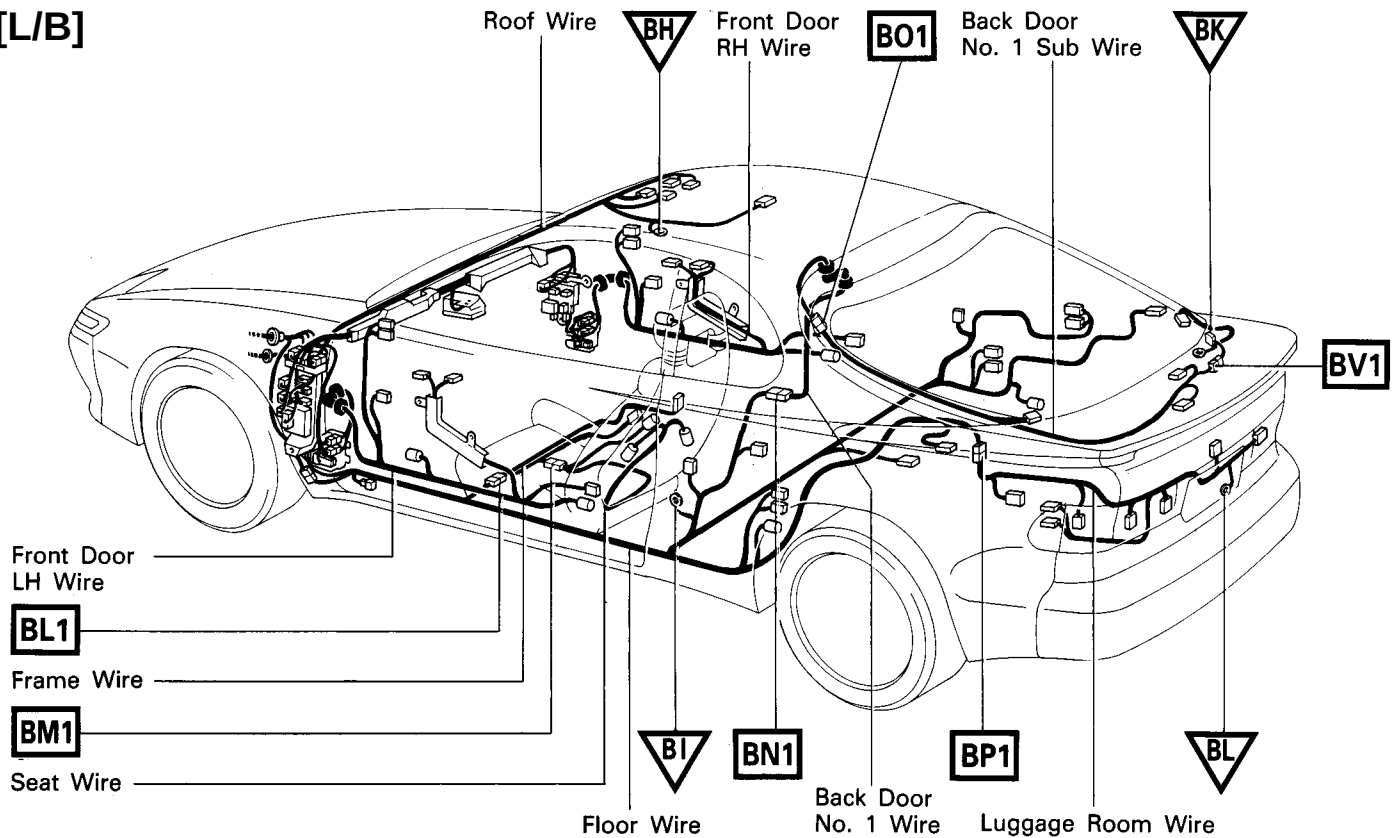
CODE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
II1	COWL WIRE AND CONSOLE BOX WIRE (INSTRUMENT PANEL CENTER)
II2	
II3	COWL WIRE AND CONSOLE BOX WIRE (RIGHT KICK PANEL OF ALL-TRACK/4WD)
	COWL WIRE AND CONSOLE BOX WIRE (RIGHT KICK PANEL OF 2WD)
IJ1	COWL WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)
IJ2	
IK1	FRONT DOOR RH WIRE AND COWL WIRE (RIGHT KICK PANEL)
IK2	
IQ2	COWL WIRE AND A/C NO. 2 WIRE (BESIDE HEATER UNIT)
IQ3	

ELECTRICAL WIRING ROUTING

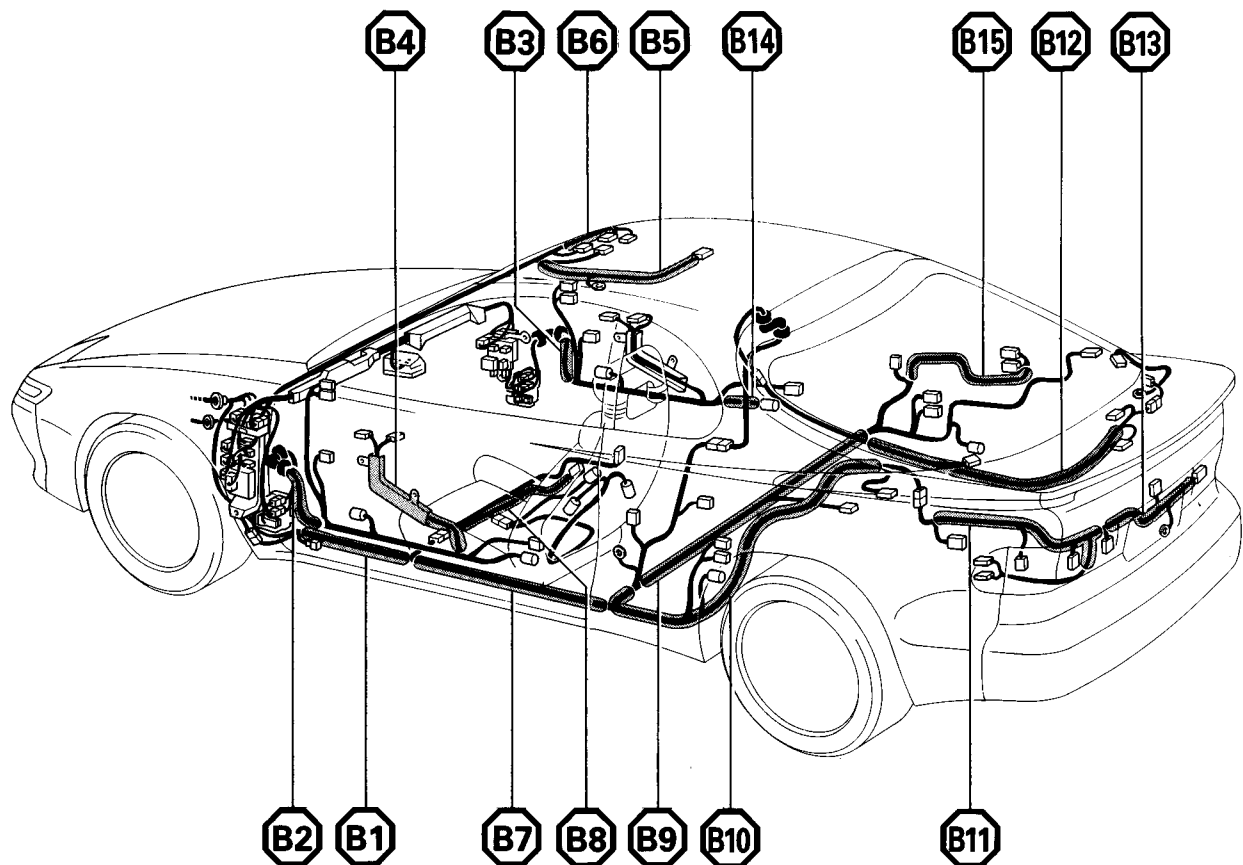
□ : Location of Connector Joining Wire Harness and Wire Harness

▽ : Location of Ground Points

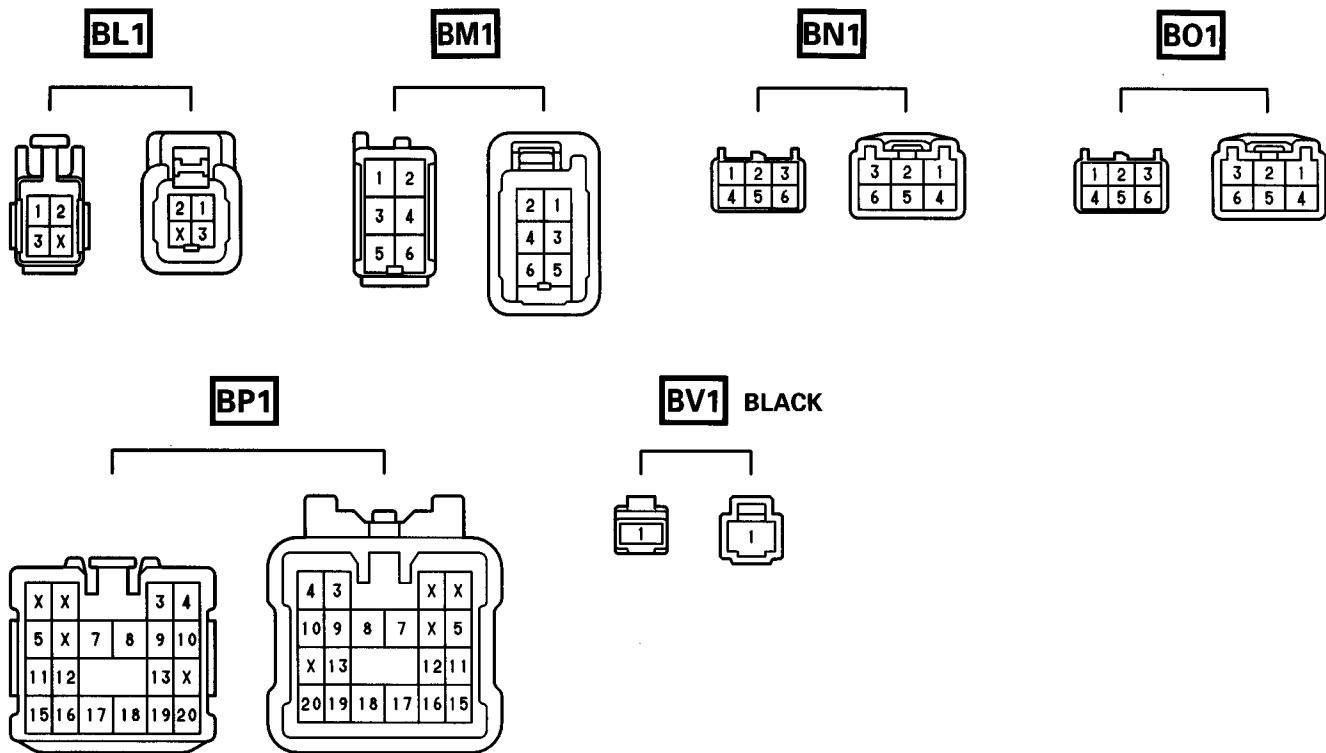
[L/B]



○ : Location of Splice Points



Connector Joining Wire Harness and Wire Harness



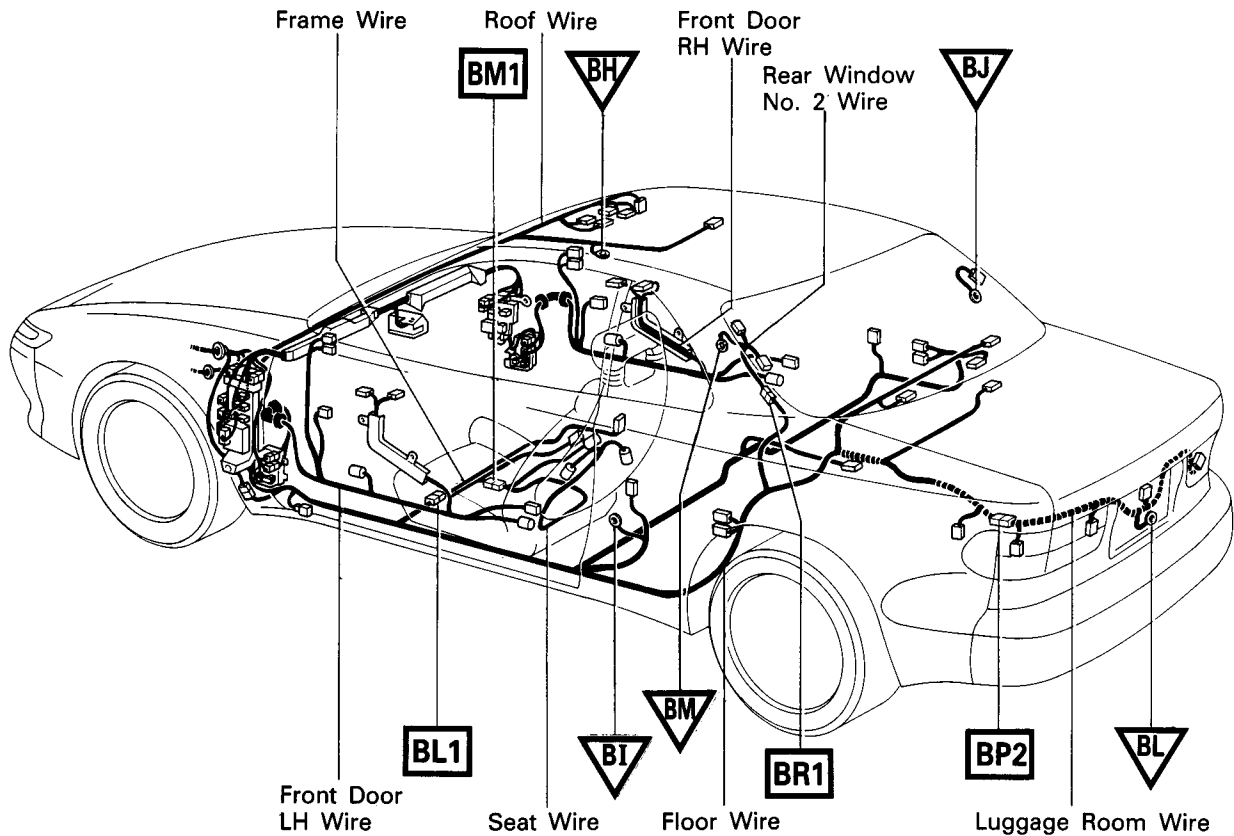
CODE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
BL1	FLOOR WIRE AND FRAME WIRE (LEFT SIDE OF FRONT FLOOR PANEL)
BM1	FRAME WIRE AND SEAT WIRE (UNDER THE DRIVER'S SEAT)
BN1	BACK DOOR NO. 1 WIRE AND FLOOR WIRE (LEFT SIDE OF PACKAGE TRAY TRIM)
BO1	BACK DOOR NO. 1 WIRE AND BACK DOOR NO. 1 SUB WIRE (BACK DOOR UPPER LEFT)
BP1	FLOOR WIRE AND LUGGAGE ROOM WIRE (LEFT QUARTER PANEL CENTER)
BV1	BACK DOOR NO. 1 SUB WIRE AND REAR WINDOW DEFOGGER WIRE (BACK DOOR RIGHT)

ELECTRICAL WIRING ROUTING

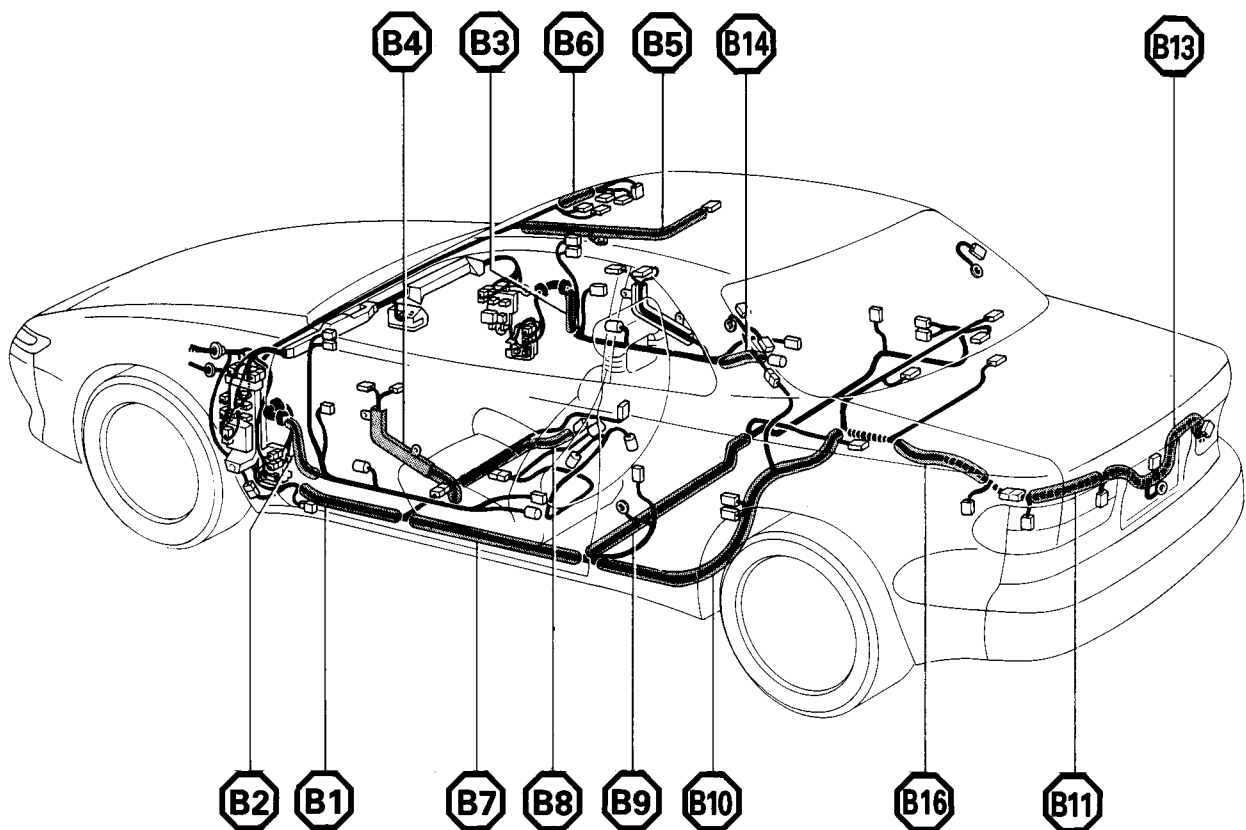
□ : Location of Connector Joining Wire Harness and Wire Harness

▽ : Location of Ground Points

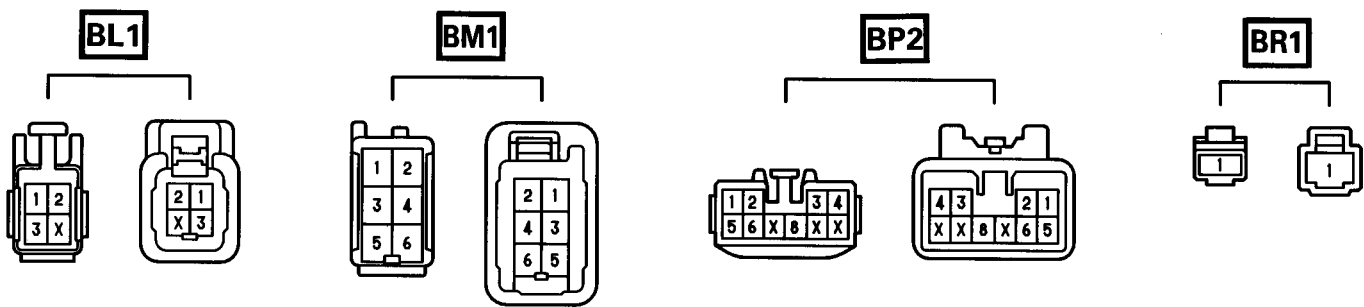
[C/P]



○ : Location of Splice Points



Connector Joining Wire Harness and Wire Harness



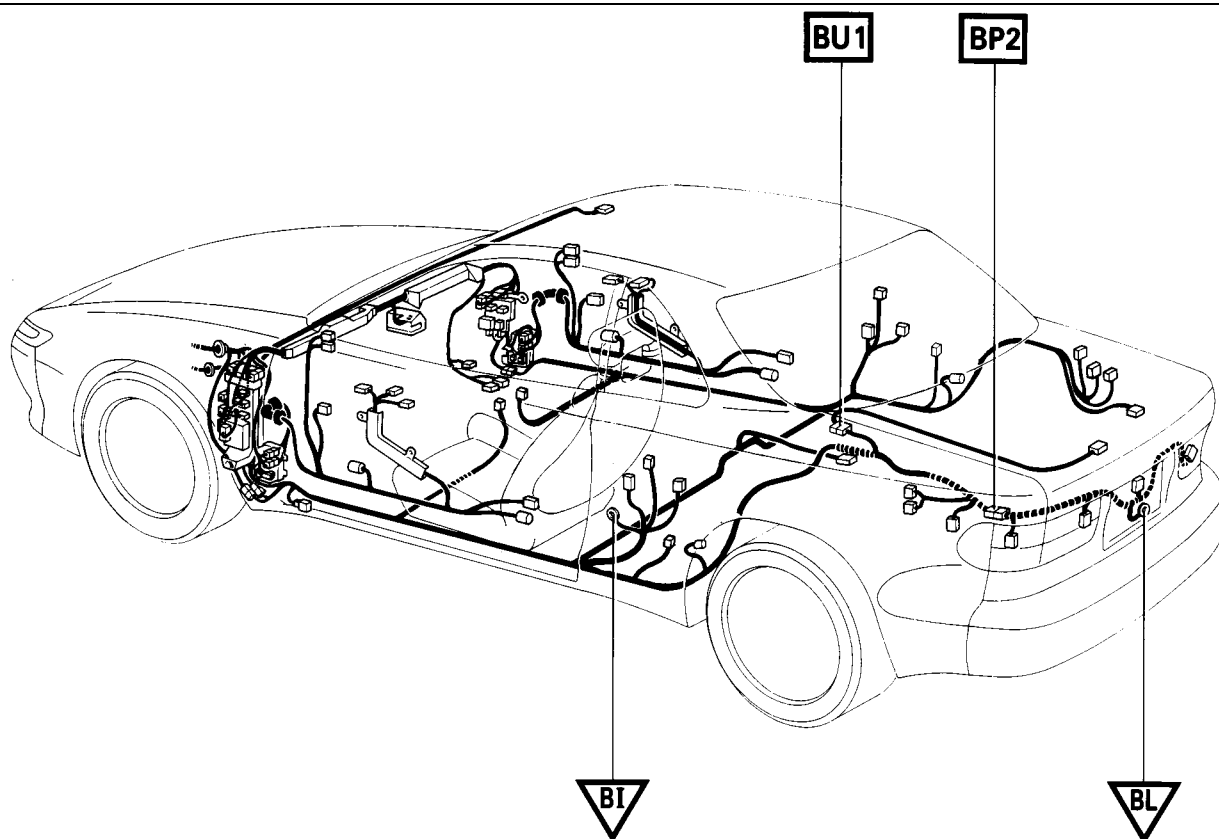
CODE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
BL1	FLOOR WIRE AND FRAME WIRE (LEFT SIDE OF FRONT FLOOR PANEL)
BM1	FRAME WIRE AND SEAT WIRE (UNDER THE DRIVER'S SEAT)
BP2	FLOOR WIRE AND LUGGAGE ROOM WIRE (LUGGAGE COMPARTMENT LEFT)
BR1	FLOOR WIRE AND REAR WINDOW NO. 2 WIRE (LEFT REAR PILLAR)

ELECTRICAL WIRING ROUTING

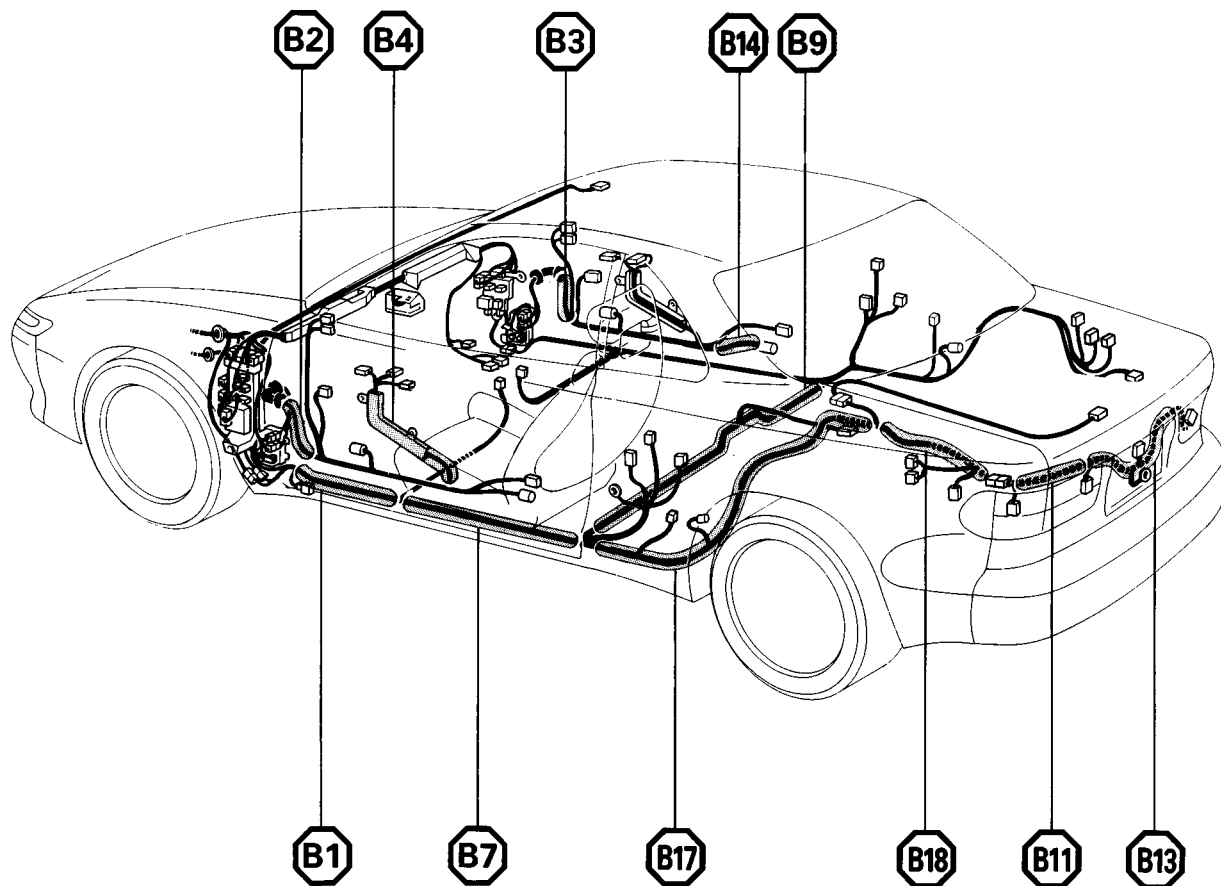
□ : Location of Connector Joining Wire Harness and Wire Harness

▽ : Location of Ground Points

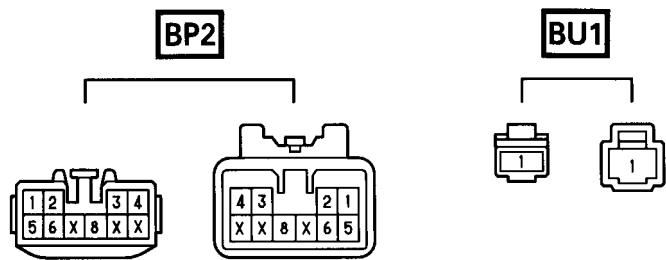
[C/P]



○ : Location of Splice Points



Connector Joining Wire Harness and Wire Harness

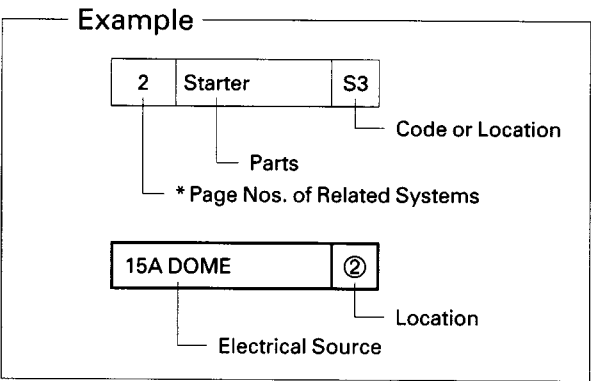
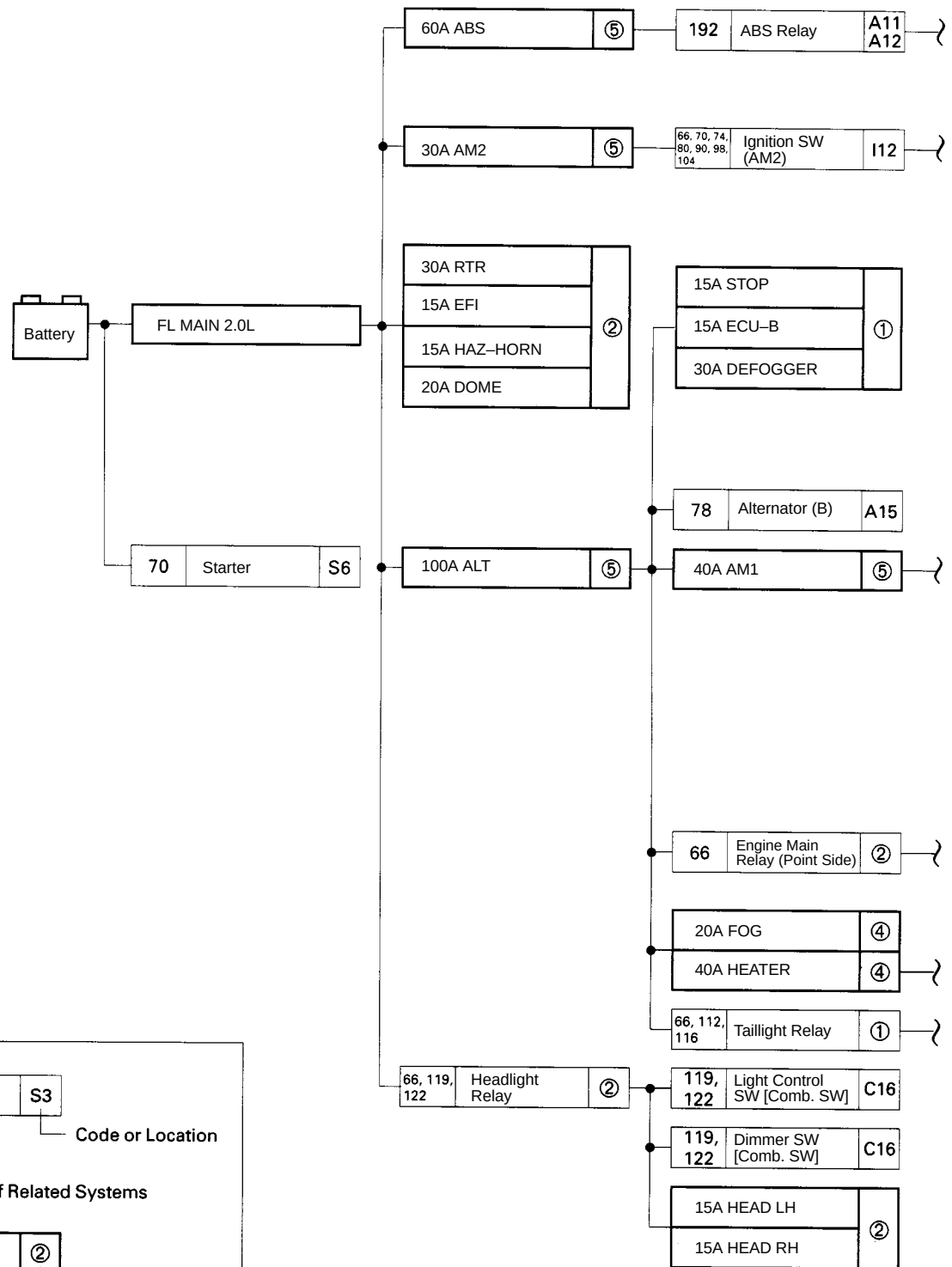


CODE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
BP2	FLOOR WIRE AND LUGGAGE ROOM WIRE (LUGGAGE COMPARTMENT LEFT)
BU1	BACK DOOR NO. 1 WIRE AND FLOOR WIRE (LUGGAGE COMPARTMENT LEFT)

POWER SOURCE (Current Flow Chart)

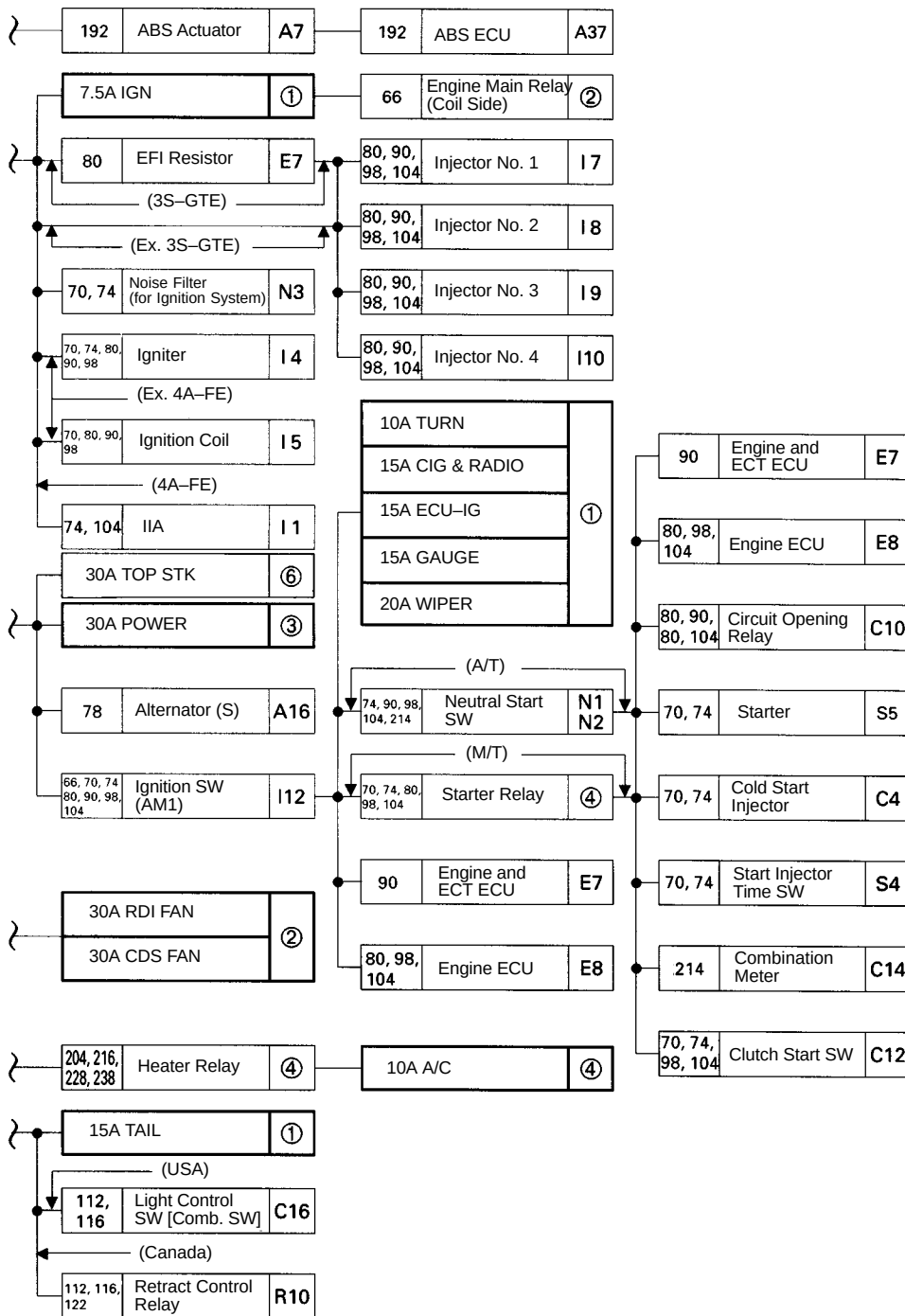
The chart below shows the route by which current flows from the battery to each electrical source (Fusible Link, Circuit Breaker, Fuse, etc.) and other parts.

The next page and following pages show the parts to which each electrical source outputs current.



* These are the page numbers of the first page on which the related system is shown.
The part indicated is located somewhere in the system, not necessarily on the page indicated here.

[LOCATION] (1) : J/B No. 1 (See page 20) (2) : J/B No. 2 (See page 22) (3) : R/B No. 2 (See page 26)



(4) : R/B No. 4 (See page 27) (2) : Fuse Box (F10 See on page 28) (3) : R/B No. 3 (See page 26)

POWER SOURCE (Current Flow Chart)

Location			*Page Nos. of Related Systems		218 230				192	78	118 230	112 230	112 218	218 230	218	218	218 230	112	218	176	192	206	192																																															
					Parts				Code or Location				CB or Fuse				A/C Condenser Fan Motor				A/C Idle-Up VSV																																																	
							A/C Magnetic Clutch				A/C Pressure SW				ABS Actuator				ABS Relay				Alternator				A/C Amplifier				A/C Control Assembly				A/C Power Transistor				A/C System Amplifier				Air Vent Mode Control Servo Motor				Ashtray Illumination				Auto A/C Amplifier				Auto Tilt Away ECU				ABS ECU				Auto Antenna Control Relay and Motor				ABS Deceleration Sensor (All-Trac/4WD)			
							A2	A3	A4	A5	A7	A11	A16	A17	A18	A19	A20	A22	A25	A30	A31	A32	A34	A36	A37	A38	A41	A43																																										
①	30A	DEFOGGER																																																																				
	20A	WIPER																																																																				
	15A	GAUGE		●	●	●	●	●		●		●	●	●		●	●	●	●	●	●																																																	
	7.5A	IGN				●			●																																																													
	15A	ECU-B																			●																																																	
	10A	TURN																																																																				
	15A	CIG & RADIO															●	●				●																																																
	15A	TAIL									●	●				●																																																						
	15A	ECU-IG							●												●	●	●																																															
	15A	STOP																			●																																																	
②	15A	HEAD LH																																																																				
	15A	HEAD RH																																																																				
	30A	RTR																																																																				
	15A	EFI																																																																				
	20A	DOME																●				●																																																
	15A	HAZ-HORN																																																																				
	30A	RDI FAN																																																																				
	30A	CDS FAN	●																																																																			
③	30A	POWER																	●																																																			
④	20A	FOG																																																																				
	10A	A/C		●		●				●		●																																																										
	40A	HEATER											●	●				●	●																																																			

* These are the page numbers of the first page on which the related system is shown.
The part indicated is located somewhere in the system, not necessarily on the page indicated here.

[LOCATION] (1) : J/B No. 1 (See page 20) (2) : J/B No. 2 (See page 22) (3) R/B No. 2 (See page 26)

[illegible]

The part indicated is located somewhere in the system, not necessarily on the page indicated here.

[LOCATION] (1) : J/B No. 1 (See page 20) (2) : J/B No. 2 (See page 22) (3) R/B No. 2 (See page 26)

[illegible]

POWER SOURCE (Current Flow Chart)

Location			*Page Nos. of Related Systems	116	136	136	136	132	132 169	160	104	214	104	112 169 174	80	162	140	112	138						
			Parts	L1	L2	L3	L4	N1	N2	N4	O1	O2	O6	O7	O8	P2	P3	P4	P5	P6	P7	P8	P9	Q1	Q2
			Code or Location	L1	L2	L3	L4	N1	N2	N4	O1	O2	O6	O7	O8	P2	P3	P4	P5	P6	P7	P8	P9	Q1	Q2
			CB or Fuse	L1	L2	L3	L4	N1	N2	N4	O1	O2	O6	O7	O8	P2	P3	P4	P5	P6	P7	P8	P9	Q1	Q2
①	30A	DEFOGGER								●															
	20A	WIPER																							
	15A	GAUGE					●	●		●	●		●												
	7.5A	IGN																							
	15A	ECU-B																							
	10A	TURN																							
	15A	CIG & RADIO																							
	15A	TAIL	●										●								●				
	15A	ECU-IG																							
	15A	STOP																							
②	15A	HEAD LH																							
	15A	HEAD RH																							
	30A	RTR																							
	15A	EFI										●		●											
	20A	DOME		●	●	●																			
	15A	HAZ-HORN																							
	30A	RDI FAN																							
	30A	CDS FAN																							
③	30A	POWER												●	●	●	●	●	●	●					
④	20A	FOG																							
	10A	A/C																							
	40A	HEATER						●																	

* These are the page numbers of the first page on which the related system is shown.
The part indicated is located somewhere in the system, not necessarily on the page indicated here.

[LOCATION] (1) : J/B No. 1 (See page 20) (2) : J/B No. 2 (See page 22) (6) R/B No. 2 (See page 26)

[illegible]

POWER SOURCE (Current Flow Chart)

Location			*Page Nos. of Related Systems		Parts		Code or Location		CB or Fuse		80	90	130	166	136	166	158	134	214	80	90	98	104	80	80	150	152	174	214	218	230	160	80	90	98	136	138	155	80	90	98	104	169	218	230	140	162	164
											80	90	130	166	136	166	158	134	214	80	90	98	104	80	80	150	152	174	214	218	230	160	80	90	98	136	138	155	80	90	98	104	169	218	230	140	162	164
											80	90	130	166	136	166	158	134	214	80	90	98	104	80	80	150	152	174	214	218	230	160	80	90	98	136	138	155	80	90	98	104	169	218	230	140	162	164
											80	90	130	166	136	166	158	134	214	80	90	98	104	80	80	150	152	174	214	218	230	160	80	90	98	136	138	155	80	90	98	104	169	218	230	140	162	164
			Stop Light SW	Sun Roof Control Relay	Personal Light [Sunroof SW]	Sun Roof Motor	Seat Belt Solenoid (Driver's)	Seat Belt Solenoid (Passenger's)	Turn Signal Flasher	Turbo Charging Pressure Sensor	VSV (for EGR System)	VSV (for Electrical Idle-Up System)	VSV (for Turbo Charging Pressure)	VSV (for T-VIS)	Water Motor	Water Temp. SW (for O/D System)	Water Temp. Sender	Water Temp. SW (for Fans Control)	Defogger Relay	Diode	Integration Relay	EFI Main Relay	Radiator Fan Relay No. 1	Power Main Relay																								
			S10	S11	S12	S14	S15	S16	T2	T3	V1	V2	V4	V5	W1	W2	W3	W4		①		②	③																									
①	30A	DEFOGGER																	●																													
	20A	WIPER													●																																	
	15A	GAUGE								●						●	●		●				●																									
	7.5A	IGN																●				●																										
	15A	ECU-B																																														
	10A	TURN							●												●																											
	15A	CIG & RADIO																			●																											
	15A	TAIL																	●		●																											
	15A	ECU-IG																																														
	15A	STOP	●																																													
②	15A	HEAD LH																																														
	15A	HEAD RH																																														
	30A	RTR																																														
	15A	EFI									●	●	●	●								●																										
	20A	DOME			●																●																											
	15A	HAZ-HORN							●																																							
	30A	RDI FAN																				●																										
	30A	CDS FAN																																														
③	30A	POWER		●		●																	●																									
④	20A	FOG																																														
	10A	A/C																																														
	40A	HEATER																																														

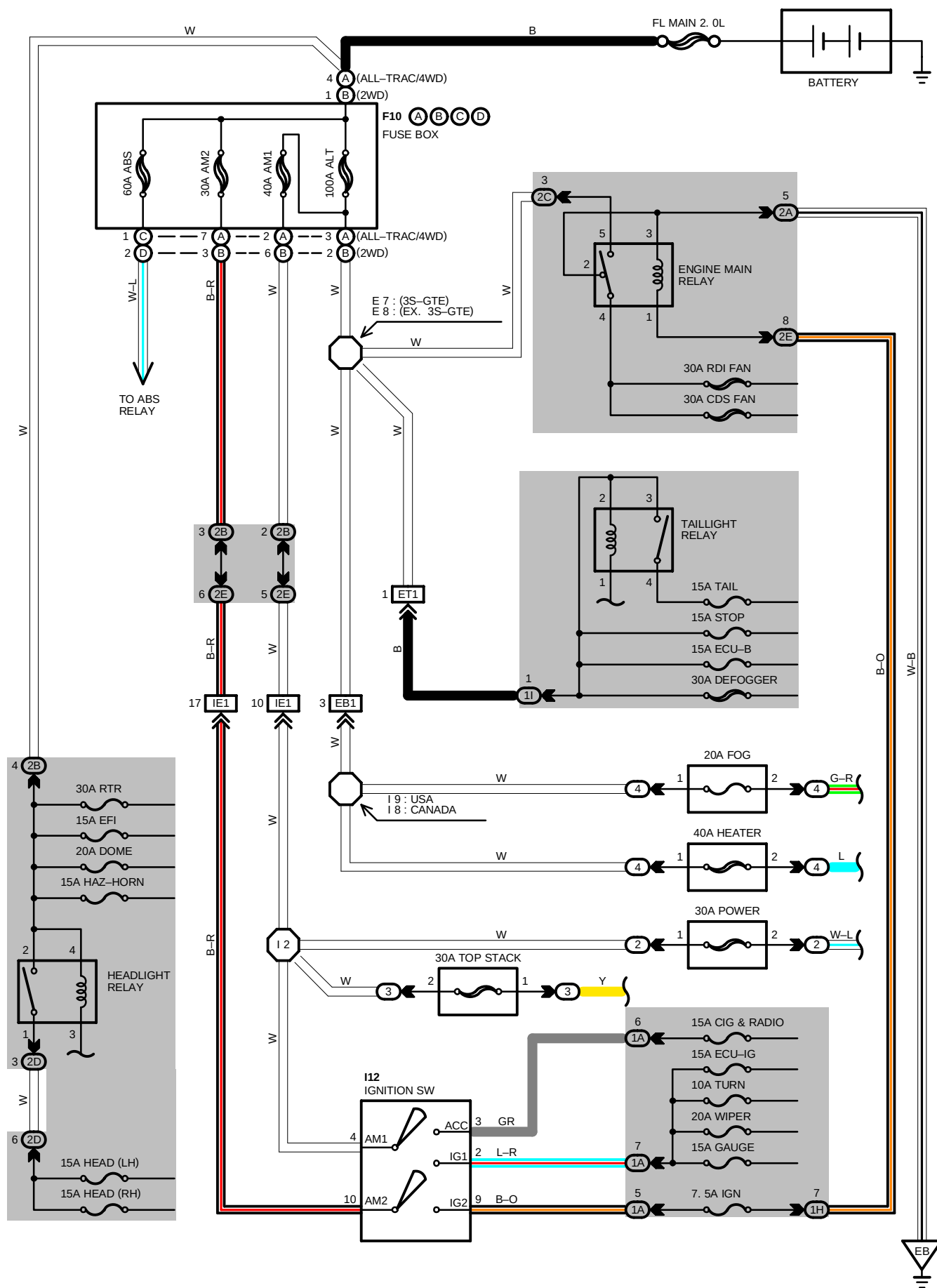
* These are the page numbers of the first page on which the related system is shown.
The part indicated is located somewhere in the system, not necessarily on the page indicated here.

[LOCATION] (1) : J/B No. 1 (See page 20) (2) : J/B No. 2 (See page 22) (3) : R/B No. 2 (See page 26)
(4) : R/B No. 4 (See page 27) (6) : R/B No. 3 (See page 26) (7) : R/B No. 5 (See page 27)

-Memo-

-Memo-

POWER SOURCE



SERVICE HINTS

TAILLIGHT RELAY

3-4 : CLOSED WITH LIGHT CONTROL SW AT **TAIL** OR **HEAD** POSITION
CLOSED WITH ENGINE RUNNING AND PARKING BRAKE LEVER RELEASED (CANADA)

ENGINE MAIN RELAY

4-5 : CLOSED WITH IGNITION SW AT **ON** OR **ST** POSITION

HEADLIGHT RELAY

1-2 : CLOSED WITH LIGHT CONTROL SW AT **HEAD** POSITION OR DIMMER SW AT **FLASH** POSITION

I12 IGNITION SW

4-3 : CLOSED WITH IGNITION KEY AT **ACC** OR **ON** POSITION
9-10 : CLOSED WITH IGNITION KEY AT **ON** OR **ST** POSITION



: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
F10	A	28 (3S-GTE)	F10	C	28 (3S-GTE)
	B	30 (5S-FE), 31 (4A-FE)		D	30 (5S-FE), 31 (4A-FE)
				I12	33



: RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
2	26	R/B NO. 2 (LEFT KICK PANEL)
3	26	R/B NO. 3 (RIGHT KICK PANEL)
4	27	R/B NO. 4 (RIGHT KICK PANEL)



: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1I		
2A	22 (2WD) 23 (ALL-TRAC/4WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2B	22 (2WD) 23 (ALL-TRAC/4WD)	
2C	22 (2WD) 23 (ALL-TRAC/4WD)	ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2D	22 (2WD) 23 (ALL-TRAC/4WD)	
2E	22 (2WD) 23 (ALL-TRAC/4WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)



: CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EB1	38 (3S-GTE)	ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER)
	40 (5S-FE)	
	42 (4A-FE)	
ET1	38 (3S-GTE)	ENGINE ROOM MAIN WIRE AND ENGINE WIRE (NEAR THE J/B NO. 2)
	40 (5S-FE)	
	42 (4A-FE)	
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)



: GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
EB	38 (3S-GTE)	FRONT LEFT FENDER
	40 (5S-FE)	
	42 (4A-FE)	

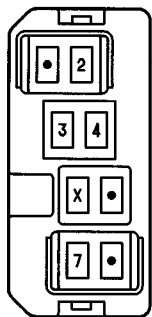


: SPLICE POINTS

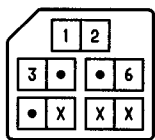
CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 7	38 (3S-GTE)	ENGINE WIRE	I 2	46	COWL WIRE
E 8	38 (3S-GTE)		I 8		
	40 (5S-FE)		I 9		
	42 (3S-GTE)				

POWER SOURCE

(ALL-TRAC/4WD) F10 (A) BLACK



(2WD) F10 (B) BLACK



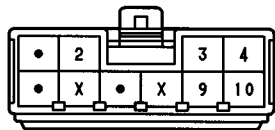
(ALL-TRAC/4WD) F10 (C) BLACK



(2WD) F10 (D) BLACK

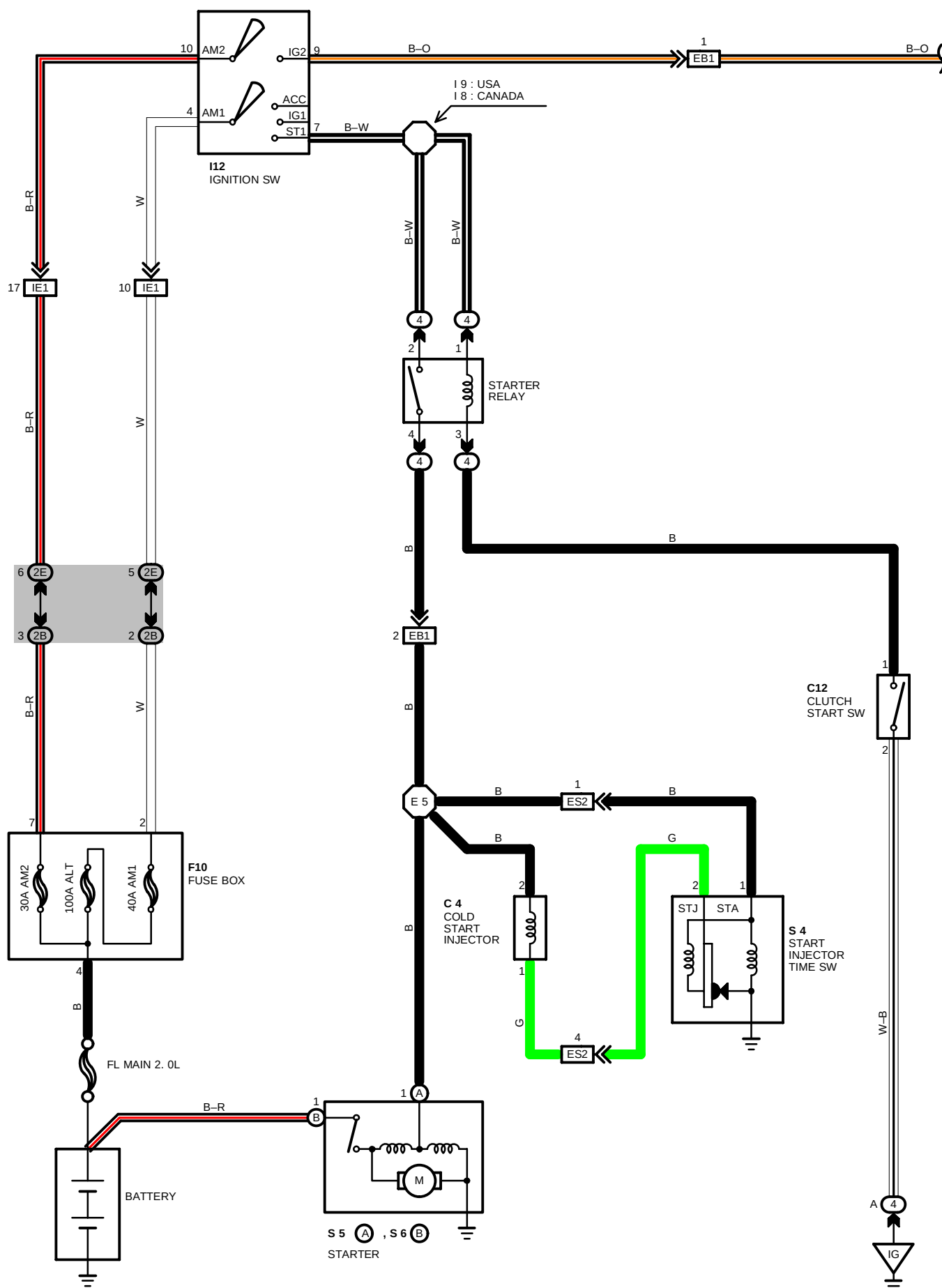


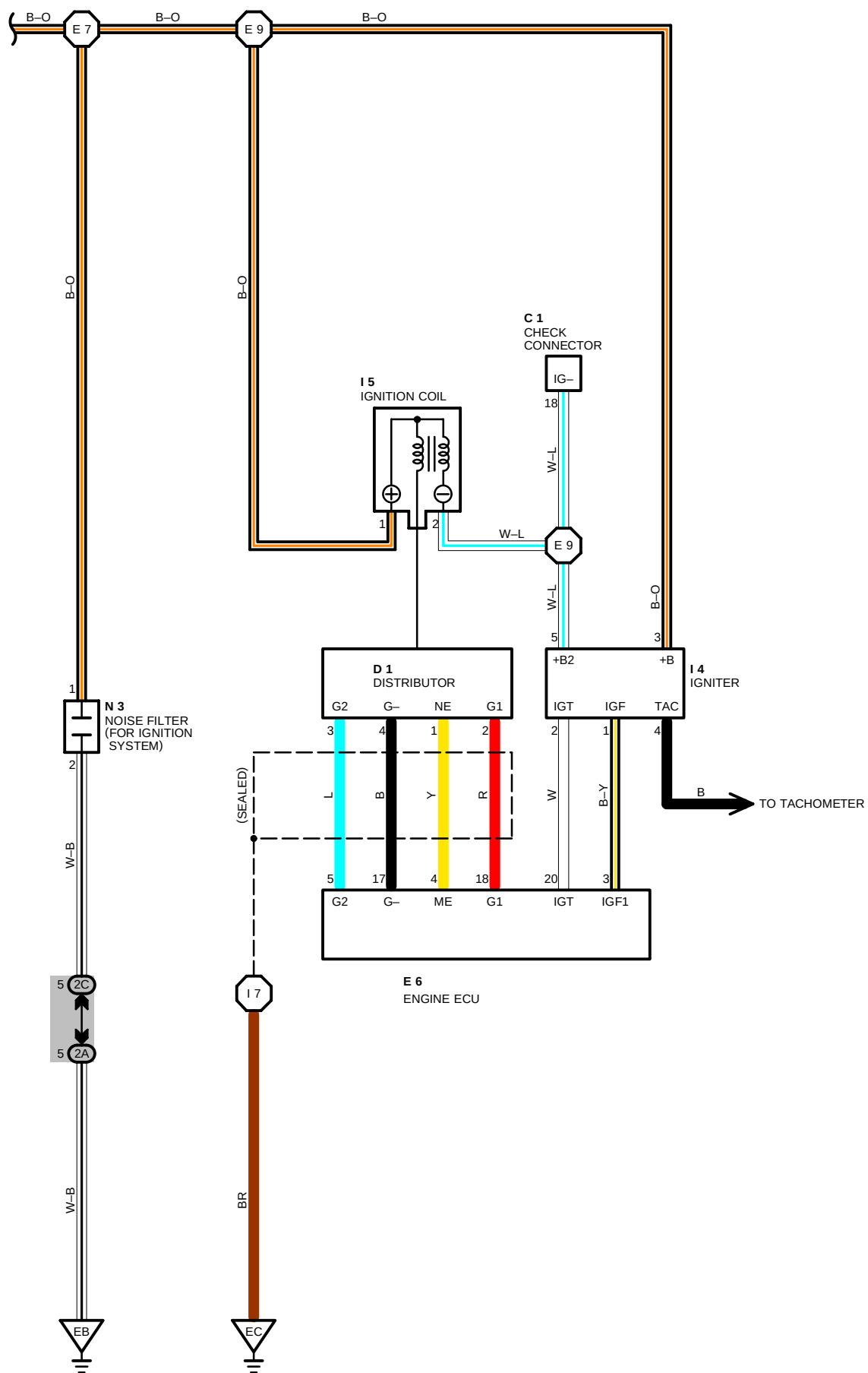
I12 BLACK



-Memo-

STARTING AND IGNITION (ALL-TRAC/4WD)





STARTING AND IGNITION (ALL-TRAC/4WD)

SERVICE HINTS

STARTER

POINTS CLOSED WITH CLUTCH START SW ON AND IGNITION SW AT **ST** POSITION

STARTER RELAY

(4) 2-(4) 4 : CLOSED WITH CLUTCH START SW ON AND IGNITION SW AT **ST** POSITION

C 4 COLD START INJECTOR

1-2 : APPROX. 12 VOLTS WHILE START INJECTOR TIME SW IS CLOSED AND STARTER CRANKING

C12 CLUTCH START SW

1-2 : CLOSED WITH CLUTCH PEDAL FULLY DEPRESSED

I12 IGNITION SW

4-7 : CLOSED WITH IGNITION SW AT **ST** POSITION

9-10 : CLOSED WITH IGNITION SW AT **ON** OR **ST** POSITION

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C 1	28 (3S-GTE)	F10	28 (3S-GTE)	S 4	29 (3S-GTE)
C 4	28 (3S-GTE)	I 4	29 (3S-GTE)	S 5	A 29 (3S-GTE)
C12	32	I 5	29 (3S-GTE)	S 6	B 29 (3S-GTE)
D 1	28 (3S-GTE)	I12	33		
E 6	33	N 3	29 (3S-GTE)		

○ : RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
4	27	R/B NO. 4 (RIGHT KICK PANEL)

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
2A	23 (ALL-TRAC/4WD)	ENGINE ROOM MAIN WIRE AND J/B NO.2 (NEAR THE BATTERY)
2B	23 (ALL-TRAC/4WD)	ENGINE WIRE AND J/B NO.2 (NEAR THE BATTERY)
2C	23 (ALL-TRAC/4WD)	ENGINE WIRE AND J/B NO.2 (NEAR THE BATTERY)
2E	23 (ALL-TRAC/4WD)	ENGINE ROOM MAIN WIRE AND J/B NO.2 (NEAR THE BATTERY)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EB1	38 (3S-GTE)	ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER)
ES2	38 (3S-GTE)	ENGINE WIRE AND ENGINE ROOM NO. 2 WIRE (NEAR THE DISTRIBUTOR)
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)

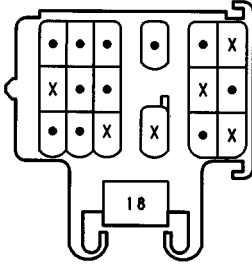
▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
EB	38 (3S-GTE)	FRONT LEFT FENDER
EC	38 (3S-GTE)	INTAKE MANIFOLD
IG	44	R/B NO. 4 SET BOLT

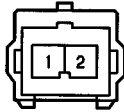
○ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 5			I 7	46	ENGINE WIRE
E 7	38 (3S-GTE)	ENGINE WIRE	I 8	46	
E 9			I 9	46	COWL WIRE

C 1 DARK GRAY



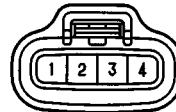
C 4 BLACK



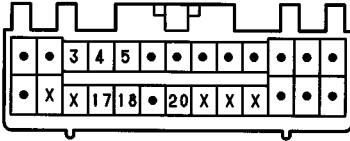
C12



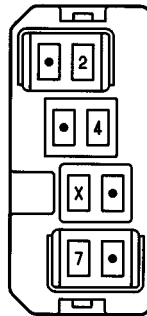
D 1 BLACK



E 6 DARK GRAY



F10 BLACK



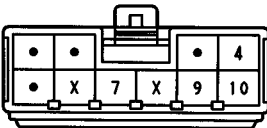
I 4 BLACK



I 5 BLACK



I12 BLACK



N 3 GRAY



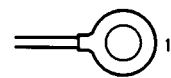
S 4 BROWN



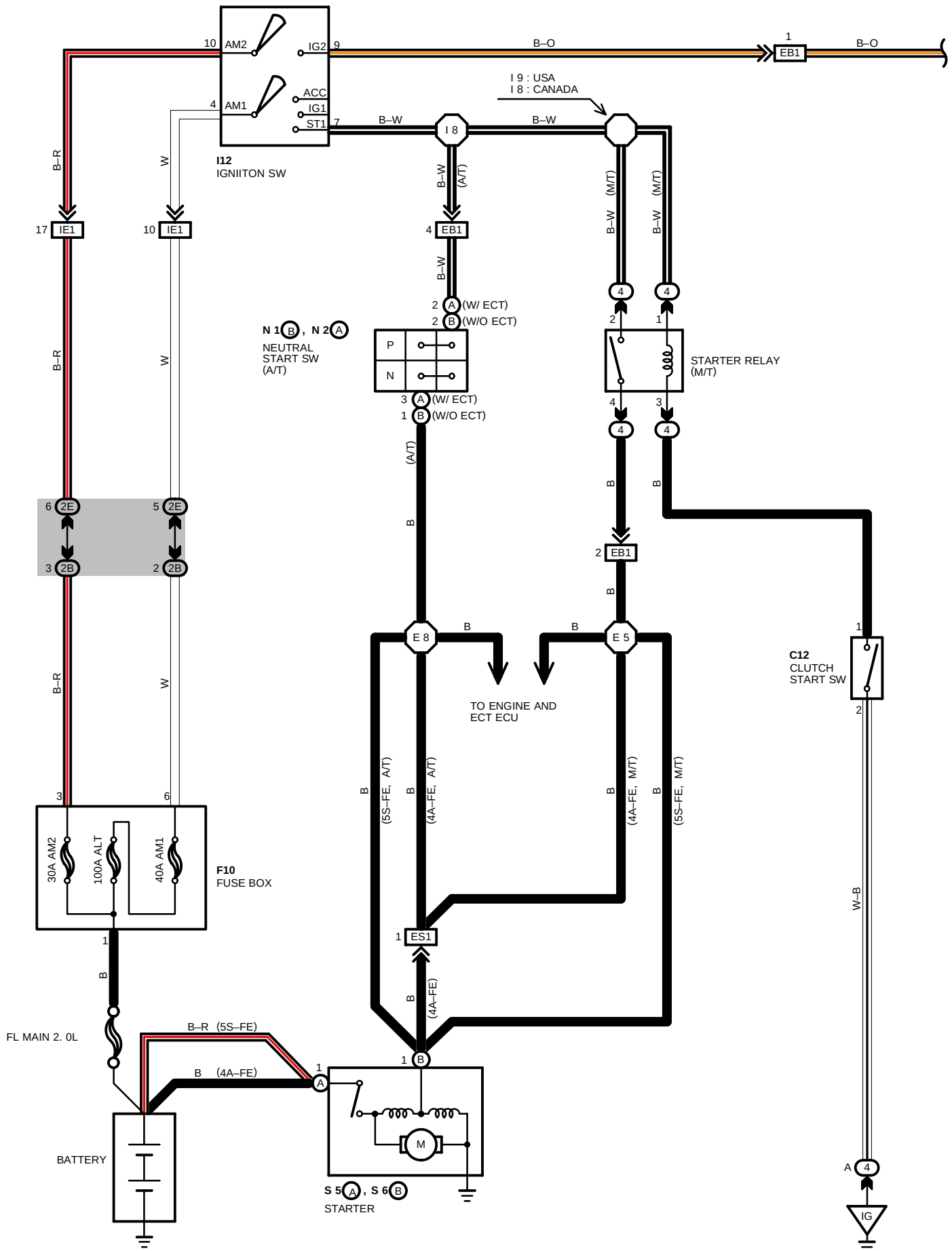
S 5 (A) BLACK



S 6 (B)



STARTING AND IGNITION (2WD)



STARTING AND IGNITION (2WD)

SERVICE HINTS

I12 IGNITION SW

4-7 : CLOSED WITH IGNITION SW AT **ST** POSITION

9-10 : CLOSED WITH IGNITION SW AT **ON** OR **ST** POSITION

C12 CLUTCH START SW

1-2 : CLOSED WITH CLUTCH PEDAL FULLY DEPRESSED

STARTER RELAY

(4) 2-(4) 4 : CLOSED WITH CLUTCH START SW ON AND IGNITION SW AT **ST** POSITION

STARTER

POINTS CLOSED WITH CLUTCH START SW ON AND IGNITION SW AT **ST** POSITION

N 1, N 2 NEUTRAL START SW (A/T)

2-3 : CLOSED WITH A/T SHIFT LEVER IN **P** OR **N** POSITION (W/ECT)

1-2 : CLOSED WITH A/T SHIFT LEVER IN **P** OR **N** POSITION (W/O ECT)

C 4 COLD START INJECTOR

1-2 : APPROX. 12 VOLTS WHILE START INJECTOR TIME SW IS CLOSED AND STARTER CRANKING



: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C 1	30 (5S-FE), 31 (4A-FE)	F10	30 (5S-FE), 31 (4A-FE)	N 2	A 30 (5S-FE)
C 4	30 (5S-FE), 31 (4A-FE)	I 1	A 31 (4A-FE)	N 3	30 (5S-FE), 31 (4A-FE)
C12	32	I 2	B 31 (4A-FE)	S 5	A 30 (5S-FE), 31 (4A-FE)
D 1	30 (5S-FE), 31 (4A-FE)	I 4	30 (5S-FE)	S 6	B 30 (5S-FE), 31 (4A-FE)
E 8	33	I 5	30 (5S-FE)		
E 8	A 33	I12	33		
	B 33	N 1	B 30 (5S-FE), 31 (4A-FE)		



: RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
4	27	R/B NO. 4 (RIGHT KICK PANEL)



: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
2A	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO.2 (NEAR THE BATTERY)
2B	22 (2WD)	ENGINE WIRE AND J/B NO.2 (NEAR THE BATTERY)
2C		
2E	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO.2 (NEAR THE BATTERY)



: CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EB1	40 (5S-FE) 42 (4A-FE)	ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER)
ES1	42 (4A-FE)	ENGINE WIRE AND ENGINE ROOM NO. 2 WIRE (NEAR THE STARTER)
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)



: GROUND POINTS

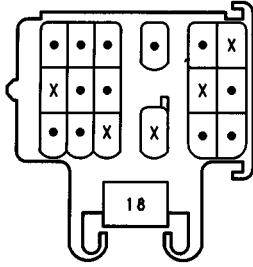
CODE	SEE PAGE	GROUND POINTS LOCATION
EB	40 (5S-FE) 42 (4A-FE)	FRONT LEFT FENDER
EC	40 (5S-FE) 42 (4A-FE)	INTAKE MANIFOLD
IG	44	R/B NO. 4 SET BOLT



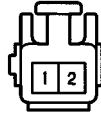
: SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 5	40 (5S-FE) 42 (4A-FE)	ENGINE WIRE	E 9	40 (5S-FE)	ENGINE WIRE
E 7	40 (5S-FE) 42 (4A-FE)		I 7	46	
			I 8	46	COWL WIRE
E 8	40 (5S-FE) 42 (4A-FE)		I 9	46	

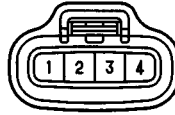
C 1 GRAY



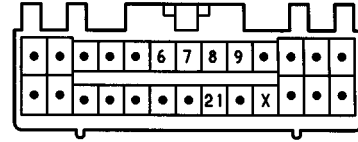
C12



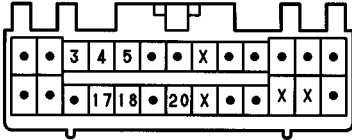
D 1 BLACK



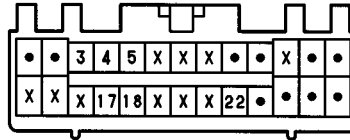
E 8 GRAY



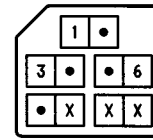
E 8(A) DARK GRAY



E 8(B) DARK GRAY



F10 BLACK



I 1(A) DARK GRAY



I 2(B) DARK GRAY



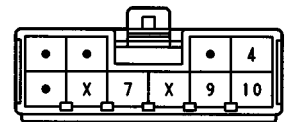
I 4 BLACK



I 5 BLACK



I12 BLACK



N 1(B) GRAY



N 2(A) GRAY



N 3 GRAY



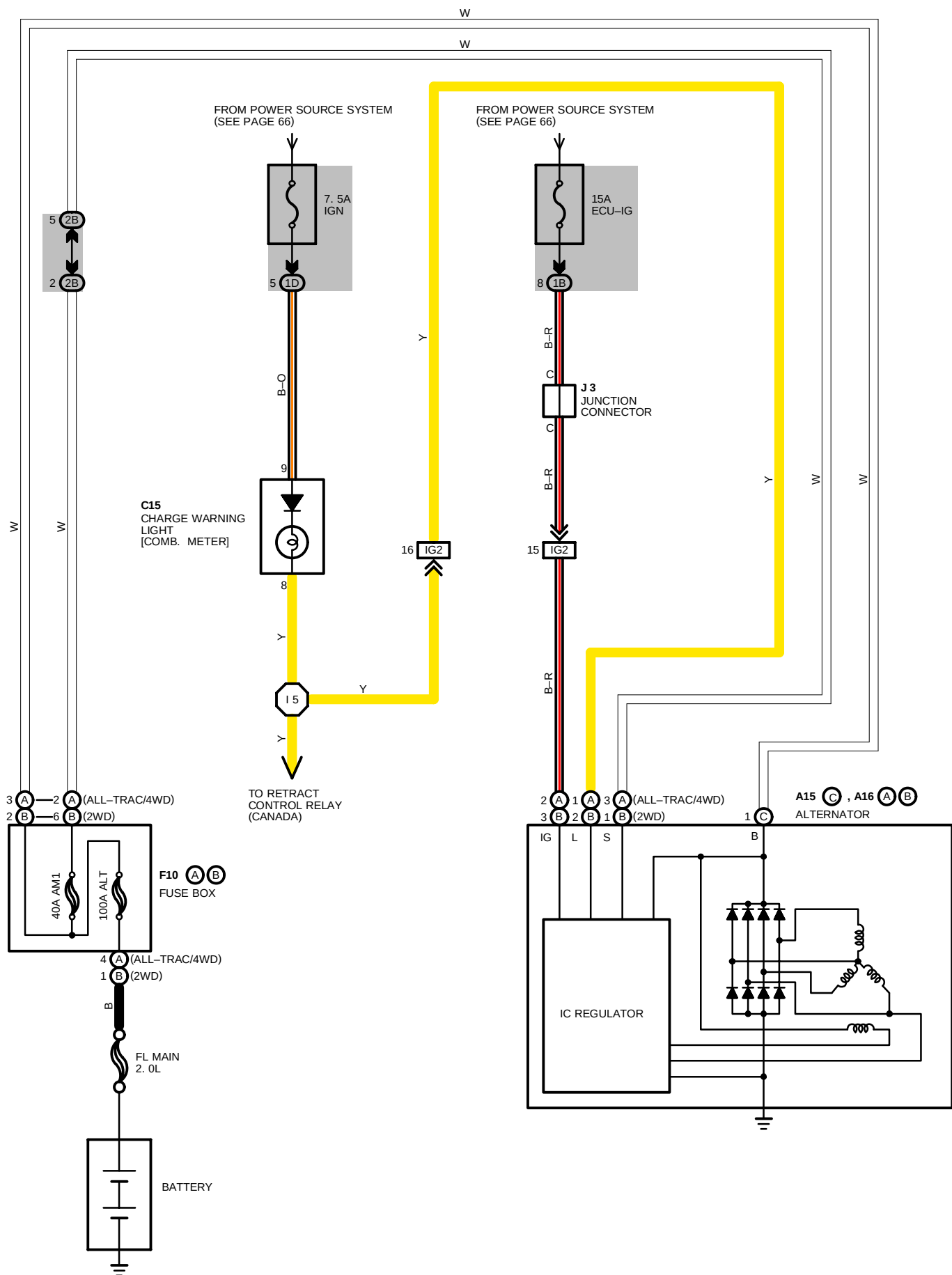
S 5(A)



S 6(B) BLACK



CHARGING



SERVICE HINTS

A15(C), A16(A) (B) ALTERNATOR

(C) 1-GROUND: 13.9–15.1 VOLTS WITH ENGINE RUNNING AT 2000 RPM AND 25°C (77°F)

13.5–14.3 VOLTS WITH ENGINE RUNNING AT 2000 RPM AND 115°C (239°F)

(A) 1, (B) 2 –GROUND : 0–4 VOLTS WITH IGNITION SW AT ON POSITION AND ENGINE NOT RUNNING

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
A15	C	28 (3S–GTE), 30 (5S–FE)	A16	B	30 (5S–FE), 31 (4A–FE)
	C	31 (4A–FE)	C15		32
A16	A	28 (3S–GTE)	F10	A	28 (3S–GTE)
			F10	B	30 (5S–FE), 31 (4A–FE)
			J 3		33

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

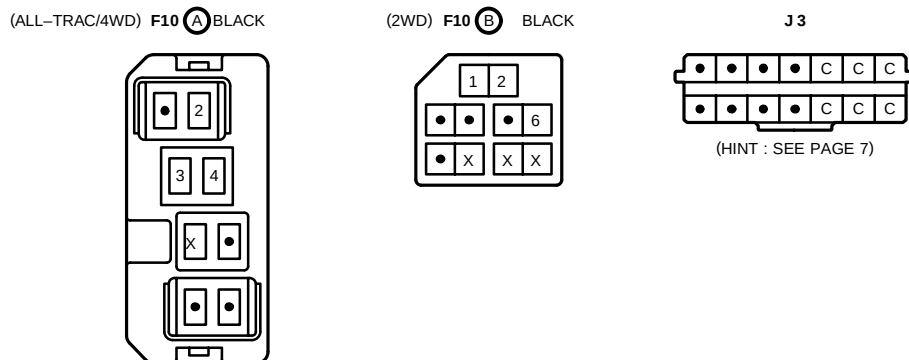
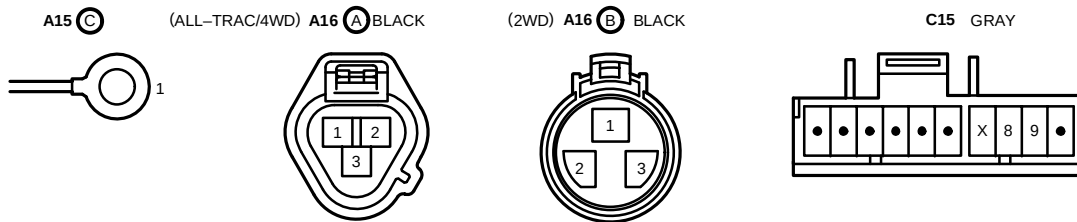
CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1B	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1D		
2B	22 (2WD)	ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)
	23 (ALL–TRAC/4WD)	

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IG2	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)

○ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I 5	46	COWL WIRE			



ENGINE CONTROL (3S-GTE)

SYSTEM OUTLINE

THE ENGINE CONTROL SYSTEM UTILIZES A MICROCOMPUTER AND MAINTAINS OVERALL CONTROL OF THE ENGINE AN OUTLINE OF ENGINE CONTROL IS GIVEN HERE.

1. INPUT SIGNALS

(1) WATER TEMP. SIGNAL SYSTEM

THE WATER TEMP. SENSOR DETECTS THE ENGINE COOLANT TEMP. AND HAS A BUILT-IN THERMISTOR WITH A RESISTANCE WHICH VARIES ACCORDING TO THE WATER TEMP. THUS THE WATER TEMP. IS INPUT IN THE FORM OF A CONTROL SIGNAL TO **TERMINAL THW** OF THE ECU.

(2) INTAKE AIR TEMP. SIGNAL SYSTEM

THE INTAKE AIR TEMP. SENSOR IS INSTALLED INSIDE THE AIR FLOW METER AND DETECTS THE INTAKE AIR TEMP. WHICH IS INPUT AS A CONTROL SIGNAL TO **TERMINAL THA** OF THE ECU.

(3) OXYGEN SENSOR SIGNAL SYSTEM

THE OXYGEN DENSITY IN THE EXHAUST EMISSIONS IS DETECTED AND INPUT AS A CONTROL SIGNAL TO **TERMINAL OX1** OF ECU. TO MAINTAIN STABLE DETECTION PERFORMANCE BY THE OXYGEN SENSOR, A HEATER IS USED FOR WARMING THE SENSOR. THE HEATER IS ALSO CONTROLLED BY THE ECU (HT).

(4) RPM SIGNAL SYSTEM

CRANKSHAFT POSITION IS DETECTED BY THE PICK-UP COIL INSTALLED INSIDE THE DISTRIBUTOR. CRANKSHAFT POSITION IS INPUT AS A CONTROL SIGNAL TO **TERMINALS G1** AND **G2** OF THE ECU, AND RPM IS INPUT TO **TERMINAL NE**.

(5) THROTTLE SIGNAL SYSTEM

THE THROTTLE POSITION SENSOR DETECTS THE THROTTLE VALVE OPENING ANGLE WHICH IS INPUT AS A CONTROL SIGNAL TO **TERMINAL VTA** OF THE ECU, OR WHEN THE VALVE IS FULLY CLOSED, TO **TERMINAL IDL**.

(6) VEHICLE SPEED SIGNAL SYSTEM

THE SPEED SENSOR, INSTALLED INSIDE THE COMBINATION METER, DETECTS THE VEHICLE SPEED AND INPUTS A CONTROL SIGNAL TO **TERMINAL SPD** OF THE ECU.

(7) A/C SW SIGNAL SYSTEM

THE OPERATING VOLTAGE OF THE A/C MAGNETIC CLUTCH IS DETECTED AND INPUT IN THE FORM OF A CONTROL SIGNAL TO **TERMINAL AC** OF THE ECU.

(8) BATTERY SIGNAL SYSTEM

VOLTAGE IS CONSTANTLY APPLIED TO **TERMINAL BATT** OF THE ECU. WHEN THE IGNITION SW IS TURNED TO ON, VOLTAGE FOR ECU OPERATION IS APPLIED VIA THE EFI MAIN RELAY TO **TERMINALS +B** AND **B1** OF THE ECU.

(9) INTAKE AIR VOLUME SIGNAL SYSTEM

INTAKE AIR VOLUME IS DETECTED BY THE POTENTIOMETER INSTALLED INSIDE THE AIR FLOW METER AND IS INPUT AS A CONTROL SIGNAL TO **TERMINAL VS** OF THE ECU.

(10) STOP LIGHT SW SIGNAL SYSTEM

THE STOP LIGHT SW IS USED TO DETECT WHETHER OR NOT THE VEHICLE IS BRAKING AND THE INFORMATION IS INPUT AS A CONTROL SIGNAL TO **TERMINAL STP** OF THE ECU.

(11) STA SIGNAL SYSTEM

TO CONFIRM THAT THE ENGINE IS CRANKING, THE VOLTAGE APPLIED TO THE STARTER MOTOR DURING CRANKING IS DETECTED AND IS INPUT AS A CONTROL SIGNAL TO **TERMINAL STA** OF THE ECU.

(12) ENGINE KNOCK CONTROL SYSTEM

ENGINE KNOCKING IS DETECTED BY THE KNOCK SENSOR AND INPUT AS A CONTROL SIGNAL TO **TERMINAL KNK** OF THE ECU.

(13) ELECTRICAL IDLE-UP SYSTEM

THE SIGNAL WHEN SYSTEMS SUCH AS THE REAR WINDOW DEFOGGER, HEADLIGHTS, ETC. WHICH CAUSE A HIGH ELECTRICAL BURDEN ARE ON IS INPUT TO **TERMINAL ELS** AS A CONTROL SIGNAL.

2. CONTROL SYSTEM

* EFI (ELECTRONIC FUEL INJECTION) SYSTEM

THE EFI SYSTEM MONITORS THE ENGINE CONDITIONS THROUGH THE SIGNALS EACH SENSOR (INPUT SIGNALS (1) TO (12)) INPUTS TO THE ECU. BASED ON THIS DATA AND THE PROGRAM MEMORIZED IN THE ECU. THE MOST APPROPRIATE FUEL INJECTION TIMING IS DECIDED AND CURRENT IS OUTPUT TO **TERMINALS #1, #2, #3** AND **#4** OF THE ECU. CAUSING THE INJECTORS TO OPERATE (TO INJECT FUEL). IT IS THIS SYSTEM. WHICH THROUGH THE WORK OF THE ECU. FINELY CONTROLS FUEL INJECTION IN RESPONSE TO DRIVING CONDITIONS.

* ESA (ELECTRONIC SPARK ADVANCE) SYSTEM

THE ESA SYSTEM MONITORS THE ENGINE CONDITIONS THROUGH THE SIGNALS (INPUT SIGNALS (1, 3, 4, 6, 7, 9, 11) INPUT TO THE ECU FROM EACH SENSOR. BASED ON THIS DATA AND THE PROGRAM MEMORIZED IN THE ECU. THE MOST APPROPRIATE INJECTION TIMING IS DECIDED AND CURRENT IS OUTPUT TO **TERMINAL IGT** OF THE ECU. THIS OUTPUT CONTROLS THE IGNITER TO PRODUCE THE MOST APPROPRIATE IGNITION TIMING FOR THE DRIVING CONDITIONS.

* FUEL PUMP CONTROL SYSTEM

COMPUTER OPERATION OUTPUTS TO **TERMINAL FPR** AND CONTROLS THE FUEL PUMP CONTROL RELAY AND THUS CONTROLS THE FUEL PUMP DRIVE SPEED IN RESPONSE TO CONDITIONS.

* OXYGEN SENSOR HEATER CONTROL SYSTEM

THE OXYGEN SENSOR HEATER CONTROL SYSTEM TURNS THE HEATER TO ON WHEN THE INTAKE AIR VOLUME IS LOW (TEMP. OF EXHAUST EMISSIONS LOW). AND WARMS UP THE OXYGEN SENSOR TO IMPROVE DETECTION PERFORMANCE OF THE SENSOR. THE ECU EVALUATES THE SIGNALS FROM EACH SENSOR (INPUT SIGNALS (1, 6, 8, 9, 11)), CURRENT IS OUTPUT TO **TERMINAL HT** AND CONTROLS THE HEATER.

* ISC (IDLE SPEED CONTROL) SYSTEM

THE ISC SYSTEM (ROTARY SOLENOID TYPE) INCREASES THE RPM AND PROVIDES IDLING STABILITY FOR FAST IDLE-UP WHEN THE ENGINE IS COLD AND WHEN THE IDLE SPEED HAS DROPPED DUE TO ELECTRICAL LOAD, ETC. THE ECU EVALUATES THE SIGNALS FROM EACH SENSOR (INPUT SIGNALS (1, 4 TO 8, 11, 13)), OUTPUTS CURRENT TO **TERMINALS ISC1** AND **ISC2** AND CONTROLS THE ISC VALVE.

* EGR CONTROL SYSTEM

WITH THE EGR CONTROL SYSTEM, THE ECU EVALUATES THE (INPUT SIGNALS (1, 4, 10)), FROM EACH SENSOR, CURRENT IS OUTPUT TO **TERMINAL EGR** AND OPERATION OF THE EGR VALVE IS CONTROLLED.

* INTAKE AIR CONTROL SYSTEM

IN THE INTAKE AIR CONTROL SYSTEM, EACH CYLINDER IN THE INTAKE MANIFOLD IS DIVIDED INTO TWO PARTS, WITH AN INTAKE AIR CONTROL VALVE INSTALLED IN THE PASSAGE ON ONE SIDE. THE OPENING AND CLOSING OF THE VALVE PROVIDES THE MOST APPROPRIATE INTAKE AIR FLOW AND, AS WELL AS PREVENTING PERFORMANCE LOSS AT LOW SPEEDS, ALSO IMPROVES FUEL ECONOMY. THE ECU EVALUATES THE SIGNALS FROM EACH SENSOR (INPUT SIGNALS (1, 4, 5)), OUTPUTS CURRENT TO **TERMINAL TVIS** CONTROLS THE VSV (FOR T-VIS) AND, CARRIES OUT OPENING AND CLOSING OF THE VALVE.

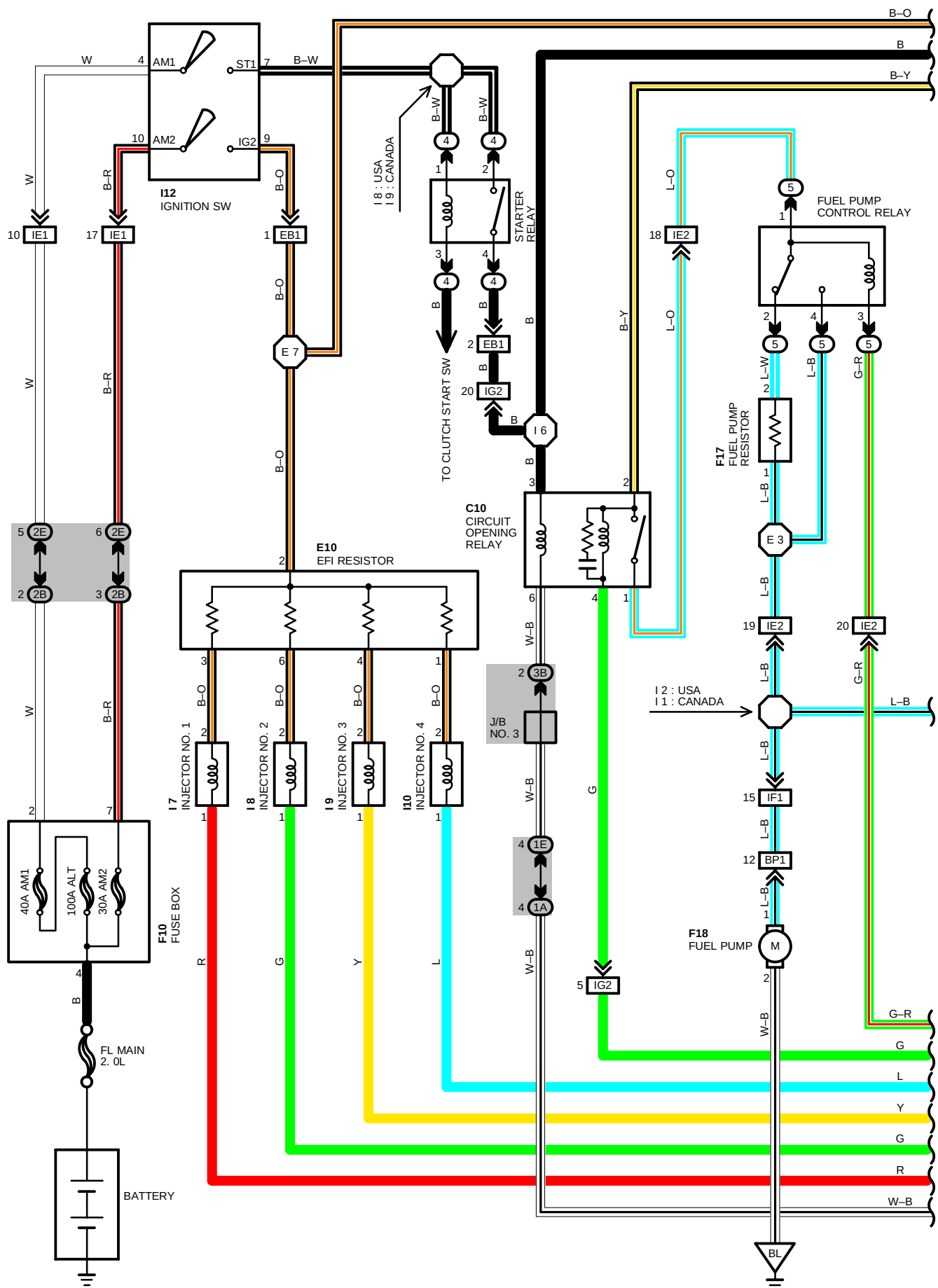
3. DIAGNOSIS SYSTEM

WITH THE DIAGNOSIS SYSTEM, WHEN THERE IS A MALFUNCTION IN THE ECU SIGNAL SYSTEM, THE MALFUNCTIONING SYSTEM IS RECORDED IN THE MEMORY. THE MALFUNCTIONING SYSTEM CAN THEN BE FOUND BY READING THE DISPLAY (CODE) OF THE CHECK ENGINE WARNING LIGHT.

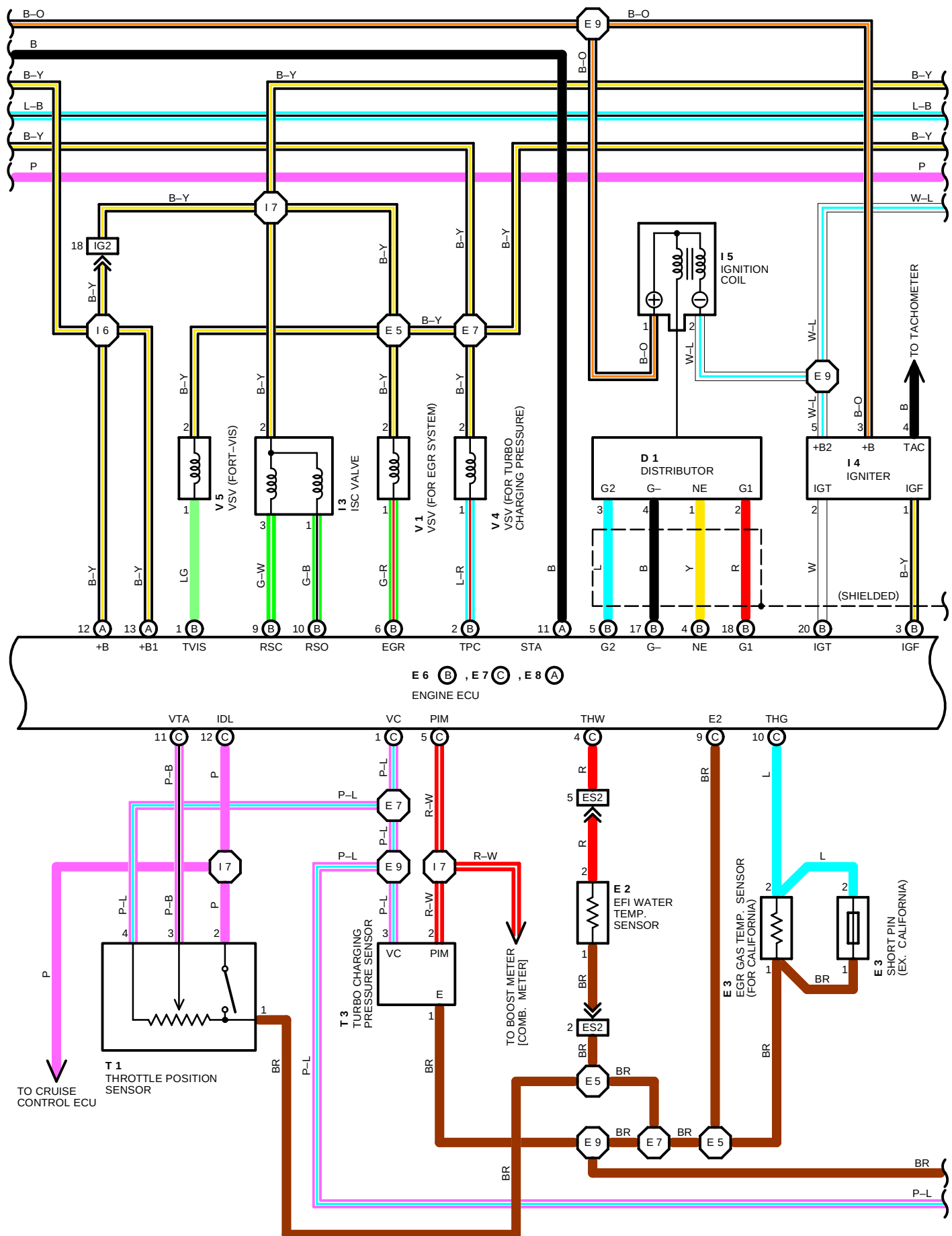
4. FAIL-SAFE SYSTEM

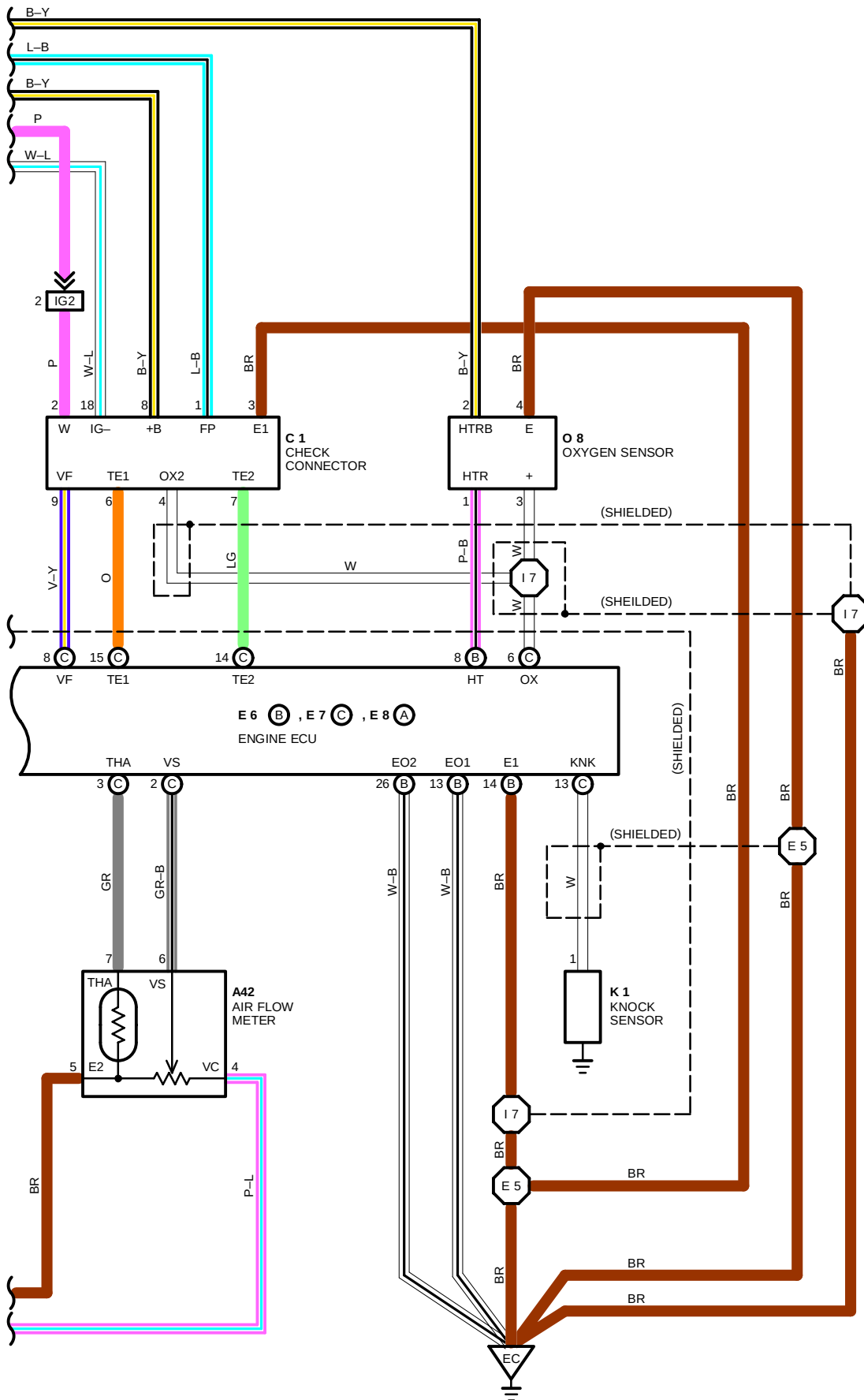
WHEN A MALFUNCTION OCCURS IN ANY SYSTEM, IF THERE IS A POSSIBILITY OF ENGINE TROUBLE BEING CAUSED BY CONTINUED CONTROL BASED ON THE SIGNALS FROM THAT SYSTEM. THE FAIL-SAFE SYSTEM EITHER CONTROLS THE SYSTEM BY USING DATA (STANDARD VALUES) RECORDED IN THE ECU MEMORY OR ELSE STOPS THE ENGINE.

ENGINE CONTROL (3S-GTE)



ENGINE CONTROL (3S-GTE)





ENGINE CONTROL (3S-GTE)

SERVICE HINTS

EFI MAIN RELAY

4-2 : CLOSED WITH IGNITION SW AT **ON** OR **ST** POSITION

E10 EFI RESISTOR

2-1, 3, 4, 5 : 5-7 Ω

I 7, I 8, I 9, I10 INJECTOR

1-2 : 2-4 Ω

F17 FUEL PUMP RESISTOR

1-2: APPROX. 73 Ω

A42 AIR FLOW METER

1-2 : CLOSED WITH STARTER RUNNING OR MEASURING PLATE OPEN

5-6 : 200-600 Ω (MEASURING PLATE CLOSED)

20-1000 Ω (MEASURING PLATE OPEN)

5-4 : 200-400 Ω

5-7 : 10-20 K Ω (-20°C, -4°F)

4-7 K Ω (0°C, 32°F)

2-3 K Ω (20°C, 68°F)

0.9-1.3 K Ω (40°C, 104°F)

0.4-0.7 K Ω (60°C, 140°F)

E 2 EFI WATER TEMP. SENSOR

1-2: 10-20 K Ω (-20°C, -4°F)

4-7 K Ω (0°C, 32°F)

2-7 K Ω (20°C, 68°F)

0.9-1.3 K Ω (40°C, 104°F)

0.4-0.7 K Ω (60°C, 140°F)

0.2-0.4 K Ω (80°C, 176°F)

T 1 THROTTLE POSITION SENSOR

2-4: 0.2-0.8 K Ω WITH CLEARANCE BETWEEN LEVER AND STOP SCREW 0 MM (0 IN)

3-4: LESS THAN 2.3 K Ω WITH CLEARANCE BETWEEN LEVER AND STOP SCREW 0.50 MM (0.020 IN)

∞ Ω WITH 0.7 MM (0.028 IN.)

2-4: 3.3-10 K Ω WITH THROTTLE VALVE FULLY OPEN

1-4: 3-8 K Ω

E 6, E 7, E 8 ENGINE ECU

VOLTAGE AT ECU CONNECTORS

BATT-E1 : 10-14 VOLTS

+B, +B1-E1 : 10-14 VOLTS (IGNITION SW ON)

IDL-E2 : 4-6 VOLTS (IGNITION SW ON AND THROTTLE VALVE OPEN)

VTA-E2 : 0.1-1.0 VOLTS (IGNITION SW ON AND THROTTLE VALVE FULLY CLOSED)

3-6 VOLTS (IGNITION SW ON AND THROTTLE VALVE OPEN)

VC-E2 : 4-6 VOLTS (IGNITION SW ON)

VS-E2 : 4-6 VOLTS (IGNITION SW ON AND MEASURING PLATE FULLY CLOSED)

1.0 VOLTS OR LESS (IGNITION SW ON AND MEASURING PLATE FULLY OPEN)

2.0-4.0 VOLTS (IDLING)

1.0-2.0 VOLTS (3000 RPM)

THA-E2 : 1.0-3.0 VOLTS (IGNITION SW ON AND INTAKE AIR TEMP. 20°C, (68°F))

THW-E2 : 0.1-1.0 VOLTS (IGNITION SW ON AND COOLANT TEMP. 80°C, (176°F))

STA-E1 : 6-14 VOLTS (CRANKING)

#1, #2, #3, #4 - E01, E02 : 10-14 VOLTS (IGNITION SW ON)

IGT-E1 : 0.7-1.0 VOLTS (CRANKING OR IDLING)

TVIS-E1 : 2.0 VOLTS OR LESS WITH IGNITION SW ON AND THROTTLE VALVE FULLY CLOSED

10-14 VOLTS WITH IGNITION SW ON AND THROTTLE VALVE OPEN

2.0 VOLTS OR LESS IDLING

10-14 VOLTS WITH 4200 RPM OR MORE

T-E1 : 10-14 VOLTS WITH IGNITION SW ON AND CHECK CONNECTOR T-E1 NO CONNECT

0.5 VOLTS OR LESS WITH IGNITION SW ON AND CHECK CONNECTOR T-E1 CONNECT

AC-E1 : 8-14 VOLTS WITH IGNITION SW ON A/C SWITCH ON

RSC, RSO -E1 : 9-14 VOLTS (IGNITION SW ON)

PIM-E2 : 2.5-4.5 VOLTS (IGNITION SW ON)

W-E1 : 10-14 VOLTS (NO TROUBLE (CHECK ENGINE WARNING LIGHT OFF) AND ENGINE RUNNING)

RESISTANCE AT ECU CONNECTORS

(DISCONNECT WIRING CONNECTOR FROM ECU)

IDL-E1 : INFINITY (THROTTLE VALVE OPEN)
LESS THAN 2300 Ω (THROTTLE VALVE FULLY CLOSED)

VTA-E2 : 3300-10000 Ω (THROTTLE VALVE OPEN)

200-800 Ω (THROTTLE VALVE FULLY CLOSED)

VS-E2 : 200-600 Ω (MEASURING PLATE FULLY CLOSED)

20-1200 Ω (MEASURING PLATE FULLY OPEN)

THA-E2 : 2000-3000 Ω (INTAKE AIR TEMP. 20°C, 68°F)

THW-E2 : 200-400 Ω (COOLANT TEMP. 80°C, 176°F)

G1, G2-G- : 140-180 Ω

NE-G- : 180-220 Ω

RSO, RSC-+B, +B1 : 17.7-23.9 Ω

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
A17	32	E10	28 (3S-GTE)	J 2	33
A42	28 (3S-GTE)	F10	28 (3S-GTE)	J 6	33
C 1	28 (3S-GTE)	F17	28 (3S-GTE)	K 1	33
C10	32	F18	34	O 8	29 (3S-GTE)
C13	B 32	I 3	29 (3S-GTE)	S10	33
C14	A 32	I 4	29 (3S-GTE)	T 1	29 (3S-GTE)
D 1	28 (3S-GTE)	I 5	29 (3S-GTE)	T 3	29 (3S-GTE)
E 2	28 (3S-GTE)	I 7	29 (3S-GTE)	V 1	29 (3S-GTE)
E 3	28 (3S-GTE)	I 8	29 (3S-GTE)	V 4	29 (3S-GTE)
E 6	B 33	I 9	29 (3S-GTE)	V 5	29 (3S-GTE)
E 7	C 33	I10	29 (3S-GTE)		
E 8	A 33	I12	33		

○ : RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
4	27	R/B NO. 4 (RIGHT KICK PANEL)
5	27	R/B NO. 5 (ENGINE COMPARTMENT FRONT RIGHT)

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1B		
1D		
1E		
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2A	23 (ALL-TRAC/4WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2B	23 (ALL-TRAC/4WD)	ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2C		
2D	23 (ALL-TRAC/4WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2E		
3B	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EB1	38 (3S-GTE)	ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER)
ES2	38 (3S-GTE)	ENGINE WIRE AND ENGINE ROOM NO. 2 WIRE (NEAR THE DISTRIBUTOR)
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)
IE2		
IF1	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
IG1	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)
IG2		
IH1	44	ENGINE WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)
IJ1	46	COWL WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)
BP1	48 (L/B)	FLOOR WIRE AND LUGGAGE ROOM WIRE (LEFT QUARTER PANEL CENTER)

▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
EB	38 (3S-GTE)	FRONT LEFT FENDER
EC	38 (3S-GTE)	INTAKE MANIFOLD
ID	44	LEFT KICK PANEL
BL	48 (L/B)	BACK PANEL CENTER

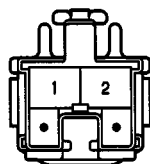
○ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 3	38 (3S-GTE)	ENGINE ROOM MAIN WIRE	I 2	46	COWL WIRE
E 5	38 (3S-GTE)	ENGINE WIRE	I 6		
E 7			I 7		ENGINE WIRE
E 9			I 8		COWL WIRE
I 1	46	COWL WIRE	I 9	46	

O 8 DARK GRAY



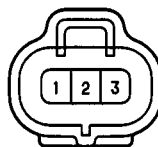
S10



T 1 BLACK



T 3 BLACK



V 1, V 5 BROWN



V 4 BLUE



ENGINE CONTROL (5S-FE)

SYSTEM OUTLINE

THE ENGINE CONTROL SYSTEM UTILIZES A MICROCOMPUTER AND MAINTAINS OVERALL CONTROL OF THE ENGINE, TRANSMISSION, ETC. AN OUTLINE OF ENGINE CONTROL IS GIVEN HERE.

1. INPUT SIGNALS

(1) WATER TEMP. SIGNAL SYSTEM

THE WATER TEMP. SENSOR DETECTS THE ENGINE COOLANT TEMP. AND HAS A BUILT-IN THERMISTOR WITH A RESISTANCE WHICH VARIES ACCORDING TO THE WATER TEMP. THUS THE WATER TEMP. IS INPUT IN THE FORM OF A CONTROL SIGNAL TO **TERMINAL THW** OF THE ECU.

(2) INTAKE AIR TEMP. SIGNAL SYSTEM

THE INTAKE AIR TEMP. SENSOR IS DETECTS THE INTAKE AIR TEMP., WHICH IS INPUT AS A CONTROL SIGNAL TO **TERMINAL THA** OF THE ECU.

(3) OXYGEN SENSOR SIGNAL SYSTEM

THE OXYGEN DENSITY IN THE EXHAUST EMISSIONS IS DETECTED AND INPUT AS A CONTROL SIGNAL TO **TERMINAL OX1** AND **OX2** (CALIFORNIA) OF THE ECU.

(4) RPM SIGNAL SYSTEM

CRANKSHAFT POSITION AND ENGINE RPM ARE DETECTED BY THE PICK-UP COIL INSTALLED INSIDE THE DISTRIBUTOR. CRANKSHAFT POSITION IS INPUT AS A CONTROL SIGNAL TO **TERMINAL G+** OF THE ECU, AND RPM IS INPUT TO **TERMINAL NE+**.

(5) THROTTLE SIGNAL SYSTEM

THE THROTTLE POSITION SENSOR DETECTS THE THROTTLE VALVE OPENING ANGLE, WHICH IS INPUT AS A CONTROL SIGNAL TO **TERMINAL PSW** OF THE ECU, OR WHEN THE VALVE IS FULLY CLOSED, TO **TERMINAL IDL**.

(6) VEHICLE SPEED SIGNAL SYSTEM

THE SPEED SENSOR, INSTALLED INSIDE THE COMBINATION METER, DETECTS THE VEHICLE SPEED AND INPUTS A CONTROL SIGNAL TO **TERMINAL SPD** OF THE ECU.

(7) NEUTRAL START SW SIGNAL SYSTEM

THE NEUTRAL START SW DETECTS WHETHER THE SHIFT POSITION IS IN NEUTRAL OR NOT, AND INPUTS A CONTROL SIGNAL TO **TERMINAL NSW** OF THE ECU.

(8) A/C SW SIGNAL SYSTEM

THE OPERATING VOLTAGE OF THE A/C MAGNETIC CLUTCH IS DETECTED AND INPUT IN THE FORM OF A CONTROL SIGNAL TO **TERMINAL ACT** OF THE ECU, AND OPERATION A/C IDLE-UP VSV IS DETECTED AND INPUT IN THE FORM OF A CONTROL SIGNAL TO **TERMINAL ACA** OF THE ECU.

(9) BATTERY SIGNAL SYSTEM

VOLTAGE IS CONSTANTLY APPLIED TO **TERMINAL BATT** OF THE ECU. WHEN THE IGNITION SW IS TURNED TO ON, VOLTAGE FOR ECU OPERATION IS APPLIED VIA THE EFI MAIN RELAY TO **TERMINALS +B** AND **+B1** OF THE ECU.

(10) INTAKE AIR VOLUME SIGNAL SYSTEM

INTAKE AIR VOLUME IS DETECTED BY THE VACUUM SENSOR AND IS INPUT AS A CONTROL SIGNAL TO **TERMINAL PIM** OF THE ECU.

(11) STA SIGNAL SYSTEM

TO CONFIRM THAT THE ENGINE IS CRANKING, THE VOLTAGE APPLIED TO THE STARTER MOTOR DURING CRANKING IS DETECTED AND IS INPUT AS A CONTROL SIGNAL TO **TERMINAL STA** OF THE ECU.

(12) ELECTRICAL LOAD SIGNAL SYSTEM

THE SIGNAL WHEN SYSTEMS SUCH AS THE REAR WINDOW DEFOGGER, HEADLIGHTS, ETC. WHICH CAUSE A HIGH ELECTRICAL BURDEN ARE ON IS INPUT TO **TERMINAL ELS** AS A CONTROL SIGNAL.

2. CONTROL SYSTEM

* EFI (ELECTRONIC FUEL INJECTION) SYSTEM

THE EFI SYSTEM MONITORS THE ENGINE CONDITIONS THROUGH THE SIGNALS EACH SENSOR (INPUT SIGNALS (1) TO (12)) INPUTS TO THE ECU. BASED ON THIS DATA AND THE PROGRAM MEMORIZED IN THE ECU, THE MOST APPROPRIATE FUEL INJECTION TIMING IS DECIDED AND CURRENT IS OUTPUT TO **TERMINALS #10** AND **#20** OF THE ECU, CAUSING THE INJECTORS TO OPERATE (TO INJECT FUEL). IT IS THIS SYSTEM WHICH, THROUGH THE WORK OF THE ECU, FINELY CONTROLS FUEL INJECTION IN RESPONSE TO DRIVING CONDITIONS.

DURING ENGINE CRANKING (SIGNAL INPUT TO **TERMINAL STA**) OR FOR APPROX. 2 SECONDS AFTER NE SIGNAL INPUT, ECU OPERATION ENERGIZES (POINT CLOSED) THE FUEL PUMP CIRCUIT INSIDE THE CIRCUIT OPENING RELAY, CAUSING THE FUEL PUMP TO OPERATE.

* ESA (ELECTRONIC SPARK ADVANCE) SYSTEM

THE ESA SYSTEM MONITORS THE ENGINE CONDITIONS USING THE SIGNALS (INPUT SIGNALS (1, 4, 5, 10, 11)) INPUT TO THE ECU FROM EACH SENSOR. BASED ON THIS DATA AND THE PROGRAM MEMORIZED IN THE ECU, THE MOST APPROPRIATE IGNITION TIMING IS DECIDED AND CURRENT IS OUTPUT TO **TERMINAL IGT** OF THE ECU. THIS OUTPUT CONTROLS THE IGNITER TO PRODUCE THE MOST APPROPRIATE IGNITION TIMING FOR THE DRIVING CONDITIONS.

* ISC (IDLE AIR CONTROL) SYSTEM

THE ISC SYSTEM (ROTARY SOLENOID TYPE) INCREASES THE RPM AND PROVIDES IDLING STABILITY FOR FAST IDLE-UP WHEN THE ENGINE IS COLD AND WHEN THE IDLE SPEED HAS DROPPED DUE TO ELECTRICAL LOAD, ETC. THE ECU EVALUATES THE SIGNALS FROM EACH SENSOR (INPUT SIGNALS (1, 4 TO 8, 11, 12)), OUTPUTS CURRENT TO **TERMINAL ISCC** AND **ISCO**, AND CONTROLS THE ISC VALVE.

* EGR CONTROL SYSTEM

THE EGR CUT CONTROL SYSTEM CONTROLS THE VSV (FOR EGR) BY EVALUATING THE SIGNALS FROM EACH SENSOR INPUT TO THE ECU (INPUT SIGNALS (1, 5, 6, 9)) AND BY SENDING OUTPUT TO **TERMINAL EGR** OF THE ECU.

* A/C CUT CONTROL SYSTEM

WHEN THE VEHICLE SUDDENLY ACCELERATES FROM LOW ENGINE SPEED, THIS SYSTEM CUTS OFF AIR CONDITIONER OPERATION FOR A FIXED PERIOD OF TIME IN RESPONSE TO THE VEHICLE SPEED AND THROTTLE VALVE OPENING ANGLE IN ORDER TO MAINTAIN ACCELERATION PERFORMANCE.

THE ECU RECEIVES INPUT SIGNALS (5, 6), AND OUTPUTS SIGNALS TO **TERMINAL ACT**.

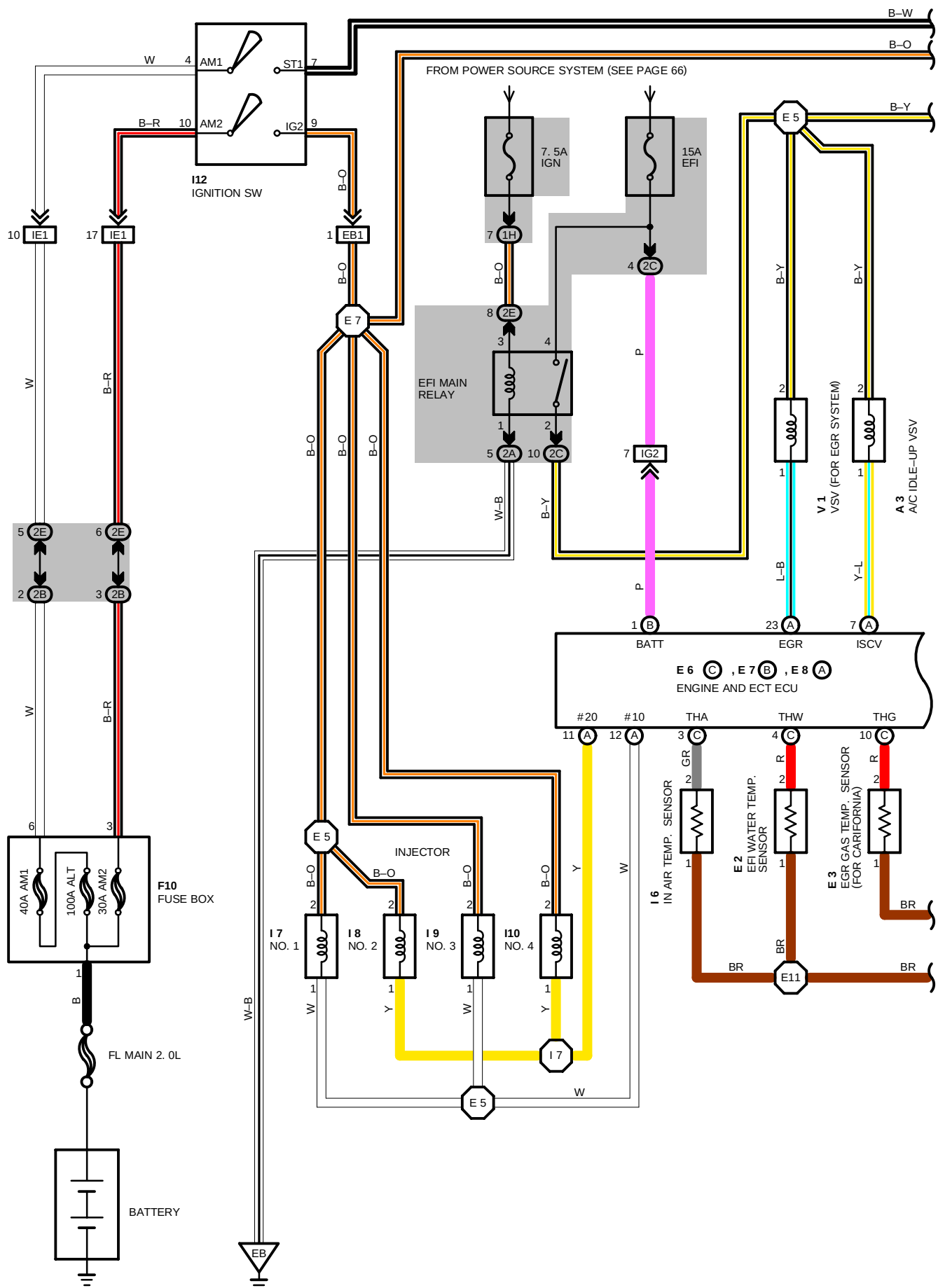
3. DIAGNOSIS SYSTEM

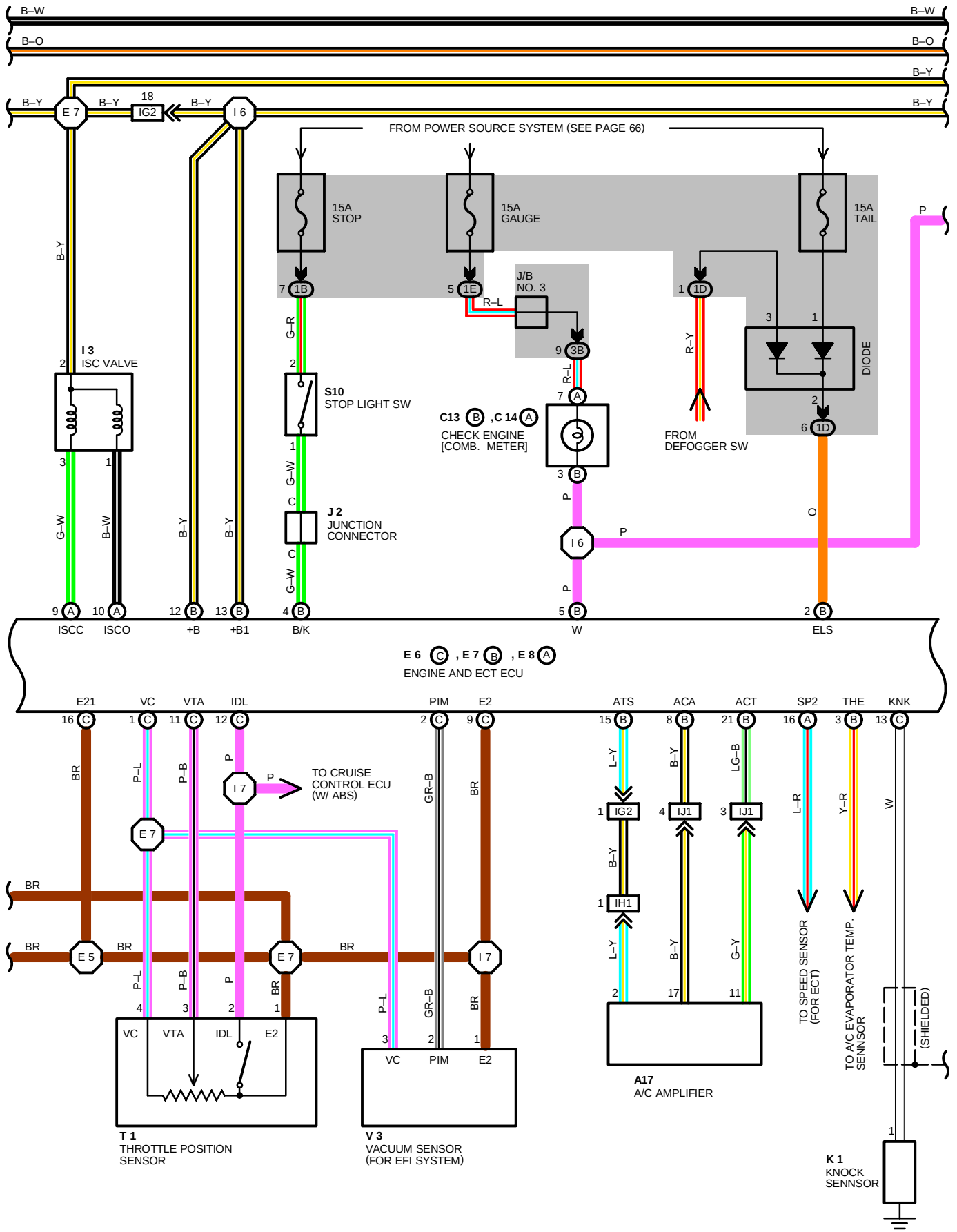
WITH THE DIAGNOSIS SYSTEM, WHEN THERE IS A MALFUNCTIONING IN THE ECU SIGNAL SYSTEM, THE MALFUNCTION SYSTEM IS RECORDED IN THE MEMORY. THE MALFUNCTIONING SYSTEM CAN THEN BE FOUND BY READING THE DISPLAY (CODE) OF THE CHECK ENGINE WARNING LIGHT.

4. FAIL-SAFE SYSTEM

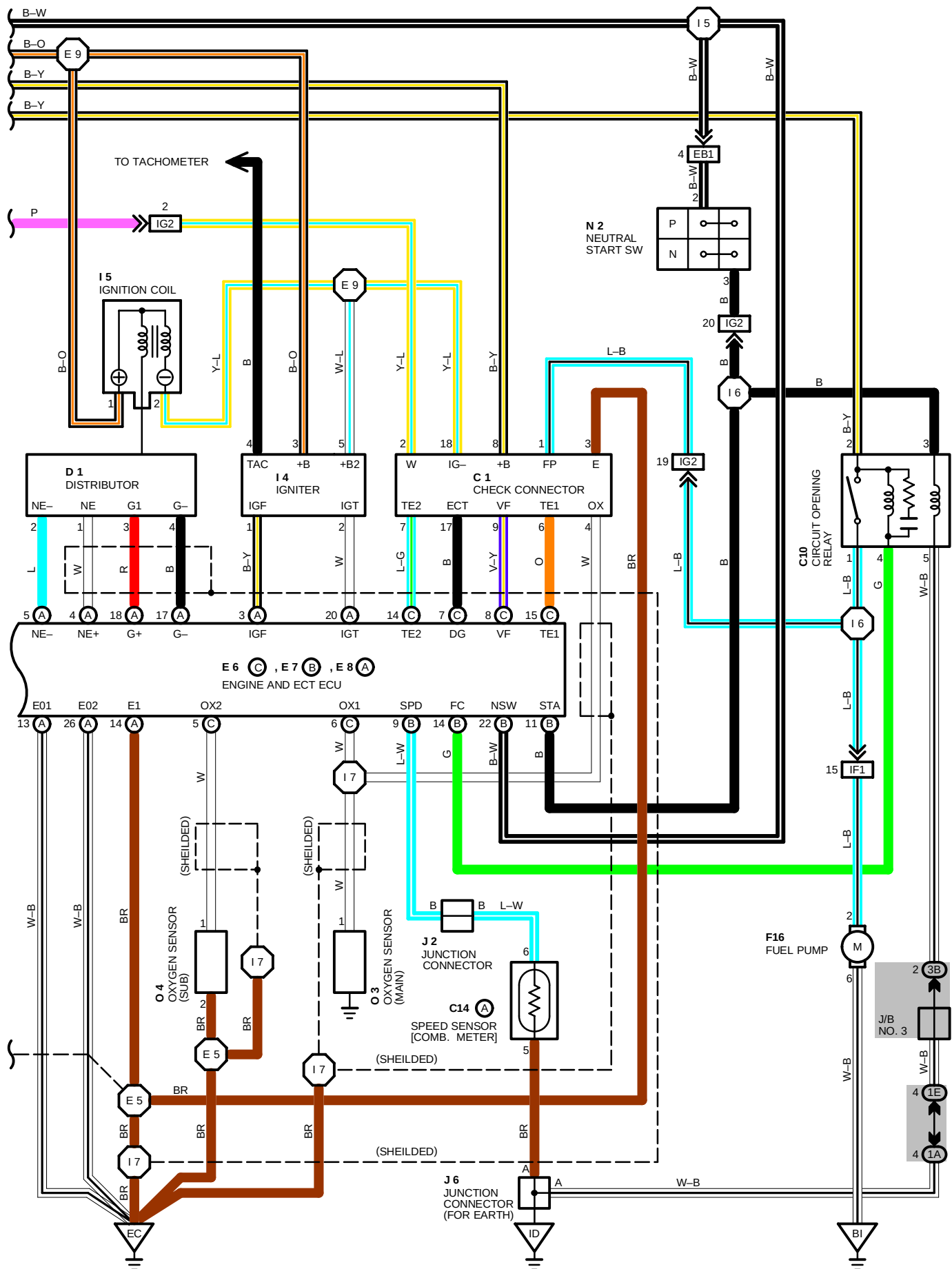
WHEN A MALFUNCTION OCCURS IN ANY SYSTEM, IF THERE IS A POSSIBILITY OF ENGINE TROUBLE BEING CAUSED BY CONTINUED CONTROL BASED ON THE SIGNALS FROM THAT SYSTEM, THE FAIL-SAFE SYSTEM EITHER CONTROLS THE SYSTEM BY USING DATA (STANDARD VALUES) RECORDED IN THE ECU MEMORY OR ELSE STOPS THE ENGINE.

ENGINE CONTROL (5S-FE A/T)

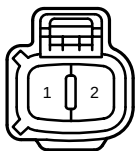




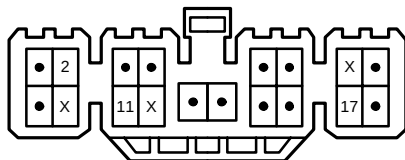
ENGINE CONTROL (5S-FE A/T)



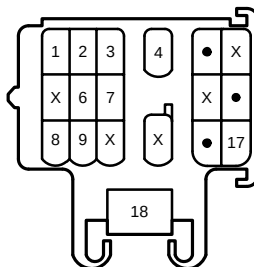
A 3 BLACK



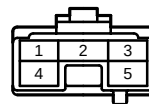
A17



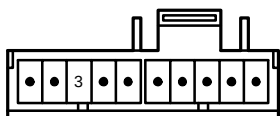
C 1 DARK GRAY



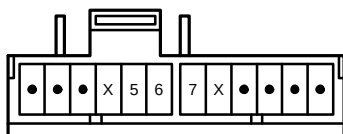
C10 DARK GRAY



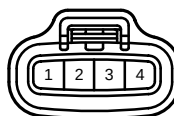
C13 (B) BROWN



C14 (A)



D 1 BLACK



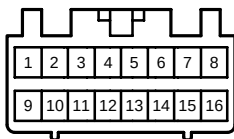
E 2 DARK GREEN



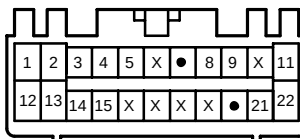
E 3 GRAY



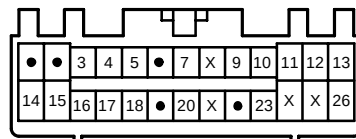
E 6 (C) DARK GRAY



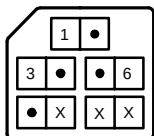
E 7 (B) DARK GRAY



E 8 (A) DARK GRAY



F10 BLACK



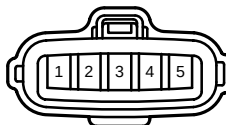
F16 DARK GRAY



I 3 GRAY



I 4 BLACK



I 5 BLACK



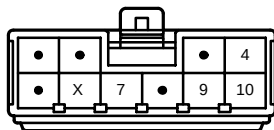
I 6 BLACK



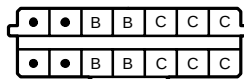
I 7, I 8, I 9, I10 GRAY



I12 BLACK

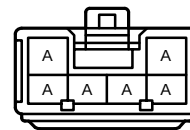


J 2



(HINT : SEE PAGE 7)

J 6



(HINT : SEE PAGE 7)

K 1 DARK GRAY



N 2 GRAY



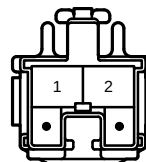
O 3 DARK GRAY



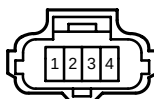
O 4 DARK GRAY



S10



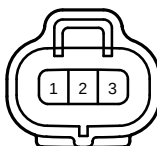
T 1 BLACK



V 1 BLUE



V 3 BLACK



ENGINE CONTROL (5S-FE A/T)

SERVICE HINTS

E 6, E7, E8 ENGINE AND ECT ECU

VOLTAGE AT ECU WIRING CONNECTOR

BATT-E1	: ALWAYS 10.0-14.0 VOLTS
+B-E1	: 10-14.0 VOLTS (IGNITION SW AT ON POSITION)
+B1-E1	: 10-14.0 VOLTS (IGNITION SW AT ON POSITION)
IDL-E2	: 8.0-14.0 VOLTS (IGNITION SW ON AND THROTTLE VALVE OPEN)
VC-E2	: 4.5-5.5 VOLTS (IGNITION SW AT ON POSITION)
VTA-E2	: 0.8-1.2 VOLTS (IGNITION SW ON AND THROTTLE VALVE FULLY CLOSED)
	: 3.2-4.2 VOLTS (IGNITION SW ON AND THROTTLE VALVE OPEN)
PIM-E2	: 3.3-3.9 VOLTS (IGNITION SW AT ON POSITION)
#10, #20 - E01, E02	: 10-14.0 VOLTS (IGNITION SW AT ON POSITION)
THA-E2	: 1.9-2.9 VOLTS (IGNITION SW ON AND INTAKE AIR TEMP. 20°C, 68°F)
THW-E2	: 0.1-1.1 VOLTS (IGNITION SW ON AND COOLANT TEMP. 80°C, 176°F)
STA-E1	: 6.0-14.0 VOLTS (ENGINE CRANKING)
IGT-E1	: 0.8-1.2 VOLTS (ENGINE CRANKING OR IDLING)
W-E1	: 8.0-14.0 VOLTS (NO TROUBLE AND ENGINE RUNNING)
ACT-E1	: 4.5-5.5 VOLTS (IGNITION SW ON AND AIR CONDITIONING ON)
ACA-E1	: 2.0 VOLTS OR LESS (IGNITION SW ON AND AIR CONDITIONING ON)
TE1-E1	: 10-14.0 VOLTS (IGNITION SW ON AND CHECK CONNECTOR T-E1 NOT CONNECTED)
	: 1.0 VOLTS OR LESS (IGNITION SW ON AND CHECK CONNECTOR T-E1 CONNECTED)
NSW-E1	: 0-2.0 VOLTS (IGNITION SW ON AND NEUTRAL START SW POSITION P OR N RANGE)
	: 6.0-14.0 VOLTS (IGNITION SW ON AND EX. NEUTRAL START SW POSITION P OR N RANGE)
B/K-E1	: 10-14.0 VOLTS (BRAKE PEDAL DEPRESSED)

RESISTANCE AT ECU WIRING CONNECTORS

(DISCONNECT WIRING CONNECTOR)

IDL-E2	: INFINITY (THROTTLE VALVE OPEN)
	: 2.3 KΩ OR LESS (THROTTLE VALVE FULLY CLOSED)
VTA-E2	: 3.3-10.0 KΩ (THROTTLE VALVE FULLY OPEN)
	: 0.2-0.8 KΩ (THROTTLE VALVE FULLY CLOSED)
VC-E2	: 3.0-7.0 KΩ
THA-E2	: 2.0-3.0 KΩ (INTAKE AIR TEMP. 20°C, 68°F)
THW-E2	: 0.2-0.4 KΩ (COOLANT TEMP. 80°C, 176°F)
G+, NE+ - G-	: 0.17-0.21 KΩ
ISCC, ISCO-B, +B1	: 19.3-22.3 Ω

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
A 3	30	E 8 A	33	I12	33
A17	32	F10	30 (5S-FE)	J 2	33
C 1	30 (5S-FE)	F16	34 (L/B)	J 6	33
C10	32	I 3	30 (5S-FE)	K 1	30
C13 B	32	I 4	30 (5S-FE)	N 2	30 (5S-FE)
C14 A	32	I 5	30 (5S-FE)	O 3	30 (5S-FE)
D 1	30 (5S-FE)	I 6	30 (5S-FE)	O 4	30 (5S-FE)
E 2	30 (5S-FE)	I 7	30 (5S-FE)	S10	33
E 3	30 (5S-FE)	I 8	30 (5S-FE)	T 1	30 (5S-FE)
E 6 C	33	I19	30 (5S-FE)	V 1	30 (5S-FE)
E 7 B	33	I10	30 (5S-FE)	V 3	30 (5S-FE)

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1B		
1D		
1E		
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2A	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2B	22 (2WD)	ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2C		
2E	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
3B	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

 : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EB1	40 (5S-FE)	ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER)
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)
IF1	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
IG2	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)
IH1	44	ENGINE WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)
IJ1	46	COWL WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)

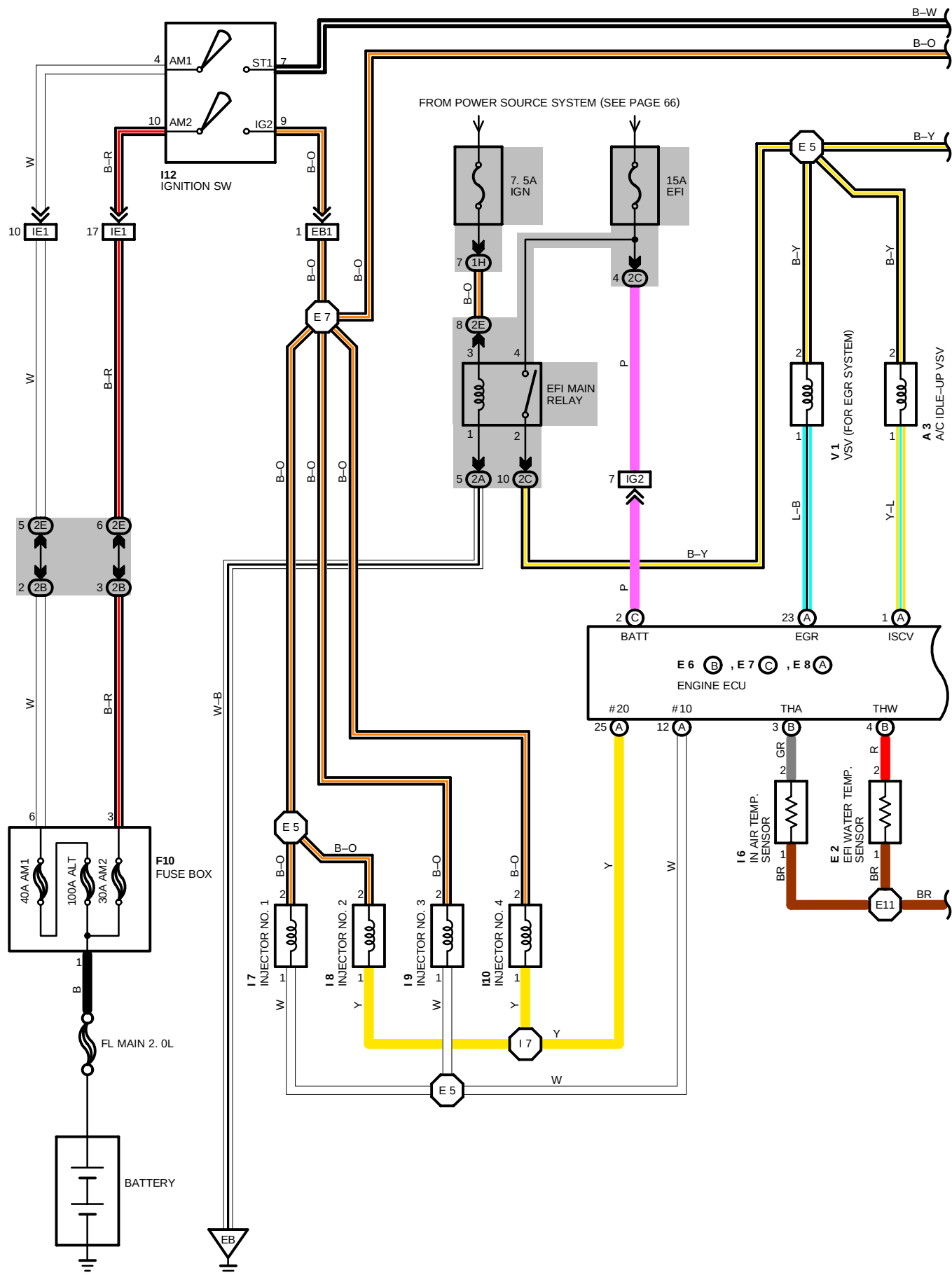
 : GROUND POINTS

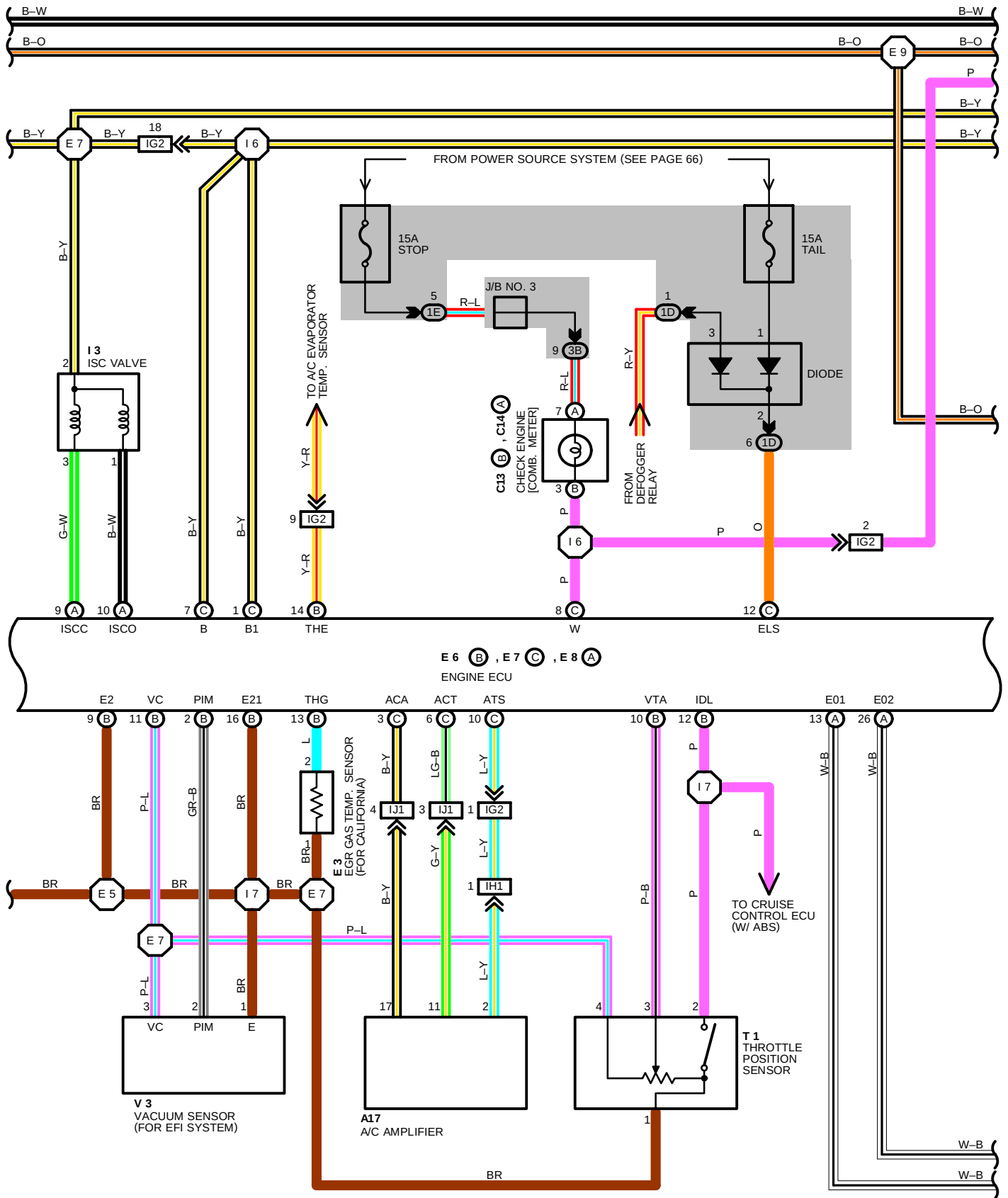
CODE	SEE PAGE	GROUND POINTS LOCATION
EB	40 (5S-FE)	FRONT LEFT FENDER
EC	40 (5S-FE)	INTAKE MANIFOLD
ID	44	LEFT KICK PANEL
BI	48 (L/B)	UNDER THE LEFT CENTER PILLAR

 : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 5	40 (5S-FE)	ENGINE WIRE	I 6	46	COWL WIRE
E 7			I 7	46	ENGINE WIRE
E 9					
E11					
I 5	46	COWL WIRE			

ENGINE CONTROL (5S-FE M/T)







ENGINE CONTROL (5S-FE M/T)

SERVICE HINTS

E 6, E 7, E 8 ENGINE ECU

VOLTAGE AT ECU WIRING CONNECTOR

BATT-E1	: ALWAYS 10.0-14.0 VOLTS
+B-E1	: 10.0-14.0 VOLTS (IGNITION SW AT ON POSITION)
+B1-E1	: 10.0-14.0 VOLTS (IGNITION SW AT ON POSITION)
IDL-E1	: 8.0-14.0 VOLTS (IGNITION SW ON AND THROTTLE VALVE OPEN)
PSW-E1	: 4.5-5.5 VOLTS (IGNITION SW ON AND THROTTLE VALVE FULLY CLOSED)
PIM-E2	: 3.3-3.9 VOLTS (IGNITION SW AT ON POSITION)
VC-E2	: 4.5-5.5 VOLTS (IGNITION SW AT ON POSITION)
#10, #20 - E01, E02	: 10.0-14.0 VOLTS (IGNITION SW AT ON POSITION)
THA-E2	: 1.9-2.9 VOLTS (IGNITION SW ON AND INTAKE AIR TEMP. 20°C, 68°F)
THW-E2	: 0.1-1.1 VOLTS (IGNITION SW ON AND COOLANT TEMP. 80°C, 176°F)
STA-E1	: 6.0-14.0 VOLTS (ENGINE CRANKING)
IGT-E1	: 0.8-1.2 VOLTS (ENGINE CRANKING OR IDLING)
ISCC, ISCO-E1	: 8.0-14.0 VOLTS (IGNITION SW AT ON POSITION)
W-E1	: 10.0-14.0 VOLTS (IGNITION SW ON, NO TROUBLE AND ENGINE RUNNING)
ACT-E1	: 4.5-5.5 VOLTS (IGNITION SW ON AND AIR CONDITIONING ON)
ACA-E1	: 2.0 VOLTS OR LESS (IGNITION SW ON AND AIR CONDITIONING ON)
TE1-E1	: 10.0-14.0 VOLTS (IGNITION SW ON AND CHECK CONNECTOR T-E1 NOT CONNECTED)
	: 1.0 VOLTS OR LESS (IGNITION SW ON AND CHECK CONNECTOR T-E1 CONNECTOR)
NSW-E1	: 0-2.0 VOLTS (IGNITION SW ON AND NEUTRAL START SW POSITION P OR N RANGE)
	: 6.0-14.0 VOLTS (IGNITION SW ON AND EX. NEUTRAL START SW POSITION P OR N RANGE)

RESISTANCE AT ECU WIRING CONNECTORS

(DISCONNECT WIRING CONNECTOR)

IDL-E1	: INFINITY (THROTTLE VALVE OPEN)
	: 0 Ω (THROTTLE VALVE FULLY CLOSED)
PSW-E1	: 0 Ω (THROTTLE VALVE FULLY OPEN)
	: INFINITY (THROTTLE VALVE FULLY CLOSED)
THA-E2	: 2.0-3.0 KΩ (INTAKE AIR TEMP. 20°C, 68°F)
THW-E2	: 0.2-0.4 KΩ (COOLANT TEMP. 80°C, 176°F)
G+ - G-	: 0.17-0.21 KΩ
ISCC, ISCO-+B, +B1	: 19.3-22.3 KΩ

: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
A 3	30	E 7	C 33	I10	30 (5S-FE)
A17	32	E 8	A 33	I12	33
C 1	30 (5S-FE)	F10	30 (5S-FE)	J 2	33
C10	32	F16	34 (L/B), 35 (C/P)	J 6	33
C12	32	I 3	30 (5S-FE)	K 1	30
C13	B 32	I 4	30 (5S-FE)	O 3	30 (5S-FE)
C14	A 32	I 5	30 (5S-FE)	O 4	30 (5S-FE)
D 1	30 (5S-FE)	I 6	30 (5S-FE)	T 1	30 (5S-FE)
E 2	30 (5S-FE)	I 7	30 (5S-FE)	V 1	30 (5S-FE)
E 3	30 (5S-FE)	I 8	30 (5S-FE)	V 3	30 (5S-FE)
E 6	B 33	I 9	30 (5S-FE)		

: RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
4	27	R/B NO. 4 (RIGHT KICK PANEL)

: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1D		
1E		
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2A	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2B	22 (2WD)	ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2C		
2E	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
3B	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

 : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EB1	40 (5S-FE)	ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER)
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)
IF1	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
IG2	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)
IH1	44	ENGINE WIRE AND A/C NO. 1 WIRE (BEHIND GLOVE BOX WIRE)
IJ1	46	COWL WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)

 : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
EB	40 (5S-FE)	FRONT LEFT FENDER
EC	40 (5S-FE)	INTAKE MANIFOLD
ID	44	LEFT KICK PANEL
IG	44	R/B NO. 4 SET BOLT
BI	48 (L/B)	UNDER THE LEFT CENTER PILLAR
	50 (C/P)	

 : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 5	40 (5S-FE)	ENGINE WIRE	I 6	46	COWL WIRE
E 7			I 7	46	ENGINE WIRE
E 9			I 8	46	COWL WIRE
E11			I 9	46	

ENGINE CONTROL (4A-FE)

SYSTEM OUTLINE

THE ENGINE CONTROL SYSTEM UTILIZES A MICROCOMPUTER AND MAINTAINS OVERALL CONTROL OF THE ENGINE ETC. AN OUTLINE OF ENGINE CONTROL IS GIVEN HERE.

1. INPUT SIGNALS

- (1) WATER TEMP. SIGNAL SYSTEM
THE WATER TEMP. SENSOR DETECTS THE ENGINE COOLANT TEMP. AND HAS A BUILT-IN THERMISTOR WITH A RESISTANCE WHICH VARIES ACCORDING TO THE WATER TEMP. THUS THE WATER TEMP. IS INPUT IN THE FORM OF A CONTROL SIGNAL TO **TERMINAL THW** OF THE ECU.
- (2) INTAKE AIR TEMP. SIGNAL SYSTEM
THE INTAKE AIR TEMP. SENSOR IS DETECTS THE INTAKE AIR TEMP., WHICH IS INPUT AS A CONTROL SIGNAL TO **TERMINAL THA** OF THE ECU.
- (3) OXYGEN SENSOR SIGNAL SYSTEM
THE OXYGEN DENSITY IN THE EXHAUST EMISSIONS IS DETECTED AND INPUT AS A CONTROL SIGNAL TO **TERMINAL OX** OF THE ECU.
- (4) RPM SIGNAL SYSTEM
CRANKSHAFT POSITION AND ENGINE RPM ARE DETECTED BY THE PICK-UP COIL INSTALLED INSIDE THE DISTRIBUTOR. CRANKSHAFT POSITION IS INPUT AS A CONTROL SIGNAL TO **TERMINAL G1** OF THE ECU, AND RPM SIGNAL IS INPUT TO **TERMINAL NE**.
- (5) THROTTLE SIGNAL SYSTEM
THE THROTTLE POSITION SENSOR DETECTS THE THROTTLE VALVE OPENING ANGLE, WHICH IS INPUT AS A CONTROL SIGNAL TO **TERMINAL PSW** OF THE ECU, OR WHEN THE VALVE IS FULLY CLOSED, TO **TERMINAL IDL**.
- (6) VEHICLE SPEED SIGNAL SYSTEM
THE SPEED SENSOR, INSTALLED INSIDE THE COMBINATION METER, DETECTS THE VEHICLE SPEED AND INPUTS A CONTROL SIGNAL TO **TERMINAL SPD** OF THE ECU.
- (7) NEUTRAL START SW SIGNAL SYSTEM
THE NEUTRAL START SW DETECTS WHETHER THE SHIFT POSITION IS IN NEUTRAL OR NOT, AND INPUTS A CONTROL SIGNAL TO **TERMINAL NSW** OF THE ECU.
- (8) A/C SW SIGNAL SYSTEM
THE OPERATING VOLTAGE OF THE A/C MAGNETIC CLUTCH IS DETECTED AND INPUT IN THE FORM OF A CONTROL SIGNAL TO **TERMINAL AC1** OF THE ECU.
- (9) BATTERY SIGNAL SYSTEM
VOLTAGE IS CONSTANTLY APPLIED TO **TERMINAL BATT** OF THE ECU. WHEN THE IGNITION SW IS TURNED TO ON, VOLTAGE FOR ECU OPERATION IS APPLIED VIA THE EFI MAIN RELAY TO **TERMINALS +B** AND **+B1** OF THE ECU.
- (10) INTAKE AIR VOLUME SIGNAL SYSTEM
INTAKE AIR VOLUME IS DETECTED BY THE VACUUM SENSOR AND IS INPUT AS A CONTROL SIGNAL TO **TERMINAL PIM** OF THE ECU.
- (11) STA SIGNAL SYSTEM
TO CONFIRM THAT THE ENGINE IS CRANKING, THE VOLTAGE APPLIED TO THE STARTER MOTOR DURING CRANKING IS DETECTED AND IS INPUT AS A CONTROL SIGNAL TO **TERMINAL STA** OF THE ECU.

2. CONTROL SYSTEM

* EFI (ELECTRONIC FUEL INJECTION) SYSTEM

THE EFI SYSTEM MONITORS THE ENGINE CONDITIONS THROUGH THE SIGNALS EACH SENSOR (INPUT SIGNALS (1) TO (11)) INPUTS TO THE ECU. BASED ON THIS DATA AND THE PROGRAM MEMORIZED IN THE ECU, THE MOST APPROPRIATE FUEL INJECTION TIMING IS DECIDED AND CURRENT IS OUTPUT TO **TERMINALS #10** AND **#20** OF THE ECU, CAUSING THE INJECTORS TO OPERATE IT (TO INJECT FUEL). IT IS THIS SYSTEM WHICH, THROUGH THE WORK OF THE ECU, FINELY CONTROLS FUEL INJECTION IN RESPONSE TO DRIVING CONDITIONS.

DURING ENGINE CRANKING (SIGNAL INPUT TO **TERMINAL STA**) OR FOR APPROX. 2 SECONDS AFTER NE SIGNAL INPUT, ECU OPERATION ENERGIZES (POINT CLOSED) THE FUEL PUMP CIRCUIT INSIDE THE CIRCUIT OPENING RELAY, CAUSING THE FUEL PUMP TO OPERATE.

* ESA (ELECTRONIC SPARK ADVANCE) SYSTEM

THE ESA SYSTEM MONITORS THE ENGINE CONDITIONS USING THE SIGNALS (INPUT SIGNALS (1, 4, 5, 10, 11)) INPUT TO THE ECU FROM EACH SENSOR. BASED ON THIS DATA AND THE PROGRAM MEMORIZED IN THE ECU, THE MOST APPROPRIATE IGNITION TIMING IS DECIDED AND CURRENT IS OUTPUT TO **TERMINAL IGT** OF THE ECU. THIS OUTPUT CONTROLS THE IGNITER TO PRODUCE THE MOST APPROPRIATE IGNITION TIMING FOR THE DRIVING CONDITIONS.

* IDLE-UP SPEED CONTROL SYSTEM

THE IDLE-UP SYSTEM USES THE AIR CONTROL VALVE FOR IDLE-UP TO INCREASE THE RPM AND PROVIDE STABLE IDLING WHEN THE IDLE SPEED DROPS DUE TO THE ELECTRICAL LOAD, ETC. THE ECU EVALUATES THE SIGNALS FROM EACH SENSOR (INPUT SIGNALS (1, 4 TO 8)), CURRENT IS OUTPUT TO **TERMINAL V-ISC** AND CONTROLS THE VSV.

* EGR CUT CONTROL SYSTEM

THE EGR CUT CONTROL SYSTEM CONTROLS THE VSV (FOR EGR) BY EVALUATING THE SIGNALS FROM EACH SENSOR INPUT TO THE ECU (INPUT SIGNALS (1, 5, 6, 9)) AND BY SENDING OUTPUT TO **TERMINAL EGR** OF THE ECU.

* A/C CUT CONTROL SYSTEM

WHEN THE VEHICLE SUDDENLY ACCELERATES FROM LOW ENGINE SPEED, THIS SYSTEM CUTS OFF AIR CONDITIONER OPERATION FOR A FIXED PERIOD OF TIME IN RESPONSE TO THE VEHICLE SPEED, THROTTLE VALVE OPENING ANGLE AND INTAKE MANIFOLD PRESSURE IN ORDER TO MAINTAIN ACCELERATION PERFORMANCE.

THE ECU RECEIVES INPUT SIGNALS (4, 5, 6, 7 AND 10), AND OUTPUTS SIGNALS TO **TERMINAL ACT**.

* OVERDRIVE CONTROL SYSTEM

THE ECU CONTROLS THE O/D SOLENOID OF THE AUTOMATIC TRANSAXLE IN RESPONSE TO SIGNALS FROM EACH SENSOR (INPUT SIGNALS (1, 4, 5, 6 AND 10)) IN ORDER TO MAINTAIN DRIVABILITY AND ACCELERATION PERFORMANCE.

THE ECU OUTPUTS A SIGNAL FROM **TERMINAL OD** TO CONTROL THE O/D (CALIFORNIA), **TERMINAL ODT** (EX. CALIFORNIA) TO CONTROL THE O/D SOLENOID.

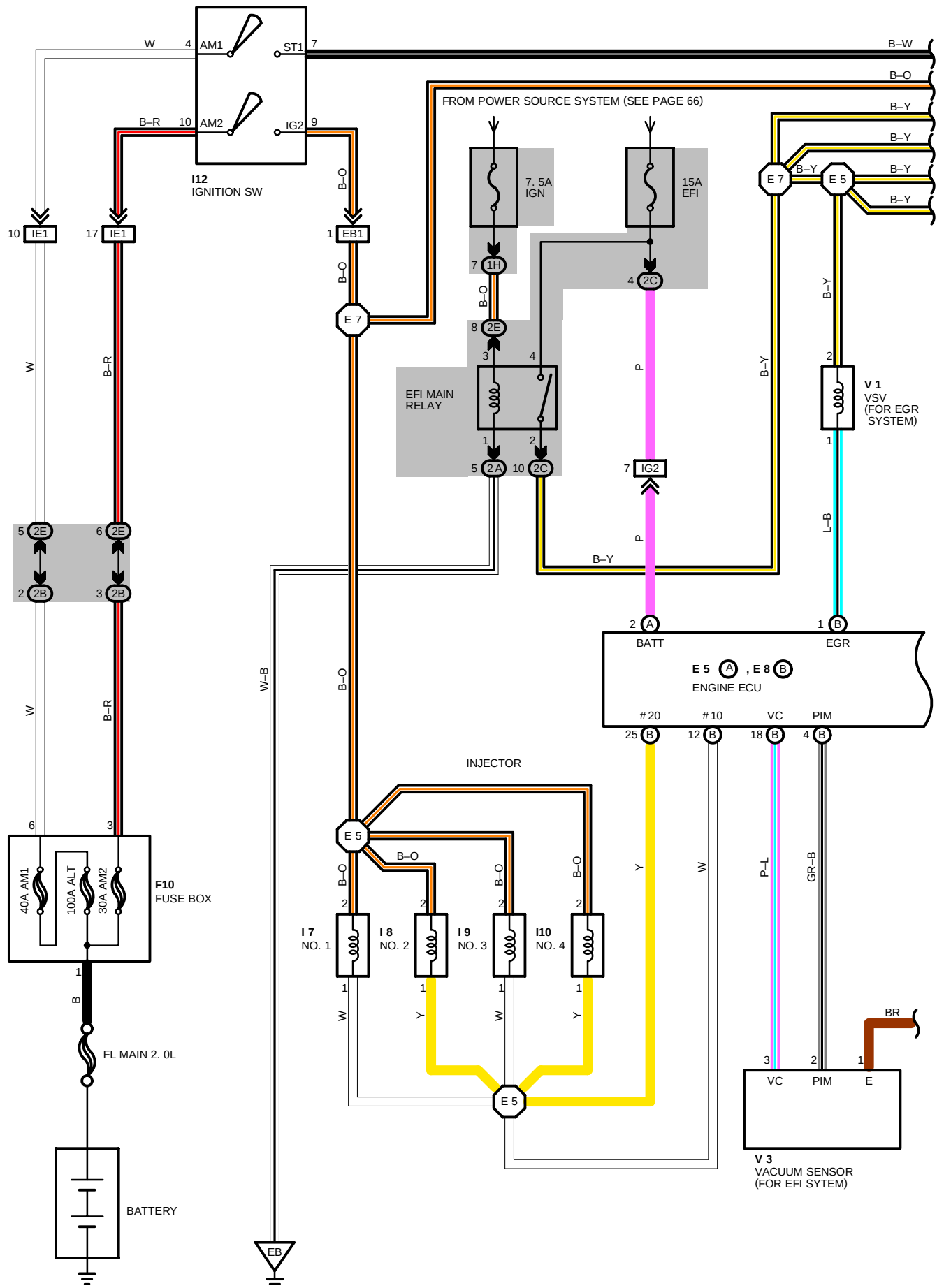
3. DIAGNOSIS SYSTEM

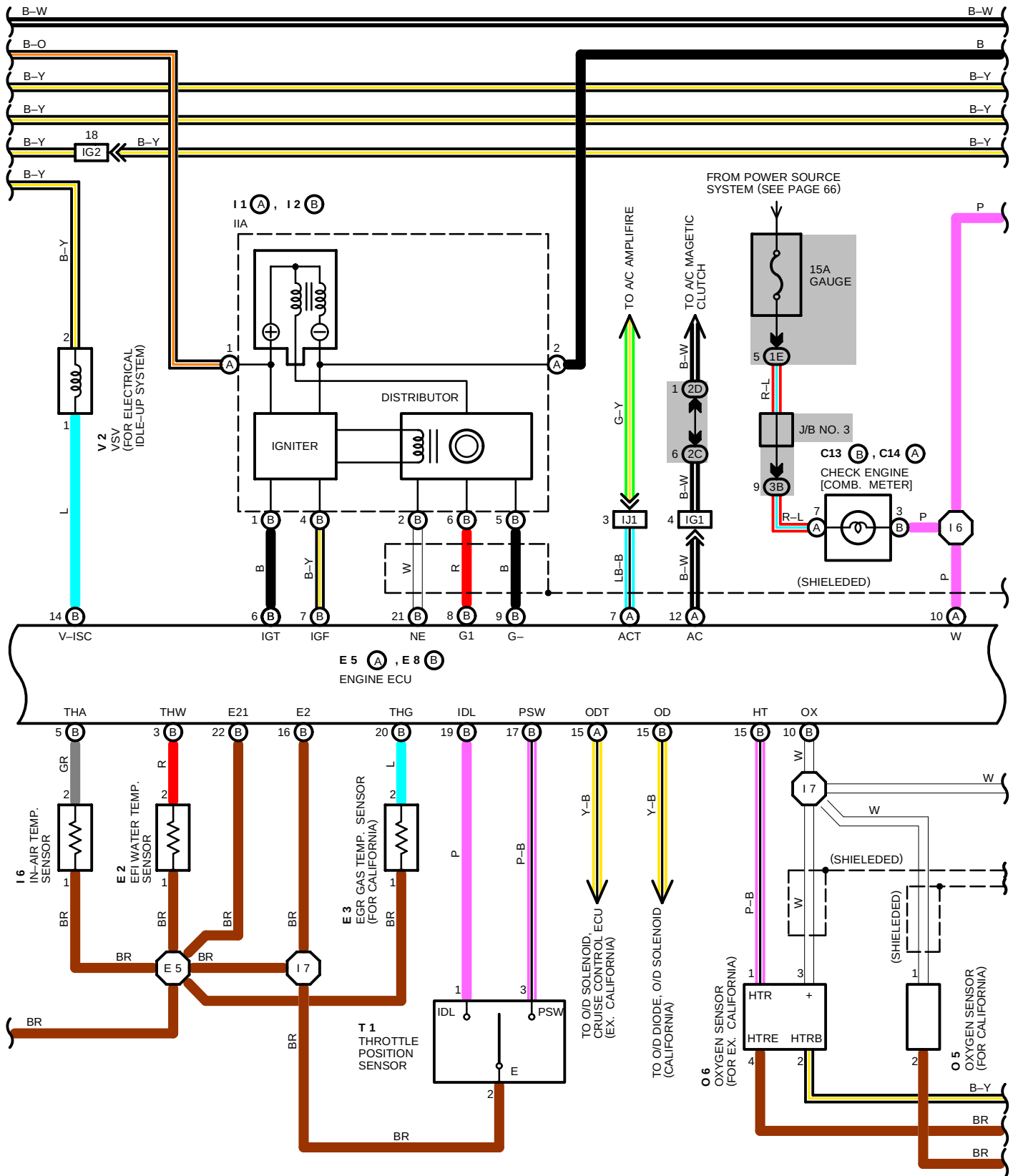
WITH THE DIAGNOSIS SYSTEM, WHEN THERE IS A MALFUNCTIONING IN THE ECU SIGNAL SYSTEM, THE MALFUNCTION SYSTEM IS RECORDED IN THE MEMORY. THE MALFUNCTIONING SYSTEM CAN THEN BE FOUND BY READING THE DISPLAY (CODE) OF THE CHECK ENGINE WARNING LIGHT.

4. FAIL-SAFE SYSTEM

WHEN A MALFUNCTION OCCURS IN ANY SYSTEM, IF THERE IS A POSSIBILITY OF ENGINE TROUBLE BEING CAUSED BY CONTINUED CONTROL BASED ON THE SIGNALS FROM THAT SYSTEM, THE FAIL-SAFE SYSTEM EITHER CONTROLS THE SYSTEM BY USING DATA (STANDARD VALUES) RECORDED IN THE ECU MEMORY OR ELSE STOPS THE ENGINE.

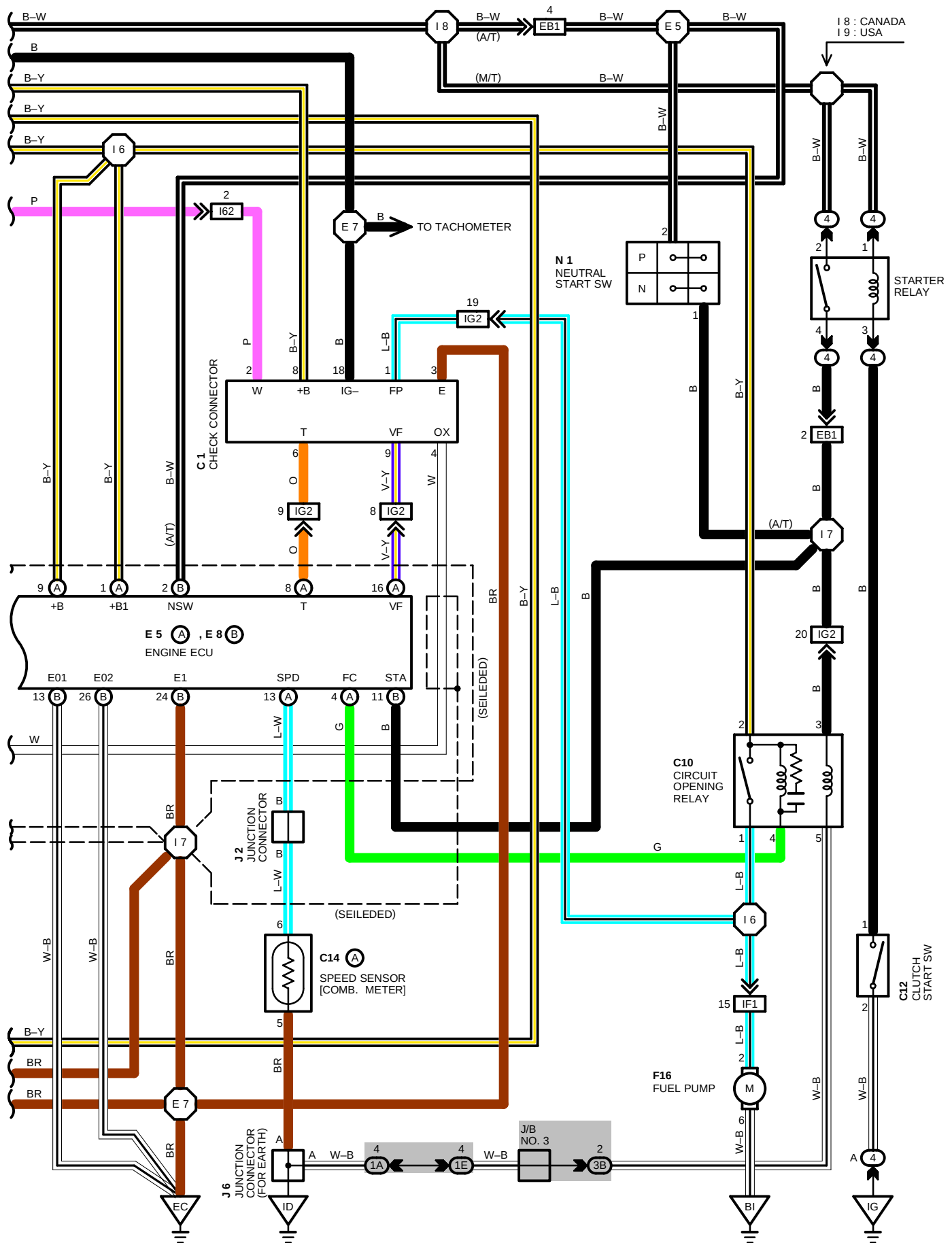
ENGINE CONTROL (4A-FE)



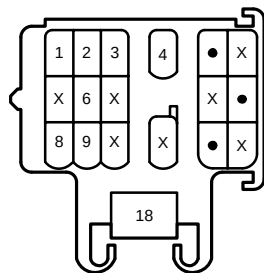


ENGINE CONTROL (4A-FE)

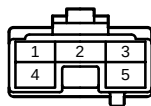
* 1: A/T EX. CALIFORNIA
* 2: M/T, FOR CALIFORNIA A/T



C 1 DAEK GRAY



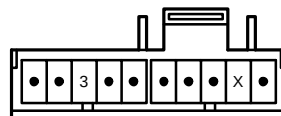
C10 DARK GRAY



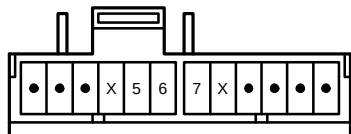
C12



C13 (B) BROWN



C14 (A)



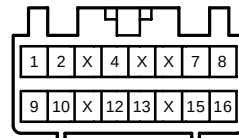
E 2 DARK GREEN



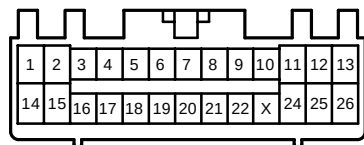
E 3 DARK GRAY



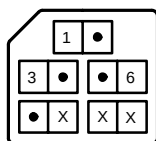
E 5 (A) DARK GRAY



E 8 (B) DARK GRAY



F10 BLACK



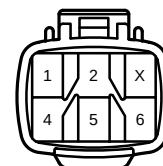
F16 DARK GRAY



I 1 (A) DARK GRAY



I 2 (B) DARK GRAY

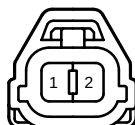


I 6 BLACK

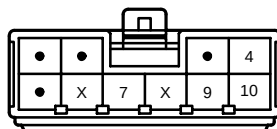


I 7, I 8, I 9, I 10

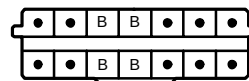
GRAY



I12 BLACK

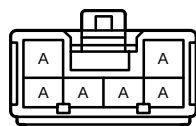


J 2



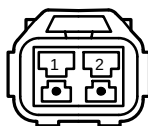
(HINT : SEE PAGE 7)

J 6



(HINT : SEE PAGE 7)

N 1 GRAY



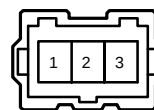
O 5 DARK GRAY



O 6 DARK GRAY



T 1 BLACK



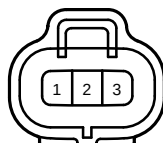
V 1 BROWN



V 2 BLUE



V 3 BLACK



ENGINE CONTROL (4A-FE)

SERVICE HINTS

E 5, E 8 ENGINE ECU

VOLTAGE AT ECU WIRING CONNECTOR

BATT-E1	: ALWAYS 10.0-14.0 VOLTS
+B-E1	: 10.0-14.0 VOLTS (IGNITION SW AT ON POSITION)
+B1-E1	: 10.0-14.0 VOLTS (IGNITION SW AT ON POSITION)
IDL-E1	: 8.0-14.0 VOLTS (IGNITION SW ON AND THROTTLE VALVE OPEN)
PSW-E1	: 4.0-5.0 VOLTS (IGNITION SW ON AND THROTTLE VALVE FULLY CLOSED)
PIM-E2	: 3.3-3.9 VOLTS (IGNITION SW AT ON POSITION)
VC-E2	: 4.5-5.5 VOLTS (IGNITION SW AT ON POSITION)
#10, #20-E01, E02	: 10.0-14.0 VOLTS (IGNITION SW AT ON POSITION)
THA-E2	: 1.0-3.0 VOLTS (IGNITION SW ON AND INTAKE AIR TEMP. 20°C (68°F))
THW-E2	: 0.1-1.0 VOLTS (IGNITION SW ON AND COOLANT TEMP. 80°C (176°F))
STA-E1	: 6.0-14.0 VOLTS (ENGINE CRANKING)
IGT-E1	: 0.7-1.0 VOLTS (ENGINE CRANKING OR IDLING)
WRN-E1	: 10.0-14.0 VOLTS (IGNITION SW ON, NO TROUBLE AND ENGINE RUNNING)
AC-E1	: 8.0-14.0 VOLTS (IGNITION SW ON AND AIR CONDITIONING ON)
ACT-E1	: 4.0-6.0 VOLTS (IGNITION SW ON AND HEATER BLOWER SW ON)
T-E1	: 10.0-14.0 VOLTS (IGNITION SW ON AND CHECK CONNECTOR T-E1 NOT CONNECTED)
	: 0.5 OR LESS (IGNITION SW ON AND CHECK CONNECTOR T-E1 CONNECTOR)
NSW-E1	: 0-2.0 VOLTS (IGNITION SW ON AND NEUTRAL START SW POSITION P OR N RANGE)
	: 10.0-14.0 VOLTS (IGNITION SW ON AND EX. NEUTRAL START SW POSITION P OR N RANGE)
V-ISC-E1	: 10.0-14.0 VOLTS (CRANKING FOR 10 SECONDS AFTER STARTING)

RESISTANCE AT ECU WIRING CONNECTORS

(DISCONNECT WIRING CONNECTOR)

IDL-E2	: INFINITY (THROTTLE VALVE OPEN) 0 Ω (THROTTLE VALVE FULLY CLOSED)
PSW-E2	: 0 Ω (THROTTLE VALVE FULLY OPEN) INFINITY (THROTTLE VALVE FULLY CLOSED)
THA-E1	: 2.0 - 3.0 KΩ (INTAKE AIR TEMP. 20°C, 68°F)
THW-E1	: 0.2 - 0.4 KΩ (COOLANT TEMP. 80°C, 176°F)
G1, NE-G-	: 0.17-0.21 KΩ

: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C 1	31 (4A-FE)	F16	35 (C/P)	J 6	33
C10	32	I 1 A	31 (4A-FE)	N 1	31 (4A-FE)
C12	32	I 2 B	31 (4A-FE)	O 5	31 (4A-FE)
C13 B	32	I 6	31 (4A-FE)	O 6	31 (4A-FE)
C14 A	32	I 7	31 (4A-FE)	T 1	31 (4A-FE)
E 2	31 (4A-FE)	I 8	31 (4A-FE)	V 1	31 (4A-FE)
E 3	31 (4A-FE)	I 9	31 (4A-FE)	V 2	31 (4A-FE)
E 5 A	33	I10	31 (4A-FE)	V 3	31 (4A-FE)
E 8 B	33	I12	33		
F10	31 (4A-FE)	J 2	33		

: RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
4	27	R/B NO. 4 (RIGHT KICK PANEL)

: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2A	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2B	22 (2WD)	ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2C		
2D	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2E		
3B	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

 : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EB1	42 (4A-FE)	ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER)
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)
IF1	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
IG1	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)
IG2		
IJ1	46	COWL WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)

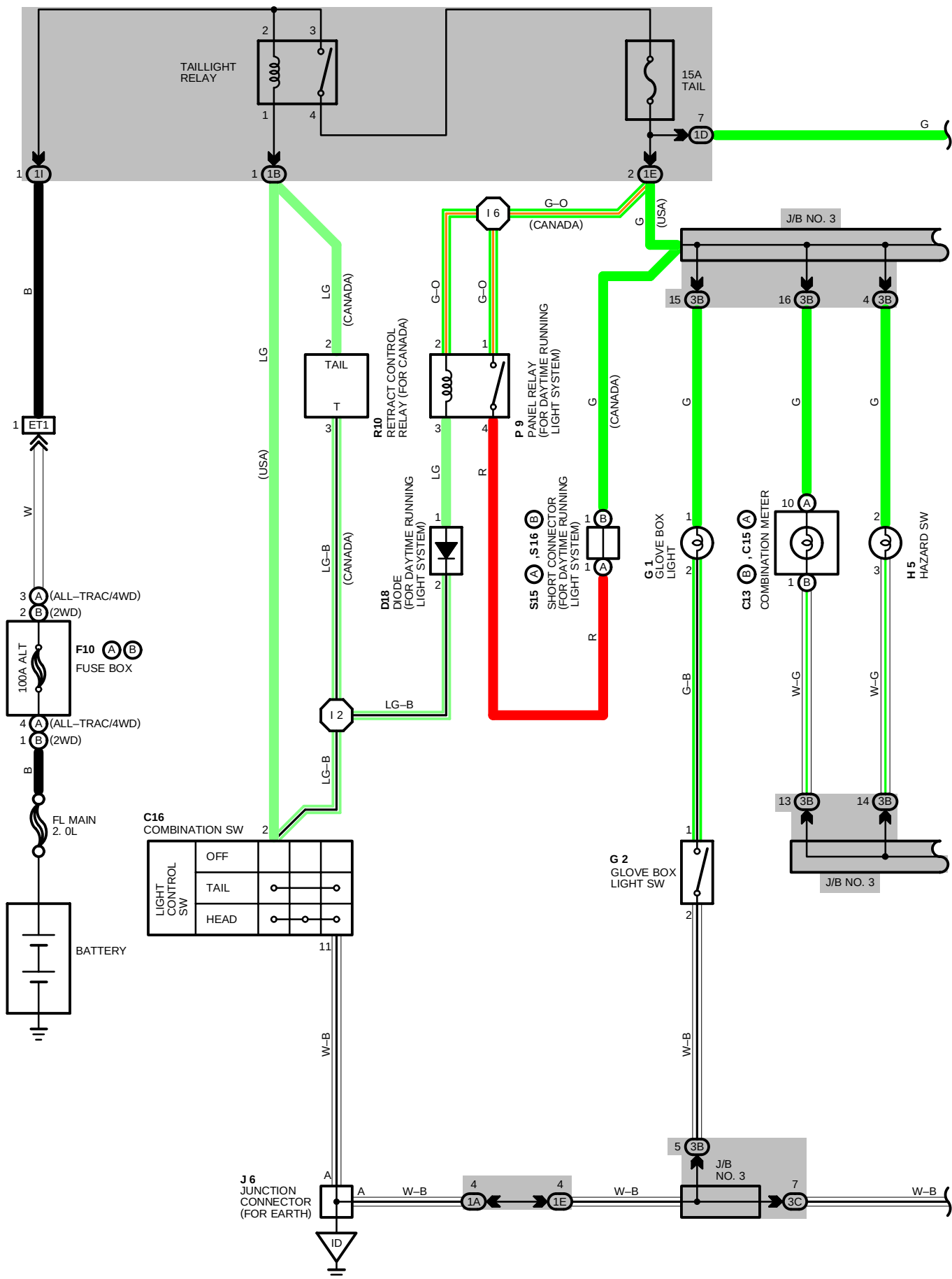
 : GROUND POINTS

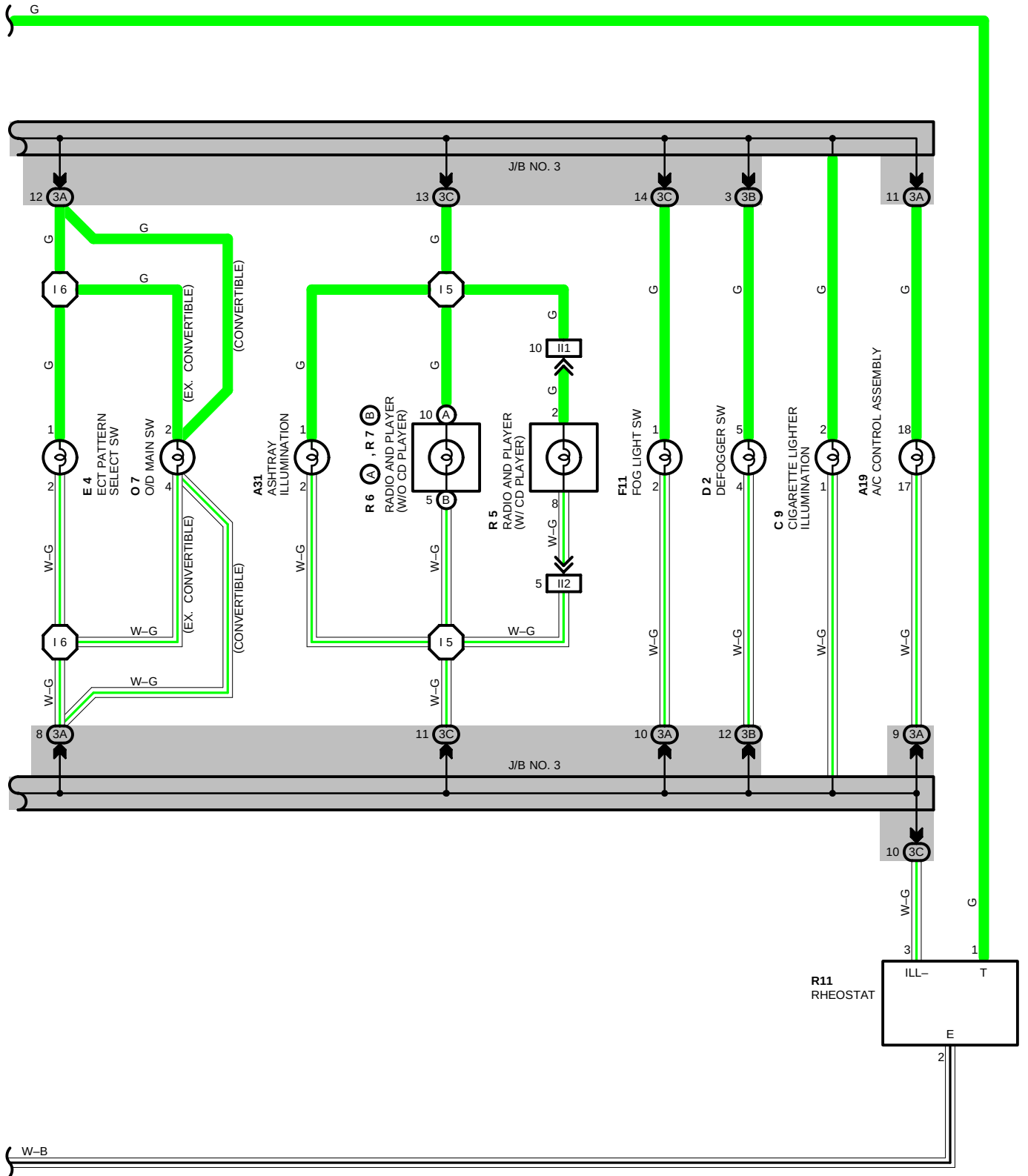
CODE	SEE PAGE	GROUND POINTS LOCATION
EB	42 (4A-FE)	FRONT LEFT FENDER
EC	42 (4A-FE)	INTAKE MANIFOLD
ID	44	LEFT KICK PANEL
IG	44	R/B NO. 4 SET BOLT
BI	50 (C/P)	UNDER THE LEFT CENTER PILLAR

 : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 5	42 (4A-FE)	ENGINE WIRE	I 7	46	ENGINE WIRE
E 7			I 8	46	COWL WIRE
I 6	46	COWL WIRE	I 9	46	

ILLUMINATION





ILLUMINATION

SERVICE HINTS

TAILLIGHT RELAY

3-4 : CLOSED WITH LIGHT CONTROL SW AT **TAIL** OR **HEAD** POSITION (WHEN LIGHT AUTO TURN OFF SYSTEM IS OFF)
CLOSED WITH ENGINE RUNNING AND PARKING LEVER RELEASED (CANADA)

R11 RHEOSTAT

1-2 : APPROX. **12** VOLTS WITH RHEOSTAT FULLY TURNED COUNTERCLOCKWISE AND **0** VOLTS WITH FULLY TURNED CLOCKWISE

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
A19	32	F10	A 28 (3S-GTE)	R 5	33
A31	32		B 30 (5S-FE), 31 (4A-FE)	R 6 A	33
C 9	32	F11	33	R 7 B	33
C13 B	32	G 1	33	R10	33
C15 A	32	G 2	33	R11	33
C16	32	H 5	33	S15 A	33
D 2	32	J 6	33	S16 B	33
D18	33	O 7	33		
E 4	33	P 9	33		

□ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1B		
1D		
1E		
1I	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
3A	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)
3B		
3C		

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

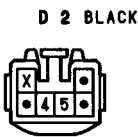
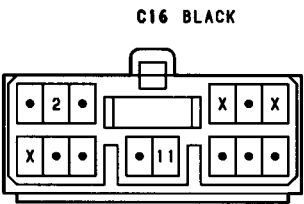
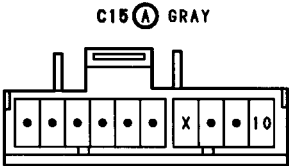
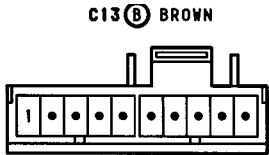
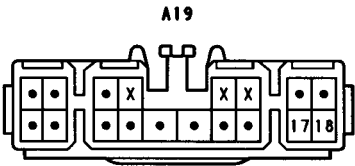
CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
ET1	38 (3S-GTE)	ENGINE ROOM MAIN WIRE AND ENGINE WIRE (NEAR THE J/B NO. 2)
	40 (5S-FE)	
	42 (4A-FE)	
II1	46	COWL WIRE AND CONSOLE BOX WIRE (INSTRUMENT PANEL CENTER)
II2		

▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL

○ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I 2	46	COWL WIRE	I 6	46	COWL WIRE
I 5					



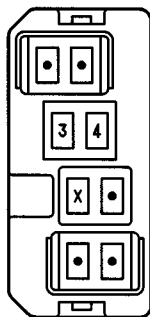
D18



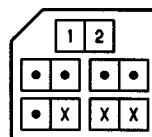
E 4



(ALL-TRAC/4WD) F10 Ⓐ BLACK



(2WD) F10 Ⓑ BLACK



F11 BLACK



G 1 GRAY



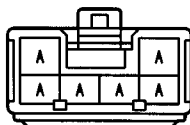
G 2



H 5 BLACK



J 6

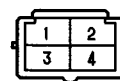


(HINT:SEE PAGE 7)

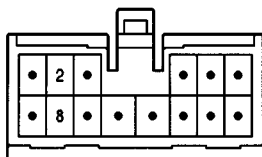
O 7



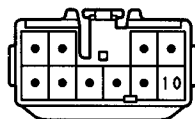
P 9



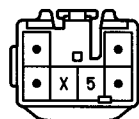
R 5



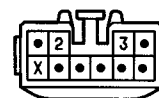
R 6 Ⓐ



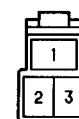
R 7 Ⓑ



R10



R11



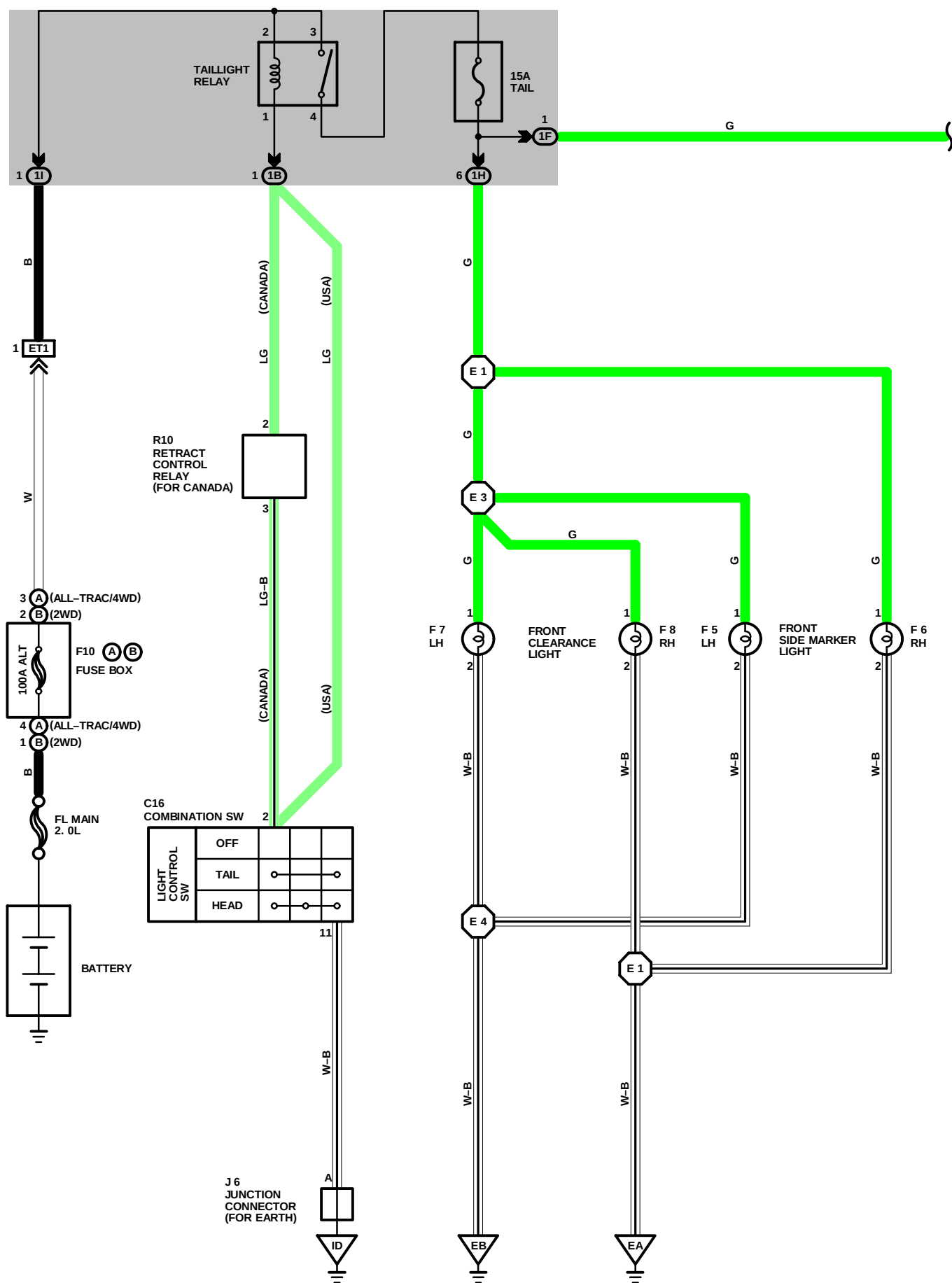
S15 Ⓐ GRAY

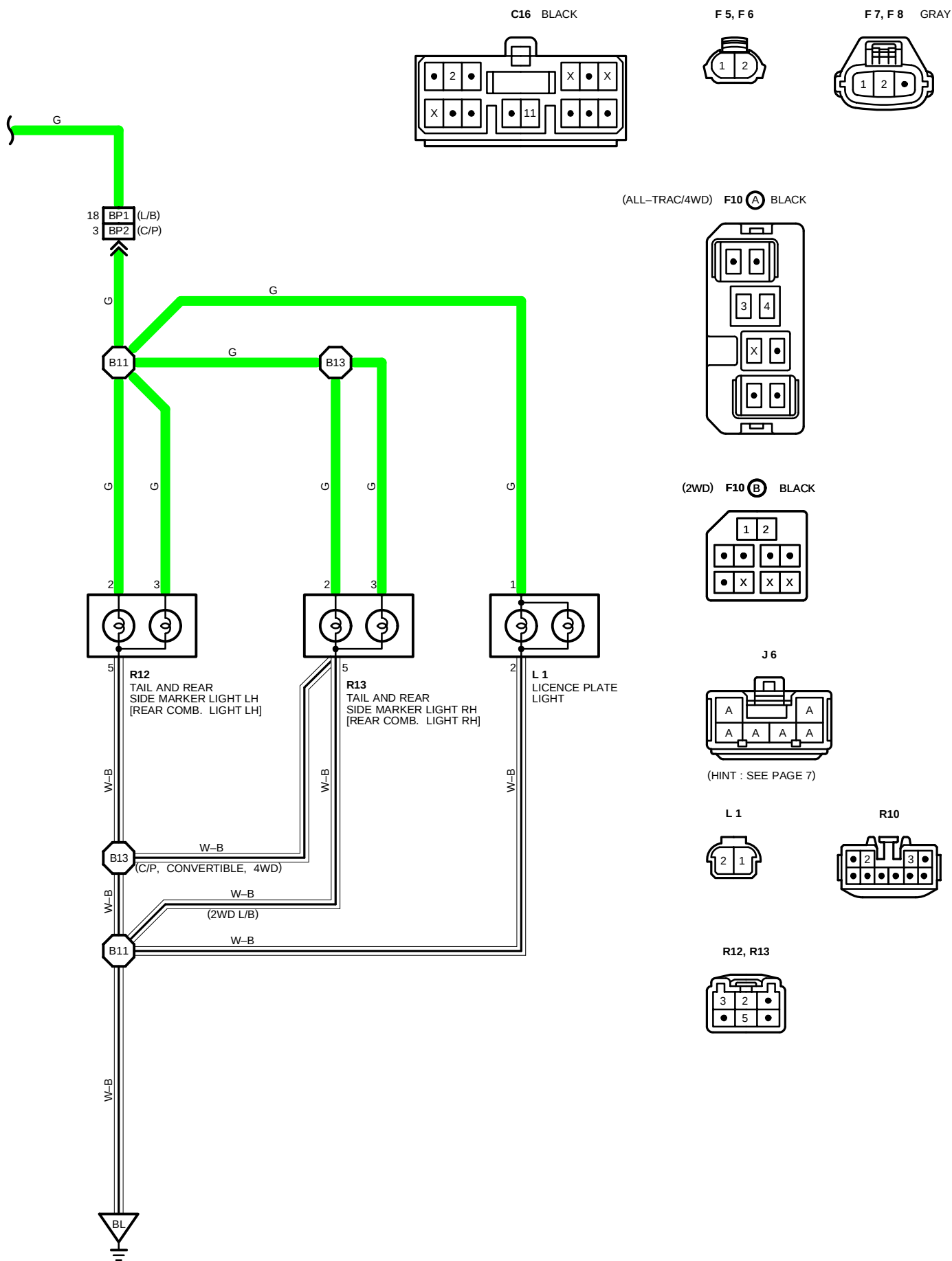


S16 Ⓑ GRAY



TAILLIGHT





TAILLIGHT

SERVICE HINTS

TAILLIGHT RELAY

3-4 : CLOSED WITH LIGHT CONTROL SW AT **TAIL** OR **HEAD** POSITION
CLOSED WITH ENGINE RUNNING AND BRAKE LEVER RELEASED (CANADA)



: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C16	32	F 7	31 (4S-FE)	L 1	34 (L/B), 35 (C/P)
F 5	28 (3S-GTE), 30 (5S-FE)	F 8	28 (3S-GTE), 30 (5S-FE)	R10	33
	31 (4S-FE)		31 (4S-FE)	R12	34 (L/B), 35 (C/P)
F 6	28 (3S-GTE), 30 (5S-FE)	F10	28 (3S-GTE)	R13	34 (L/B), 35 (C/P)
	31 (4S-FE)		30 (5S-FE), 31 (4S-FE)		
F 7	28 (3S-GTE), 30 (5S-FE)	J 6	33		



: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1B	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1F	20	FLOOR WIRE AND J/B NO.1 (LEFT KICK PANEL)
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1I		



: CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
ET1	38 (3S-GTE)	ENGINE ROOM MAIN WIRE AND ENGINE WIRE (NEAR THE J/B NO. 2)
	40 (5S-FE)	
	42 (4A-FE)	
BP1	48 (L/B)	FLOOR WIRE AND LUGGAGE ROOM WIRE (LEFT QUARTER PANEL CENTER)
BP2	50 (C/P)	FLOOR WIRE AND LUGGAGE ROOM WIRE (LUGGAGE COMPARTMENT LEFT)



: GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
EA	38 (3S-GTE)	FRONT RIGHT FENDER
	40 (5S-FE)	
	42 (4A-FE)	
EB	38 (3S-GTE)	FRONT LEFT FENDER
	40 (5S-FE)	
	42 (4A-FE)	
ID	44	LEFT KICK PANEL
BL	48 (L/B)	BACK PANEL CENTER
	50 (C/P)	



: SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 1	38 (3S-GTE)	ENGINE ROOM MAIN WIRE	E 4	42 (4A-FE)	ENGINE ROOM MAIN WIRE
	40 (5S-FE)		B11	48 (L/B)	LUGGAGE ROOM WIRE
	42 (4A-FE)			50 (C/P)	
E 3	38 (3S-GTE)			52 (CONVERTIBLE)	
	40 (5S-FE)		B13	48 (L/B)	
	42 (4A-FE)			50 (C/P)	
E 4	38 (3S-GTE)			52 (CONVERTIBLE)	
	40 (5S-FE)				

SERVICE HINTS

HEADLIGHT RELAY

1-2 : CLOSED WITH LIGHT CONTROL SW AT **HEAD** POSITION OR DIMMER SW AT **FLASH** POSITION

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C14	32	H 1	28 (3S-GTE), 30 (5S-FE)	R 2	28 (3S-GTE), 30 (5S-FE)
C16	A 32		31 (4A-FE)		31 (4A-FE)
C17	B 32	H 2	28 (3S-GTE), 30 (5S-FE)	R 3	28 (3S-GTE), 30 (5S-FE)
F10	A 28 (3S GTE)		31 (4A-FE)		31 (4A-FE)
	B 30 (5S-FE), 31 (4A-FE)	J 6	33	R 9	33

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
2A	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2B	23 (ALL-TRAC/4WD)	
	22 (2WD)	ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2D	23 (ALL-TRAC/4WD)	
	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2E	23 (ALL-TRAC/4WD)	
3B	23 (ALL-TRAC/4WD)	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)
	25	

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)

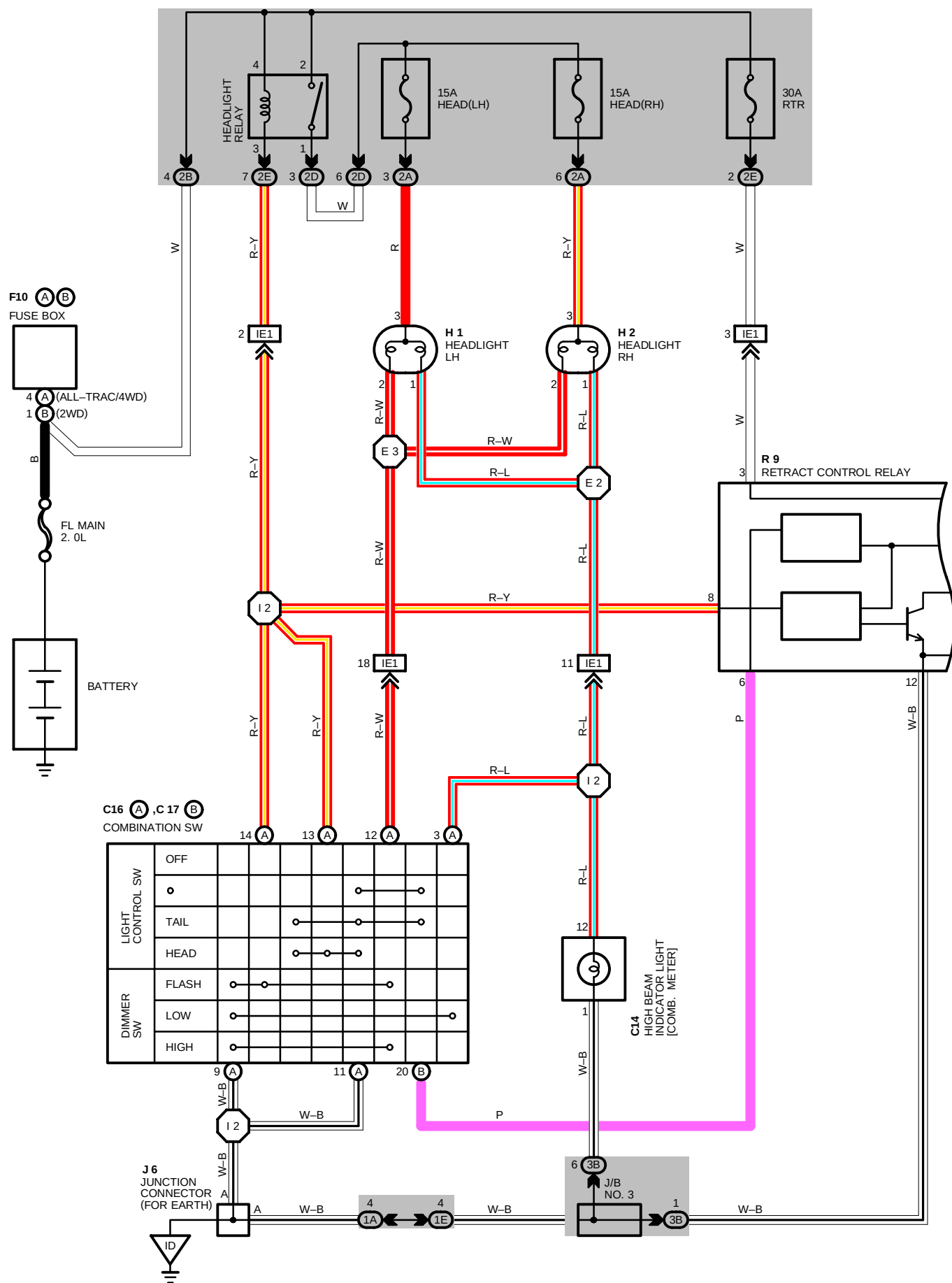
▽ : GROUND POINTS

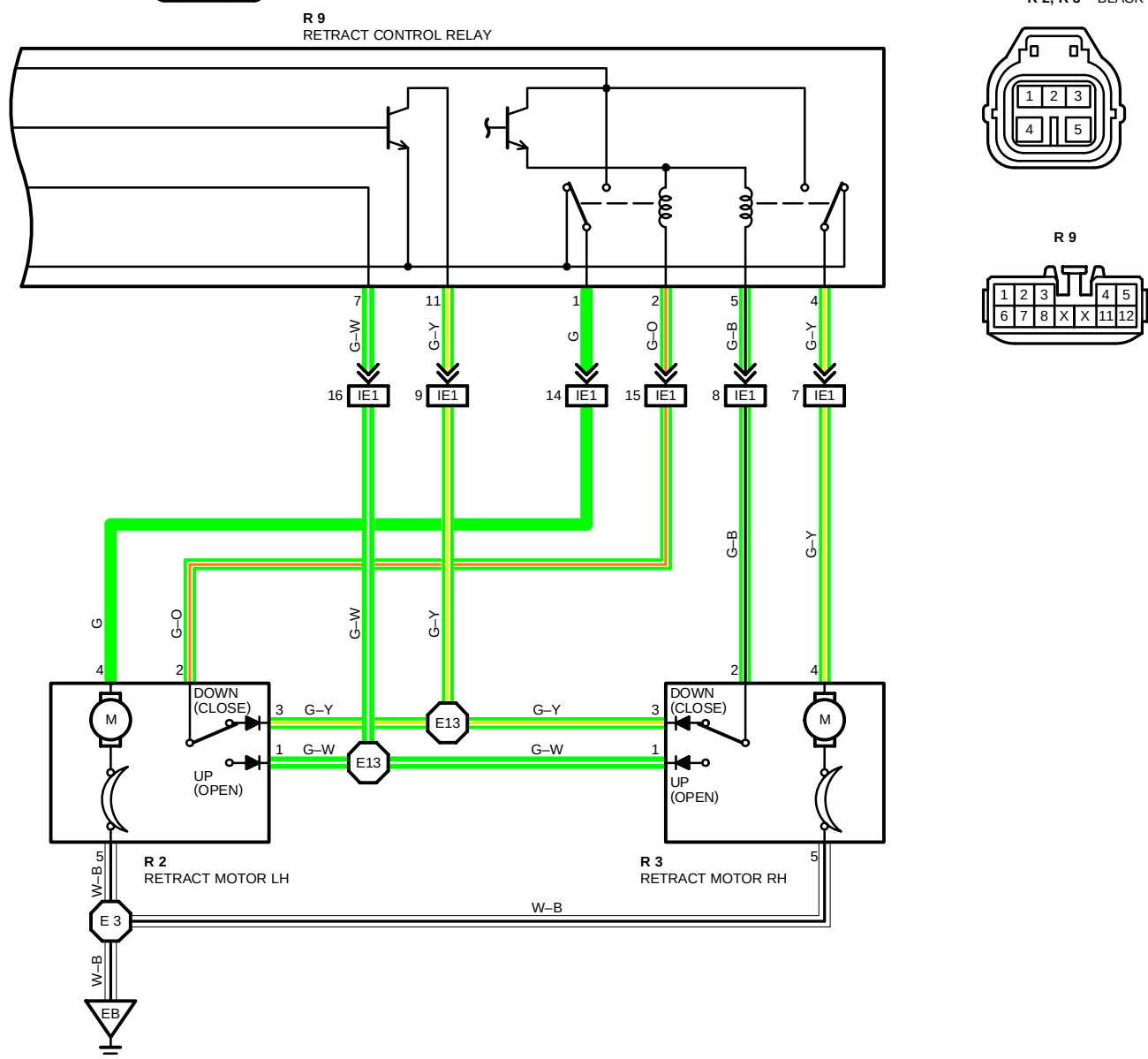
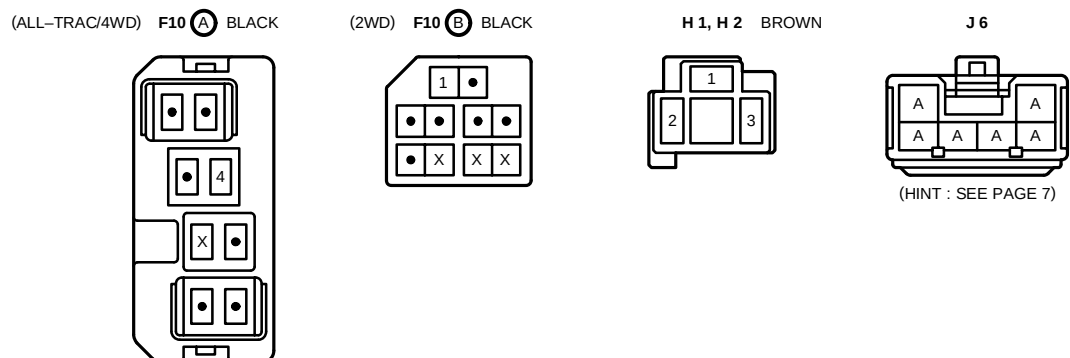
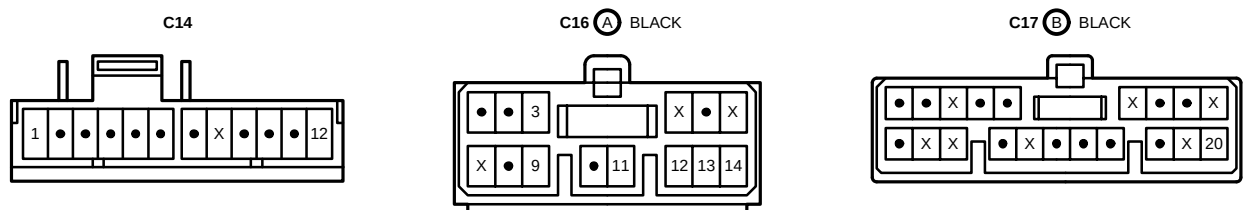
CODE	SEE PAGE	GROUND POINTS LOCATION
EB	38 (3S-GTE)	FRONT LEFT FENDER
	40 (5S-FE)	
	42 (4A-FE)	
ID	44	LEFT KICK PANEL

○ : SPLICE POINTS

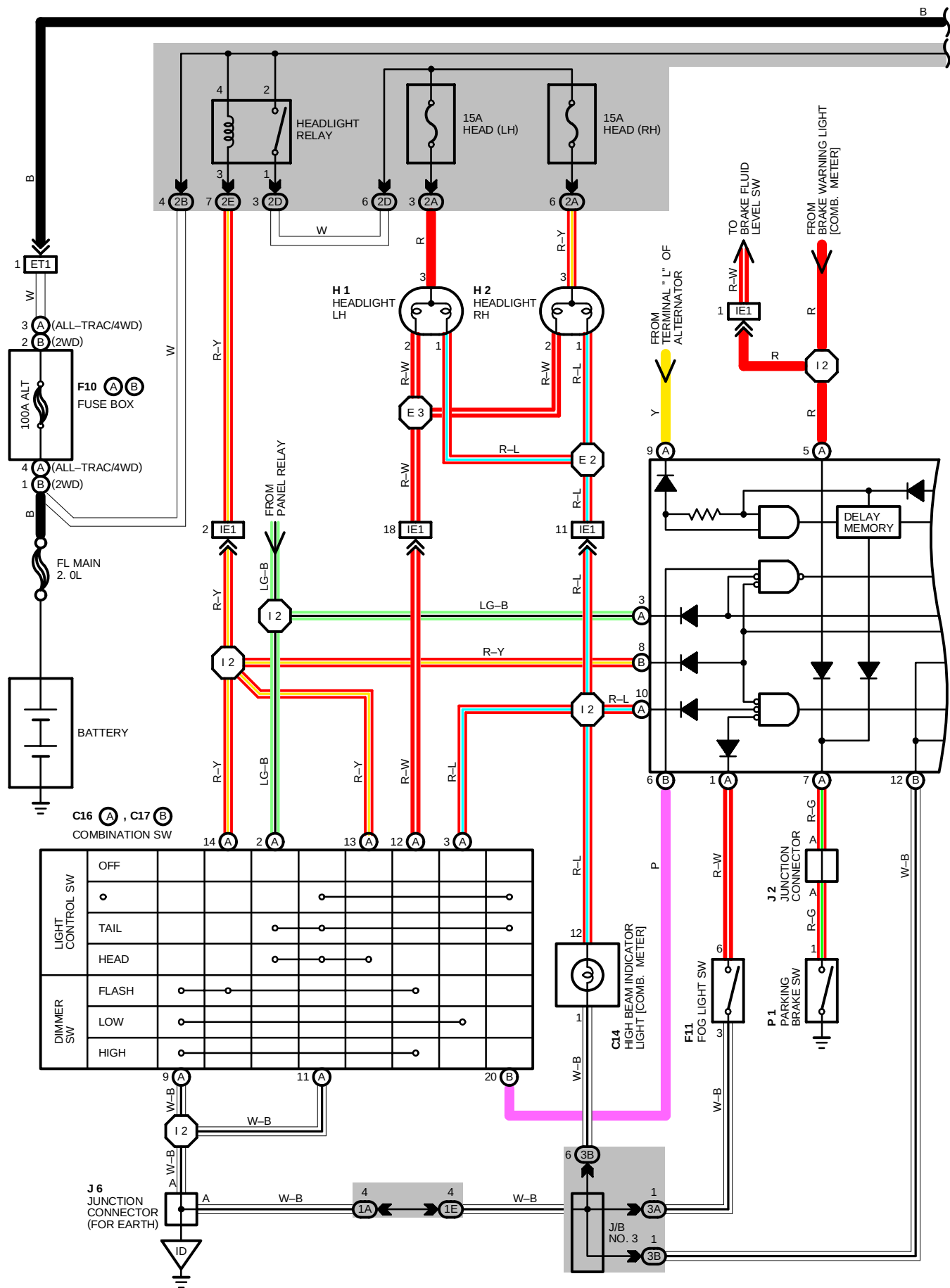
CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 2	38 (3S-GTE)	ENGINE ROOM MAIN WIRE	E 3	42 (4A-FE)	ENGINE ROOM MAIN WIRE
	40 (5S-FE)		E13	38 (3S-GTE)	
	42 (4A-FE)			40 (5S-FE)	
E 3	38 (3S-GTE)		I 2	42 (4A-FE)	COWL WIRE
	40 (5S-FE)			46	

HEADLIGHT (FOR USA)





HEADLIGHT AND FOG LIGHT (FOR CANADA)



HEADLIGHT AND FOG LIGHT (FOR CANADA)

SYSTEM OUTLINE

CURRENT FROM THE BATTERY FLOWS CONTINUOUSLY FROM FL MAIN 2.0L → ALT FUSE → TAILLIGHT RELAY (COIL SIDE) → **TERMINAL (A) 2** OF RETRACT CONTROL RELAY, AND FL MAIN 2.0L → HEADLIGHT RELAY (COIL SIDE) → **TERMINAL (B) 8** OF RETRACT CONTROL RELAY.

WHEN THE IGNITION SW IS TURNED ON, THE CURRENT FLOWING THROUGH THE ECU-IG FUSE FLOWS TO **TERMINAL (A) 4** OF THE RETRACT CONTROL RELAY.

DAYTIME RUNNING LIGHT OPERATION

WHEN THE ENGINE IS STARTED, VOLTAGE IS GENERATED AT **TERMINAL L** OF THE ALTERNATOR AND VOLTAGE IS APPLIED TO **TERMINAL (A) 9** OF THE RETRACT CONTROL RELAY.

IF THE PARKING BRAKE LEVER IS PULLED UP (PARKING BRAKE SW ON) AT THIS TIME, THE RELAY IS NOT ENERGIZED, SO THE DAYTIME RUNNING LIGHTS SYSTEM DOSE NOT OPERATE.

IF THE PARKING BRAKE LEVER IS THEN RELEASED (PARKING BRAKE SW OFF), THE SIGNAL IS INPUT TO **TERMINAL (A) 7** OF THE RELAY.

THIS ACTIVATES THE RELAY AND CURRENT FROM ALT FUSE FLOWS TO TAILLIGHT RELAY (POINT SIDE) → TAIL FUSE → TAIL, LICENCE AND FRONT CLEARANCE (SIDEMARKER) LIGHTS → **GROUND**.

ALSO, CURRENT FROM ALT FUSE FLOWS TO FOG FUSE → FOG LIGHT RELAY (POINT SIDE) → FOG LIGHTS → **GROUND**, SO BOTH TAIL AND FOG LIGHTS LIGHT UP. THIS IS HOW THE DAYTIME RUNNING LIGHT SYSTEM OPERATES AND TAIL, FOG LIGHT HAVE LIGHTED UP. THE TAIL AND FOG LIGHT REMAIN ON EVEN IF THE PARKING BRAKE LEVER IS PULLED UP (PARKING BRAKE SW ON).

EVEN IF THE ENGINE STALLS WITH THE IGNITION SW ON AND THERE IS NO VOLTAGE FROM **TERMINAL “L”** OF ALTERNATOR, THE TAIL AND FOG LIGHTS REMAIN ON.

IF THE IGNITION SW IS THEN TURNED OFF, THE TAIL AND FOG LIGHTS ARE TURNED OFF.

IF THE ENGINE IS STARTED WHILE THE PARKING BRAKE IS RELEASED (PARKING BRAKE SW OFF), THE DAYTIME RUNNING LIGHT SYSTEM OPERATES AND THE TAIL AND FOG LIGHTS LIGHT UP AS THE ENGINE STARTS.

SERVICE HINTS

R 9(B), R10(A) RETRACT CONTROL RELAY

(A) 2, (B) 3, (B) 8—GROUND: ALWAYS APPROX. 12 VOLTS

(A) 4—GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION

(B) 6—GROUND : NO CONTINUITY WITH LIGHT CONTROL SW AT **OFF** OR **HEAD** POSITION
CONTINUITY WITH LIGHT CONTROL SW AT **HOLD** OR **TAIL** POSITION

(A) 3—GROUND : NO CONTINUITY WITH LIGHT CONTROL SW AT **OFF** OR **HOLD** POSITION
CONTINUITY WITH LIGHT CONTROL SW AT **TAIL** OR **HEAD** POSITION

(B) 8—GROUND : NO CONTINUITY WITH LIGHT CONTROL SW AT **OFF**, **HOLD** OR **TAIL** POSITION
CONTINUITY WITH LIGHT CONTROL SW AT **HEAD** POSITION OR DIMMER SW AT **FLASH** POSITION

(B)12—GROUND : ALWAYS CONTINUITY

(A) 5—GROUND : CONTINUITY WITH PARKING BRAKE LEVER PULLED UP (PARKING BRAKE SW ON)

(B) 2, (B) 5—(B)11 : NO CONTINUITY WITH RETRACT MOTOR AT **LOWERMOST** POSITION
CONTINUITY WITH RETRACT MOTOR AT ANY POSITION EXCEPT **LOWERMOST** POSITION

(B) 2, (B) 5—(B) 7 : NO CONTINUITY WITH RETRACT MOTOR AT **UPPERMOST** POSITION
CONTINUITY WITH RETRACT MOTOR AT ANY POSITION EXCEPT **UPPERMOST** POSITION

R 2, R 3 RETRACT MOTOR

2-3 : OPEN WITH RETRACT MOTOR AT **LOWERMOST** POSITION

2-1 : OPEN WITH RETRACT MOTOR AT **UPPERMOST** POSITION

C16 (A) DIMMER SW [COMB. SW]

9-12 : CLOSED WITH DIMMER SW AT **HIGH** OR **FLASH** POSITION

FOG LIGHT RELAY

1-4 : CLOSED WITH LIGHT CONTROL SW AT **HEAD** POSITION, DIMMER SW AT **LOW** POSITION AND FOG LIGHT SW ON
CLOSED WITH ENGINE RUNNING AND PARKING BRAKE LEVER RELEASED

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C14	32	F10	B 30 (5S-FE), 31 (4A-FE)	J 6	33
C16	A 32	F11	33	P 1	33
C17	B 32	H 1	29 (3S-GTE), 30 (5S-FE)	R 2	29 (3S-GTE), 30 (5S-GTE)
F 1	28 (3S-GTE), 30 (5S-FE)		31 (4A-FE)		31 (4A-FE)
	31 (4A-FE)	H 2	29 (3S-GTE), 30 (5S-FE)	R 3	29 (3S-GTE), 30 (5S-GTE)
F 2	28 (3S-GTE), 30 (5S-FE)		31 (4A-FE)		31 (4A-FE)
	31 (4A-FE)	J 2	33	R 9	B 33
F10	A 28 (3S-GTE)	J 3	33	R10	A 33

 : RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
3	26	R/B NO. 3 (RIGHT KICK PANEL)
4	27	R/B NO. 4 (RIGHT KICK PANEL)

 : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1B		
1E		
1I	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2A	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
	23 (ALL-TRAC/4WD)	
2B	22 (2WD)	ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)
	23 (ALL-TRAC/4WD)	
2D	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
	23 (ALL-TRAC/4WD)	
	22 (2WD)	
2E	23 (ALL-TRAC/4WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
	22 (2WD)	
3A	23 (ALL-TRAC/4WD)	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)
3B	25	

 : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EA1	38 (3S-GTE)	COWL WIRE AND ENGINE ROOM MAIN WIRE (FRONT SIDE OF RIGHT FRONT FENDER)
	40 (5S-FE)	
	42 (4A-FE)	
ET1	38 (3S-GTE)	ENGINE ROOM MAIN WIRE AND ENGINE WIRE (NEAR THE J/B NO. 2)
	40 (5S-FE)	
	42 (4A-FE)	
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)

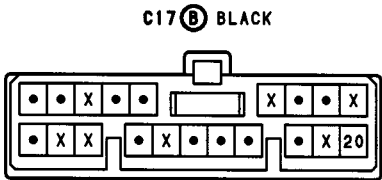
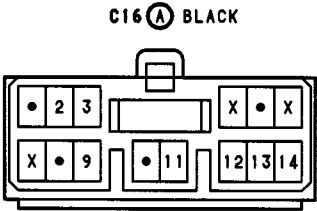
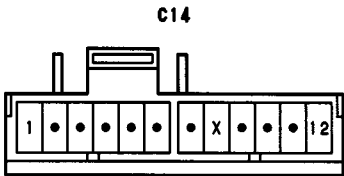
 : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
EA	38 (3S-GTE)	FRONT RIGHT FENDER
	40 (5S-FE)	
	42 (4A-FE)	
EB	38 (3S-GTE)	FRONT LEFT FENDER
	40 (5S-FE)	
	42 (4A-FE)	
ID	44	LEFT KICK PANEL

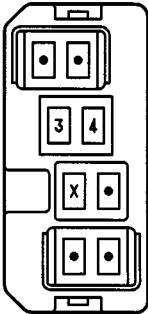
 : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 1	38 (3S-GTE)	ENGINE ROOM MAIN WIRE	E 3	40 (5S-FE)	ENGINE ROOM MAIN WIRE
	40 (5S-FE)			42 (4A-FE)	
	42 (4A-FE)		E13	38 (3S-GTE)	
E 2	38 (3S-GTE)			40 (5S-FE)	
	40 (5S-FE)			42 (4A-FE)	
	42 (4A-FE)		I 2	46	COWL WIRE
E 3	38 (3S-GTE)		I 8		

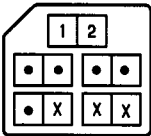
HEADLIGHT AND FOGLIGHT (FOR CANADA)



F 1, F 2 (ALL-TRAC/4WD) F10(A) BLACK



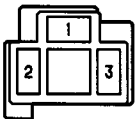
(2WD) F10(B) BLACK



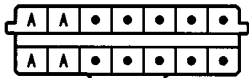
F11 BLACK



H 1, H 2 BROWN

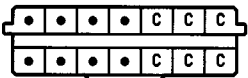


J 2



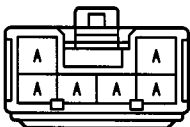
(HINT:SEE PAGE 7)

J 3



(HINT:SEE PAGE 7)

J 6



(HINT:SEE PAGE 7)

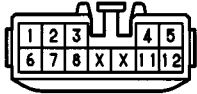
P 1 BLACK



R 2, R 3 BLACK



R 9(B)

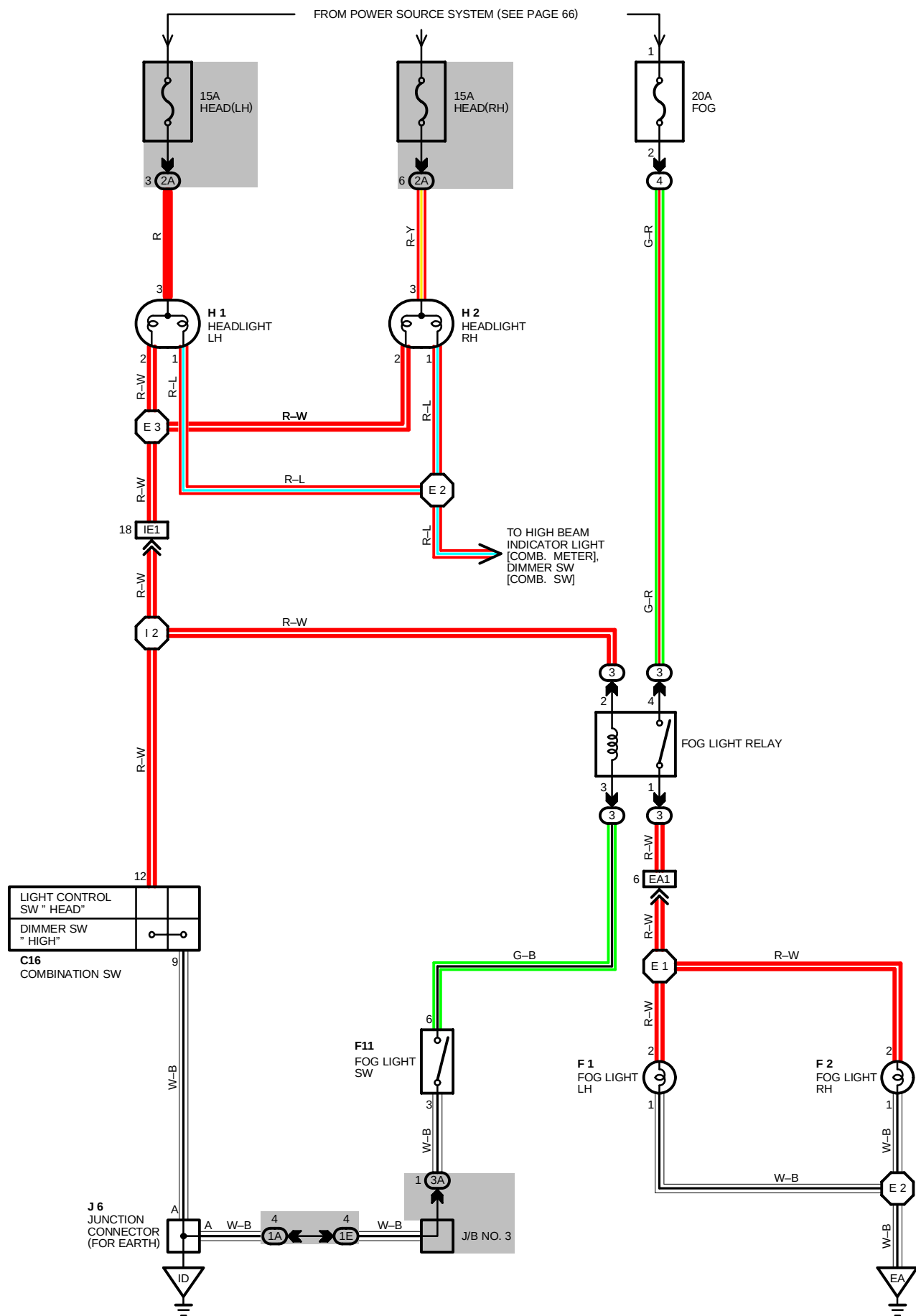


R10(A)



-Memo-

FOG LIGHT (FOR USA)



SERVICE HINTS

FOG LIGHT RELAY

(3) 1–(3) 4 : CLOSED WITH LIGHT CONTROL SW AT **HEAD** POSITION, DIMMER SW AT **LOW** POSITION AND FOG LIGHT SW ON

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C16	32	F 2	31 (4A–FE)	H 2	28 (3S–GTE), 30 (5S–FE)
F 1	28 (3S–GTE), 30 (5S–FE)	F11	33		31 (4A–FE)
	31 (4A–FE)	H 1	28 (3S–GTE), 30 (5S–FE)	J 6	33
F 2	28 (3S–GTE), 30 (5S–FE)		31 (4A–FE)		

○ : RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
3	26	R/B NO. 3 (RIGHT KICK PANEL)
4	27	R/B NO. 4 (RIGHT KICK PANEL)

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
2A	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
	23 (ALL–TRAC/4WD)	
3A	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EA1	38 (3S–GTE)	COWL WIRE AND ENGINE ROOM MAIN WIRE (FRONT SIDE OF RIGHT FRONT FENDER)
	40 (5S–FE)	
	42 (4A–FE)	
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)

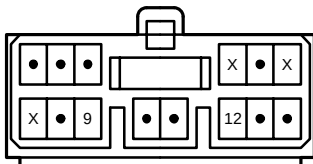
▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
EA	38 (3S–GTE)	FRONT RIGHT FENDER
	40 (5S–FE)	
	42 (4A–FE)	
ID	44	LEFT KICK PANEL

○ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 1	38 (3S–GTE)	ENGINE ROOM MAIN WIRE	E 2	42 (4A–FE)	ENGINE ROOM MAIN WIRE
	40 (5S–FE)		E 3	38 (3S–GTE)	
	42 (4A–FE)			40 (5S–FE)	
E 2	38 (3S–GTE)			42 (4A–FE)	
	40 (5S–FE)		I 2	46	COWL WIRE

C16 BLACK



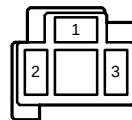
F 1, F 2



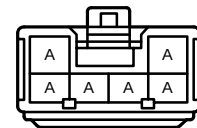
F11 BLACK



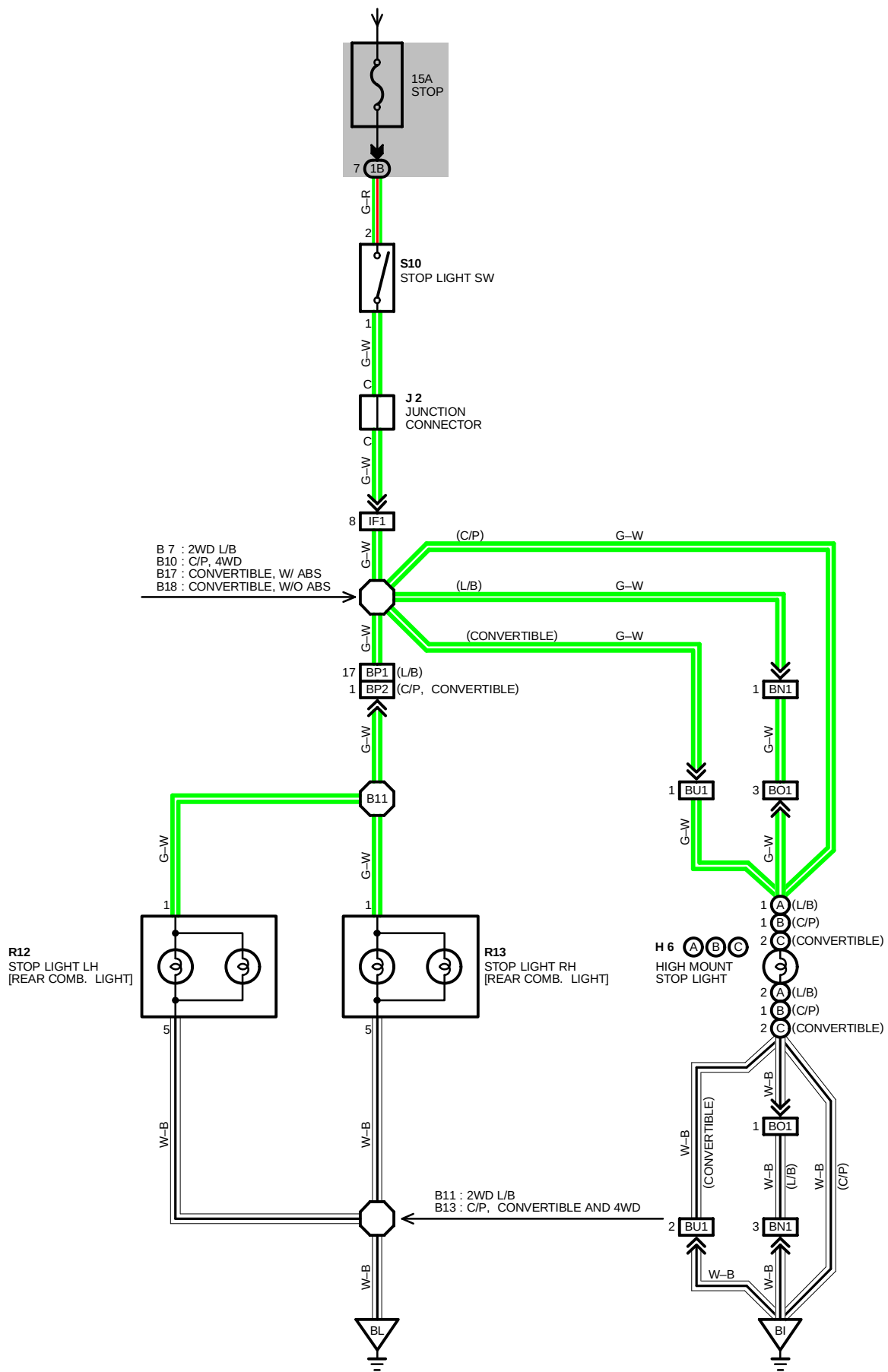
H 1, H 2 BROWN



J 6



(HINT : SEE PAGE 7)



SERVICE HINTS

S10 STOP LIGHT SW

1-2: CLOSED WITH BRAKE PEDAL DEPRESSED

○ : PARTS LOCATION

CODE		SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
H 6	A	34 (L/B)	J 2	33	R10	33
	B	35 (C/P)	R12	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)		
	C	36 (CONVERTIBLE)	R13	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)		

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1B	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IF1	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
BN1	48 (L/B)	BACK DOOR NO. 1 WIRE AND FLOOR WIRE (LEFT SIDE OF PACKAGE TRAY TRIM)
BO1	48 (L/B)	BACK DOOR NO. 1 WIRE AND BACK DOOR NO. 1 SUB WIRE (BACK DOOR UPPER LEFT)
BP1	48 (L/B)	FLOOR WIRE AND LUGGAGE ROOM WIRE (LEFT QUARTER PANEL CENTER)
BP2	50 (C/P)	FLOOR WIRE AND LUGGAGE ROOM WIRE (LUGGAGE COMPARTMENT LEFT)
	52 (CONVERTIBLE)	
BU1	50 (C/P)	FLOOR WIRE AND LUGGAGE ROOM WIRE (LUGGAGE COMPARTMENT LEFT)
	52 (CONVERTIBLE)	

▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
BI	48 (L/B)	UNDER THE LEFT CENTER PILLAR
	50 (C/P)	
BL	48 (L/B)	BACK PANEL CENTER
	50 (C/P)	
	52 (CONVERTIBLE)	

○ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
B 7	48 (L/B)	FLOOR WIRE	B13	48 (L/B)	LUGGAGE ROOM WIRE
B10	48 (L/B)			50 (C/P)	
	50 (C/P)			52 (CONVERTIBLE)	
B11	48 (L/B)	LUGGAGE ROOM WIRE	B17	52 (CONVERTIBLE)	FLOOR WIRE
	50 (C/P)		B18		
	52 (CONVERTIBLE)				

(L/B) H 6 A



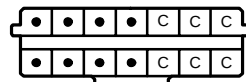
(C/P) H 6 B



(CONVERTIBLE) H 6 C

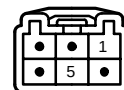


J 2

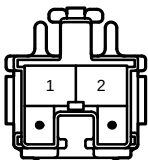


(HINT : SEE PAGE 7)

R12, R13

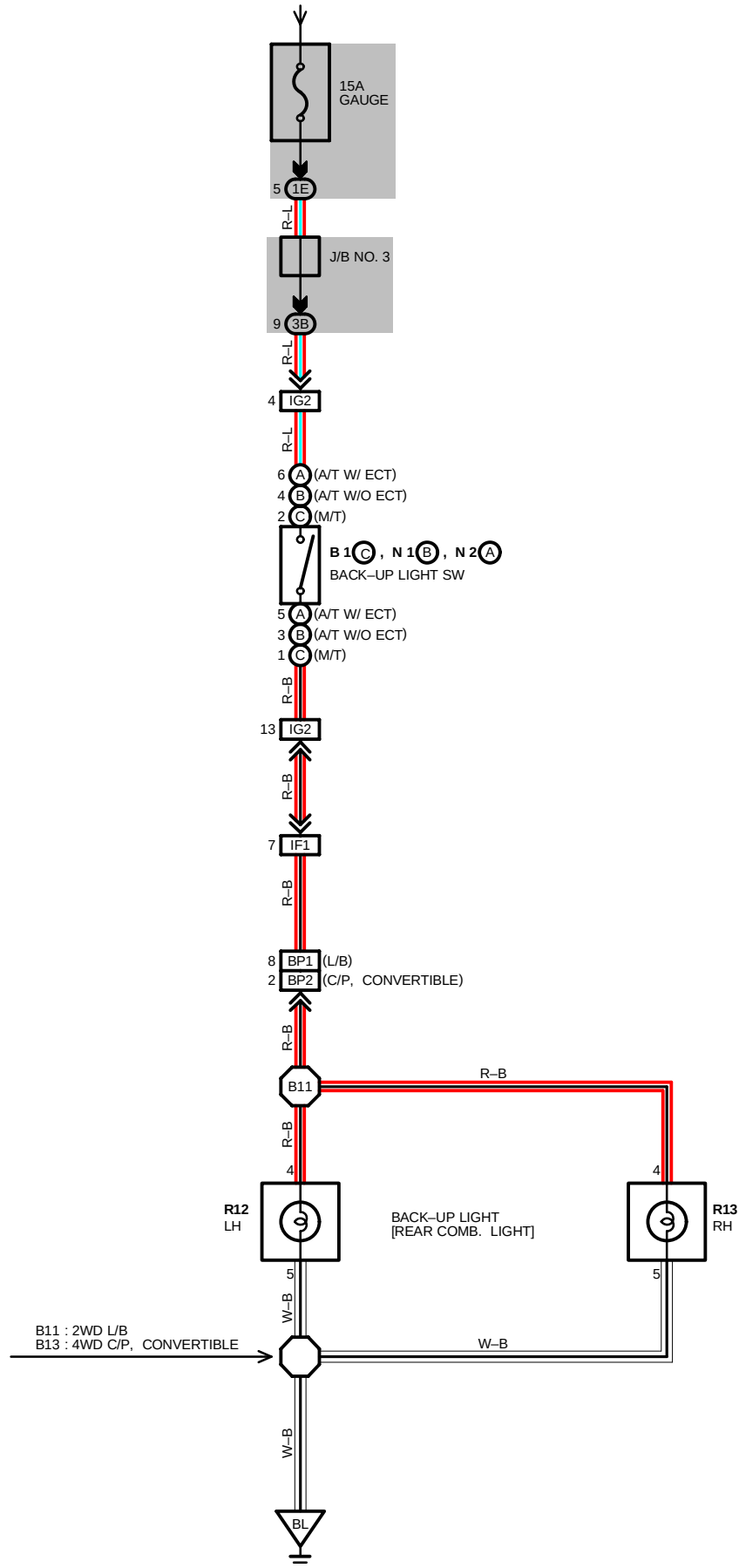


S10



BACK-UP LIGHT

FROM POWER SOURCE SYSTEM (SEE PAGE 66)



SERVICE HINTS

B 1(C), N 1(B), N 2(A) BACK-UP LIGHT SW

(A) 5-6, (B) 3-4, (C) 1-2 : CLOSED WITH SHIFT LEVER IN R POSITION

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
B 1	C	28 (3S-GTE), 30 (5S-FE)	N 1 B	31 (4A-FE)	R12 34 (L/B), 35 (C/P), 36 (CONVERTIBLE)
		31 (4A-FE)	N 2 A	30 (5S-FE)	R13 34 (L/B), 35 (C/P), 36 (CONVERTIBLE)

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1E	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
3B	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

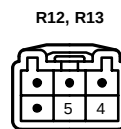
CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IF1	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
IG2	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)
BP1	48 (L/B)	FLOOR WIRE AND LUGGAGE ROOM WIRE (LEFT QUARTER PANEL CENTER)
BP2	50 (C/P)	FLOOR WIRE AND LUGGAGE ROOM WIRE (LUGGAGE COMPARTMENT LEFT)
	50 (CONVERTIBLE)	

▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
BL	48 (L/B)	BACK PANEL CENTER
	50 (C/P)	
	50 (CONVERTIBLE)	

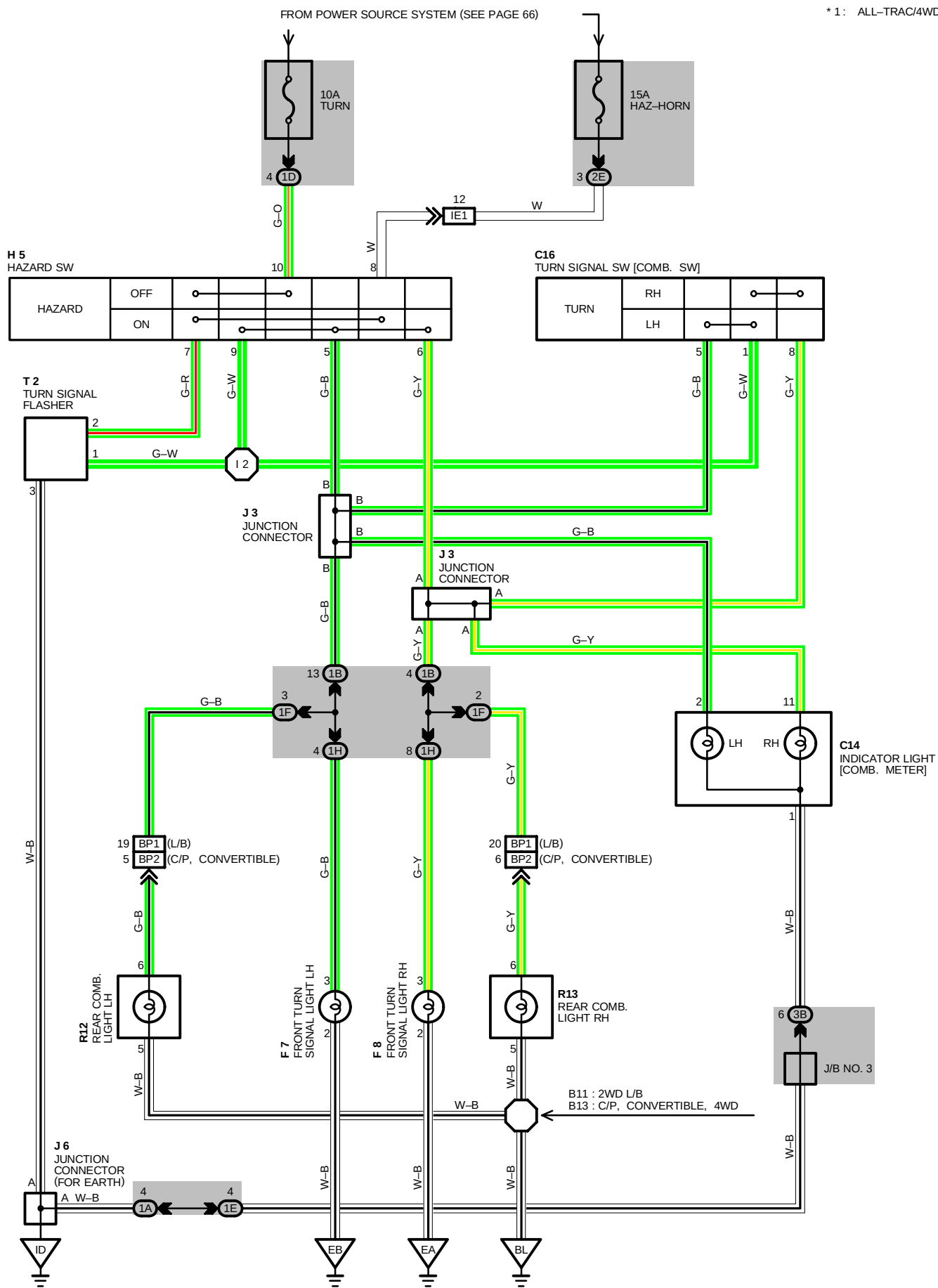
○ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
B11	48 (L/B), 50 (C/P)	LUGGAGE ROOM WIRE	B13	48 (L/B), 50 (C/P)	LUGGAGE ROOM WIRE
	52 (CONVERTIBLE)			52 (CONVERTIBLE)	



TURN SIGNAL AND HAZARD WARNING LIGHT

* 1: ALL-TRAC/4WD



SERVICE HINTS

T 2 TURN SIGNAL FLASHER

- 2-GROUND : APPROX. 12 VOLTS WITH IGNITION SW ON OR HAZARD SW ON
- 1-GROUND : CHANGES FROM 12 TO 0 VOLTS WITH IGNITION SW ON AND TURN SIGNAL SW LEFT OR RIGHT, OR WITH HAZARD SW ON
- 3-GROUND : ALWAYS CONTINUITY

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C14	32	F 8	28 (3S-GTE), 30 (5S-FE)	J 6	33
C16	32		31 (4A-FE)	R12	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)
F 7	28 (3S-GTE), 30 (5S-FE)	H 5	33	R13	34 (L/B), 35 (C/P) 36 (CONVERTIBLE)
	31 (4A-FE)	J 3	33	T 2	33

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1B		
1D		
1E		
1F	20	FLOOR WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2E	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO.2 (NEAR THE BATTERY)
	23 (ALL-TRAC/4WD)	
3B	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)
BP1	48 (L/B)	FLOOR WIRE AND LUGGAGE ROOM WIRE (LEFT QUARTER PANEL CENTER)
BP2	50 (C/P)	FLOOR WIRE AND LUGGAGE ROOM WIRE (LUGGAGE COMPARTMENT LEFT)
	52 (CONVERTIBLE)	

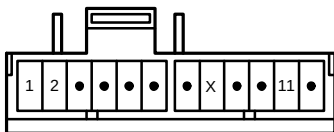
▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
EA	38 (3S-GTE)	FRONT RIGHT FENDER
	40 (5S-FE)	
	42 (4A-FE)	
EB	38 (3S-GTE)	FRONT LEFT FENDER
	40 (5S-FE)	
	42 (4A-FE)	
ID	44	LEFT KICK PANEL
BL	48 (L/B)	BACK PANEL CENTER
	50 (C/P)	
	52 (CONVERTIBLE)	

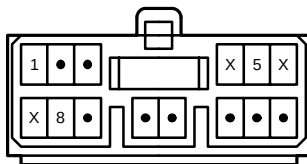
○ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I 2	46	COWL WIRE	B13	46 (L/B), (C/P)	LUGGAGE ROOM WIRE
B11	48 (L/B)	LUGGAGE ROOM WIRE		52 (CONVERTIBLE)	

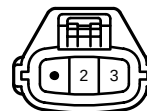
C14



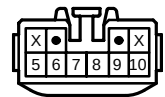
C16 BLACK



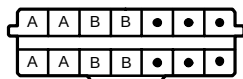
F 7, F 8 GRAY



H 5 BLACK

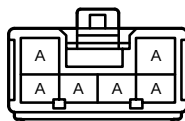


J 3



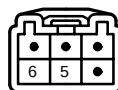
(HINT : SEE PAGE 7)

J 6

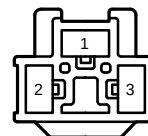


(HINT : SEE PAGE 7)

R12, R13



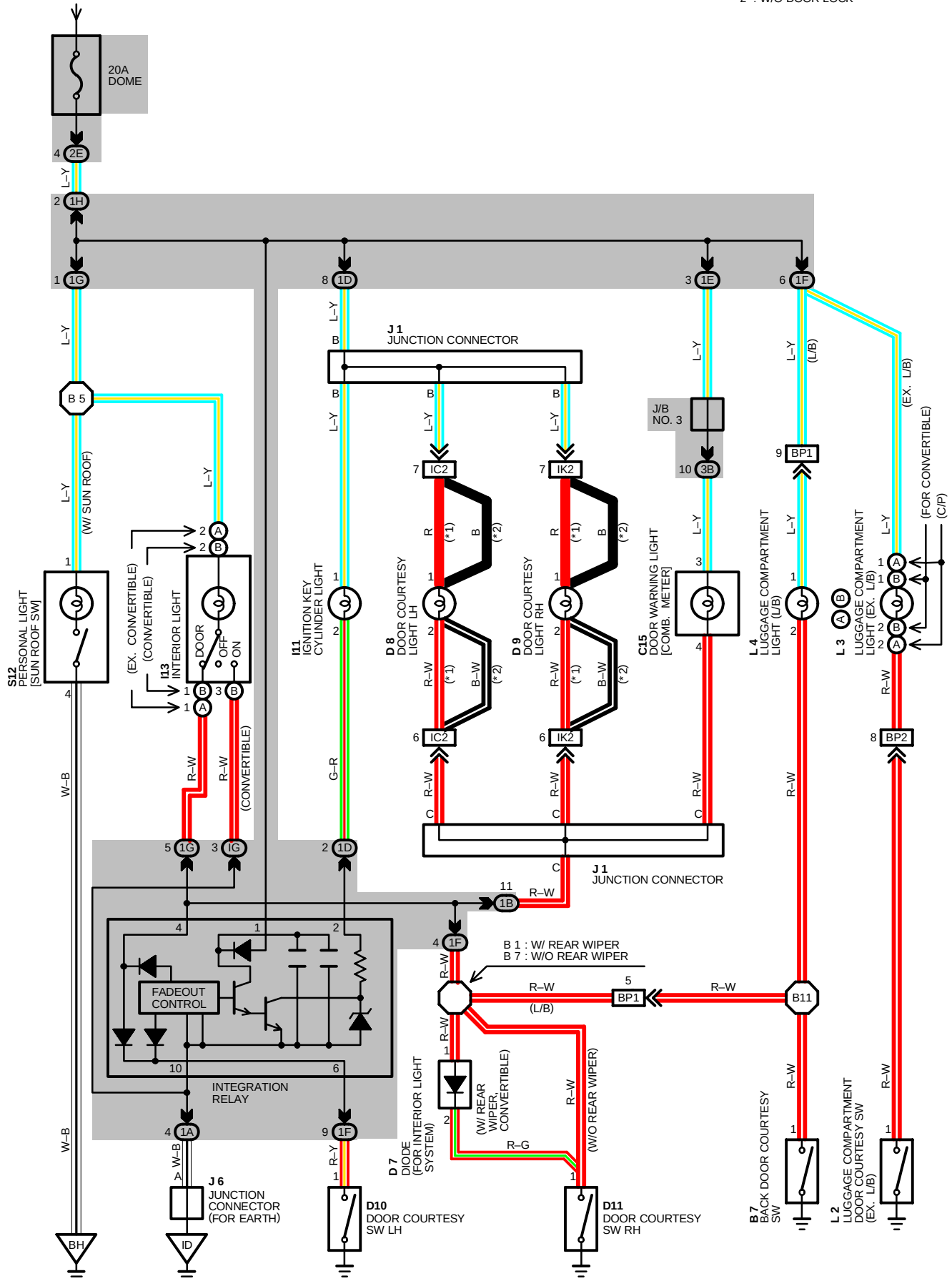
T 2



INTERIOR LIGHT

FROM POWER SOURCE SYSTEM (SEE PAGE 66)

*1 : W/ DOOR LOCK
*2 : W/O DOOR LOCK



SERVICE HINTS

INTEGRATION RELAY

(1B) 11-GROUND : APPROX. 12 VOLTS WITH DOOR CLOSED
0 VOLTS WITH EACH DOOR OPEN

D10, D11 DOOR COURTESY SW

1-GROUND : CLOSED WITH DOOR OPEN

L 2 LUGGAGE COMPARTMENT LIGHT SW

1-GROUND : CLOSED WITH LUGGAGE COMPARTMENT DOOR OPEN

B 7 BACK DOOR COURTESY SW

1-GROUND : CLOSED WITH BACK DOOR OPEN

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
B 7	34 (L/B)	D11	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)	L 2	35 (C/P), 36 (CONVERTIBLE)
C15	32	I11	33	L 3	A
D 7	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)	I13	A		B
D 8	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)		B	L 4	34 (L/B)
D 9	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)	J 1	33	S12	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)
D10	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)	J 6	33		

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1B		
1D		
1E		
1F	20	FLOOR WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1G	20	ROOF WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2E	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
	23 (ALL-TRAC/4WD)	
3B	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

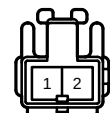
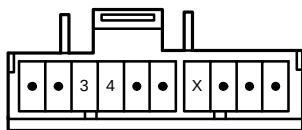
CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IC2	44	FRONT DOOR LH WIRE AND COWL WIRE (LEFT KICK PANEL)
IK2	46	FRONT DOOR RH WIRE AND COWL WIRE (RIGHT KICK PANEL)
BP1	48 (L/B)	FLOOR WIRE AND LUGGAGE ROOM WIRE (LEFT QUARTER PANEL CENTER)
BP2	50 (C/P)	FLOOR WIRE AND LUGGAGE ROOM WIRE (LUGGAGE COMPARTMENT LEFT)

▽ : GROUND POINTS

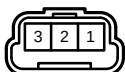
CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL
BH	48 (L/B), 50 (C/P)	ROOF LEFT
	52 (CONVERTIBLE)	

○ : SPLICE POINTS

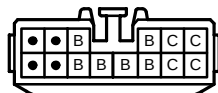
CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
B 1	48 (L/B)	FLOOR WIRE	B 7	48 (L/B), 50 (C/P)	FLOOR WIRE
B 5	48 (L/B), 50 (C/P)	ROOF WIRE	B11	52 CONVERTIBLE	
B 7, L 2	52 (CONVERTIBLE)			48 (L/B)	LUGGAGE ROOM WIRE



I13 (B) BLACK

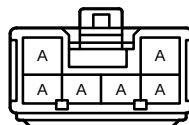


J 1



(HINT : SEE PAGE 7)

J 6

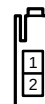


(HINT : SEE PAGE 7)

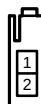
L 3 (A) GRAY



L 3 (B)



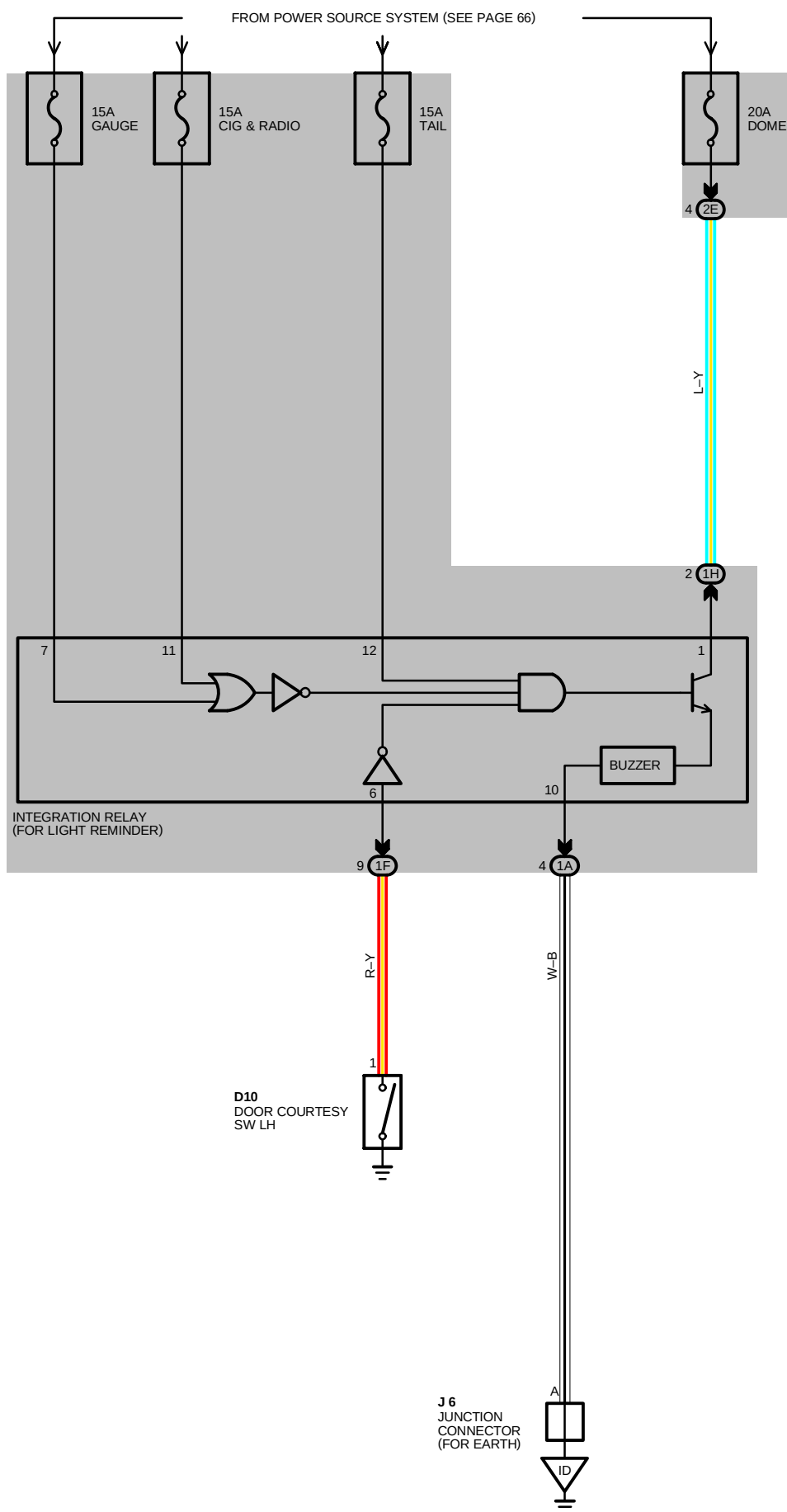
L 4



S12



LIGHT REMINDER BUZZER



SYSTEM OUTLINE

CURRENT ALWAYS FLOWS TO **TERMINAL 1** OF THE LIGHT REMINDER RELAY THROUGH THE DOME FUSE.

WITH THE IGNITION SW IN **ACC** POSITION, CURRENT FLOWS TO **TERMINAL 11** OF THE LIGHT REMINDER RELAY. WHEN THE IGNITION SW IS TURNED TO **ON** POSITION, CURRENT FLOWS TO **TERMINAL 7** THE LIGHT REMINDER RELAY. WHEN THE LIGHT CONTROL SW IS TURNED TO **TAIL** OR **HEAD** POSITION, CURRENT IS APPLIED TO **TERMINAL 12** THE LIGHT REMINDER RELAY.

LIGHT REMINDER SYSTEM

WHEN THE LIGHT CONTROL SW IN **TAIL** OR **HEAD** POSITION, THE IGNITION SW IS TURNED TO **OFF** FROM **ON** POSITION, AND THE DRIVER'S DOOR IS OPENED (DOOR COURTESY SW ON), THE CURRENT FLOW TO **TERMINAL 7** AND **11** OF THE LIGHT REMINDER RELAY STOPS. AS A RESULT, THE RELAY IS ACTIVATED AND CURRENT FLOWS FROM **TERMINAL 1** OF THE RELAY → **TERMINAL 10** → TO **GROUND**, SOUNDING THE LIGHT REMINDER BUZZER.

SERVICE HINTS

LIGHT REMINDER RELAY (INTEGRATION RELAY)

- 7-GROUND : APPROX. **12** VOLTS WITH IGNITION AT **ON** POSITION
- 11-GROUND : APPROX. **12** VOLTS WITH IGNITION AT **ACC** OR **ON** POSITION
- 12-GROUND : APPROX. **12** VOLTS WITH LIGHT CONTROL SW AT **TAIL** OR **HEAD** POSITION
- 6-GROUND : CONTINUITY WITH DRIVER'S DOOR OPEN
- 1-GROUND : ALWAYS APPROX. **12** VOLTS
- 10-GROUND : ALWAYS CONTINUITY



: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
D10	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)	J 6	33		



: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1F	20	FLOOR WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2E	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
	23 (ALL-TRAC/4WD)	



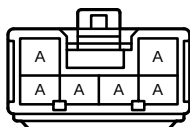
: GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL

D10

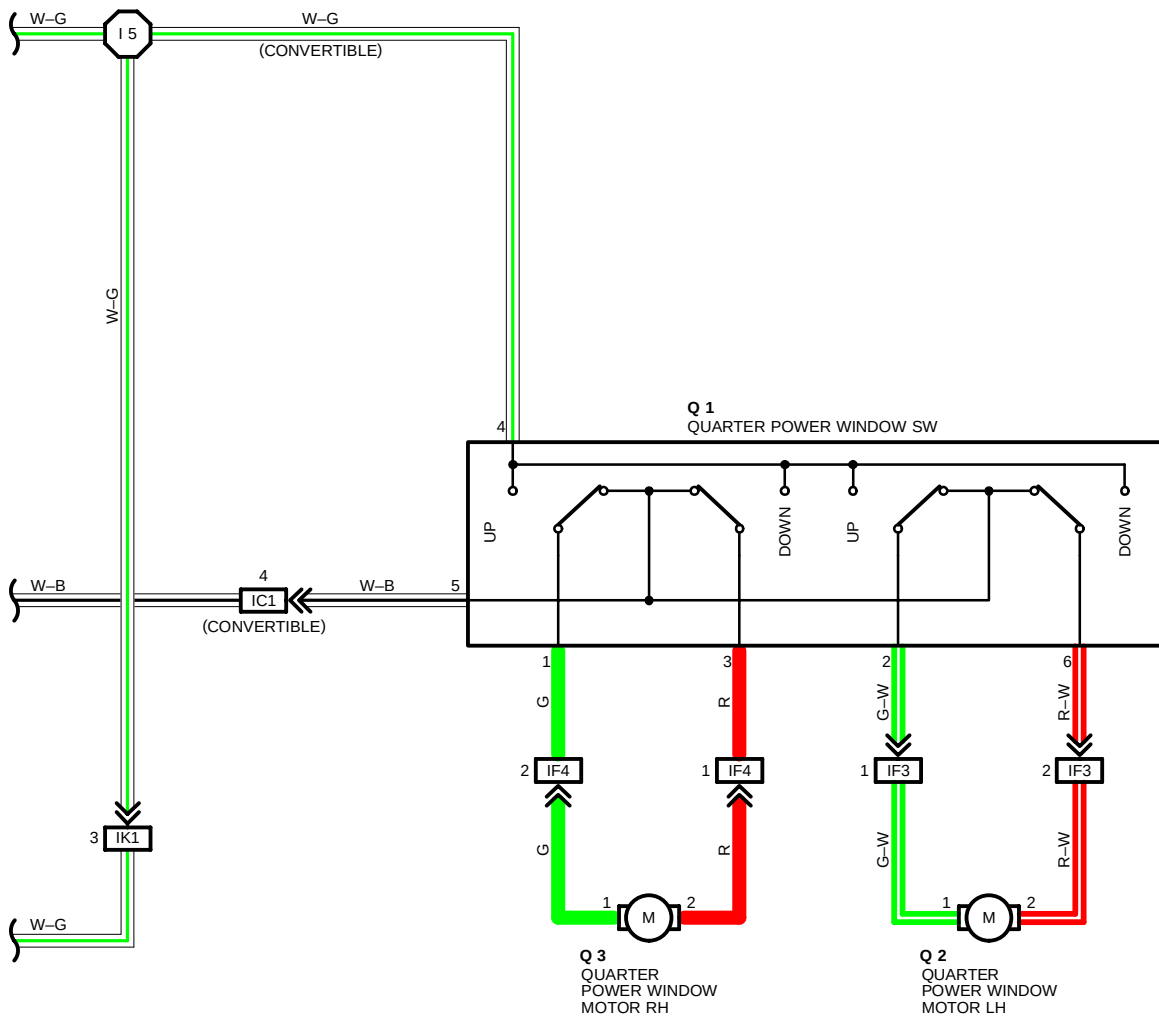


J 6



(HINT : SEE PAGE 7)





POWER WINDOW

SYSTEM OUTLINE

CURRENT ALWAYS FLOWS **TERMINAL 4** OF THE POWER MAIN RELAY THROUGH THE POWER FUSE. WITH THE IGNITION SW TURNED ON, CURRENT FLOWS THROUGH THE GAUGE FUSE TO **TERMINAL 2** OF THE POWER MAIN RELAY → **TERMINAL 3** → TO **GROUND**. THIS ACTIVATES THE RELAY AND CURRENT FLOWING TO **TERMINAL 1** OF THE POWER MAIN RELAY → TO **TERMINAL (A)6** OF THE POWER WINDOW MASTER SW AND **TERMINAL 6** OF THE POWER WINDOW SW RH (PASSENGER'S) AND **TERMINAL 4** OF THE QUARTER POWER WINDOW SW.

1. MANUAL OPERATION (DRIVER'S WINDOW)

WITH THE IGNITION SW TURNED ON AND WITH THE POWER WINDOW MASTER SW IN **UP** POSITION, THE CURRENT FLOWING TO **TERMINAL (A)6** OF THE POWER WINDOW MASTER SW FLOWS TO **TERMINAL (A)5** OF THE MASTER SW → **TERMINAL 2** OF THE POWER WINDOW MOTOR (DRIVER'S) → MOTOR → **TERMINAL 1** → **TERMINAL (A)1** OF THE MASTER SW → **TERMINAL (A)3** → TO **GROUND** AND CAUSES THE POWER WINDOW MOTOR TO ROTATE IN THE UP DIRECTION. THE WINDOW ASCENDS ONLY WHILE THE SW IS BEING PUSHED. IN DOWN OPERATION, THE FLOW OF CURRENT FROM **TERMINAL (A)6** OF THE POWER WINDOW MASTER SW TO **TERMINAL (A)1** OF THE MASTER SW CAUSES THE FLOW OF CURRENT FROM **TERMINAL 1** OF THE MOTOR → MOTOR → **TERMINAL 2** → **TERMINAL (A)5** OF THE MASTER SW → **TERMINAL (A)3** → TO **GROUND**, FLOWING IN THE OPPOSITE DIRECTION TO MANUAL UP OPERATION AND CAUSING THE MOTOR TO ROTATE IN REVERSE, LOWERING THE WINDOW.

2. AUTO DOWN OPERATION

WITH THE IGNITION SW ON AND WITH THE DRIVER'S SW OF THE POWER WINDOW MASTER SW IN **DOWN** POSITION, CURRENT FLOWING TO **TERMINAL (A)6** OF THE MASTER SW FLOWS TO **TERMINAL (A)1** OF THE MASTER SW → **TERMINAL 1** OF THE POWER WINDOW MOTOR → MOTOR → **TERMINAL 2** → **TERMINAL (A)5** OF THE MASTER SW → **TERMINAL 3** → TO **GROUND**, CAUSING THE MOTOR TO ROTATE TOWARDS THE DOWN SIDE. THEN THE SOLENOID IN THE MASTER SW IS ACTIVATED AND IT LOCKS THE DRIVER'S SW BEING PUSHED, CAUSING THE MOTOR TO CONTINUE TO ROTATE IN AUTO DOWN OPERATION.

WHEN THE WINDOW HAS COMPLETELY DESCENDED, THE CURRENT FLOW BETWEEN **TERMINAL (A)5** OF THE MASTER SW AND **TERMINAL (A)3** INCREASES. AS A RESULT, THE SOLENOID STOPS OPERATING, THE DRIVER'S SW TURNS OFF AND FLOW FROM **TERMINAL (A)6** OF THE MASTER SW TO **TERMINAL (A)1** IS CUT OFF, STOPPING THE MOTOR SO THAT AUTO STOP OCCURS.

3. STOPPING OF AUTO DOWN AT DRIVER'S WINDOW

WHEN THE DRIVER'S SW IS PUSHED TO THE UP SIDE DURING AUTO DOWN OPERATION, A GROUND CIRCUIT OPENS IN THE MASTER SW AND CURRENT DOES NOT FLOW FROM **TERMINAL (A)5** OF THE MASTER SW → TO **TERMINAL (A)3**, SO THE MOTOR STOPS, CAUSING AUTO DOWN OPERATION TO STOP. IF THE DRIVER'S SW IS PUSHED CONTINUOUSLY, THE MOTOR ROTATES IN THE UP DIRECTION IN MANUAL UP OPERATION.

4. MANUAL OPERATION BY POWER WINDOW SW (PASSENGER'S WINDOW)

WITH POWER WINDOW SW (PASSENGER'S) PUSHED TO THE UP SIDE, CURRENT FLOWING FROM **TERMINAL 6** OF THE POWER WINDOW SW FLOWS TO **TERMINAL 5** OF THE POWER WINDOW SW → **TERMINAL 2** OF THE WINDOW MOTOR → MOTOR → **TERMINAL 1** → **TERMINAL 1** OF THE POWER WINDOW SW → **TERMINAL 3** → **TERMINAL (A)4** OF THE MASTER SW → **TERMINAL (A)3** → TO **GROUND** AND CAUSES THE POWER WINDOW MOTOR (PASSENGER'S) TO ROTATE IN THE UP DIRECTION. UP OPERATION CONTINUES ONLY WHILE THE POWER WINDOW SW IS PUSHED TO THE UP SIDE. WHEN THE WINDOW DESCENDS, THE CURRENT FLOWING TO THE MOTOR FLOWS IN THE OPPOSITE DIRECTION, FROM **TERMINAL 1** → MOTOR → TO **TERMINAL 2**, AND THE MOTOR ROTATES IN REVERSE. WHEN THE WINDOW LOCK SW IS PUSHED TO THE LOCK SIDE, THE GROUND CIRCUIT TO THE PASSENGER'S WINDOW BECOMES OPEN.

AS A RESULT, EVEN IF OPEN/CLOSE OPERATION OF THE PASSENGER'S WINDOW IS TRIED, THE CURRENT FROM **TERMINAL (A)6** OF THE POWER WINDOW MASTER SW IS NOT GROUNDED AND THE MOTOR DOES NOT ROTATE, SO THE PASSENGER'S WINDOW CAN NOT BE OPERATED AND WINDOW LOCK OCCURS.

5. MANUAL OPERATION BY QUARTER WINDOW (CONVERTIBLE)

WITH THE QUARTER POWER WINDOW SW PUSHED TO UP SIDE, FLOWING FROM **TERMINAL 4** OF THE QUARTER POWER WINDOW SW FLOWS TO **TERMINAL 1** AND/OR **2** → **TERMINAL 1** OF THE QUARTER POWER WINDOW MOTOR → MOTOR → **TERMINAL 2** OF THE MOTOR → **TERMINAL 3** AND/OR **6** OF THE QUARTER POWER WINDOW SW → **TERMINAL 5** → **TERMINAL (B)1** OF THE POWER WINDOW MASTER SW → **TERMINAL (A)3** → TO **GROUND** AND CAUSES THE QUARTER POWER WINDOW MOTOR TO ROTATE IN THE UP DIRECTION. UP OPERATION CONTINUES ONLY WHILE THE QUARTER POWER WINDOW SW IS PUSHED TO UP SIDE. WHEN THE WINDOW DESCENDS, THE CURRENT FLOWING TO THE MOTOR FLOWS IN THE OPPOSITE DIRECTION, FROM **TERMINAL 2** → MOTOR → TO **TERMINAL 1**, AND THE MOTOR ROTATES IN REVERSE.

WHEN THE WINDOW LOCK SW IS PUSHED TO THE LOCK SIDE, THE **GROUND** CIRCUIT TO THE QUARTER POWER WINDOW BECOMES OPEN. AS A RESULT, EVEN IF OPEN/CLOSED OPERATION OF QUARTER WINDOW IS TRIED, THE CURRENT FROM **TERMINAL 5** OF THE QUARTER POWER WINDOW SW IS NOT GROUNDED AND THE MOTOR DOES NOT ROTATE, SO THE QUARTER WINDOW CAN NOT BE OPERATED AND WINDOW LOCK OCCURS.

6. KEY OFF POWER WINDOW OPERATION

WITH THE IGNITION SW TURNED FROM ON TO OFF, THE DOOR LOCK ECU OPERATES AND CURRENT FLOWS FROM POWER FUSE → **TERMINAL 8** OF THE DOOR LOCK ECU → **TERMINAL 15** → **TERMINAL 2** OF THE POWER MAIN RELAY → **TERMINAL 3** → TO **GROUND** FOR ABOUT **60** SECONDS. THE SAME AS NORMAL OPERATION, THE CURRENT FLOWS FROM POWER FUSE → **TERMINAL 4** OF THE POWER MAIN RELAY → **TERMINAL 1** → **TERMINAL (A)6** OF THE POWER WINDOW MASTER SW AND **TERMINAL 6** OF POWER WINDOW SW RH (PASSENGER'S) AND **TERMINAL 4** OF THE QUARTER POWER WINDOW SW. AS A RESULT, FOR ABOUT **60** SECONDS AFTER THE IGNITION SW IS TURNED OFF, IT IS POSSIBLE TO RAISE AND LOWER THE POWER WINDOW BY THE FUNCTIONING OF THIS RELAY. ALSO, BY OPENING THE DOOR (DOOR COURTESY SW ON) WITHIN ABOUT **60** SECONDS AFTER TURNING THE IGNITION SW TO OFF, A SIGNAL IS INPUT TO **TERMINAL 2** OF DOOR LOCK ECU . AS A RESULT, THE ECU TURNS OFF AND UP AND DOWN OF THE MOVEMENT OF THE WINDOW STOPS.

SERVICE HINTS

D 6 DOOR LOCK ECU

8-GROUND : ALWAYS APPROX. **12** VOLTS

16-GROUND: ALWAYS CONTINUITY

1-GROUND : APPROX. **12** VOLTS WITH IGNITION SW AT **ON** POSITION

2-GROUND : CONTINUITY WITH DOOR OPENED

15-GROUND: APPROX. **12** VOLTS WITH IGNITION SW ON AND STAYS AT **12** VOLTS FOR **60** SECONDS AFTER THE IGNITION SW IS TURNED OFF, BUT IF THE DOOR IS OPENED IN THIS **60** SECONDS PERIOD, VOLTAGE WILL DROP TO **0** VOLTS

D10 DOOR COURTESY SW

1-GROUND: CONTINUITY WITH DOOR OPEN

P 8 POWER WINDOW SW (PASSENGER'S)

6-GROUND: APPROX. **12** VOLTS WITH IGNITION SW ON AND STAYS AT **12** VOLTS FOR **60** SECONDS AFTER THE IGNITION SW IS TURNED OFF, BUT IF A DOOR IS OPENED IN THIS **60** SECONDS PERIOD, VOLTAGE WILL DROP TO **0** VOLTS

Q 1 QUARTER POWER WINDOW MOTOR

4-GROUND: APPROX. **12** VOLTS WITH IGNITION SW ON AND STAYS AT **12** VOLTS FOR **60** SECONDS AFTER THE IGNITION SW IS TURNED OFF, BUT IF A DOOR IS OPENED IN THIS **60** SECONDS PERIOD, VOLTAGE WILL DROP TO **0** VOLTS

P 1 POWER WINDOW MASTER SW

(A)3-GROUND: ALWAYS CONTINUITY

(A)6-GROUND: APPROX. **12** VOLTS WITH IGNITION SW ON AND STAYS AT **12** VOLTS FOR **60** SECONDS AFTER THE IGNITION SW IS TURNED OFF, BUT IF A DOOR IS OPENED IN THIS **60** SECONDS PERIOD, VOLTAGE WILL DROP TO **0** VOLTS

(A)5-GROUND: APPROX. **12** VOLTS WITH IGNITION SW AT **ON** POSITION AND MASTER SW AT **UP** POSITION

(A)1-GROUND: APPROX. **12** VOLTS WITH IGNITION SW AT **ON** POSITION AND MASTER SW AT **DOWN** OR **AUTO DOWN** POSITION

WINDOW LOCK SW

OPEN WITH WINDOW LOCK SW AT **LOCK** POSITION

POWER WINDOW

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
D 5	32	P 5	A 34 (L/B), 35 (C/P), 36 (CONVERTIBLE)	Q 1	36
D 6	32	P 6	B 36	Q 2	36
D10	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)	P	7 34 (L/B), 35 (C/P), 36 (CONVERTIBLE)	Q 3	36
J 4	33	P 8	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)		
J 6	33	P 9	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)		

○ : RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
2	26	R/B NO. 2 (LEFT KICK PANEL)

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1B	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
1F	20	FLOOR WIRE AND J/B NO. 1 (LEFT KICK PANEL)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

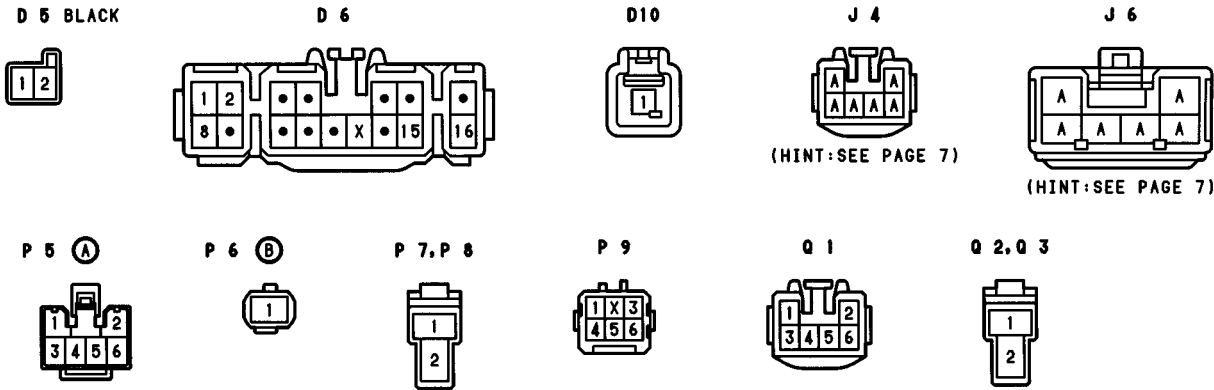
CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IC1	44	FRONT DOOR LH WIRE AND COWL WIRE (LEFT KICK PANEL)
IF3	46	FRONT WIRE AND COWL WIRE (LEFT KICK PANEL)
IF4	46	
IK1	46	FRONT DOOR RH WIRE AND COWL WIRE (RIGHT KICK PANEL)

▽ : GROUND POINTS

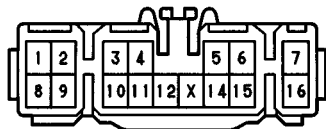
CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL

○ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I 1	46	COWL WIRE	I 5	46	COWL WIRE
I 2					



D 6



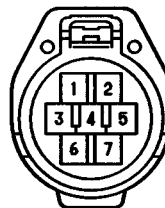
D10, D11



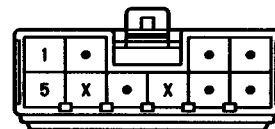
D12, D13



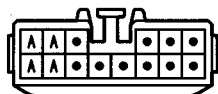
D14, D15 GRAY



I12 BLACK



J 1



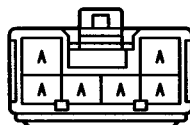
(HINT:SEE PAGE 7)

J 4



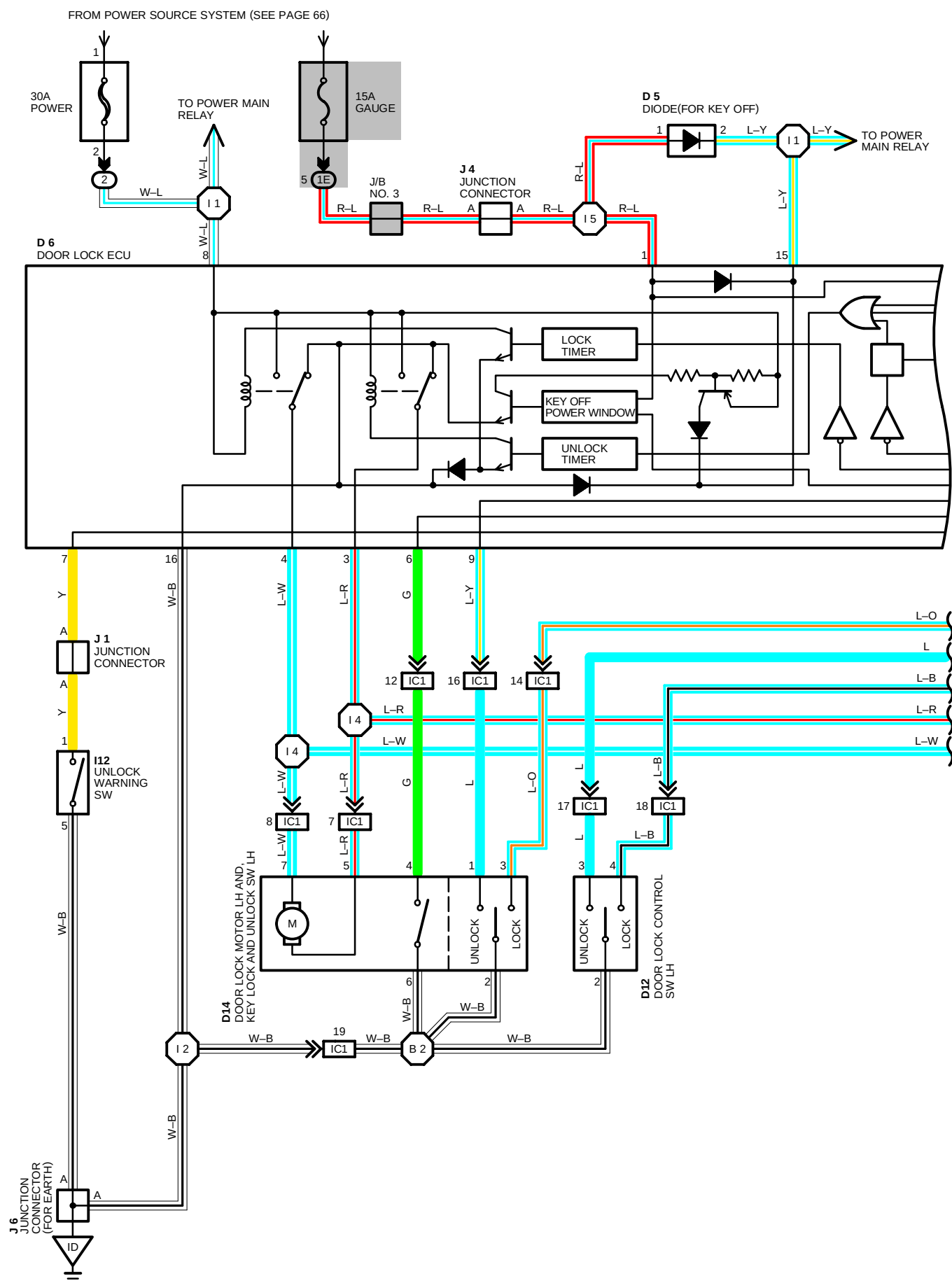
(HINT:SEE PAGE 7)

J 6



(HINT:SEE PAGE 7)

146





DOOR LOCK

SYSTEM OUTLINE

CURRENT ALWAYS FLOWS TO **TERMINAL 8** OF THE DOOR LOCK ECU.

1. MANUAL LOCK OPERATION

TO CHANGE DOOR LOCK SW AND KEY SW TO **LOCK** POSITION, A LOCK SIGNAL IS INPUT TO **TERMINAL 10, 12** OF THE DOOR LOCK ECU AND CAUSES THE ECU TO FUNCTION. CURRENT FLOWS FROM **TERMINAL 8** OF THE ECU → **TERMINAL 4** → **TERMINAL 7** OF THE DOOR LOCK MOTOR → **TERMINAL 5** → **TERMINAL 3** OF THE ECU → **TERMINAL 16** → TO **GROUND** AND DOOR LOCK MOTOR CAUSES THE DOOR TO LOCK.

2. MANUAL UNLOCK OPERATION

TO CHANGE DOOR LOCK CONTROL SW AND KEY SW TO **UNLOCK** POSITION, AN UNLOCK SIGNAL IS INPUT TO **TERMINAL 9, 11** OF THE DOOR LOCK ECU AND CAUSES THE ECU TO FUNCTION. CURRENT FLOWS FROM **TERMINAL 8** OF THE ECU → **TERMINAL 3** → **TERMINAL 5** OF THE DOOR LOCK MOTOR → **TERMINAL 7** → **TERMINAL 4** OF THE ECU → **TERMINAL 16** → TO **GROUND** AND DOOR LOCK MOTOR CAUSES DOOR TO UNLOCK.

3. DOUBLE OPERATION UNLOCK OPERATION

WHEN THE DOOR LOCK KEY SW (DRIVER'S) IS TURNED TO THE UNLOCK SIDE, ONLY THE DRIVER'S DOOR IS MECHANICALLY UNLOCKED. TURNING THE DOOR LOCK KEY SW (DRIVER'S) TO THE UNLOCK SIDE CAUSES A SIGNAL TO BE INPUT TO **TERMINAL 9** OF THE ECU, AND IF THE SIGNAL IS INPUT AGAIN WITHIN **3** SECONDS BY TURNING THE SWITCH TO THE UNLOCK SIDE AGAIN, CURRENT FLOWS **TERMINAL 3** → **TERMINAL 5** OF DOOR LOCK MOTOR → **TERMINAL 7** → **TERMINAL 4** OF THE ECU → **TERMINAL 16** → **GROUND**, CAUSING THE DOOR LOCK MOTOR TO OPERATE AND UNLOCK THE PASSENGER'S DOOR.

4. IGNITION KEY REMINDER OPERATION

* OPERATING DOOR LOCK KNOB (IN DOOR LOCK SOLENOIDS OPERATION)

WITH IGNITION KEY IN CYLINDER (UNLOCK WARNING SW ON), WHEN THE DOOR IS OPENED AND LOCKED USING DOOR LOCK KNOB (DOOR LOCK MOTOR), THE DOOR IS LOCKED ONCE BUT EACH DOOR IS UNLOCKED SOON BY THE FUNCTION OF ECU. AS A RESULT, THE CURRENT FLOWS FROM **TERMINAL 8** OF THE ECU → **TERMINAL 3** → **TERMINAL 5** OF THE DOOR LOCK MOTOR → **TERMINAL 7** → **TERMINAL 4** OF THE ECU → **TERMINAL 16** → TO **GROUND** AND CAUSES ALL THE DOORS TO UNLOCK.

* OPERATING DOOR LOCK CONTROL SW OR DOOR LOCK KEY SW

WITH IGNITION KEY IN CYLINDER (UNLOCK WARNING SW ON), WHEN THE DOOR IS OPENED AND LOCKED USING DOOR LOCK CONTROL SW OR KEY SW, THE DOOR IS LOCKED ONCE BUT EACH DOOR IS UNLOCK BY THE FUNCTION OF SW CONTAINED IN SOLENOIDS, WHICH THE SIGNAL IS INPUT TO **TERMINAL 6** (DRIVER'S) OR **5** (PASSENGER'S) OF THE ECU. ACCORDING TO THIS INPUT SIGNAL, THE CURRENT IN ECU FLOWS FROM **TERMINAL 8** OF THE ECU → **TERMINAL 3** → **TERMINAL 5** OF THE DOOR LOCK MOTOR → **TERMINAL 7** → **TERMINAL 4** OF THE ECU → **TERMINAL 16** → TO **GROUND** AND CAUSES ALL THE DOOR TO UNLOCK.

* IN CASE OF KEY LESS LOCK

WITH IGNITION KEY IN CYLINDER (UNLOCK WARNING SW ON), WHEN THE UNLOCK FUNCTION IS DISTURBED MORE THAN **0.2** SECONDS, FOR EXAMPLE PUSHING THE DOOR LOCK KNOB ETC., THE DOOR HOLDS ON LOCK CONDITION. CLOSING THE DOOR AFTER, DOOR COURTESY SW INPUTS THE SIGNAL INTO **TERMINAL 2** OR **14** OF THE ECU. BY THIS INPUT SIGNAL, THE ECU WORKS AND CURRENT FLOWS FROM **TERMINAL 8** OF THE ECU → **TERMINAL 3** → **TERMINAL 5** OF THE DOOR LOCK MOTOR → **TERMINAL 7** → **TERMINAL 4** OF THE ECU → **TERMINAL 16** → TO **GROUND** AND CAUSES ALL THE DOORS TO UNLOCK.

SERVICE HINTS

I12 UNLOCK WARNING SW

1-5 : CLOSED WITH IGNITION KEY IN CYLINDER

D14, D15 KEY LOCK AND UNLOCK SW

1-2 : CLOSED WITH DOOR LOCK CYLINDER UNLOCKED WITH KEY

2-3 : CLOSED WITH DOOR LOCK CYLINDER LOCKED WITH KEY

D10, D11 DOOR COURTESY SW

1-GROUND : CLOSED WITH DOOR OPEN

- 1-GROUND : APPROX. **12** VOLTS WITH IGNITION SW AT **ON** POSITION
 9-GROUND : **0** VOLTS WITH DRIVER'S DOOR LOCK CYLINDER UNLOCKED WITH KEY
 15-GROUND : APPROX. **12** VOLTS WITH IGNITION SW AT **ON** POSITION
 12-GROUND : **0** VOLTS WITH DRIVER'S, PASSENGER'S DOOR LOCK CYLINDER LOCKED WITH KEY

D 6 DOOR LOCK ECU

- 16-GROUND : ALWAYS CONTINUOUS
 2-GROUND : CONTINUITY WITH DRIVER'S DOOR OPEN
 8-GROUND : ALWAYS APPROX. **12** VOLTS
 3-GROUND : APPROX. **12** VOLTS **0.2** SECONDS WITH FOLLOWING OPERATION
 * DOOR LOCK CONTROL SW UNLOCKED
 * DOOR LOCK CONTROL SW LOCKED WITH IGNITION KEY IN CYLINDER AND DRIVER'S DOOR OPEN (IGNITION KEY REMINDER FUNCTION)
 * DOOR LOCK KNOB LOCKED WITH IGNITION KEY IN CYLINDER AND DRIVER'S DOOR OPEN (IGNITION KEY REMINDER FUNCTION)
 * UNLOCKING THE DRIVER'S, PASSENGER'S DOOR CYLINDER WITH KEY
 4-GROUND : APPROX. **12** VOLTS **0.2** SECONDS WITH FOLLOWING OPERATION
 * DOOR LOCK CONTROL SW LOCKED
 * LOCKING THE DRIVER'S, PASSENGER'S DOOR CYLINDER WITH KEY
 10-GROUND : **0** VOLTS WITH DOOR LOCK CONTROL SW LOCKED
 14-GROUND : CONTINUITY WITH PASSENGER'S DOOR OPEN
 6-GROUND : CONTINUITY WITH DRIVER'S DOOR LOCK KNOB UNLOCKED
 5-GROUND : CONTINUITY WITH PASSENGER'S DOOR LOCK KNOB UNLOCKED
 11-GROUND : **0** VOLTS WITH DOOR LOCK CONTROL SW UNLOCKED, PASSENGER'S DOOR LOCK CYLINDER UNLOCKED WITH KEY

: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
D 6	32	D13	34 (L/B), 35 (C/P), 36 (CONVERT.)	J 1	33
D10	34 (L/B), 35 (C/P), 36 (CONVERT.)	D14	34 (L/B), 35 (C/P), 36 (CONVERT.)	J 4	33
D11	34 (L/B), 35 (C/P), 36 (CONVERT.)	D15	34 (L/B), 35 (C/P), 36 (CONVERT.)	J 6	33
D12	34 (L/B), 35 (C/P), 36 (CONVERT.)	I12	33		

: RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
2	26	R/B NO. 2 (LEFT KICK PANEL)
4	27	R/B NO. 4 (RIGHT KICK PANEL)

: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1B	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
1F	20	FLOOR WIRE AND J/B NO. 1 (LEFT KICK PANEL)

: CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IC1	44	FRONT DOOR LH WIRE AND COWL WIRE (LEFT KICK PANEL)
IF1	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
IK1	46	FRONT DOOR RH WIRE AND COWL WIRE (RIGHT KICK PANEL)

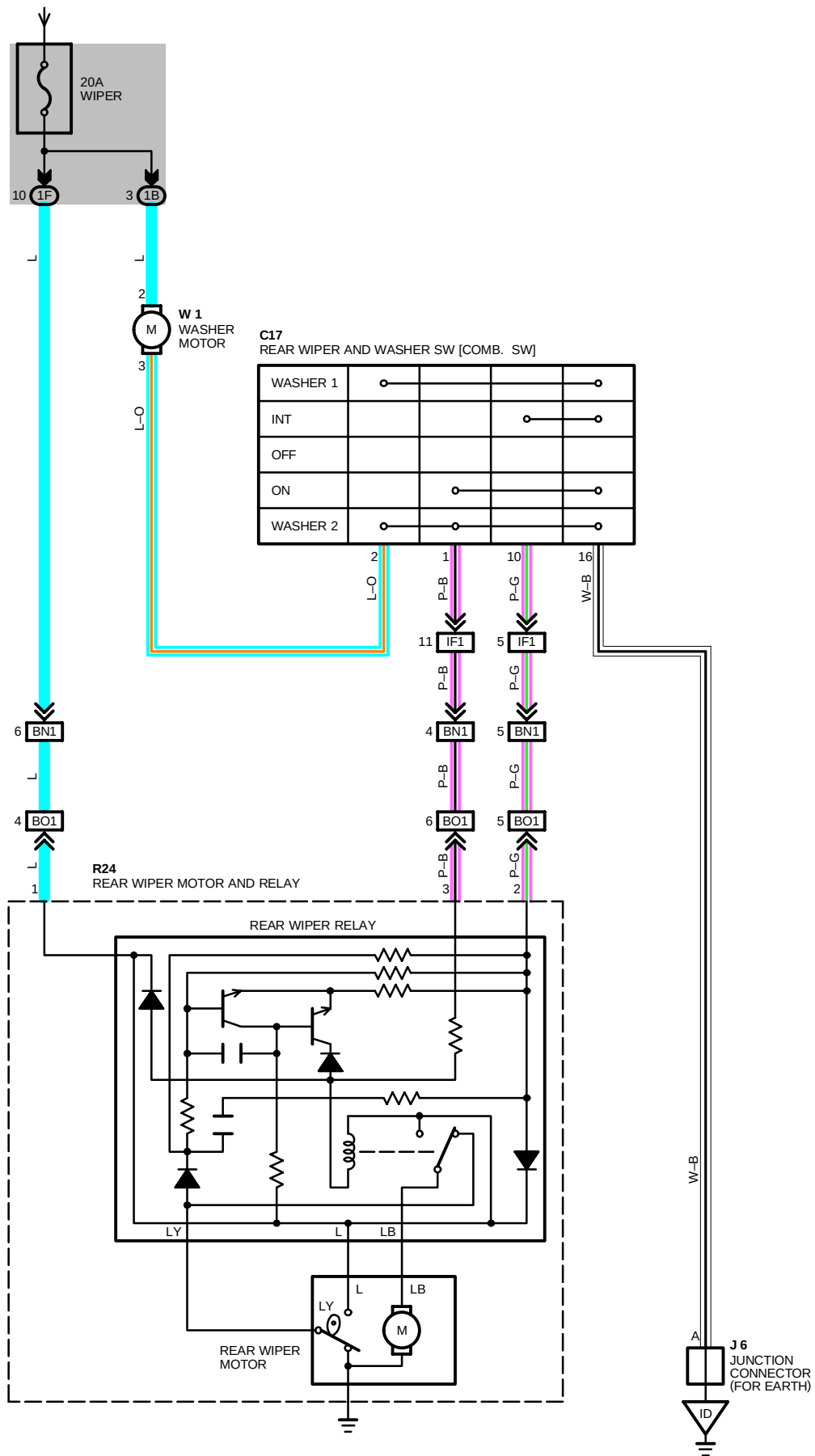
: GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL
IG	44	R/B NO. 4 SET BOLT

: SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I 1	46	COWL WIRE	B 3	48 (L/B)	FRONT DOOR RH WIRE
I 2				50 (C/P)	
I 4				52 (CONVERTIBLE)	
I 5			B14	48 (L/B)	
B 2	50 (C/P)				
	52 (CONVERTIBLE)				
		FRONT DOOR LH WIRE			

FROM POWER SOURCE SYSTEM (SEE PAGE 66)



SYSTEM OUTLINE

WHEN THE IGNITION SW IS TURNED ON, CURRENT FLOWS TO **TERMINAL 2** OF THE REAR WASHER MOTOR, **TERMINAL 1** OF THE REAR WIPER MOTOR AND RELAY THROUGH THE WIPER FUSE.

1. REAR WIPER NORMAL OPERATION

WITH THE IGNITION SW TURNED ON AND REAR WIPER AND WASHER SW TURNED ON, THE CURRENT FLOWING TO **TERMINAL 1** OF THE REAR WIPER RELAY FLOWS TO **TERMINAL 3** OF THE RELAY → **TERMINAL 1** OF THE REAR WIPER AND WASHER SW → **TERMINAL 16** → TO **GROUND**. THUS, THE RELAY COIL IS ACTIVATED AND THE CURRENT TO **TERMINAL 1** OF THE REALY FLOWS TO **TERMINAL LB** → **TERMINAL LB** OF THE REAR WIPER MOTOR → MOTOR → TO **GROUND** AND CAUSES THE MOTOR TO OPERATE THE WIPER.

2. REAR WIPER INTERMITTENT OPERATION

WHEN THE IGNITION SW IS ON AND THE REAR WIPER AND WASHER SW IS TURNED TO **INT** POSITION, CURRENT FLOWING TO **TERMINAL 1** OF THE REAR WIPER MOTOR AND RELAY FLOWS TO **TERMINAL 2** OF THE RELAY → **TERMINAL 10** OF THE REAR WIPER AND WASHER SW → **TERMINAL 16** → **GROUND**.

THIS CAUSES THE MOTOR TO OPERATE (THE POINT CHANGES) AND THE INTERMITTENT CIRCUIT OF THE RELAY OPERATES. INTERMITTENT OPERATION OF THE CIRCUIT IS CONTROLLED BY THE CHARGING AND DISCHARGING OF THE CONDENSER INSTALLED INSIDE THE RELAY.

3. WASHER OPERATION

WITH THE IGNITION SW TURNED ON AND THE REAR WIPER AND WASHER SW TURNED TO **ON** POSITION, WHEN THE WIPER SW IS TURNED FURTHER, THE CURRENT FLOWING TO **TERMINAL 2** OF THE REAR WASHER MOTOR FLOWS TO **TERMINAL 3** OF THE MOTOR → **TERMINAL 2** OF THE REAR WIPER AND WASHER SW → **TERMINAL 16** → TO **GROUND** SO THAT THE WASHER MOTOR ROTATES AND THE WINDOW WASHER EJECTS THE SPRAY, ONLY WHILE THE SWITCH IS FULLY TURNED.

WHEN THE WIPER SW IS OFF AND THEN TURNED TO WASHER ON (WIPER OFF SIDE), ONLY THE WASHER OPERATES.

SERVICE HINTS

W1 WASHER MOTOR

- 2-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION
- 3-GROUND : CONTINUITY WITH WASHER SW TURNED ON

R24 REAR WIPER MOTOR AND RELAY

- 1-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION
- 2-GROUND : CONTINUITY WITH REAR WIPER SW AT **INT** POSITION
- 3-GROUND : CONTINUITY WITH REAR WIPER SW AT **ON** POSITION

: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C17	32	R24	34 (L/B)		
J 6	33	W 1	29 (3S-GTE), 30 (5S-FE)		

: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1B	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1F	20	FLOOR WIRE AND J/B NO. 1 (LEFT KICK PANEL)

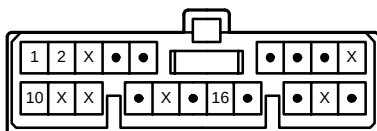
: CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IF1	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
BN1	48 (L/B)	BACK DOOR NO. 1 WIRE AND FLOOR WIRE (LEFT SIDE OF PACKAGE TRAY TRIM)
BO1	48 (L/B)	BACK DOOR NO. 1 WIRE AND BACK DOOR NO. 1 SUB WIRE (BACK DOOR UPPER LEFT)

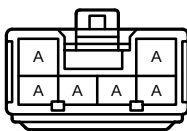
: GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL

C17 BLACK

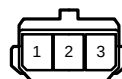


J 6



(HINT : SEE PAGE 7)

R24



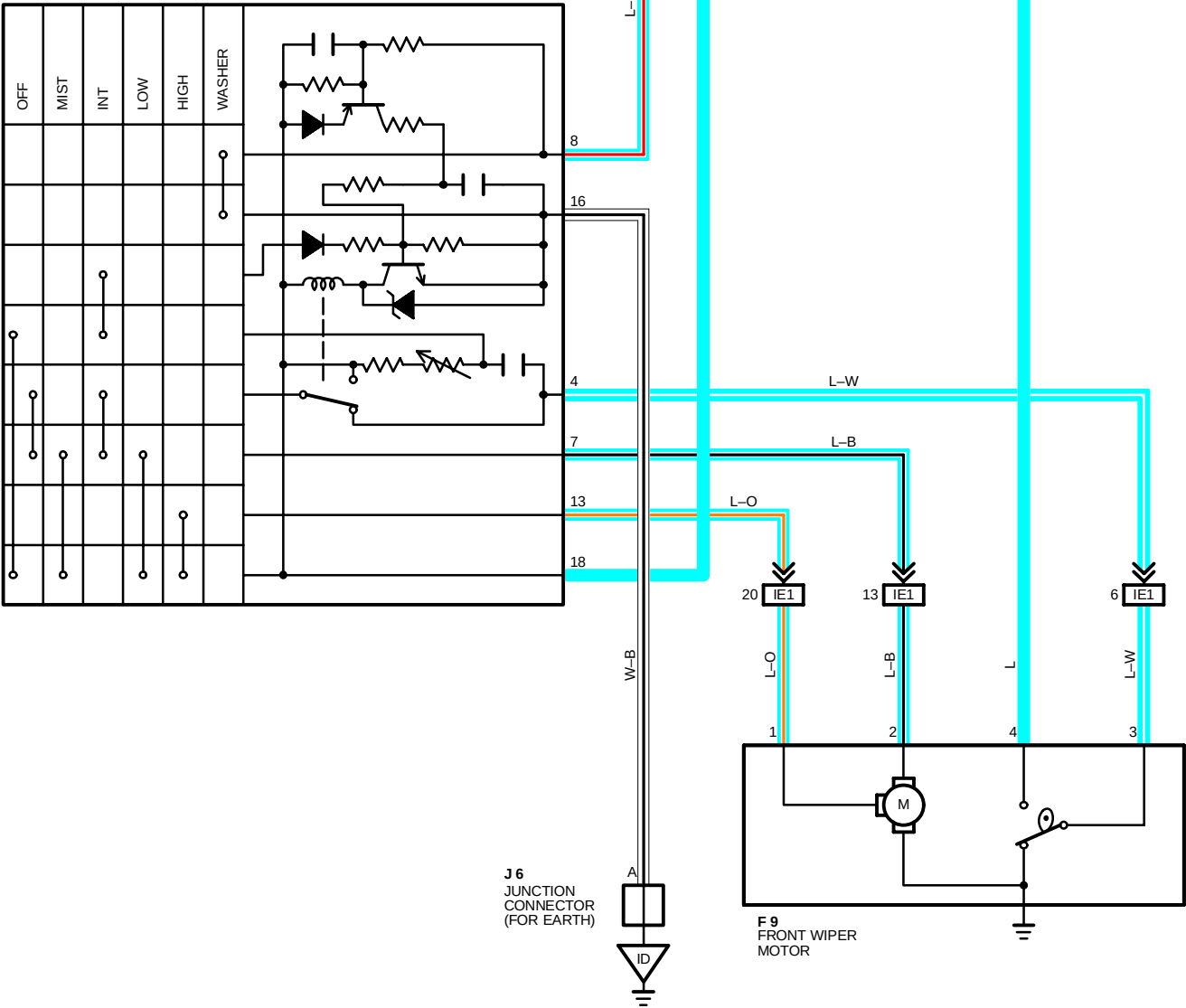
W 1 GRAY



FRONT WIPER AND WASHER

FROM POWER SOURCE SYSTEM (SEE PAGE 66)

C17
FRONT WIPER AND WASHER SW (W/ WIPER RELAY)
[COMB. SW]



SYSTEM OUTLINE

WITH THE IGNITION SW TURNED ON, THE CURRENT FLOWS TO **TERMINAL 18** OF THE WIPER AND WASHER SW, **TERMINAL 2** OF THE WASHER MOTOR AND **TERMINAL 4** OF THE FRONT WIPER MOTOR THROUGH THE WIPER FUSE.

1. LOW SPEED POSITION

WITH WIPER SW TURNED TO **LOW** POSITION, THE CURRENT FLOWS FROM **TERMINAL 18** OF THE WIPER AND WASHER SW → **TERMINAL 7** → **TERMINAL 2** OF THE FRONT WIPER MOTOR → FRONT WIPER MOTOR → TO **GROUND** AND CAUSES TO THE WIPER MOTOR TO RUN AT LOW SPEED.

2. HIGH SPEED POSITION

WITH WIPER SW TURNED TO **HIGH** POSITION, THE CURRENT FLOWS FROM **TERMINAL 18** OF THE WIPER AND WASHER SW → **TERMINAL 13** → **TERMINAL 1** OF THE FRONT WIPER MOTOR → FRONT WIPER MOTOR → TO **GROUND** AND CAUSES TO THE WIPER MOTOR TO RUN AT HIGH SPEED.

3. INT POSITION (W/ INT SW)

WITH WIPER SW TURNED TO **INT** POSITION, THE RELAY OPERATES AND THE CURRENT WHICH IS CONNECTED BY RELAY FUNCTION FLOWS FROM **TERMINAL 18** OF THE WIPER AND WASHER SW → **TERMINAL 16** → TO **GROUND**. THIS FLOW OF CURRENT OPERATES THE INTERMITTENT CIRCUIT AND THE CURRENT FLOWS FROM **TERMINAL 18** OF THE WIPER AND WASHER SW → **TERMINAL 7** → **TERMINAL 2** OF THE FRONT WIPER MOTOR → FRONT WIPER MOTOR → TO GROUND AND FUNCTIONS.

THE INTERMITTENT OPERATION IS CONTROLLED BY A CONDENSER'S CHARGED AND DISCHARGED FUNCTION INSTALLED IN RELAY AND THE INTERMITTENT TIME IS CONTROLLED BY A TIME CONTROL SW TO CHANGE THE CHARGING TIME OF THE CONDENSER.

4. MIST POSITION (W/ MIST SW)

WITH WIPER SW TURNED TO **MIST** POSITION, THE CURRENT FLOWS FROM **TERMINAL 18** OF THE WIPER AND WASHER SW → **TERMINAL 7** → **TERMINAL 2** OF THE FRONT WIPER MOTOR → FRONT WIPER MOTOR → TO GROUND AND CAUSES TO THE WIPER MOTOR TO RUN AT LOW SPEED.

5. WASHER CONTINUOUS OPERATION (W/ INT CONTROL)

WITH WASHER SW TURNED TO ON, THE CURRENT FLOWS FROM **TERMINAL 2** OF THE WASHER MOTOR → **TERMINAL 1** → **TERMINAL 8** OF THE WIPER AND WASHER SW → **TERMINAL 16** → TO GROUND AND CAUSES TO THE WASHER MOTOR TO RUN. AND WINDOW WASHER IS JET. THIS CAUSES THE CURRENT TO FLOW WASHER CONTINUOUS OPERATION CIRCUIT (W/ INT SW) IN **TERMINAL 18** OF THE WIPER AND WASHER SW → **TERMINAL 7** → **TERMINAL 2** OF THE FRONT WIPER MOTOR → FRONT WIPER MOTOR → TO **GROUND** AND FUNCTION.

SERVICE HINTS

C17 FRONT WIPER AND WASHER SW (W/ WIPER RELAY)

16-GROUND: ALWAYS CONTINUITY

18-GROUND: APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION

7-GROUND : APPROX. 12 VOLTS WITH WIPER AND WASHER SW AT **LOW** POSITION

APPROX. 12 VOLTS EVERY APPROX. 1 TO 10 SECONDS INTERMITTENTLY WITH WIPER SW AT **INT** POSITION

4-GROUND : APPROX. 12 VOLTS WITH IGNITION SW ON UNLESS WIPER MOTOR AT **STOP** POSITION

13-GROUND: APPROX. 12 VOLTS WITH IGNITION SW ON AND AFTER WIPER SW OFF UNTIL WIPER MOTOR STOPS

F 9 FRONT WIPER MOTOR

3-4 : CLOSED UNLESS WIPER MOTOR AT **STOP** POSITION



: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C17	32	F 9	31 (4A-FE)	W 1	29 (3S-GTE), 30 (5S-FE)
F 9	28 (3S-GTE), 30 (5S-FE)	J 6	33		31 (4A-FE)



: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1B	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)



: CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)



: GROUND POINTS

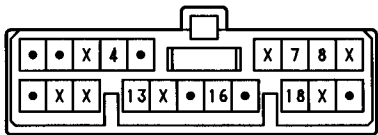
CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL

FRONT WIPER AND WASHER

 : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I 2	46	COWL WIRE			

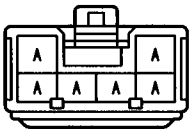
C17 BLACK



F 9 GRAY

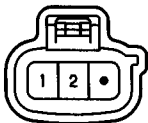


J 6



(HINT:SEE PAGE 7)

W 1 GRAY

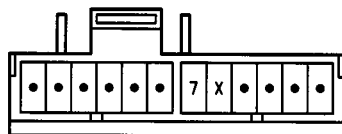


UNLOCK AND SEAT BELT WARNING

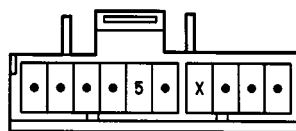
B 8, B 9, B10 ⑧



C14 ①



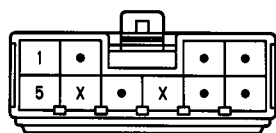
C15 ⑧ GRAY



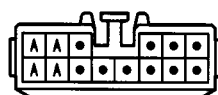
D10



I12 BLACK

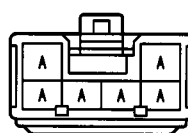


J 1



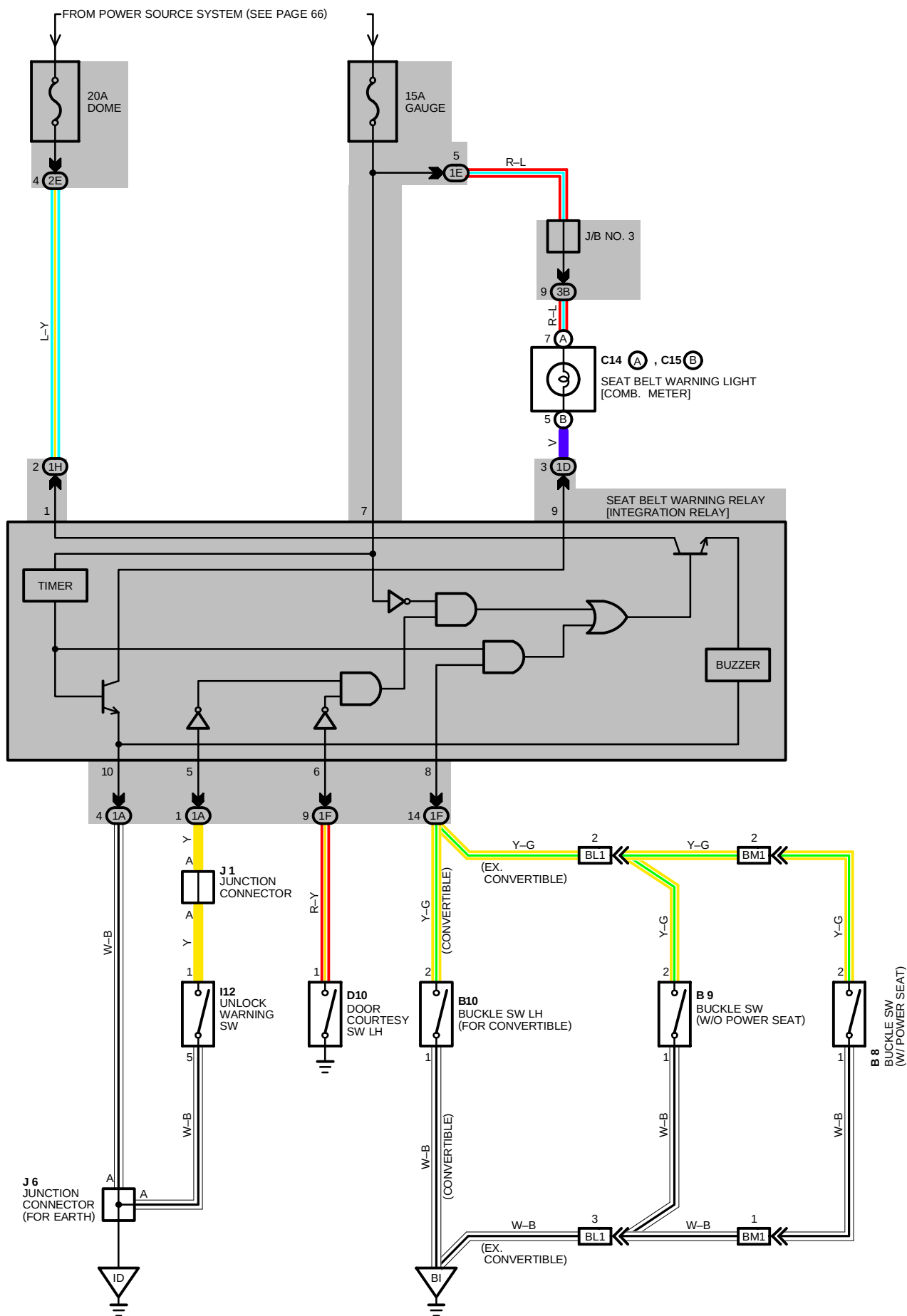
(HINT:SEE PAGE 7)

J 6



(HINT:SEE PAGE 7)

UNLOCK AND SEAT BELT WARNING



SYSTEM OUTLINE

CURRENT ALWAYS FLOWS TO **TERMINAL 1** OF THE SEAT BELT WARNING RELAY (INTEGRATION RELAY) THROUGH DOME FUSE.

1. SEAT BELT WARNING SYSTEM

WHEN THE IGNITION SW IS TURN ON, CURRENT FLOWS FROM THE GAUGE FUSE TO **TERMINAL 7** OF THE SEAT BELT WARNING RELAY. AT THE SAME TIME, CURRENT FLOWS TO **TERMINAL 9** OF THE RELAY FROM THE GAUGE FUSE THROUGH THE SEAT BELT WARNING LIGHT. THIS CURRENT ACTIVATES THE SEAT BELT WARNING RELAY AND, FOR APPROX. **4-8** SECONDS, CURRENT FLOWING THROUGH THE WARNING LIGHT FLOWS FROM **TERMINAL 9** OF THE RELAY → **TERMINAL 10** → **GROUND**, CAUSING THE WARNING LIGHT TO LIGHT UP. AT THE SAME AS THE WARNING LIGHT LIGHTS UP, A BUCKLE SW OFF SIGNAL IS INPUT TO **TERMINAL 8** OF THE RELAY, THE CURRENT FLOWING TO **TERMINAL 1** OF THE RELAY FLOWS FROM **TERMINAL 10** → **GROUND** AND THE SEAT BELT WARNING BUZZER SOUNDS FOR APPROX. **4-8** SECONDS. HOWEVER, IF THE SEAT BELT IS PUT ON (BUCKLE SW ON) DURING THIS PERIOD (WHILE THE BUZZER IS SOUNDING), SIGNAL INPUT TO **TERMINAL 8** OF RELAY STOPS AND THE CURRENT FLOW FROM **TERMINAL 1** OF THE RELAY → **TERMINAL 10** → **GROUND** IS CUT, CAUSING THE BUZZER TO STOP.

2. UNLOCK WARNING SYSTEM

WITH THE IGNITION KEY INSERTED IN THE KEY CYLINDER (UNLOCK SW ON). THE IGNITION SW STILL OFF AND DOOR OPEN (DOOR COURTESY SW ON), WHEN A SIGNAL IS INPUT TO **TERMINAL 6** OF THE RELAY, THE SEAT BELT WARNING RELAY OPERATES, CURRENT FLOWS FROM **TERMINAL 1** OF THE RELAY → **TERMINAL 10** → **GROUND** AND THE UNLOCK WARNING BUZZER SOUNDS.

SERVICE HINTS

I12 UNLOCK WARNING SW

1-5 : CLOSED WITH IGNITION KEY IN CYLINDER

SEAT BELT WARNING RELAY [INTEGRATION RELAY]

10-GROUND : ALWAYS CONTINUITY

6-GROUND : CONTINUITY WITH DRIVER'S DOOR OPEN

5-GROUND : CONTINUITY WITH IGNITION KEY IN CYLINDER

8-GROUND : CONTINUITY WITH DRIVER'S LAP BELT IN USE

9-GROUND : **0** VOLTS FOR **4-8** SECONDS WITH IGNITION SW ON AND APPROX. **12** VOLTS **4-8** SECONDS AFTER IGNITION SW ON

1-GROUND : ALWAYS APPROX. **12** VOLTS

D10 DOOR COURTESY SW

1-GROUND : CLOSED WITH DRIVER'S DOOR OPEN

B 8, B 9, B10 BUCKLE SW (DRIVER'S)

2-1 : CLOSED WITH DRIVER'S LAP BELT IN USE



: PARTS LOCATION

CODE	SEE PAGE	CODE		SEE PAGE	CODE	SEE PAGE
B 8	34 (L/B) , 35 (C/P)	C14	A	32	I12	33
B 9	34 (L/B) , 35 (C/P)	C15	B	32	J 1	33
B10	36	D10		34 (LB/) , 35 (C/P) , 36 (CONVERTIBLE)	J 6	33



: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1D		
1E		
1F	20	FLOOR WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2E	22 (2WD) 23 (ALL-TRAC/4WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
3B	25	
		COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)



: CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
BL1	48 (L/B) 50 (C/P)	FLOOR WIRE AND FRAME WIRE (LEFT SIDE OF FRONT FLOOR PANEL)
BM1	48 (L/B) 50 (C/P)	
		FRAME WIRE AND SEAT WIRE (UNDER THE DRIVER'S SEAT)

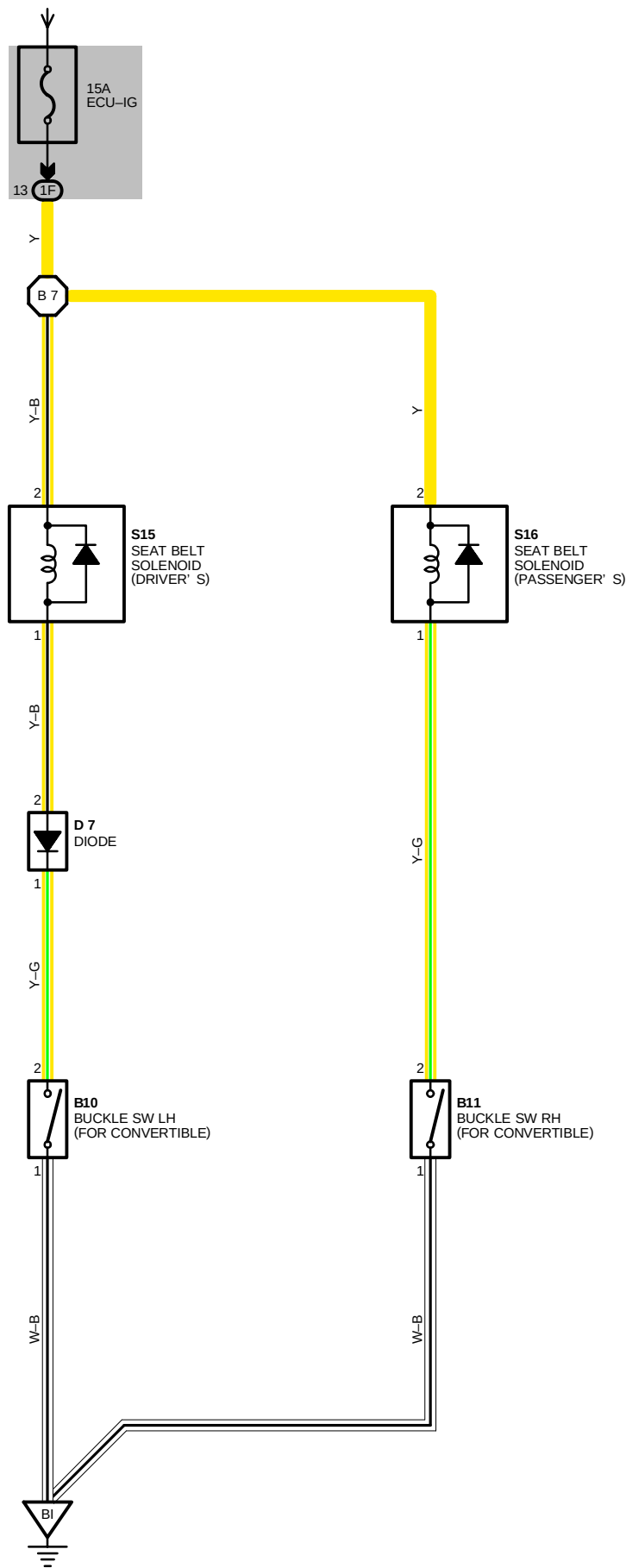


: GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL
BI	48 (L/B) 50 (C/P) 52 (CONVERTIBLE)	UNDER THE LEFT CENTER PILLAR

ELECTRIC TENSION REDUCER

FROM POWER SOURCE SYSTEM (SEE PAGE 66)



SERVICE HINTS

B10 BUCKLE SW LH

1-2 : CLOSED WITH DRIVER'S LAP BELT IN USE

B11 BUCKLE SW RH

1-2 : CLOSED WITH PASSENGER'S LAP BELT IN USE

S15, S16 SEAT BELT SOLENOID LH, RH

2-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
B10	36	D 7	36 (CONVERTIBLE)	S16	36
B11	36	S15	36		

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1F	20	FLOOR WIRE AND J/B NO. 1 (LEFT KICK PANEL)

▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
BI	52 (CONVERTIBLE)	UNDER THE LEFT CENTER

○ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
B 7	52 (CONVERTIBLE)	FLOOR WIRE			

B10, B11, S15, S16



D 7



*1 : W/ DIVERSITY ANTENNA
*2 : W/O DIVERSITY ANTENNA

The diagram illustrates the electrical circuit for the rear window defogger system. Key components and their connections are as follows:

- Power Source:** A 15A gauge and a 30A defogger fuse are connected to the main power line.
- Defogger Relay:** The relay controls the defogger system. It has a coil (2, 4) and a switch (3, 1). The switch is connected to the 30A defogger fuse and the defogger heaters.
- Defogger Switch (D 2):** A manual switch that controls the defogger system. It is connected to the 15A gauge and the defogger relay.
- Noise Filter (N 4):** A 4-terminal noise filter (for defogger system) connected to the defogger relay and the defogger heaters.
- Rear Window Defogger Heaters:** Three heaters are shown: R20 (A, R23 B), R21 (C, R22 D), and R22 (D, R23 B). They are connected to the defogger relay and the defogger switch.
- Wiring:** The diagram shows various wiring paths, including W-B (Wiring Bundle), B (Black), and R-Y (Red-Yellow). It also includes terminal blocks (ID, BK, BM, BJ) and junction boxes (J/B NO. 3, J 6 JUNCTION CONNECTOR FOR EARTH).

SERVICE HINTS

DEFOGGER RELAY

1-3 : CLOSED WITH IGNITION SW ON AND DEFOGGER SW ON

D 2 DEFOGGER SW

3-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION

2-GROUND : ALWAYS CONTINUITY

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
D 2	32	R20	A 34 (L/B), 35 (C/P)	R23	B 34 (L/B), 35 (C/P)
J 6	33	R21	C 34 (L/B), 35 (C/P)		
N 4	33	R22	D 35 (C/P)		

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1B		
1D		
1E		
1F	20	FLOOR WIRE AND J/B NO. 1 (LEFT KICK PANEL)
3B	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
BN1	48 (L/B)	BACK DOOR NO. 1 WIRE AND FLOOR WIRE (LEFT SIDE OF PACKAGE TRAY TRIM)
BO1	48 (L/B)	BACK DOOR NO. 1 WIRE AND BACK DOOR NO. 1 SUB WIRE (BACK DOOR UPPER LEFT)
BR1	50 (C/P)	FLOOR WIRE AND REAR WINDOW NO. 2 WIRE (LEFT REAR PILLAR)
BV1	50 (C/P)	FLOOR WIRE AND LUGGAGE ROOM NO. 4 WIRE

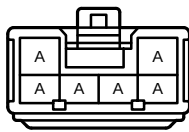
▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL
BJ	50 (C/P)	RIGHT REAR PILLAR (C/P)
BK	48 (L/B)	BACK DOOR RIGHT (L/B)
BM	50 (C/P)	LEFT REAR PILLAR (C/P)

D 2 BLACK



J 6



(HINT : SEE PAGE 7)

N 4



R20 (A) , R21 (E) BLACK



R22 (D) , R23 (B) BLACK



R21 (C)





SERVICE HINTS

POWER MAIN RELAY

(2) 1-(2) 4 : CLOSED WITH IGNITION SW AT **ON** POSITION

P 4 POWER SEAT SW

7- GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION

8- GROUND : ALWAYS CONTINUITY

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
D 5	32	J 6	33	P 3	34 (L/B), 35 (C/P)
J 4	33	P 2	34 (L/B), 35 (C/P)	P 4	34 (L/B), 35 (C/P)

○ : RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
2	26	R/B NO. 2 (LEFT KICK PANEL)

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1E	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IF1	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
BL1	48 (L/B) 50 (C/P)	FLOOR WIRE AND FRAME WIRE (LEFT SIDE OF FRONT FLOOR PANEL)
BM1	48 (L/B) 50 (C/P)	FRAME WIRE AND SEAT WIRE (UNDER THE DRIVER'S SEAT)

▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL
BI	48 (L/B) 50 (C/P)	UNDER THE LEFT CENTER PILLAR

○ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I 1	46	COWL WIRE	I 5	46	COWL WIRE

D 5 BLACK

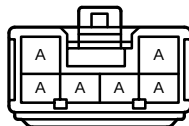


J 4



(HINT : SEE PAGE 7)

J 6



(HINT : SEE PAGE 7)

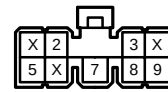
P 2



P 3 BLUE

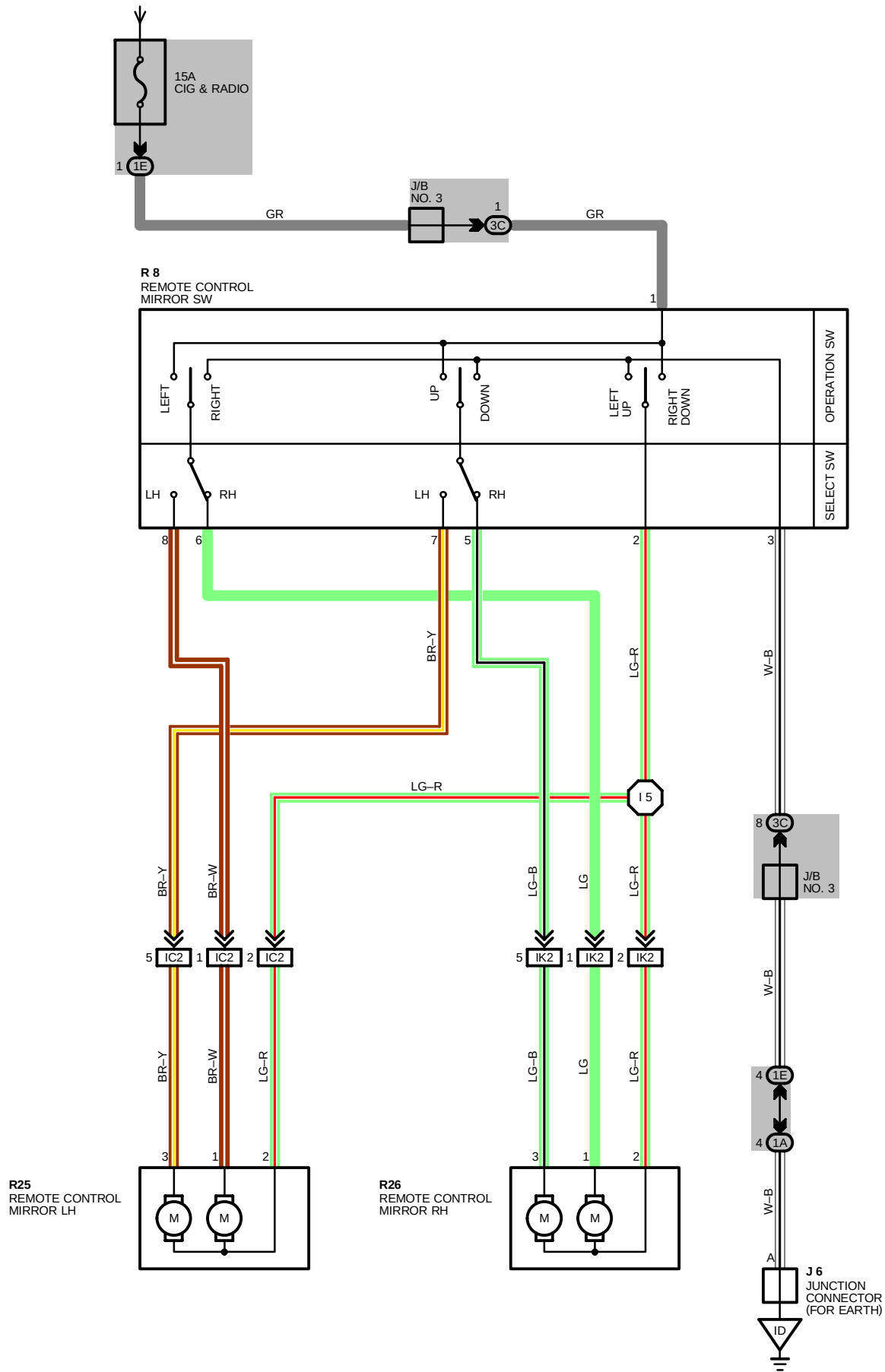


P 4



REMOTE CONTROL MIRROR

FROM POWER SOURCE SYSTEM (SEE PAGE 66)



SERVICE HINTS

R 8 REMOTE CONTROL MIRROR SW

- 1-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ACC** OR **ON** POSITION
- 2-3 : CONTINUITY WITH OPERATION SW AT **UP** OR **LEFT** POSITION
- 1-2 : CONTINUITY WITH OPERATION SW AT **DOWN** OR **RIGHT** POSITION

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
J 6	33	R25	34 (L/B), 35 (C/P)		
R 8	33	R26	34 (L/B), 35 (C/P)		

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
3C	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IC2	44	FRONT DOOR LH WIRE AND COWL WIRE (LEFT KICK PANEL)
IK2	46	FRONT DOOR RH WIRE AND COWL WIRE (RIGHT KICK PANEL)

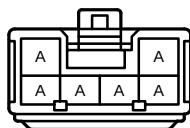
▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL

○ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I 5	46	COWL WIRE			

J 6



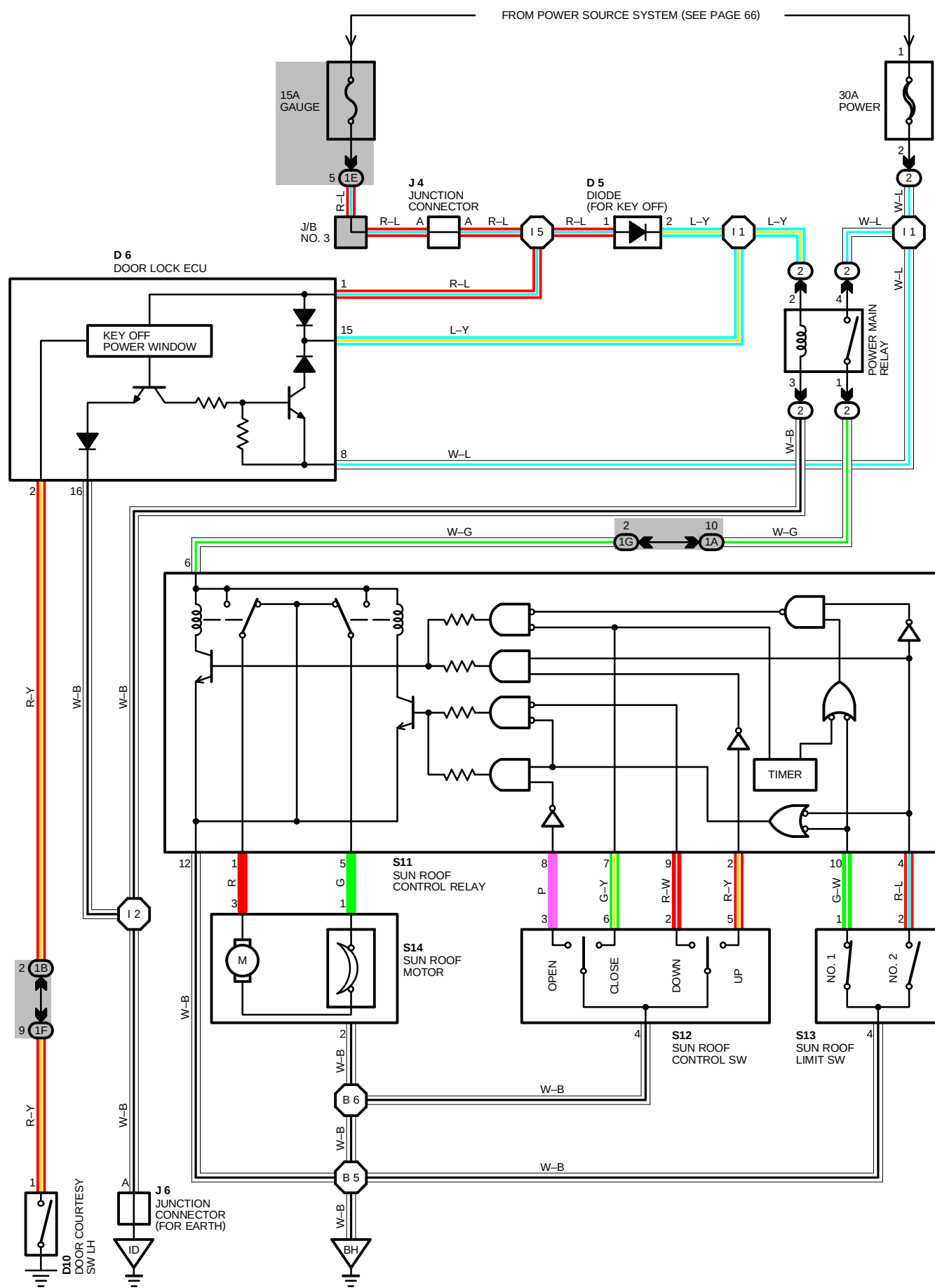
(HINT : SEE PAGE 7)

R 8



R25, R26





SYSTEM OUTLINE

CURRENT ALWAYS FLOWS TO **TERMINAL 4** OF THE POWER MAIN RELAY THROUGH THE POWER FUSE, WITH THE IGNITION SW TURNED ON, CURRENT FLOWS THROUGH THE GAUGE FUSE TO **TERMINAL 2** OF THE POWER MAIN RELAY. THIS ACTIVATES THE RELAY AND CURRENT FLOWING TO **TERMINAL 4** OF THE POWER MAIN RELAY FLOWS TO **TERMINAL 1** OF THE POWER MAIN RELAY → TO **TERMINAL 6** OF THE SUN ROOF CONTROL RELAY.

1. SLIDE OPEN OPERATION

WHEN THE IGNITION SW IS ON AND THE SUN ROOF SW IS PUSHED TO THE OPEN SIDE, A SIGNAL IS INPUT TO **TERMINAL 8** OF THE SUN ROOF CONTROL RELAY. WHEN THIS OCCURS ACTIVATING THE RELAY SO THAT CURRENT FLOWS FROM **TERMINAL 6** OF THE RELAY → **TERMINAL 5** → **TERMINAL 1** OF THE SUN ROOF MOTOR → MOTOR → **TERMINAL 3** → **TERMINAL 1** OF THE RELAY → **TERMINAL 12** → **GROUND**, THE MOTOR ROTATES TO THE OPEN SIDE AND THE SUN ROOF SLIDES OPEN AS LONG AS THE SUN ROOF CONTROL SW IS PUSHED TO THE OPEN SIDE.

WHEN THE SUN ROOF IS OPENED COMPLETELY, EVEN IF THE SUN ROOF SW IS PUSHED CONTINUOUSLY, THE CURRENT TO THE SUN ROOF MOTOR INCREASES.

IN THIS CASE, THE CIRCUIT BREAKER BUILT INTO THE MOTOR OPENS AND CUTS OUT THE CURRENT TO THE MOTOR, PREVENTING THE MOTOR FROM BURNING OUT.

2. SLIDE CLOSE OPERATION

WHEN THE IGNITION SW IS ON AND THE SUN ROOF CONTROL SW IS PUSHED TO THE CLOSE SIDE, A SIGNAL IS INPUT FROM **TERMINAL 6** TO **TERMINAL 7** OF THE SUN ROOF CONTROL RELAY. THIS ACTIVATES THE RELAY AND THE CURRENT FLOWING TO **TERMINAL 6** FLOWS TO **TERMINAL 1** → **TERMINAL 3** OF THE SUN ROOF MOTOR → MOTOR → **TERMINAL 10** → **TERMINAL 5** OF THE RELAY → **TERMINAL 12** → **GROUND**. THIS CAUSES THE MOTOR TO ROTATE TO THE CLOSE SIDE AND SLIDE CLOSE OPERATION CONTINUES AS LONG AS THE SUN ROOF CONTROL SW IS PUSHED TO THE CLOSE SIDE.

100 MM BEFORE THE FULLY **CLOSED** POSITION THE SUN ROOF LIMIT NO. 1 SW TURN OFF. THIS SIGNAL IS INPUT INTO THE RELAY, SO THE RELAY STOPS OPERATION. THUS CURRENT DOES NOT FLOW TO THE SUN ROOF MOTOR AND THE SUN ROOF AUTOMATICALLY STOPS.

IF THE SUN ROOF SW IS THEN PUSH AGAIN, THE TIMER INSTALLED IN THE SUN ROOF CONTROL TURNS ON AND THE RELAY OPERATES FOR **0.65 SEC.** TO RE-OPERATE THE MOTOR SO THAT THE SUN ROOF LIMIT SW NO. 1 TURNS ON. AS A RESULT, AS LONG AS THE SUN ROOF SW IS PUSHED, SLIDE CLOSE OPERATION OCCURS AND THE SUN ROOF IS ABLE TO FULLY CLOSE.

3. TILT UP OPERATION

WHEN THE SUN ROOF CONTROL SW IS PUSHED TO **TILT UP** POSITION, WITH THE IGNITION SW TURNED ON AND THE SLIDE ROOF COMPLETELY CLOSED A SIGNAL IS INPUT TO **TERMINAL 2** OF THE SUN ROOF CONTROL RELAY AND SUN ROOF LIMIT SW NO. 2 IS TURNED OFF SIMULTANEOUSLY, CAUSING THE SUN ROOF CONTROL RELAY TO OPERATE. AS A RESULT, THE RELAY IS ACTIVATED AND CURRENT FLOWS FROM **TERMINAL 6** OF THE RELAY → **TERMINAL 1** → **TERMINAL 3** OF THE SUN ROOF MOTOR → MOTOR → **TERMINAL 1** → **TERMINAL 5** OF THE RELAY → **TERMINAL 12** → **GROUND**, ROTATING THE MOTOR FOR TILT UP OPERATION.

4. TILT DOWN OPERATION

WHEN THE SUN ROOF CONTROL SW IS PUSHED TO THE **TILT DOWN** POSITION, WITH THE IGNITION SW TURNED ON AND THE SLIDE ROOF TILTED UP, A SIGNAL IS INPUT TO **TERMINAL 9** OF THE SUN ROOF CONTROL RELAY SIGNALS THAT SUN ROOF LIMIT SW NO. 1 AND NO. 2 ARE OFF ARE INPUT SEPARATELY TO **TERMINAL 10** AND **TERMINAL 4**.

AS A RESULT, RELAY ACTIVATES AND THE CURRENT FLOWS FROM **TERMINAL 6** OF THE RELAY → **TERMINAL 5** → **TERMINAL 1** OF THE SUN ROOF MOTOR → MOTOR → **TERMINAL 3** → **TERMINAL 1** OF THE RELAY → **TERMINAL 12** → **GROUND**, ROTATING THE MOTOR FOR TILT DOWN OPERATION.

5. KEY OFF OPERATION

WITH THE IGNITION SW TURNED ON TO OFF, THE DOOR LOCK ECU OPERATES AND CURRENT FLOWS FROM POWER FUSE → **TERMINAL 8** OF THE DOOR LOCK ECU → **TERMINAL 15** → **TERMINAL 2** OF THE POWER MAIN RELAY → **TERMINAL 3** → TO **GROUND** FOR ABOUT **60 SECONDS**. THE SAME NORMAL OPERATION, THE CURRENT FLOWS FROM POWER FUSE → **TERMINAL 1** → **TERMINAL 6** OF THE SUN ROOF CONTROL RELAY. AS A RESULT, FOR ABOUT **60 SECONDS** AFTER THE IGNITION SW IS TURNED OFF, IT IS POSSIBLE TO OPEN AND CLOSE THE SUN ROOF BY THE FUNCTIONING OF THIS RELAY. ALSO, BY OPENING THE DOOR (DOOR COURTESY SW ON) WITHIN **60 SECONDS** AFTER TURNING THE IGNITION SW TO OFF, A SIGNAL IS INPUT **TERMINAL 2** OF THE DOOR LOCK ECU. AS A RESULT, THE ECU TURNS OFF.

SERVICE HINTS

S11 SUN ROOF CONTROL RELAY

12-GROUND : ALWAYS CONTINUITY

6-GROUND : APPROX. **12 VOLTS** WITH IGNITION SW AT **ON** POSITION

1-GROUND : APPROX. **12 VOLTS** WITH IGNITION SW ON AND SUN ROOF SW **CLOSED** OR **UP** POSITION

5-GROUND : APPROX. **12 VOLTS** WITH IGNITION SW ON AND SUN ROOF SW **OPEN** OR **DOWN** POSITION

(DISCONNECT WIRING CONNECTOR FROM RELAY)

8-GROUND : CONTINUITY WITH SUN ROOF SW AT **OPEN** POSITION

7-GROUND : CONTINUITY WITH SUN ROOF SW AT **CLOSE** POSITION

9-GROUND : CONTINUITY WITH SUN ROOF SW AT **DOWN** POSITION

2-GROUND : CONTINUITY WITH SUN ROOF SW AT **UP** POSITION

SUN ROOF

: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
D 5	32	J 4	33	S12	34 (L/B), 35 (C/P)
D 6	32	J 6	33	S13	34 (L/B), 35 (C/P)
D10	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)	S11	34 (L/B), 35 (C/P)	S14	34 (L/B), 35 (C/P)

: RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
2	26	R/B NO. 2 (LEFT KICK PANEL)

: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1B		
1E		
1F	20	FLOOR WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1G	20	ROOF WIRE AND J/B NO. 1 (LEFT KICK PANEL)

: GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL
BH	48 (L/B) 50 (C/P)	ROOF LEFT

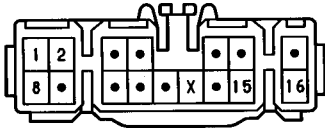
: SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
B 5	48 (L/B)	ROOF WIRE	I 1	46	COWL WIRE
	50 (C/P)		I 2		
B 6	48 (L/B) 50 (C/P)		I 5		

D 5 BLACK



D 6



D10

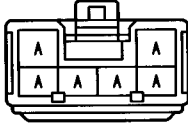


J 4



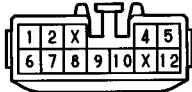
(HINT:SEE PAGE 7)

J 6



(HINT:SEE PAGE 7)

S11



S12



S13

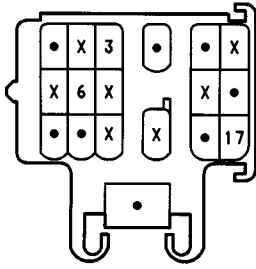


S14

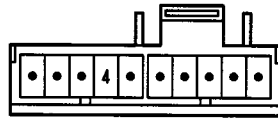


ECT (ELECTRONIC CONTROLLED TRANSMISSION)

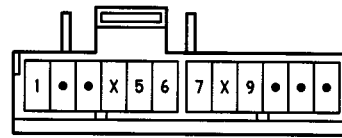
C 1 DARK GRAY



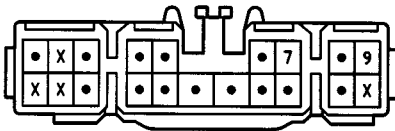
C13 (B) BROWN



C14 (A)



C19 (B) BLUE



C20 (A) BLUE



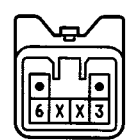
E 1 BLACK



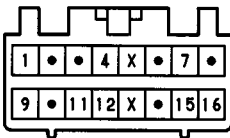
E 2 DARK GREEN



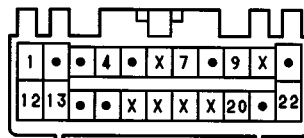
E 4



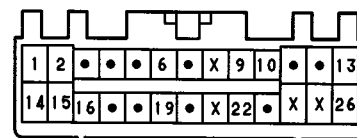
E 6 (C) GRAY



E 7 (A) DARK GRAY



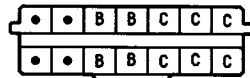
E 8 (B) DARK GRAY



I 3 GRAY

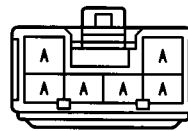


J 2



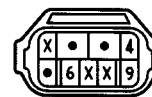
(HINT:SEE PAGE 7)

J 6



(HINT:SEE PAGE 7)

N 2 GRAY



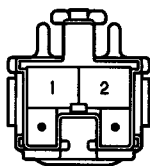
O 7



S 2 GRAY



S10

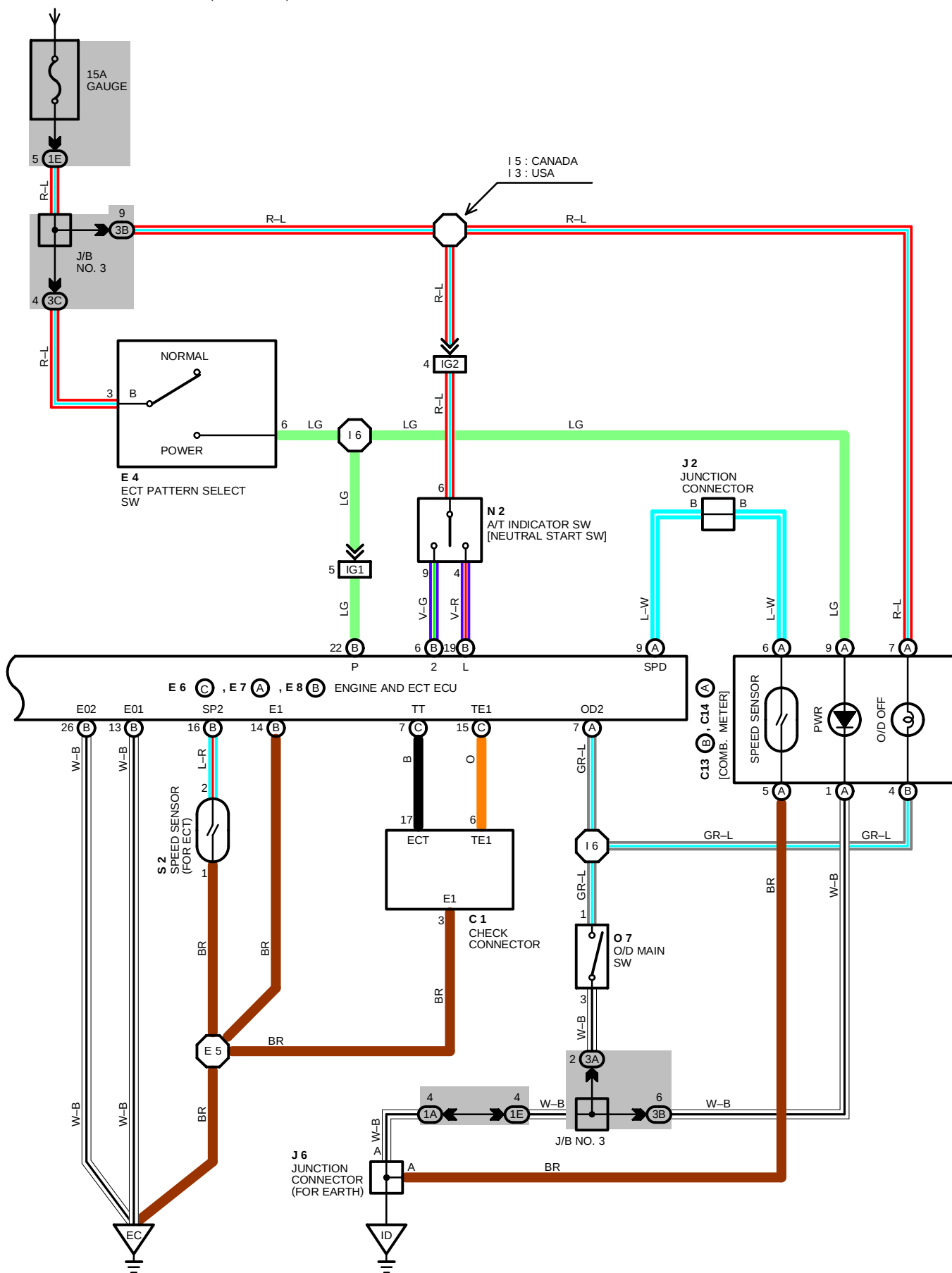


T 1 BLACK





FROM POWER SOURCE SYSTEM (SEE PAGE 66)



ECT (ELECTRONIC CONTROLLED TRANSMISSION)

SYSTEM OUTLINE

PREVIOUS AUTOMATIC TRANSMISSIONS HAVE SELECTED EACH GEAR SHIFT USING MECHANICALLY CONTROLLED THROTTLE HYDRAULIC PRESSURE, GOVERNOR HYDRAULIC PRESSURE AND LOCK-UP HYDRAULIC PRESSURE. THE ECT, HOWEVER, ELECTRICALLY CONTROLS THE LINE PRESSURE AND LOCK-UP PRESSURE ETC., THROUGH THE SOLENOID VALVE. ECT ECU CONTROL OF THE SOLENOID VALVE BASED ON THE INPUT SIGNALS FROM EACH SENSOR MAKES SMOOTH DRIVING POSSIBLE BY SHIFT SELECTION FOR EACH GEAR WHICH IS MOST APPROPRIATE TO THE DRIVING CONDITIONS AT THAT TIME.

1. GEAR SHIFT OPERATION

DURING DRIVING, THE ECU SELECTS THE SHIFT FOR EACH GEAR WHICH IS MOST APPROPRIATE TO THE DRIVING CONDITIONS, BASED ON INPUT SIGNALS FROM THE EFI WATER TEMP. SENSOR TO **TERMINAL THW** OF THE ECU, AND ALSO THE INPUT SIGNALS TO **TERMINAL SPD2** OF THE ECU FROM THE SPEED SENSOR DEVOTED TO THE ECT. CURRENT IS THEN OUTPUT TO THE ECT SOLENOIDS. WHEN SHIFTING TO 1ST SPEED, CURRENT FLOWS FROM **TERMINAL S1** OF THE ECU → **TERMINAL 3** OF THE ECT SOLENOIDS → **GROUND**, AND CONTINUITY TO THE NO. 1 SOLENOID CAUSES THE SHIFT.

FOR 2ND SPEED, CURRENT FLOWS FROM **TERMINAL S1** OF THE ECU → **TERMINAL 3** OF THE ECT SOLENOIDS → **GROUND**, AND FROM **TERMINAL S2** OF THE ECU → **TERMINAL 1** OF THE ECT SOLENOIDS → **GROUND**, AND CONTINUITY TO SOLENOIDS NO. 1 AND NO. 2 CAUSES THE SHIFT.

FOR 3RD SPEED, THERE IS NO CONTINUITY TO NO. 1 SOLENOID, ONLY TO NO. 2, CAUSING THE SHIFT.

SHIFTING INTO 4TH SPEED (OVERDRIVE) TAKES PLACE WHEN THERE IS NO CONTINUITY TO EITHER NO. 1 OR NO. 2 SOLENOID.

2. LOCK-UP OPERATION

WHEN THE ECT ECU JUDGES FROM EACH SIGNAL THAT LOCK-UP OPERATION CONDITIONS HAVE BEEN MET, CURRENT FLOWS FROM **TERMINAL SL** OF THE ECT ECU → **TERMINAL 2** OF THE ECT SOLENOID → **GROUND**, CAUSING CONTINUITY TO THE LOCK-UP SOLENOID AND CAUSING LOCK-UP OPERATION.

3. STOP LIGHT SW CIRCUIT

IF THE BRAKE PEDAL IS DEPRESSED (STOP LIGHT SW ON) WHEN DRIVING IN LOCK-UP CONDITION, A SIGNAL IS INPUT TO **TERMINAL STP** OF THE ECU, THE ECU OPERATES AND CONTINUITY TO THE LOCK-UP SOLENOID IS CUT.

4. OVERDRIVE CIRCUIT

* O/D MAIN SW ON

WHEN THE O/D MAIN SW IS TURNED ON (O/D OFF INDICATOR LIGHT TURNS OFF), A SIGNAL IS INPUT TO **TERMINAL OD2** OF THE ECU AND ECU OPERATION CAUSES GEAR SHIFT WHEN THE CONDITIONS FOR OVERDRIVE ARE MET.

* O/D MAIN SW OFF

WHEN THE O/D MAIN SW IS TURNED TO OFF, THE CURRENT FLOWING THROUGH THE O/D OFF INDICATOR LIGHT FLOWS THROUGH THE O/D MAIN SW TO **GROUND**. CAUSING THE INDICATOR LIGHT TO LIGHT UP. AT THE SAME TIME, A SIGNAL IS INPUT TO **TERMINAL OD2** OF THE ECU AND ECU OPERATION PREVENTS SHIFT INTO OVERDRIVE.

5. ECT PATTERN SELECT SW CIRCUIT

IF THE ECT PATTERN SELECT SW IS CHANGED FROM NORMAL TO POWER, THE CURRENT FLOWING THROUGH THE POWER INDICATOR FLOWS TO **GROUND**, CURRENT FLOWS TO **TERMINAL P** OF THE ECT ECU, THE ECU OPERATES, AND SHIFT UP AND SHIFT DOWN OCCUR AT HIGHER VEHICLE SPEEDS THAN WHEN THE SW IS IN **NORMAL** POSITION.

SERVICE HINTS

E 6(C), E 7(A), E 8(B) ENGINE AND ECT ECU

(B) 6-(B)14 : 10-14 VOLTS (IGNITION SW ON)

(B)4, (B)5-(B)14 : UNDER 1 VOLTS (IGNITION SW ON)

(B)22-(B)14 : 10-14 VOLTS (IGNITION SW AND ECT PATTERN SELECT SW POWER)

UNDER 1 VOLTS (IGNITION SW ON AND ECT PATTERN SELECT SW NORMAL)

(A) 4-(B)14 : 10-14 VOLTS (BRAKE PEDAL IS DEPRESSED)

UNDER 1 VOLTS (BRAKE PEDAL IS NOT DEPRESSED)

(C) 4-(C) 9 : 0.1-0.8 VOLTS (IGNITION SW ON AND COOLANT TEMP. 80°C (176°F))

(C)12-(C) 9 : UNDER 0.5 VOLTS (IGNITION SW ON AND THROTTLE VALVE FULLY CLOSED)

4.5-5.5 VOLTS (IGNITION SW ON AND THROTTLE VALVE FULLY OPEN)

(C)11-(C) 9 : 0.1-0.8 VOLTS (IGNITION SW ON AND THROTTLE VALVE FULLY CLOSED)

4.5-5.5 VOLTS (IGNITION SW ON AND THROTTLE VALVE FULLY OPEN)

(C) 1-(C) 9 : 4.5-5.5 VOLTS (IGNITION SW ON)

(A)20-(B)14 : 10-14 VOLTS (IGNITION SW ON)

(A) 7-(B)14 : 10-14 VOLTS (IGNITION SW ON AND O/D MAIN SW TURNED ON)

UNDER 1 VOLTS (IGNITION SW ON AND O/D MAIN SW TURNED OFF)

(A) 9-(B)14 : UNDER 1 VOLTS (IGNITION SW ON, CRUISE CONTROL SW OFF AND STARTING STILL)

0↔4-10 VOLTS REPEAT (IGNITION SW ON, CRUISE CONTROL SW OFF AND VEHICLE MOVING)

(B)17-(B)14 : UNDER 1 VOLTS (IGNITION SW ON AND STARTING STILL)

0↔4.5-5.5 VOLTS REPEAT (IGNITION SW ON AND VEHICLE MOVING)

(B)22-(B)14 : 10-14 VOLTS (IGNITION SW ON AND NEUTRAL START SW P OR N POSITION)

UNDER 1 VOLTS (IGNITION SW ON AND EX. NEUTRAL START SW P OR N POSITION)

(B)18–(B)14 : **10–14 VOLTS** (IGNITION SW ON AND NEUTRAL START SW **2** POSITION)
 UNDER **1** VOLTS (IGNITION SW ON AND EX. NEUTRAL START SW **2** POSITION)
 (B)19–(B)14 : **10–14 VOLTS** (IGNITION SW ON AND NEUTRAL START SW **L** POSITION)
 UNDER **1** VOLTS (IGNITION SW ON AND EX. NEUTRAL START SW **L** POSITION)
 (A)12, (B)13–(B)14 : **10–14 VOLTS** (IGNITION SW ON)
 (A) 1–(B)14 : **10–14 VOLTS** (ALL CONDITIONS)

RESISTANCE AT ECU WIRING CONNECTORS

(DISCONNECT WIRING CONNECTOR)

(C)12–(C) 9 : INFINITY (THROTTLE VALVE OPEN)
 2.3 K Ω OR LESS (THROTTLE VALVE FULLY CLOSED)
 (C)11–(C) 9 : **3.3–10.0 K Ω** (THROTTLE VALVE FULLY OPEN)
 0.2–0.8 K Ω (THROTTLE VALVE FULLY CLOSED)
 (C) 1–(C) 9 : **3.0–7.0 K Ω**
 (C) 4–(C) 9 : **0.2–0.4 K Ω** (COOLANT TEMP. **80°C, 176°F**)
 (B) 4, (B) 5, (B) 6–GROUND : **11–15 Ω** (ALL CONDITIONS)

: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C 1	30 (5S–FE)	E 4	33	N 2	30 (5S–FE)
C13	32	E 6	33	O 7	33
C14	32	E 7	33	S 2	30 (5S–FE)
C19	32	E 8	33	S10	33
C20	32	I 3	30 (5S–FE)	T 1	30 (5S–FE)
E 1	30 (5S–FE)	J 2	33		
E 2	30 (5S–FE)	J 6	33		

: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1B		
1E		
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2A	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2C	22 (2WD)	ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2E	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
3A	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)
3B		
3C		

: CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IG1	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)
IG2		

: GROUND POINTS

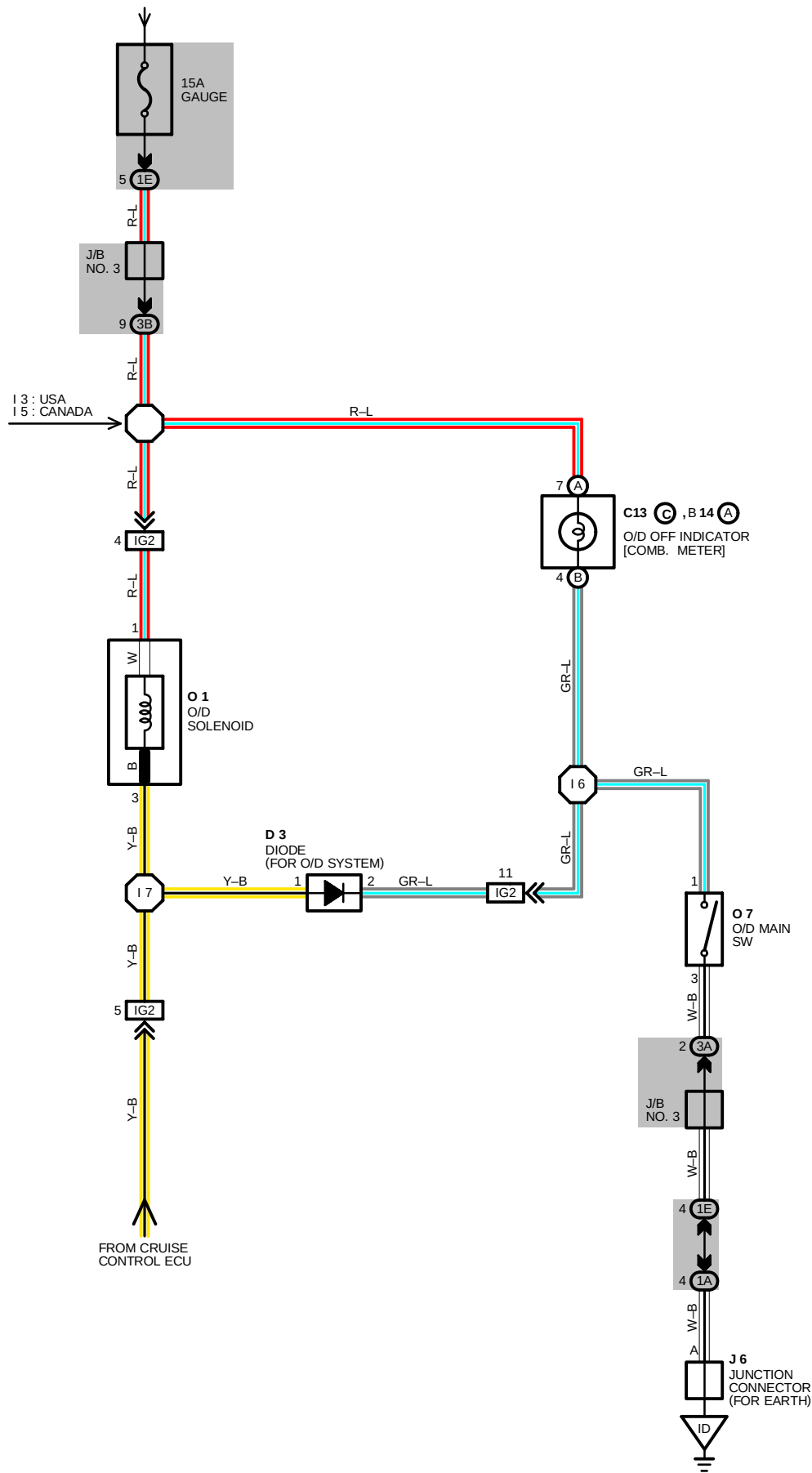
CODE	SEE PAGE	GROUND POINTS LOCATION
EB	40 (5S–FE)	FRONT LEFT FENDER
EC	40 (5S–FE)	INTAKE MANIFOLD
ID	44	LEFT KICK PANEL

: SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 5	40 (5S–FE)	ENGINE WIRE	I 5	46	COWL WIRE
E 7			I 6		
I 3	46	COWL WIRE	I 7	46	ENGINE WIRE

OVERDRIVE (4A-FE, A/T)

FROM POWER SOURCE SYSTEM (SEE PAGE 66)



SERVICE HINTS

o 7 O/D MAIN SW

1-3 : CLOSED WITH O/D MAIN SW OFF, OPEN WITH O/D MAIN SW ON

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C13	B 32	D 3	32	O 1	31 (4A-FE)
C14	A 32	J 6	33	O 7	33

□ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
3A	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)
3B		

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

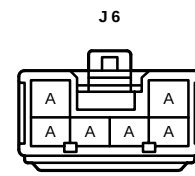
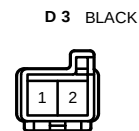
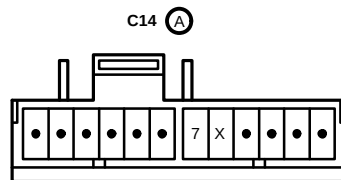
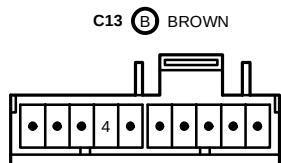
CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IG2	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)

▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL

○ : SPLICE POINTS

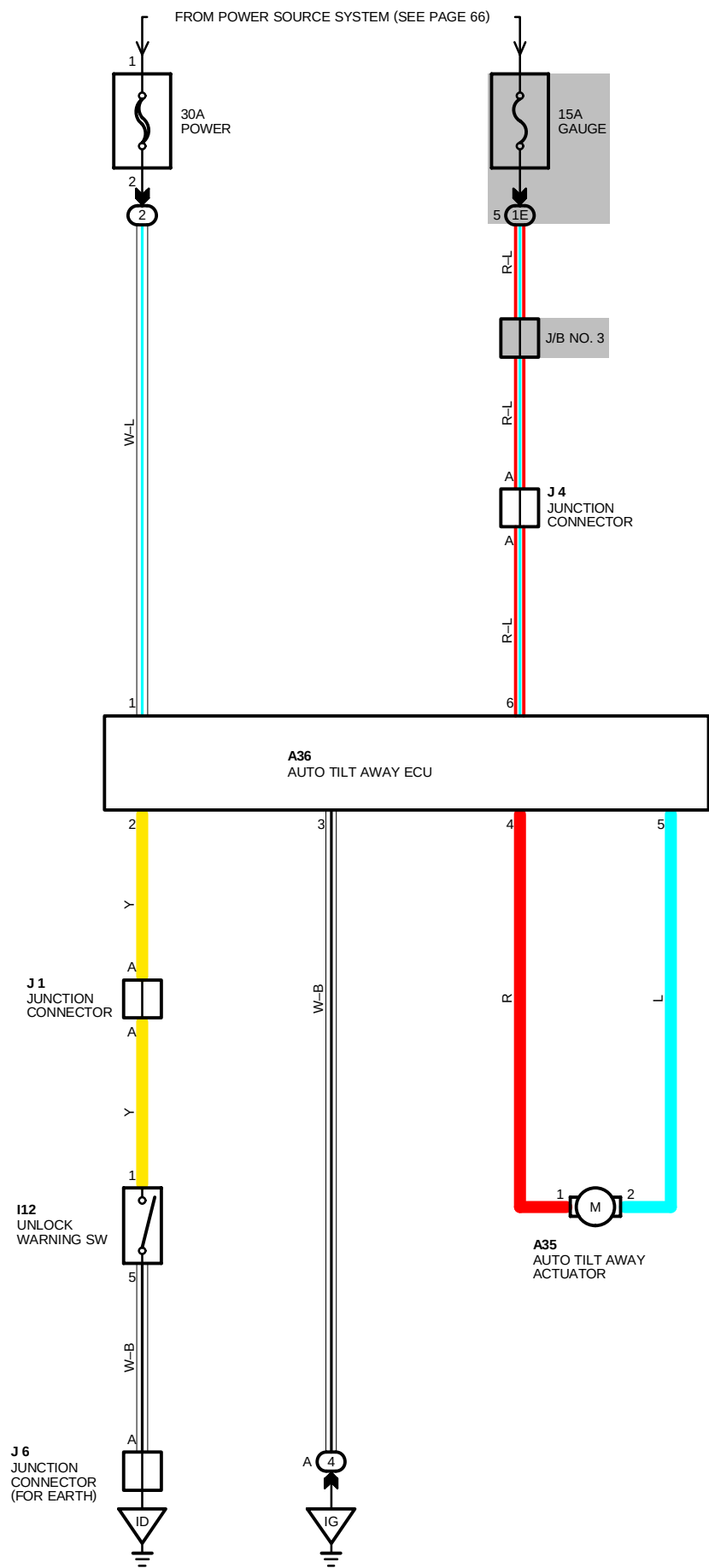
CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I 3	46	COWL WIRE	I 6	46	COWL WIRE
I 5			I 7	46	ENGINE WIRE



(HINT : SEE PAGE 7)



AUTO TILT AWAY STEERING



SYSTEM OUTLINE

THIS SYSTEM ALLOWS EASIER ENTRY AND EXIT BY THE DRIVER BY MOVING THE STEERING POSITION TO THE **UPPER** POSITION WHEN THE IGNITION KEY IS REMOVED.

CURRENT ALWAYS FLOWS FROM THE POWER FUSE TO **TERMINAL 1** OF AUTO TILT AWAY ECU. WHEN THE IGNITION SW IS TURNED ON, CURRENT FLOWS FROM THE GAUGE FUSE AS FAR AS **TERMINAL 6**.

1. TILT-UP OPERATION

WHEN IGNITION KEY IS REMOVED FROM THE IGNITION KEY CYLINDER (UNLOCK WARNING SW OFF), A WARNING SW OFF SIGNAL IS INPUT TO **TERMINAL 2** OF THE AUTO TILT AWAY ECU. AS A RESULT, THE AUTO TILT AWAY OPERATES AND CURRENT FLOWS FROM **TERMINAL 4** OF THE AUTO TILT AWAY ECU → **TERMINAL 1** OF THE AUTO TILT AWAY ACTUATOR → ACTUATOR → **TERMINAL 2** → **TERMINAL 5** OF THE AUTO TILT AWAY ECU, CAUSING THE ACTUATOR TO ROTATE AND RELEASE THE LOCK MECHANISM LOCKED AT THE **MEMORY** POSITION (THE POSITION SELECTED USING THE TILT ADJUSTMENT LEVER). THE STEERING WHEEL THE MOVES FROM THE **MEMORY** POSITION TO THE **UPPER** POSITION USING SPRING FORCE, THEN MECHANICAL LOCK OCCURS.

2. TILT RETURN OPERATION

WHEN THE IGNITION KEY IS INSERTED INTO THE IGNITION KEY CYLINDER (UNLOCK WARNING SW ON), A WARNING SW ON SIGNAL IS INPUT TO **TERMINAL 2** OF THE AUTO TILT AWAY ECU. AS A RESULT, THE AUTO TILT AWAY ECU OPERATES AND CURRENT FLOWS FROM **TERMINAL 5** OF THE AUTO TILT AWAY ECU → **TERMINAL 2** OF THE AUTO TILT AWAY ACTUATOR → ACTUATOR → **TERMINAL 1** → **TERMINAL 4** OF THE AUTO TILT AWAY ECU, CAUSING THE ACTUATOR TO ROTATE AND RELEASE THE LOCK MECHANISM LOCKED AT THE **UPPER** POSITION. THE STEERING WHEEL CAN THEN BE RETURNED BY HAND TO THE **MEMORY** POSITION, WHERE MECHANICAL LOCK OCCURS.

SERVICE HINTS

A36 AUTO TILT AWAY ECU

- 1-GROUND : ALWAYS APPROX. 12 VOLTS
- 6-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION
- 5-GROUND : CONTINUITY WITH IGNITION KEY IN CYLINDER
- 3-GROUND : ALWAYS CONTINUITY



: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
A35	32	I12	33	J 4	33
A36	32	J 1	33	J 6	33



: RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
2	26	R/B NO. 2 (LEFT KICK PANEL)
4	27	R/B NO. 4 (RIGHT KICK PANEL)



: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1E	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)



: GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL
IG	44	R/B NO. 4 SET BOLT

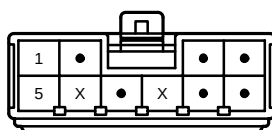
A35



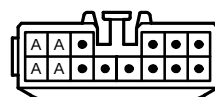
A36



I12 BLACK



J 1

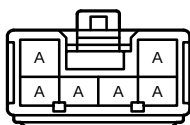


(HINT : SEE PAGE 7)

J 4

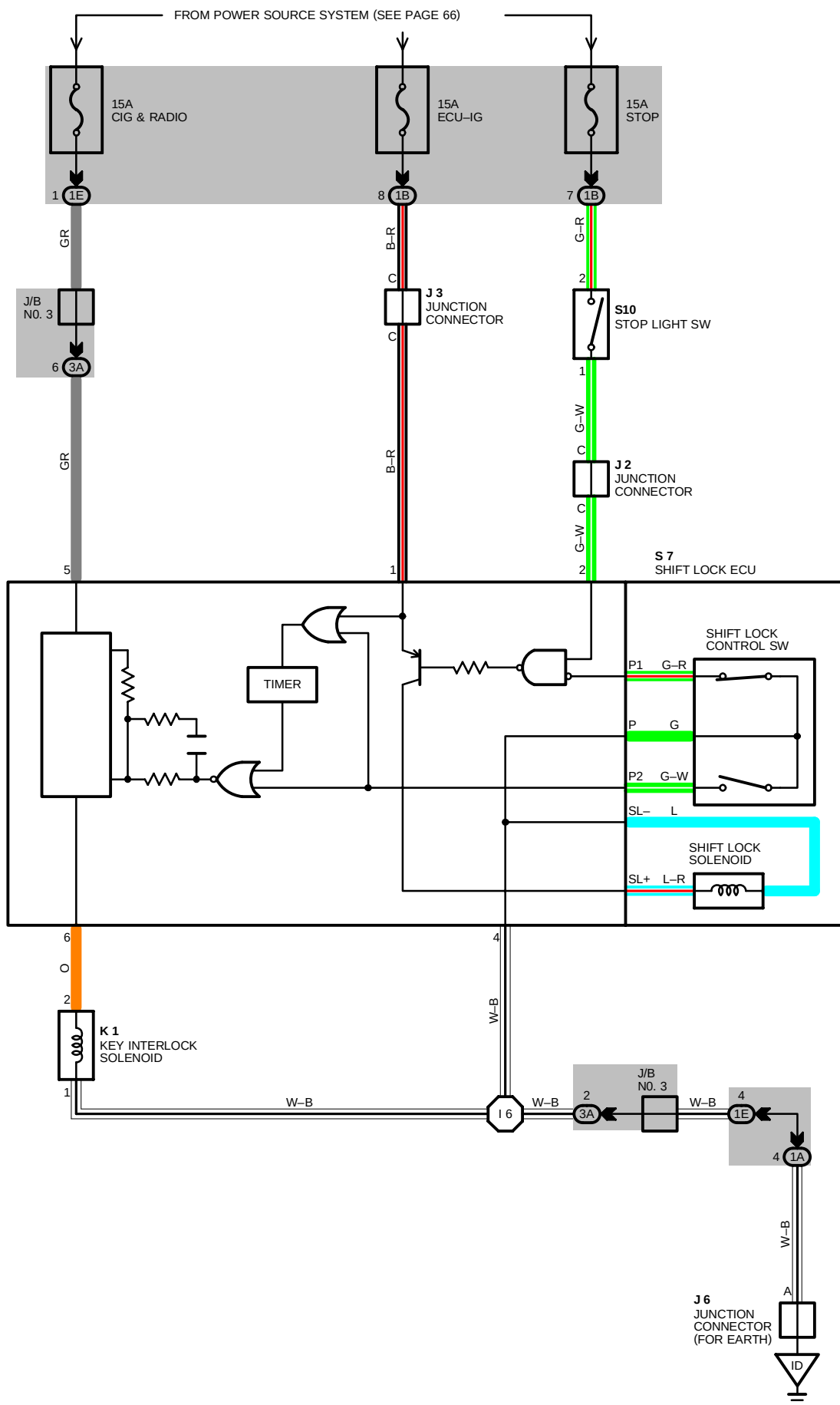


J 6



(HINT : SEE PAGE 7)

SHIFT LOCK



SYSTEM OUTLINE

WHEN THE IGNITION SW IS TURNED TO **ACC** POSITION THE CURRENT FROM THE CIG FUSE FLOWS TO **TERMINAL 5** OF THE SHIFT LOCK ECU. IN THE **ON** POSITION, THE CURRENT FROM THE ECU-IG FUSE FLOWS TO **TERMINAL 1** OF THE ECU.

1. SHIFT LOCK MECHANISM

WITH THE IGNITION SW ON, WHEN A SIGNAL THAT THE BRAKE PEDAL IS DEPRESSED (STOP LIGHT SW ON) AND A SIGNAL THAT THE SHIFT LEVER IS PUT IN "P" RANGE (CONTINUITY BETWEEN P1 AND P OF THE SHIFT POSITION SW) IS INPUT TO THE ECU. THE ECU OPERATES AND CURRENT FLOWS FROM **TERMINAL 1** OF THE ECU → **TERMINAL SL+** OF THE SHIFT LOCK SOLENOID → SOLENOID → **TERMINAL SL-** → **TERMINAL 4** OF THE ECU → **GROUND**. THIS CAUSES THE SHIFT LOCK SOLENOID TO TURN ON (PLATE STOPPER DISENGAGES) AND THE SHIFT LEVER CAN SHIFT INTO OTHER RANGE THAN THE "P" RANGE.

2. KEY INTERLOCK MECHANISM

WITH THE IGNITION SW IN **ON** OR **ACC** POSITION, WHEN THE SHIFT LEVER IS PUT IN "P" RANGE (NO CONTINUITY BETWEEN P2 AND P OF LOCK CONTROL SW). THE CURRENT FLOWING FROM **TERMINAL 5** OF THE ECU → KEY INTERLOCK SOLENOID IS CUT OFF. THIS CAUSES THE KEY INTERLOCK SOLENOID TO TURN OFF (LOCK LEVER DISENGAGES FROM LOCK POSITION) AND THE IGNITION KEY CAN BE TURNED FROM **ACC** TO **LOCK** POSITION. IF THE IGNITION IS LEFT IN **ACC** OR **ON** POSITION WITH THE SHIFT LEVER IN OTHER THAN "P" RANGE, THEN AFTER APPROX. ONE HOUR THE ECU OPERATES TO RELEASE THE LOCK.

SERVICE HINTS

S7 SHIFT LOCK ECU

- 5-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ACC** OR **ON** POSITION
- 1-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION
- 4-GROUND : ALWAYS CONTINUITY
- 2-GROUND : APPROX. 12 VOLTS WITH BRAKE PEDAL DEPRESSED

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
J 2	33	J 6	33	S 7	33
J 3	33	K 1	33	S10	33

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1B		
1E		
3A	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

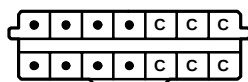
▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL

○ : SPLICE POINTS

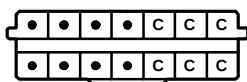
CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I 6	46	COWL WIRE			

J 2



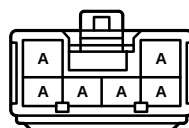
(HINT : SEE PAGE 7)

J 3



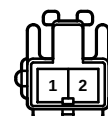
(HINT : SEE PAGE 7)

J 6

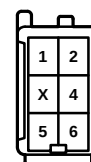


(HINT : SEE PAGE 7)

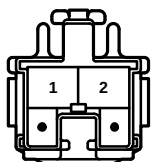
K 1 BLUE



S 7



S10







CRUISE CONTROL (MOTOR TYPE)

SYSTEM OUTLINE

CURRENT IS APPLIED AT ALL TIMES THROUGH STOP FUSE TO **TERMINAL (A) 2** OF THE CONTROL ECU AND **TERMINAL 2** OF STOP LIGHT SWITCH.

WITH THE IGNITION SWITCH TURNED TO ON, THE CURRENT FLOWS THROUGH GAUGE FUSE TO **TERMINAL (A) 7** OF CRUISE CONTROL INDICATOR LIGHT. THE CURRENT THROUGH ECU-IG FUSE FLOWS TO **TERMINAL (A) 1** OF CRUISE CONTROL ECU AND **TERMINAL 1** OF CRUISE CONTROL VEHICLE SPEED SENSOR.

WHEN THE IGNITION SWITCH IS ON AND THE CRUISE CONTROL MAIN SWITCH IS TURNED ON, A SIGNAL IS INPUT FROM **TERMINAL 15** OF CRUISE CONTROL MAIN SWITCH TO **TERMINAL (A) 8** OF CRUISE CONTROL ECU. AS A RESULT, THE CRUISE CONTROL ECU FUNCTIONS AND THE CURRENT TO **TERMINAL (A) 1** OF CRUISE CONTROL ECU TO **TERMINAL (A) 9** OF CRUISE CONTROL ECU → **GROUND**, AND THE CRUISE CONTROL SYSTEM IS IN A CONDITION READY FOR OPERATION.

AT THE SAME TIME, THE CURRENT THROUGH THE GAUGE FUSE FLOWS FROM **TERMINAL (A) 7** OF CRUISE CONTROL INDICATOR LIGHT → **TERMINAL (B) 2** → **TERMINAL (B) 6** OF CRUISE CONTROL ECU → **TERMINAL (A) 9** → TO **GROUND**, CAUSING THE CRUISE CONTROL INDICATOR LIGHT TO LIGHT UP, INDICATING THAT THE CRUISE CONTROL IS READY FOR OPERATION.

1. SET OPERATION

WHEN THE CRUISE CONTROL MAIN SWITCH IS TURNED ON AND THE SET SWITCH IS PUSHED WITH THE VEHICLE SPEED WITHIN THE SET LIMIT (APPROX. 40KM/H, 25MPH TO 200KM/H, 124MPH), A SIGNAL IS INPUT TO **TERMINAL (B) 8** OF THE CRUISE CONTROL ECU AND THE VEHICLE SPEED AT THE TIME THE SET SWITCH IS RELEASED IS MEMORIZED IN THE ECU AS THE SET SPEED.

2. SET SPEED CONTROL

DURING CRUISE CONTROL DRIVING, THE ECU COMPARES THE SET SPEED MEMORIZED IN THE ECU WITH THE ACTUAL VEHICLE SPEED INPUT INTO **TERMINAL (A) 7** OF THE CRUISE CONTROL ECU FROM THE VEHICLE SPEED SENSOR, AND CONTROLS THE CRUISE CONTROL ACTUATOR TO MAINTAIN THE SET SPEED.

WHEN THE ACTUAL SPEED IS LOWER THAN THE SET SPEED, THE ECU CAUSES THE CURRENT TO THE CRUISE CONTROL ACTUATOR TO FLOW FROM **TERMINAL (B) 4** → **TERMINAL 6** OF CRUISE CONTROL ACTUATOR → **TERMINAL 7** → **TERMINAL (B) 10** OF CRUISE CONTROL ECU. AS A RESULT, THE MOTOR IN THE CRUISE CONTROL ACTUATOR IS ROTATED TO OPEN THE THROTTLE VALVE AND THE THROTTLE CABLE IS PULLED TO INCREASE THE VEHICLE SPEED. WHEN THE ACTUAL DRIVING SPEED IS HIGHER THAN THE SET SPEED, THE CURRENT TO CRUISE CONTROL ACTUATOR FLOWS FROM **TERMINAL (B) 10** OF ECU → **TERMINAL 7** OF CRUISE CONTROL ACTUATOR → **TERMINAL 6** → **TERMINAL (B) 4** OF CRUISE CONTROL ECU.

THIS CAUSES THE MOTOR IN THE CRUISE CONTROL ACTUATOR TO ROTATE TO CLOSE THE THROTTLE VALVE AND RETURN THE THROTTLE CABLE TO DECREASE THE VEHICLE SPEED.

3. COAST CONTROL

DURING THE CRUISE CONTROL DRIVING, WHILE THE COAST SWITCH IS ON, THE CRUISE CONTROL ACTUATOR RETURNS THE THROTTLE CABLE TO CLOSE THE THROTTLE VALVE AND DECREASE THE DRIVING SPEED. THE VEHICLE SPEED WHEN THE COAST SWITCH IS TURNED OFF AND IS MEMORIZED AND THE VEHICLE CONTINUES AT THE NEW SET SPEED.

4. ACCEL CONTROL

DURING CRUISE CONTROL DRIVING, WHILE THE ACCEL SWITCH IS TURNED ON, THE CRUISE CONTROL ACTUATOR PULLS THE THROTTLE CABLE TO OPEN THE THROTTLE VALVE AND INCREASE THE DRIVING SPEED. THE VEHICLE SPEED WHEN THE ACCEL SWITCH IS TURNED OFF IS MEMORIZED AND THE VEHICLE CONTINUES AT THE NEW SET SPEED.

5. RESUME CONTROL

UNLESS THE VEHICLE SPEED FALLS BELOW THE MINIMUM SPEED LIMIT (APPROX. 40KM/H, 25MPH) AFTER CANCELING THE SET SPEED BY THE CANCEL SWITCH, PUSHING THE RESUME SWITCH WILL CAUSE THE VEHICLE TO RESUME THE SPEED SET BEFORE CANCELLATION.

6. MANUAL CANCEL MECHANISM

IF ANY OF THE FOLLOWING OPERATIONS OCCURS DURING CRUISE CONTROL OPERATION, THE SAFETY MAGNET CLUTCH OF THE ACTUATOR MOTOR TURNS OFF AND THE MOTOR ROTATES TO CLOSE THE THROTTLE VALVE AND THE CRUISE CONTROL IS RELEASED.

- * PLACING THE SHIFT LEVER IN "N" RANGE NEUTRAL START SW) ON. "SIGNAL INPUT TO **TERMINAL (A) 2** OF ECU"
- * DEPRESSING THE BRAKE PEDAL (STOP LIGHT SWITCH ON). "SIGNAL INPUT TO **TERMINAL (A) 2** OF ECU"
- * PULL UP THE PARKING BRAKE LEVER (PARKING BRAKE SWITCH ON). "SIGNAL INPUT TO **TERMINAL (A) 5** OF ECU"
- * PUSH THE CANCEL SWITCH (CANCEL SWITCH ON). "SIGNAL INPUT TO **TERMINAL (B) 8**"

7. AUTO CANCEL FUNCTION

A) IF ANY OF THE FOLLOWING OPERATE CONDITIONS OCCURS DURING CRUISE CONTROL OPERATION. THE SET SPEED IS ERASED, CURRENT FLOW TO SAFETY MAGNETIC CLUTCH IS STOPPED AND THE CRUISE CONTROL IS RELEASED. (MAIN SWITCH TURNS OFF).

WHEN THIS OCCURS, THE IGNITION SWITCH MUST BE TURNED OFF ONCE BEFORE THE MAIN SWITCH WILL TURN ON.

- * OVER CURRENT TO TRANSISTOR DRIVING MOTOR AND/OR SAFETY MAGNETIC CLUTCH.
- * CURRENT TO CONTROL THE THROTTLE VALVE IN MOTOR BECOMES ALWAYS "ON".
- * OPEN CIRCUIT IN SAFETY MAGNETIC CLUTCH.
- * MOMENTARY INTERRUPTION OF VEHICLE SPEED SIGNAL.
- * THE RESUME SWITCH IS ALREADY ON WHEN THE MAIN SWITCH IS TURNED ON.
- * SHORT CIRCUIT IN CRUISE CONTROL SWITCH.
- * MOTOR DOES NOT OPERATE DESPITE THE MOTOR DRIVE SIGNAL BEING OUTPUT.

B) IF ANY OF THE FOLLOWING CONDITIONS OCCUR DURING CRUISE CONTROL OPERATION, THE SET SPEED IS ERASED AND THE CRUISE CONTROL IS RELEASED. (THE POWER OF SAFETY MAGNETIC CLUTCH IS CUT OFF UNTIL THE SET SWITCH IS "ON" AGAIN.)

- * WHEN THE VEHICLE SPEED FALLS BELOW THE MINIMUM SPEED LIMIT, APPROX. **40 KM/H (25 MPH)**.
- * WHEN THE VEHICLE SPEED FALLS MORE THAN **16 KM/H (10 MPH)** BELOW THE SET SPEED, E.G. ON AN UPWARD SLOPE.
- * WHEN POWER TO THE CRUISE CONTROL SYSTEM IS MOMENTARILY CUT OFF.

C) IF ANY OF THE FOLLOWING CONDITIONS OCCURS DURING CRUISE CONTROL OPERATION, THE CRUISE CONTROL IS RELEASED. BUT IN THIS CASE, THE SET SPEED IS NOT ERASED. IF THE VEHICLE SPEED IS MORE THAN THE MINIMUM SPEED LIMIT (APPROX. **40 KM/H 25 MPH**), CRUISE CONTROL OPERATION IS POSSIBLE USING "SET" OR "RESUME" ON THE CONTROL SWITCH.

- * OPEN CIRCUIT FOR **TERMINAL (A) 3** OF CRUISE CONTROL ECU AND **TERMINAL 3** OF STOP LIGHT SWITCH.

8. AUTOMATIC TRANSMISSION CONTROL FUNCTION

* IN OVERDRIVE. IF THE VEHICLE SPEED BECOMES LOWER THAN THE OVERDRIVE CUT SPEED (SET SPEED MINIMUM **4 KM/H, 2.5 MPH**) DURING CRUISE CONTROL OPERATION, SUCH AS DRIVING UP A HILL, THE OVERDRIVE IS RELEASED AND THE POWER INCREASED TO PREVENT A REDUCTION IN VEHICLE SPEED.

* AFTER RELEASING THE OVERDRIVE, THE VEHICLE SPEED BECOMES HIGHER THAN THE OVERDRIVE RETURN SPEED (SET SPEED MINIMUM **2 KM/H 1.2 MPH**) AND THE ECU JUDGES BY THE SIGNALS FROM POTENTIOMETER OF THE ACTUATOR THAT THE UPWARD SLOPE HAS FINISHED, OVERDRIVE IS RESUMED AFTER APPROXIMATELY **6 SECONDS**.

* DURING CRUISE CONTROL DRIVING, THE CRUISE CONTROL OPERATION SIGNAL IS OUTPUT FROM THE CRUISE CONTROL ECU TO THE ENGINE AND ECT ECU. UPON RECEIVING THIS SIGNAL, THE ENGINE AND ECT ECU CHANGES THE SHIFT PATTERN TO NORMAL.

TO MAINTAIN SMOOTH CRUISE CONTROL OPERATION (ON A DOWNWARD SLOPE ECT.), LOCK-UP RELEASE OF THE TRANSMISSION WHEN THE IDLING POINT OF THE THROTTLE POSITION IS "ON" IS FORBIDDEN.

SERVICE HINTS

C 5 CRUISE CONTROL ACTUATOR

- 1-3 : APPROX. **2 K(**
- 5-4 : APPROX. **38.5 (**

C17 CRUISE CONTROL SW [COMB. SW]

- 15-17 : CONTINUITY WITH MAIN SW ON
- 5-17 : APPROX. **413 (** WITH CANCEL SW ON
- APPROX. **68 (** WITH RESUME/ACCEL SW ON
- APPROX. **198 (** WITH SET/COAST SW ON

C19(A), C20(B) CRUISE CONTROL ECU

- (A) 1-GROUND : APPROX. **12 VOLTS** WITH IGNITION SW AT **ON** POSITION
- (A) 2, 6-GROUND : ALWAYS APPROX. **12 VOLTS**
- (A) 5-GROUND : CONTINUITY WITH PARKING BRAKE LEVER PULLED UP (ONE OF THE CANCEL SW) OR BRAKE LEVEL WARNING SW ON
- (A) 7-GROUND : **4 PULSE** WITH **1 ROTATION** OF ROTOR SHAFT
- (B) 8-GROUND : APPROX. **419 (** WITH CANCEL SW ON IN CONTROL SW
- APPROX. **68 (** WITH RESUME/ACCEL SW ON IN CONTROL SW
- APPROX. **198 (** WITH SET/COAST SW ON IN CONTROL SW
- (A) 9-GROUND : ALWAYS CONTINUITY

CRUISE CONTROL (MOTOR TYPE)

: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C 1	28 (3S-GTE), 30 (5S-FE)	C20	B 32	J 3	33
C 5	28 (3S-GTE), 30 (5S-FE)	D 4	32	J 6	33
C13	B 32	E 6	D 33	P 1	33
C14	A 32	E 7	B 33	R10	33
C17	32		C 33	S 1	29 (3S-GTE), 30 (5S-FE)
C18	32	E 8	A 33	S10	33
C19	A 32	J 2	33		

: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1B	20	COWL WIRE AND J/B NO.1 (LEFT KICK PANEL)
1E		
2A	23 (ALL-TRAC/4WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2C	23 (ALL-TRAC/4WD)	ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)
3B	25	COWL WIRE AND J/B NO.3 (BEHIND COMBINATION METER)

: CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)
IE2		
IF1	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
IG1	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)
IG2		

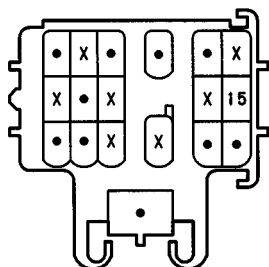
: GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
EB	38 (3S-GTE) 40 (5S-FE)	FRONT LEFT FENDER
ID	44	LEFT KICK PANEL
IE	44	INSTRUMENT PANEL BRACE LH

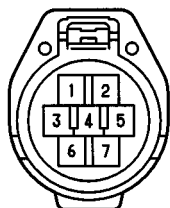
: SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I 2	46	COWL WIRE	I 4	46	COWL WIRE

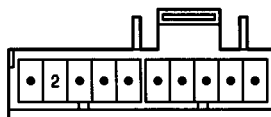
C 1 DARK GRAY



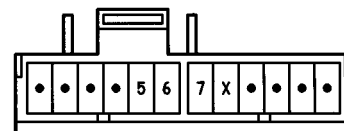
C 5 GRAY



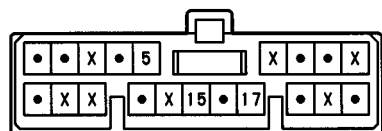
C13 (B) BROWN



C14 (A)



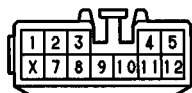
C17 BLACK



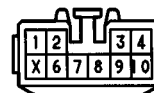
C18



C19 (A)



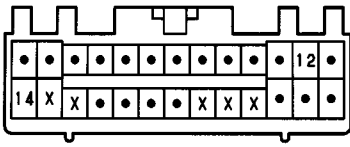
C20 (B) BLUE



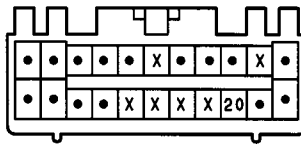
D 4



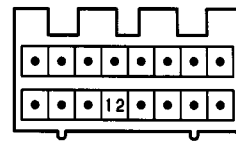
E 6 **D** DARK GRAY



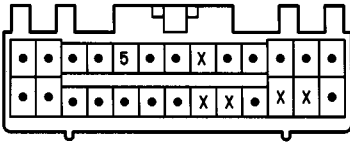
E 7 **B** DARK GRAY



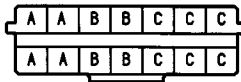
E 7 **C** DARK GRAY



E 8 **A** DARK GRAY

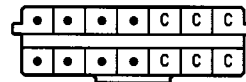


J 2



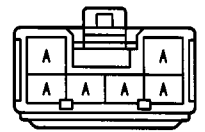
(HINT:SEE PAGE 7)

J 3



(HINT:SEE PAGE 7)

J 6

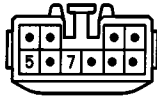


(HINT:SEE PAGE 7)

P 1 BLACK



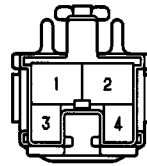
R10



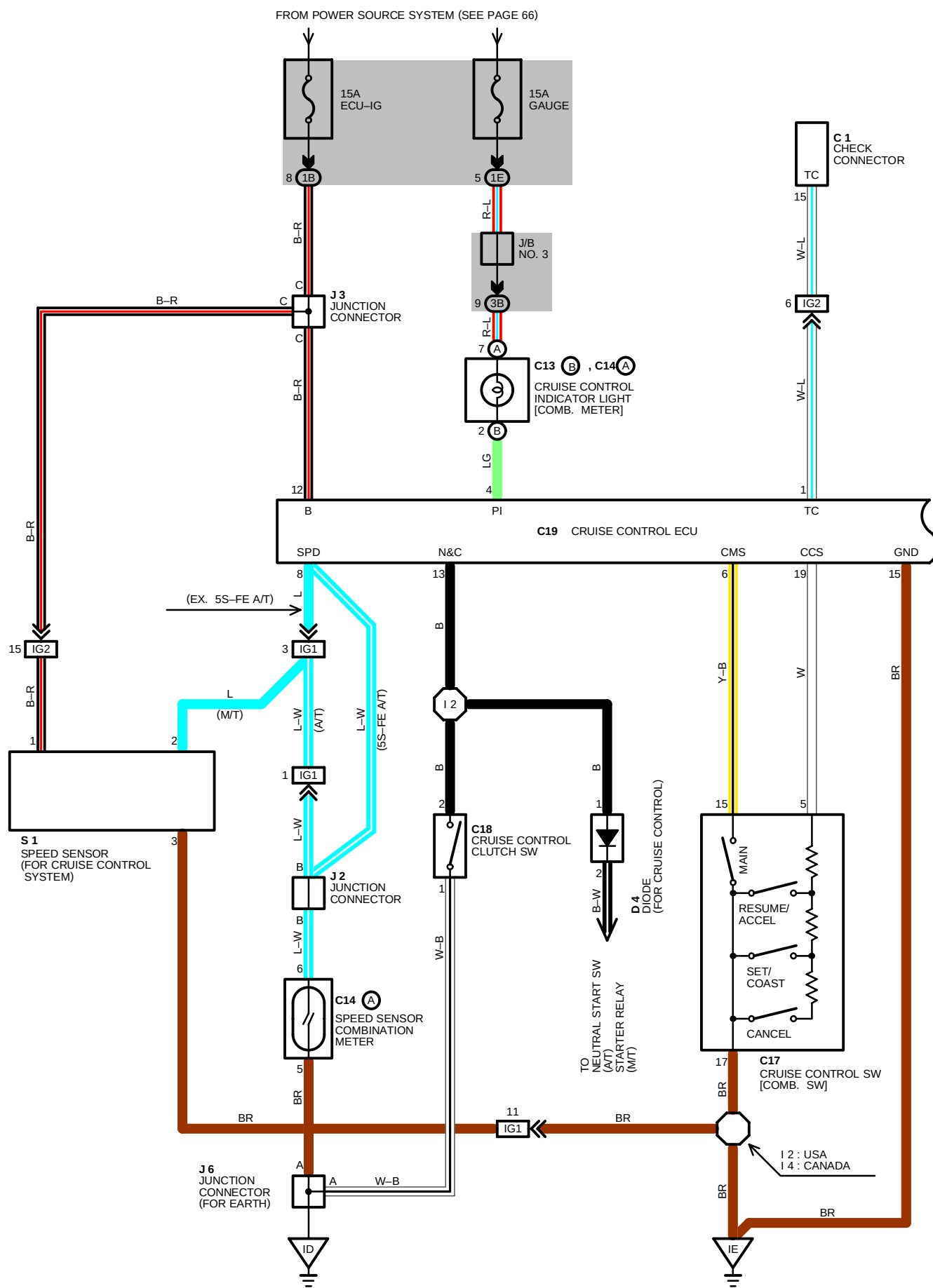
S 1 GRAY



S10



186





CRUISE CONTROL (VACUUM TYPE)

SYSTEM OUTLINE

CURRENT IS APPLIED AT ALL TIMES THROUGH STOP FUSE TO **TERMINAL 18** OF THE CRUISE CONTROL ECU AND **TERMINAL 2** OF THE STOP LIGHT SW.

WITH THE IGNITION SW TURNED TO ON, THE CURRENT FLOWS THROUGH GAUGE FUSE TO **TERMINAL (A) 7** OF THE COMBINATION METER AND THE CURRENT THROUGH ECU-IG FUSE FLOWS TO **TERMINAL 12** OF THE CRUISE CONTROL ECU.

WHEN THE IGNITION SW IS ON AND THE CRUISE CONTROL MAIN SW IS TURNED ON, A SIGNAL IS INPUT FROM **TERMINAL 15** OF THE CRUISE CONTROL MAIN SW TO **TERMINAL 6** OF THE CRUISE CONTROL ECU. AS A RESULT, THE CRUISE CONTROL ECU FUNCTIONS AND THE CURRENT TO **TERMINAL 12** OF THE CRUISE CONTROL ECU FLOWS TO **TERMINAL 15** OF THE CRUISE CONTROL ECU → **GROUND**, AND THE CRUISE CONTROL SYSTEM IS IN A CONDITION READY FOR OPERATION.

AT THE SAME TIME, THE CURRENT THROUGH GAUGE FUSE FLOWS FROM **TERMINAL (A) 7** → **TERMINAL (B) 2** → **TERMINAL 4** OF CRUISE CONTROL ECU → **TERMINAL 15** → TO **GROUND** AND CAUSING THE CRUISE CONTROL INDICATOR LIGHT TO LIGHT UP IN ORDER TO NOTE THE CRUISE CONTROL CAN START UP.

1. CRUISE CONTROL DRIVING

WHEN THE MAIN SW IS TURNED TO ON AND THE SET SW IS PUSHED IN WITH THE VEHICLE SPEED WITHIN THE SET LIMIT (APPROX. 40KM/H, 25MPH TO 200KM/H, 124MPH), A SIGNAL IS INPUT TO **TERMINAL 19** OF THE ECU AND THE VEHICLE SPEED AT THAT TIME IS RECORDED IN THE ECU MEMORY AS THE SET SPEED.

THE ECU COMPARES THE RECORDED SET SPEED WITH THE ACTUAL VEHICLE SPEED INPUT INTO **TERMINAL 8** FROM THE SPEED SENSOR AND CONTROLS THE CRUISE CONTROL ACTUATOR IN ORDER TO MAINTAIN THE SET VEHICLE SPEED.

WHEN THE ACTUAL VEHICLE SPEED IS LOWER THAN THE SET SPEED, ECU OPERATION LENGTHENS THE PERIOD OF CURRENT FLOW FROM **TERMINAL 5** OF THE ECU → **TERMINAL 2** OF THE ACTUATOR → THE CONTROL VALVE → **TERMINAL 3** → **TERMINAL 16** OF THE ECU, THE CABLE IS PULLED IN THE DIRECTION FOR OPENING THROTTLE VALVE AND THE VEHICLE SPEED INCREASES.

WHEN THE ACTUAL VEHICLE SPEED IS HIGHER THAN THE SET SPEED, A SHORTER PERIOD OF CURRENT FLOW TO THE CONTROL VALVE RETURNS THE CABLE IN THE DIRECTION FOR CLOSING THE THROTTLE VALVE AND THE VEHICLE SPEED DECREASES.

(ACTUATOR OPERATION)

WHEN THE CRUISE CONTROL SYSTEM OPERATES (THE SET SIGNAL IS INPUT), CURRENT FLOWS FROM THE ECU TO THE RELEASE VALVE, CLOSING THE ATMOSPHERIC INTAKE PORT.

WHEN THERE IS CONTINUITY TO THE CONTROL VALVE, VACUUM IS INTRODUCED INSIDE THE ACTUATOR, AND WHEN THERE IS NO CONTINUITY, VACUUM INTAKE STOPS AND ATMOSPHERE IS INTRODUCED. IN OTHER WORDS, THE ACTUATOR (THROTTLE VALVE) IS CONTROLLED BY CHANGING THE RATIO OF CONTINUITY AND NON-CONTINUITY TO THE CONTROL VALVE WITHIN A SPECIFIED PERIOD OF TIME.

2. CANCEL MECHANISM

IF ANY OF THE FOLLOWING OPERATIONS IS PERFORMED DURING CRUISE CONTROL, THEN CONTINUITY TO THE CONTROL VALVE AND THE RELEASE VALVE IS CUT OFF AND CRUISE CONTROL IS RELEASED.

- * DEPRESSING THE CLUTCH PEDAL (CLUTCH SW ON), SIGNAL INPUT TO **TERMINAL 13** OF THE ECU. (M/T)
- * PLACING THE NEUTRAL START SW IN "N" RANGE NEUTRAL START SW ON), SIGNAL INPUT TO **TERMINAL 13** OF THE ECU. (A/T)
- * DEPRESSING THE BRAKES PEDAL (STOP LIGHT SW ON), SIGNAL INPUT TO **TERMINAL 17** OF THE ECU.
- * PULLING THE PARKING BRAKE LEVER (PARKING BRAKE SW ON), SIGNAL INPUT TO **TERMINAL 14** OF THE ECU.

3. COAST CONTROL

WHILE THE COAST SW IS ON DURING CRUISE CONTROL, CURRENT FLOW TO THE CONTROL VALVE AND RELEASE VALVE IS STOPPED AND THE VEHICLE DECELERATES UNTIL THE SW IS RELEASED. THE VEHICLE SPEED WHEN THE SW IS RELEASED IS THEN RECORDED IN MEMORY.

4. RESUME CONTROL

BY TURNING THE RESUME SW TO ON AFTER CANCELLATION OF THE CRUISE CONTROL SYSTEMS, THE VEHICLE SPEED WILL RETURN TO THE SPEED SET BEFORE CANCELLATION. PROVIDED THAT THE VEHICLE SPEED IS WITHIN THE SET LIMITS.

5. ACCEL CONTROL

WHEN THE ACCEL SW IS TURNED TO ON DURING CRUISE CONTROL DRIVING, CURRENT CONTINUES TO FLOW TO THE CONTROL VALVE AND THE VEHICLE ACCELERATES. THE VEHICLE SPEED WHEN THE SW IS TURNED OFF IS RECORDED IN MEMORY.

SERVICE HINTS

C19 CRUISE CONTROL ECU

(DISCONNECT THE ECU CONNECTOR)

- 15-GROUND : ALWAYS CONTINUITY
- 4-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION
- 8-GROUND : 4 PULSE WITH 1 ROTATION OF ROTOR SHAFT
- 6-GROUND : CONTINUITY WITH MAIN SW ON
- 18-GROUND : ALWAYS APPROX. 12 VOLTS
- 17-GROUND : APPROX. 12 VOLTS WITH BRAKE PEDAL DEPRESSED (ONE OF THE CANCEL SW)
- 13-GROUND : CONTINUITY WITH CLUTCH PEDAL DEPRESSED (M/T)
APPROX. 12 VOLTS WITH IGNITION SW AT **ST** POSITION AND SHIFT LEVER IN **N** OR **P** RANGE (A/T)
(ONE OF THE CANCEL SW)
- 14-GROUND : CONTINUITY WITH PARKING BRAKE LEVER PULL UP, (ONE OF THE CANCEL SW) OR BRAKE LEVEL
WARNING SW ON



: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C 1	30 (5S-FE)	C19	32	J 6	33
C 6	30 (5S-FE)	D 4	32	P 1	33
C13 B	32	E 7 B	33	R10	33
C14 A	32	E 8 A	33	S 1	30 (5S-FE)
C17	32	J 2	33	S10	33
C18	32	J 3	33		



: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1B	20	COWL WIRE AND J/B NO.1 (LEFT KICK PANEL)
1E		
3B	25	COWL WIRE AND J/B NO.3 (BEHIND COMBINATION METER)



: CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)
IF1	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
IG1	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)
IG2		



: GROUND POINTS

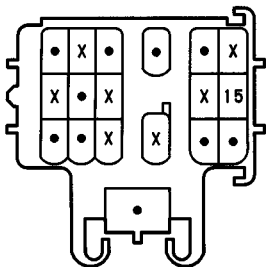
CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL
IE	44	INSTRUMENT PANEL BRACE LH



: SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I 2	46	COWL WIRE	I 4	46	COWL WIRE

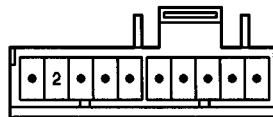
C 1 DARK GRAY



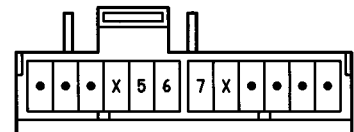
C 6 GRAY



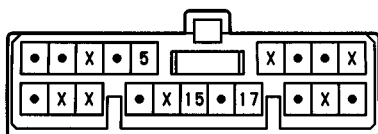
C13 (B) BROWN



C14 (A)



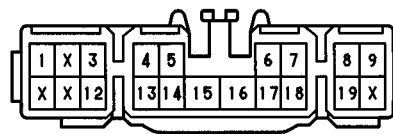
C17 BLACK



C18



C19

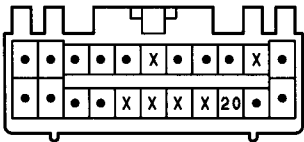


D 4

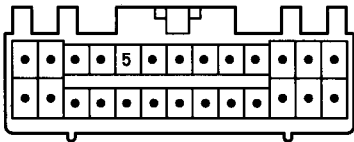


CRUISE CONTROL (VACUUM TYPE)

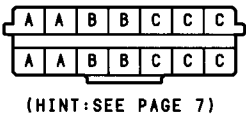
E 7 (B) DARK GRAY



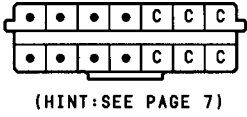
E 8 (A) DARK GRAY



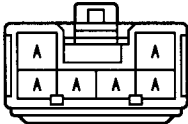
J 2



J 3



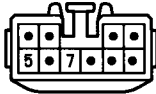
J 6



P 1 BLACK



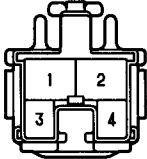
R10



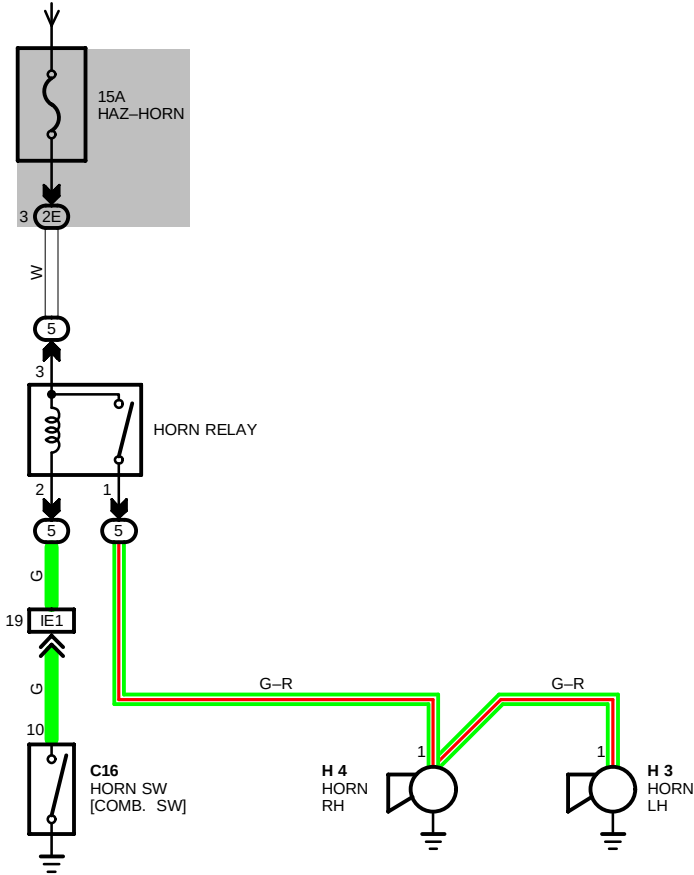
S 1 GRAY



S10



FROM POWER SOURCE SYSTEM (SEE PAGE 66)



SERVICE HINTS

HORN RELAY
(5) 3- (5) 1 : CLOSED WITH HORN SW ON

: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C16	32	H 3	31 (4A-FE)	H 4	31 (4A-FE)
H 3	29 (3S-GTE), 30 (5S-FE)	H 4	29 (3S-GTE), 30 (5S-FE)		

: RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
5	27	R/B NO. 5 (ENGINE COMPARTMENT FRONT RIGHT)

: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

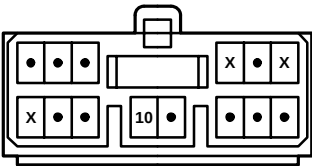
CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
2E	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO.2 (NEAR THE BATTERY)
	23 (ALL-TRAC/4WD)	

: CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

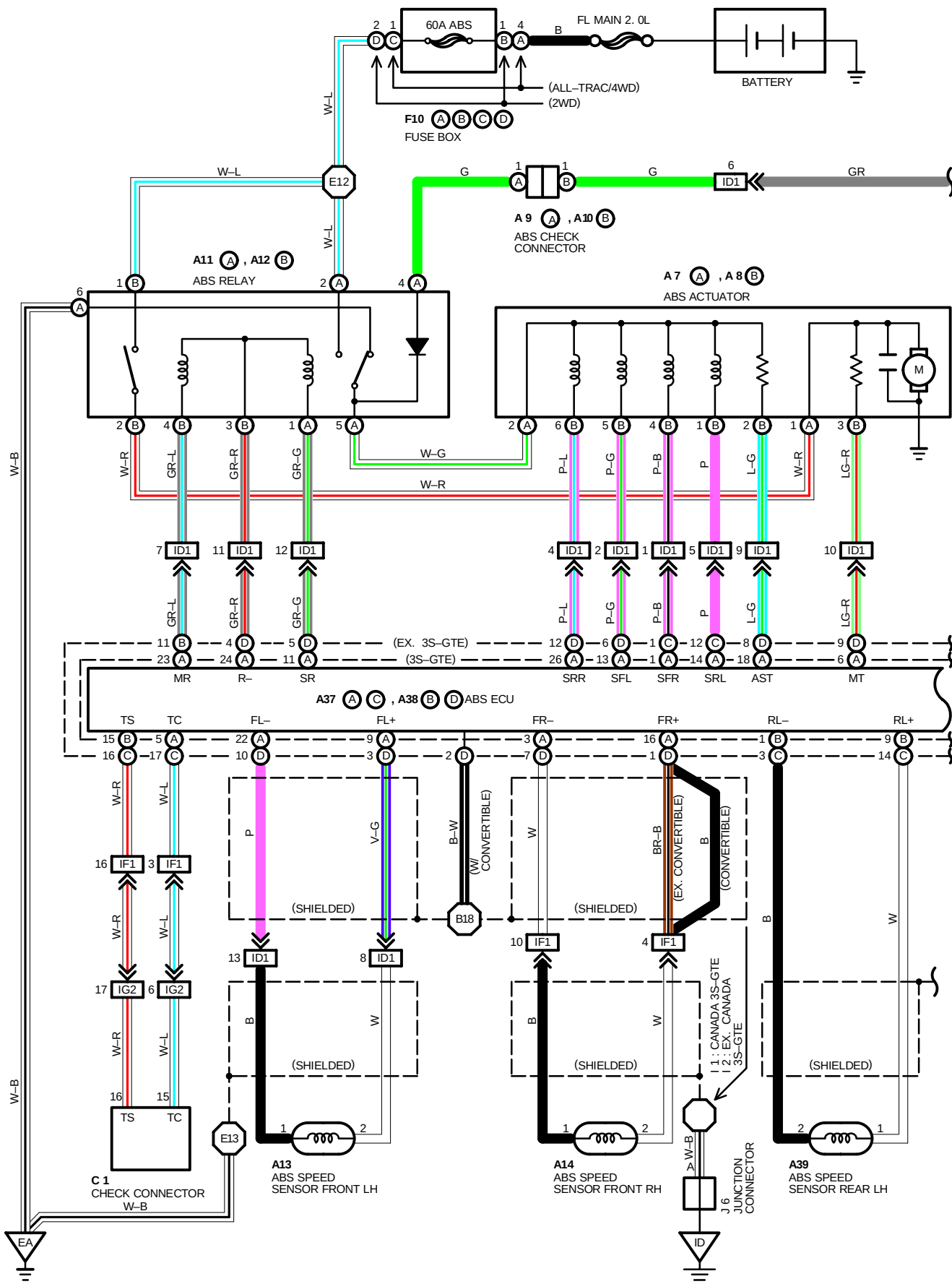
CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)

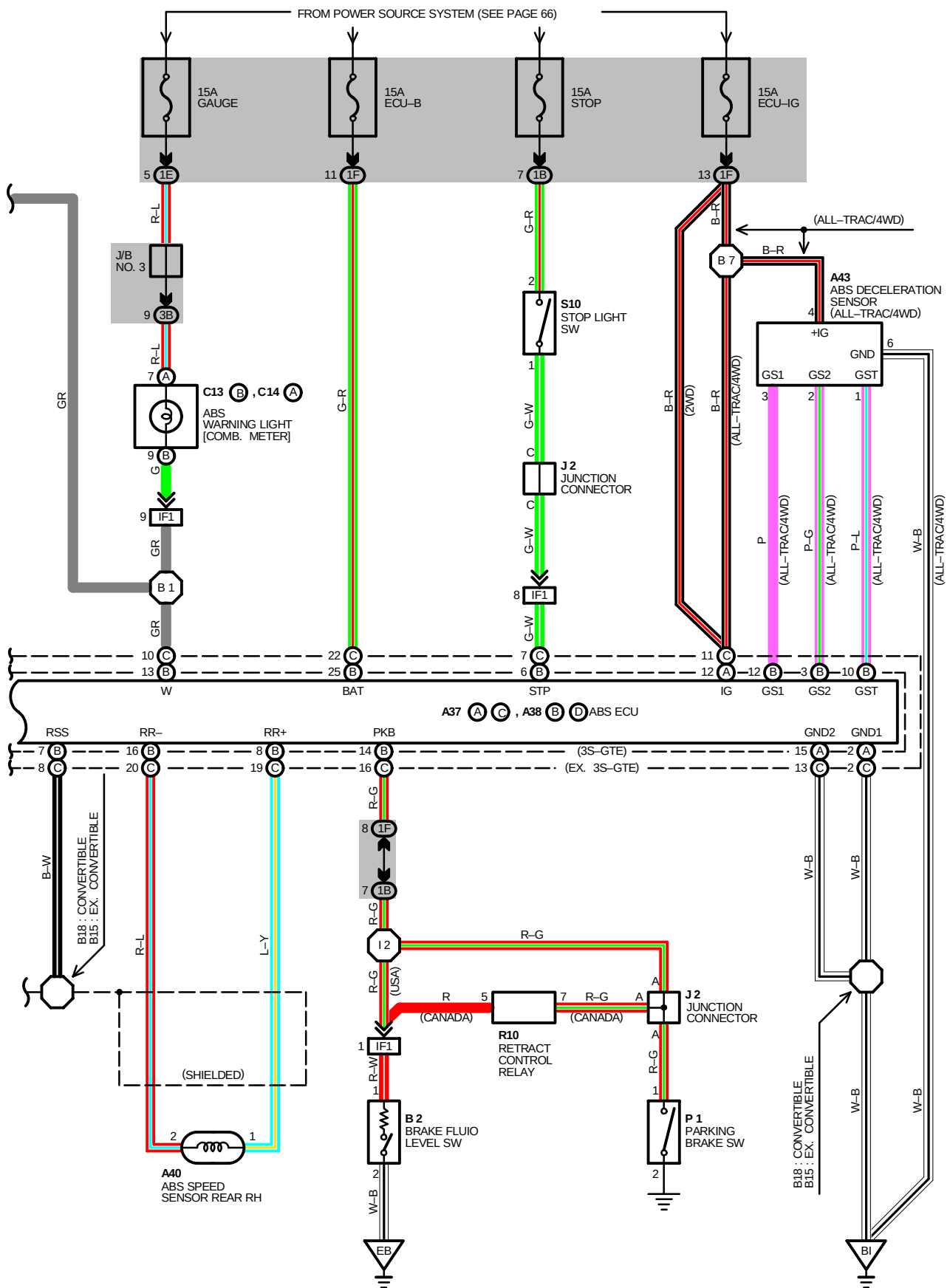
C16 BLACK

H 3, H 4 BLACK



192





ABS (ANTI-LOCK BRAKE SYSTEM)

SYSTEM OUTLINE

THIS SYSTEM CONTROLS THE RESPECTIVE BRAKE FLUID PRESSURES ACTING ON THE DISC BRAKE CYLINDERS OF THE RIGHT FRONT WHEEL, LEFT FRONT WHEEL AND REAR WHEELS WHEN THE BRAKES ARE APPLIED IN A PANIC STOP SO THAT THE WHEELS DO NOT LOCK. THIS RESULTS IN IMPROVED DIRECTIONAL STABILITY AND STEERABILITY DURING PANIC BRAKING.

1. INPUT SIGNALS

- (1) SPEED SENSOR SIGNAL
THE SPEED OF THE WHEELS IS DETECTED AND INPUT TO **TERMINALS FL+, FR+, RL+ AND RR+** OF THE ABS ECU.
- (2) STOP LIGHT SW SIGNAL
A SIGNAL IS INPUT TO **TERMINAL STP** OF THE ABS ECU WHEN BRAKE PEDAL IS OPERATED.
- (3) PARKING BRAKE SW SIGNAL
A SIGNAL IS INPUT TO **TERMINAL PKB** OF THE ABS ECU WHEN THE PARKING BRAKE IS OPERATED.
- (4) DECELERATION SENSOR SIGNAL (ALL-TRAC/4WD)
THE DEGREE OF VEHICLE DECELERATION IS DETECTED AND INPUT TO THE ABS ECU.

2. SYSTEM OPERATION

DURING SUDDEN BRAKING THE ABS ECU WHICH HAS SIGNALS INPUT FROM EACH SENSOR, CONTROLS THE CURRENT FLOWING TO THE SOLENOID INSIDE THE ACTUATOR AND LETS THE HYDRAULIC PRESSURE ACTING ON EACH WHEEL CYLINDER ESCAPE TO THE RESERVOIR. THE PUMP INSIDE THE ACTUATOR IS ALSO OPERATING AT THIS TIME AND IT RETURNS THE BRAKE FLUID FROM THE RESERVOIR TO THE MASTER CYLINDER. THUS PREVENTING LOCKING OF THE VEHICLE WHEELS.

IF THE ECU JUDGES THAT THE HYDRAULIC PRESSURE ACTING ON THE WHEEL CYLINDER IS INSUFFICIENT, THE CURRENT ACTING ON THE SOLENOID IS CONTROLLED AND THE HYDRAULIC PRESSURE IS INCREASED. HOLDING OF THE HYDRAULIC PRESSURE IS ALSO CONTROLLED BY THE ECU, BY THE SAME METHOD AS ABOVE. BY REPEATED PRESSURE. REDUCTION, HOLDING AND INCREASE ARE REPLATED TO MAINTAIN VEHICLE STABILITY AND TO IMPROVE STEERABILITY DURING SUDDEN BRAKING.

SERVICE HINTS

A37(A)(C) A38(B)(D) ABS ECU

(CONNECT THE ECU CONNECTOR)

- (B)15, (C)6-GROUND : APPROX. 12 VOLTS WITH THE IGNITION SW AT **ON** POSITION AND CHECK CONNECTOR TS-E1 CONNECTED
- (A) 5, (C)17-GROUND : APPROX. 12 VOLTS WITH THE IGNITION SW AT **ON** POSITION AND CHECK CONNECTOR TS-E1 CONNECTED
- (A)13, (D) 6-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (A)11, (D) 5-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (A)14, (C)12-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (A)18, (D) 8-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (A)26, (D)12-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (B)13, (C)10-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (A) 2, (C) 2-GROUND : ALWAYS CONTINUITY
- (A)15, (C)12-GROUND : ALWAYS CONTINUITY
- (A)12, (C)11-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION
- (B) 6, (C) 7-GROUND : APPROX. 12 VOLTS WITH BRAKE PEDAL DEPRESSED
- (A)25, (C)22-GROUND : ALWAYS APPROX. 12 VOLTS
- (B)14, (C)16-GROUND : APPROX. 12 VOLTS WITH ENGINE RUNNING AND PARKING BRAKE LEVER RELEASED

(DISCONNECT THE ECU CONNECTOR)

- (A) 1-(A)18, (C) 1-(D) 8 : APPROX. 6 Ω
- (A)13-(A)18, (D) 6-(D) 8 : APPROX. 6 Ω
- (A)14-(A)18, (C)12-(D) 8 : APPROX. 6 Ω
- (A)26-(A)18, (D)12-(D) 8 : APPROX. 6 Ω
- (A) 9-(A)22, (D) 3-(D)10 : APPROX. 0.8-1.3 Ω
- (A)16-(A) 3, (D) 1-(D) 7 : APPROX. 0.8-1.3 Ω
- (A)11-(A)24, (D) 5-(D) 4 : APPROX. 60-100 Ω
- (A)23-(A)24, (D)11-(D) 4 : APPROX. 50-80 Ω
- (B) 8-(B)16 : APPROX. 0.8-1.5 K Ω
- (C)19-(C)20 : APPROX. 1.1-1.7 K Ω

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
A 7	A	28 (3S-GTE), 30 (5S-FE)	A38	B	34 (L/B), 36 (CONVERTIBLE)
A 8	B	28 (3S-GTE), 30 (5S-FE)		D	34 (L/B), 36 (CONVERTIBLE)
A 9	A	28 (3S-GTE), 30 (5S-FE)	A39		34 (L/B), 36 (CONVERTIBLE)
A10	B	28 (3S-GTE), 30 (5S-FE)	A40		34 (L/B), 36 (CONVERTIBLE)
A11	A	28 (3S-GTE), 30 (5S-FE)	A43		34 (L/B), 36 (CONVERTIBLE)
A12	B	28 (3S-GTE), 30 (5S-FE)	B 2		28 (3S-GTE), 30 (5S-FE)
A13		28 (3S-GTE), 30 (5S-FE)	C 1		28 (3S-GTE), 30 (5S-FE)
A14		28 (3S-GTE), 30 (5S-FE)	C13	B	32
A37	A	34 (L/B), 36 (CONVERTIBLE)	C14	A	32
	C	34 (L/B), 36 (CONVERTIBLE)	F10	A	28 (3S-GTE)

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1B	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
1F	20	FLOOR WIRE AND J/B NO. 1 (LEFT KICK PANEL)
3B	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
ID1	44	ENGINE ROOM MAIN WIRE AND FLOOR WIRE (LEFT KICK PANEL)
IF1	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
IG2	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)

▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
EA	38 (3S-GTE)	FRONT RIGHT FENDER
	40 (5S-FE)	
EB	38 (3S-GTE)	FRONT LEFT FENDER
	40 (5S-FE)	
ID	44	LEFT KICK PANEL
BI	48 (L/B)	UNDER THE LEFT CENTER PILLAR
	50 (C/P)	
	52 (CONVERTIBLE)	

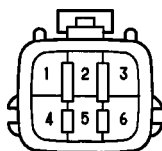
○ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E12	38 (3S-GTE)	ENGINE ROOM MAIN WIRE	B 1	50 (C/P)	FLOOR WIRE
	40 (5S-FE)			52 (CONVERTIBLE)	
E13	38 (3S-GTE)		B 7	48 (L/B)	
	40 (5S-FE)		B15	48 (L/B)	
I 1	46	COWL WIRE	B18	50 (C/P)	
I 2				52 (CONVERTIBLE)	
B1	48 (L/B)	FLOOR WIRE			

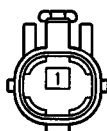
A 7 (A) GRAY



A 8 (B) GRAY



A 9 (A) DARK GRAY



A10 (B) DARK GRAY



A11 (A) GRAY



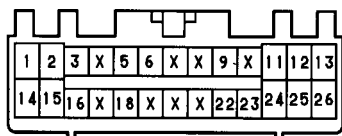
A12 (B) GRAY



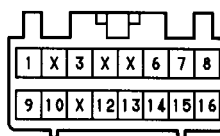
A13, A14 GRAY



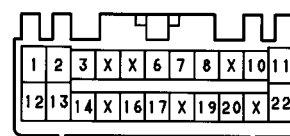
A37 (A) DARK GRAY



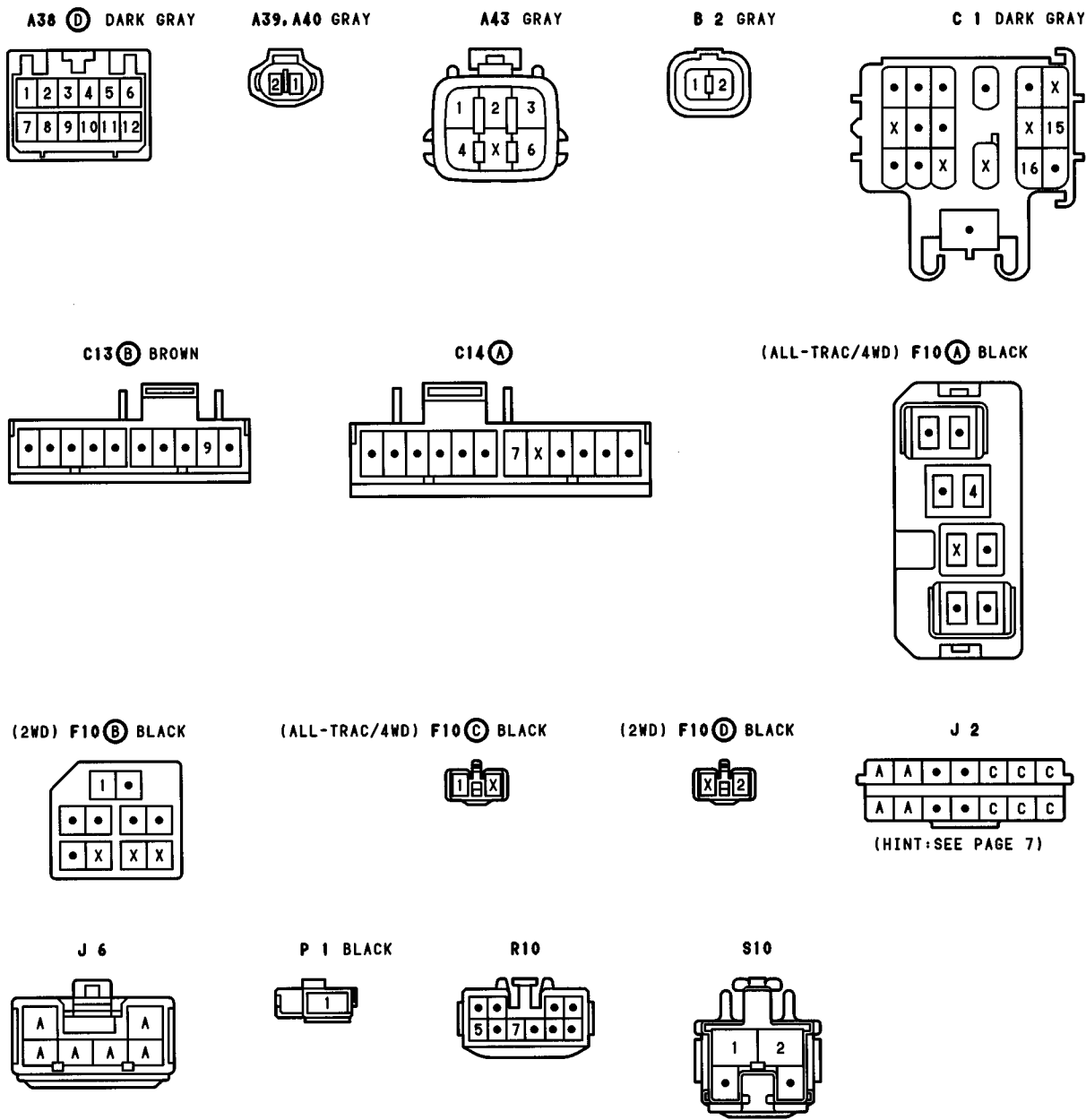
A38 (B) DARK GRAY



A37 (C) DARK GRAY



ABS (ANTI-LOCK BRAKE SYSTEM)



NOTICE: When inspecting or repairing the SRS AIRBAG, perform the operation in accordance with the following precautionary instructions and the procedure and precautions in the Repair Manual for the applicable model year.

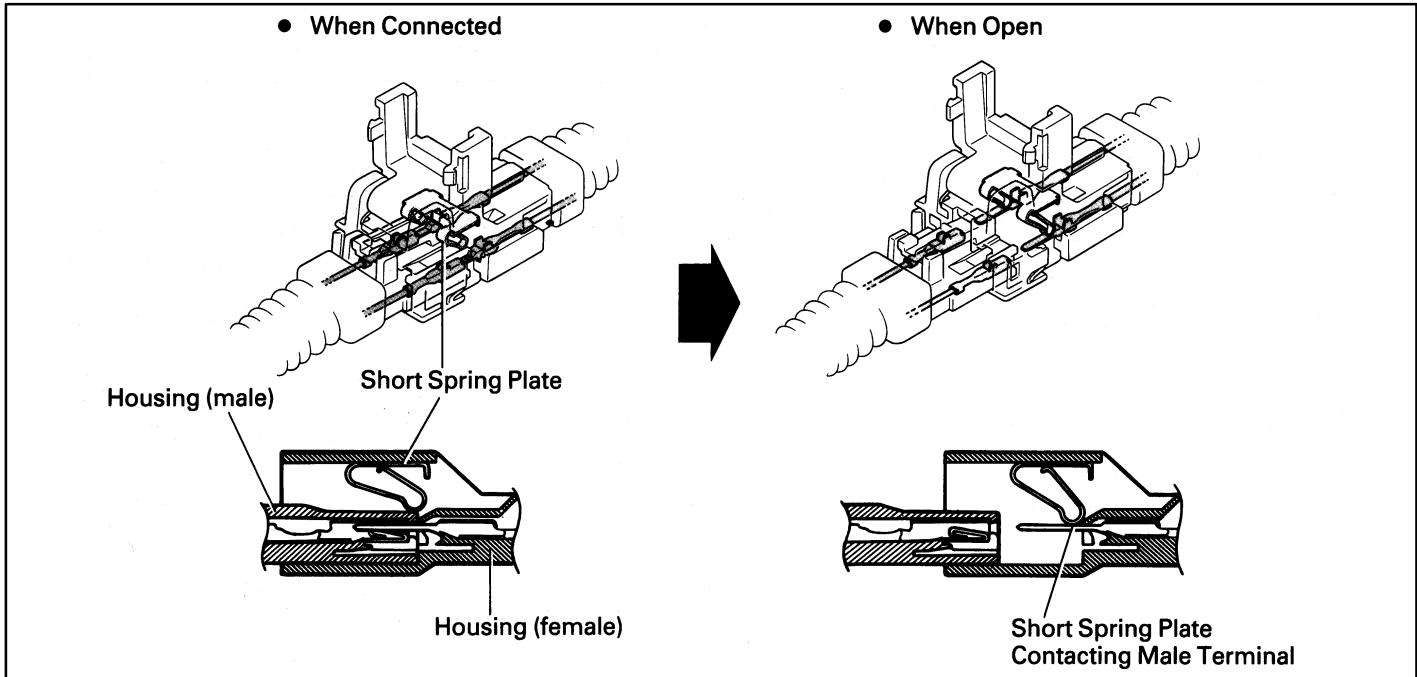
- ▼ Malfunction symptoms of the airbag system are difficult to confirm, so the diagnostic trouble codes become the most important source of information when troubleshooting.
When troubleshooting the airbag system, always inspect the diagnostic trouble codes before disconnecting the battery.
- ▼ Work must be started after 20 seconds or longer from the time the Ignition SW is set to the “LOCK” position and the negative (–) terminal cable is disconnected from the battery.
(The airbag system is equipped with a back-up power source so that if work is started within 20 seconds of disconnecting the negative (–) terminal cable of the battery, the airbag may be deployed.)
When the negative (–) terminal cable is disconnected from the battery, memory of the clock and audio systems will be cancelled. So before starting work, make a record of the contents memorized by each memory system. Then when work is finished, reset the clock and audio system as before.
To avoid erasing the memory of each memory system, never use a back-up power supply from outside the vehicle.
- ▼ When removing the steering wheel pad or handling a new steering wheel pad, keep the pad upper surface facing upward. Also, lock the lock lever of the twin lock type connector at the rear of the pad and take care not to damage the connector.
(Storing the pad with its metallic surface up may lead to a serious accident if the airbag inflates for some reason.)
- ▼ Store the steering wheel pad where the ambient temperature remains below 93°C (200°F), without high humidity and away from electrical noise.
- ▼ Never use airbag parts from another vehicle. When replacing airbag parts, replace them with new parts.
- ▼ Never disassemble and repair the steering wheel pad, center airbag sensor assembly or front airbag sensors.
- ▼ Before repairing the body, remove the airbag sensors if during repair shocks are likely to be applied to the sensors due to vibration of the body or direct tapping with tools or other parts.
- ▼ Do not reuse a steering wheel pad or front airbag sensors.
After evaluating whether the center airbag sensor assembly is damaged or not, decide whether or not to reuse it. (See the Repair Manual for the method for evaluating the center airbag sensor assembly.)
- ▼ When troubleshooting the supplemental restraint system, use a high-impedance (Min. 10k(/V) tester.
- ▼ The wire harness of the airbag system is combined with the cowl wire harness assembly.
The vehicle wiring harness exclusively for the airbag system is distinguished by corrugated yellow tubing, as are the connectors.
- ▼ Do not measure the resistance of the airbag squib.
(It is possible this will deploy the airbag and is very dangerous.)
- ▼ If the wire harness used in the supplemental restraint system is damaged, replace the whole wire harness assembly.
When the connector to the airbag front sensors can be repaired alone (when there is no damage to the wire harness), use the repair wire specially designed for the purpose.
(Refer to the Repair Manual for the applicable Model year for details of the replacement method.)
- ▼ INFORMATION LABELS (NOTICES) are attached to the periphery of the airbag components. Follow the instructions on the notices.

SRS AIRBAG

The supplemental restraint system has connectors which possess the functions described below:

1. AIRBAG ACTIVATION PREVENTION MECHANISM

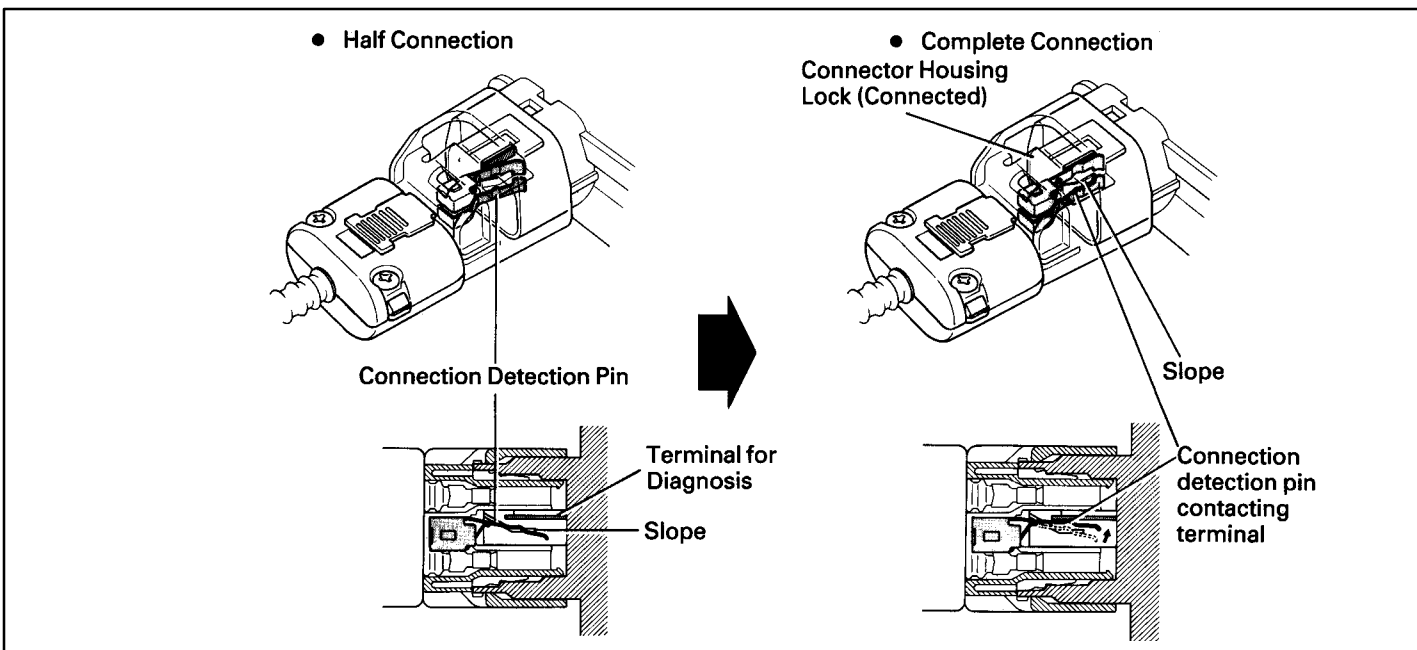
Each connector contains a short spring plate. When the connector is disconnected, the short spring plate automatically connects the power source and grounding terminals of the squib to preclude a potential difference between the terminals.



2. ELECTRICAL CONNECTION CHECK MECHANISM

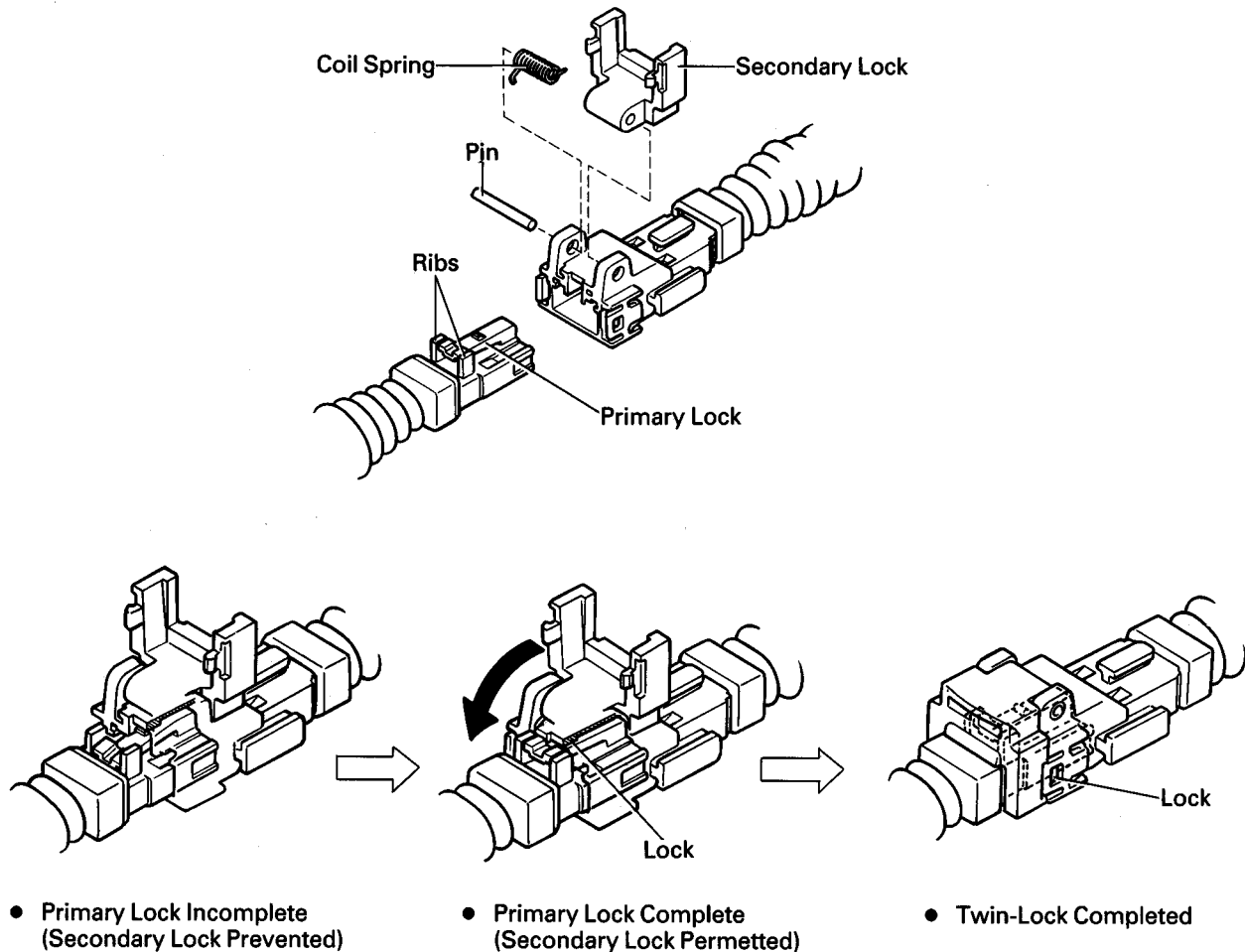
This mechanism is designed to electrically check if connectors are connected correctly and completely.

The electrical connection check mechanism is designed so that the connection detection pin connects with the diagnosis terminals when the connector housing lock is in the locked condition.

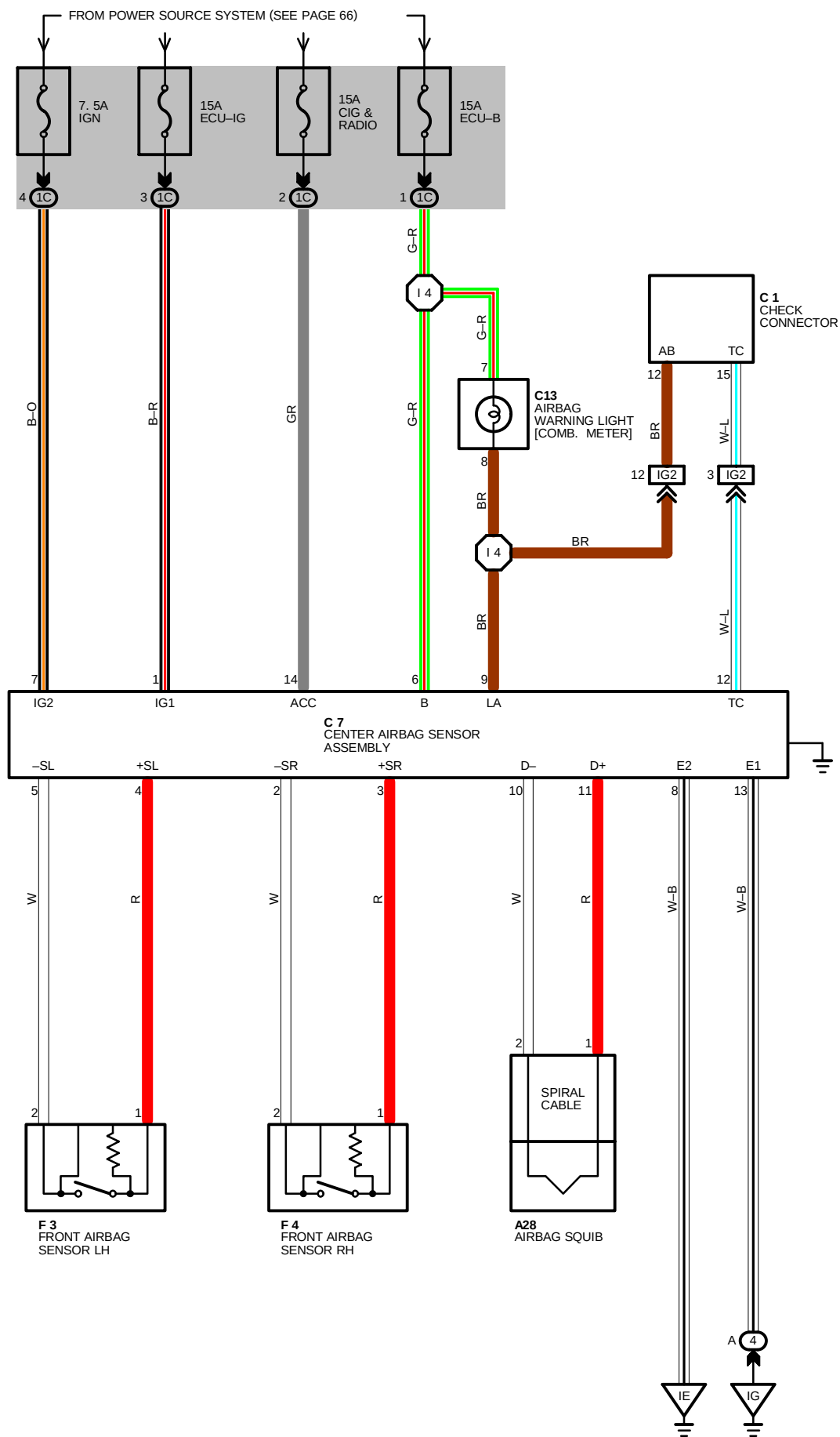


3. CONNECTOR TWIN-LOCK MECHANISM

With this mechanism connectors (male and female connectors) are locked by two locking devices to increase connection reliability. If the primary lock is incomplete, ribs interfere and prevent the secondary lock.



SRS AIRBAG



SYSTEM OUTLINE

THE SRS (SUPPLEMENTAL RESTRAINT SYSTEM) AIRBAG IS A DRIVER PROTECTION DEVICE WHICH HAS A SUPPLEMENTAL ROLE TO THE SEAT BELTS.

CURRENT FLOWS CONSTANTLY TO **TERMINAL 6** OF THE CENTER AIRBAG SENSOR ASSEMBLY. WHEN THE IGNITION SW IS TURNED TO ACC OR ON, CURRENT FROM THE CIG & RADIO FUSE FLOW TO **TERMINAL 14** OF THE CENTER AIRBAG SENSOR ASSEMBLY. ONLY WHEN THE IGNITION SW IS ON DOES THE CURRENT FROM THE ECU-IG FUSE FLOW TO **TERMINAL 1**. AND THE CURRENT FROM THE IGN FUSE TO **TERMINAL 7**.

IF AN ACCIDENT OCCURS WHILE DRIVING, DECELERATION CAUSED BY A FRONTAL IMPACT IS DETECTED BY EACH SENSOR AND SWITCH, AND WHEN THE FRONTAL IMPACT EXCEEDS A SET LEVEL (WHEN THE MERCURY SW BUILT INTO THE CENTER AIRBAG SENSOR ASSEMBLY IS ON AND THE CENTER AIRBAG SENSOR IS ON, FRONT AIRBAG SENSORS ARE OFF), CURRENT FROM THE CIG & RADIO, ECU-IG OR IGN FUSE FLOWS TO **TERMINAL 11** OF THE CENTER AIRBAG SENSOR ASSEMBLY → **TERMINAL 1** OF THE AIRBAG SQUIB → SQUIB → **TERMINAL 2** → **TERMINAL 10** OF CENTER AIRBAG SENSOR ASSEMBLY → **TERMINAL 8, TERMINAL 13** OR **BODY GROUND** → **GROUND**.

WHEN THE MERCURY SW BUILT INTO THE CENTER AIRBAG SENSOR ASSEMBLY IS ON AND THE FRONT AIRBAG SENSOR LH OR RH IS ON, CENTER AIRBAG SENSOR IS OFF CURRENT FROM THE CIG & RADIO, ECU-IG OR IGN FUSE FLOWS TO **TERMINAL 11** OF THE CENTER AIRBAG SENSOR ASSEMBLY → **TERMINAL 1** OF THE AIRBAG SQUIB → SQUIB → **TERMINAL 2** → **TERMINAL 10**, OF THE CENTER AIRBAG SENSOR ASSEMBLY → **TERMINAL 3** OR **4** → **TERMINAL 1** OF FRONT AIRBAG SENSOR → **TERMINAL 2** → **TERMINAL 2** OR **5** OF CENTER AIRBAG SENSOR ASSEMBLY → **TERMINAL 8, TERMINAL 13** OR **BODY GROUND** → **GROUND**.

WHEN THE MERCURY SW BUILT INTO THE CENTER AIRBAG SENSOR ASSEMBLY IS ON, AND THE FRONT AIRBAG SENSOR LH OR RH IS ON AND CENTER AIRBAG SENSOR IS ON ONE OF THE ABOVE-MENTIONED CIRCUITS IS ACTIVATED SO THAT CURRENT FLOWS TO THE AIRBAG SQUIB AND CAUSES IT TO OPERATE. THE BAG STORED INSIDE THE STEERING WHEEL PAD IS INSTANTANEOUSLY EXPANDED TO SOFTEN THE SHOCK TO THE DRIVER.

THE REASON WHY THERE ARE MULTIPLE POWER SOURCES AND GROUND POINTS IS SO THAT IN THE EVENT THAT ONE OR TWO OF THE POWER SOURCES AND GROUND POINTS DO NOT WORK FOR SOME REASON, THE REMAINING POWER SOURCE AND GROUND POINT WILL BE AVAILABLE TO COMPENSATE.

: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
A28	32	C 7	32	F 3	31 (4A-FE)
C 1	28 (3S-GTE), 30 (5S-FE)	C13	32	F 4	28 (3S-GTE), 30 (5S-FE)
	31 (4A-FE)	F 3	28 (3S-GTE), 30 (5S-FE)		31 (4A-FE)

: RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
4	27	R/B NO. 4 (RIGHT KICK PANEL)

: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1C	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)

: CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IG2	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)

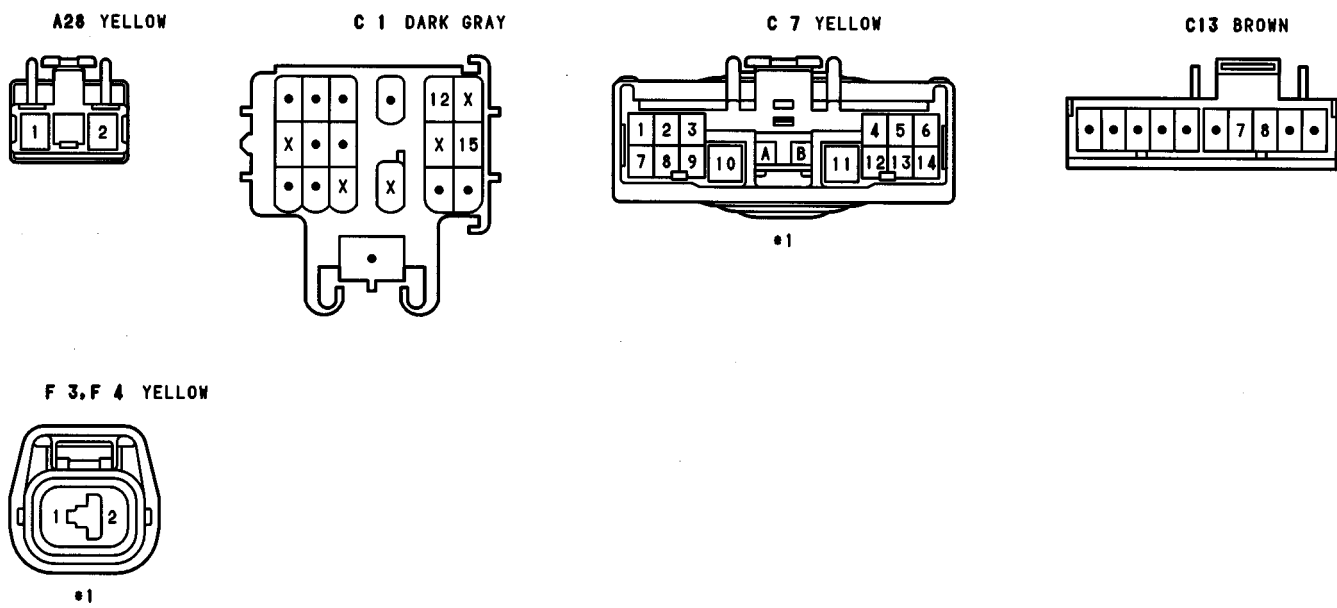
: GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
IE	44	INSTRUMENT PANEL BRACE LH
IG	44	R/B NO. 4 SET BOLT

: SPLICE POINTS

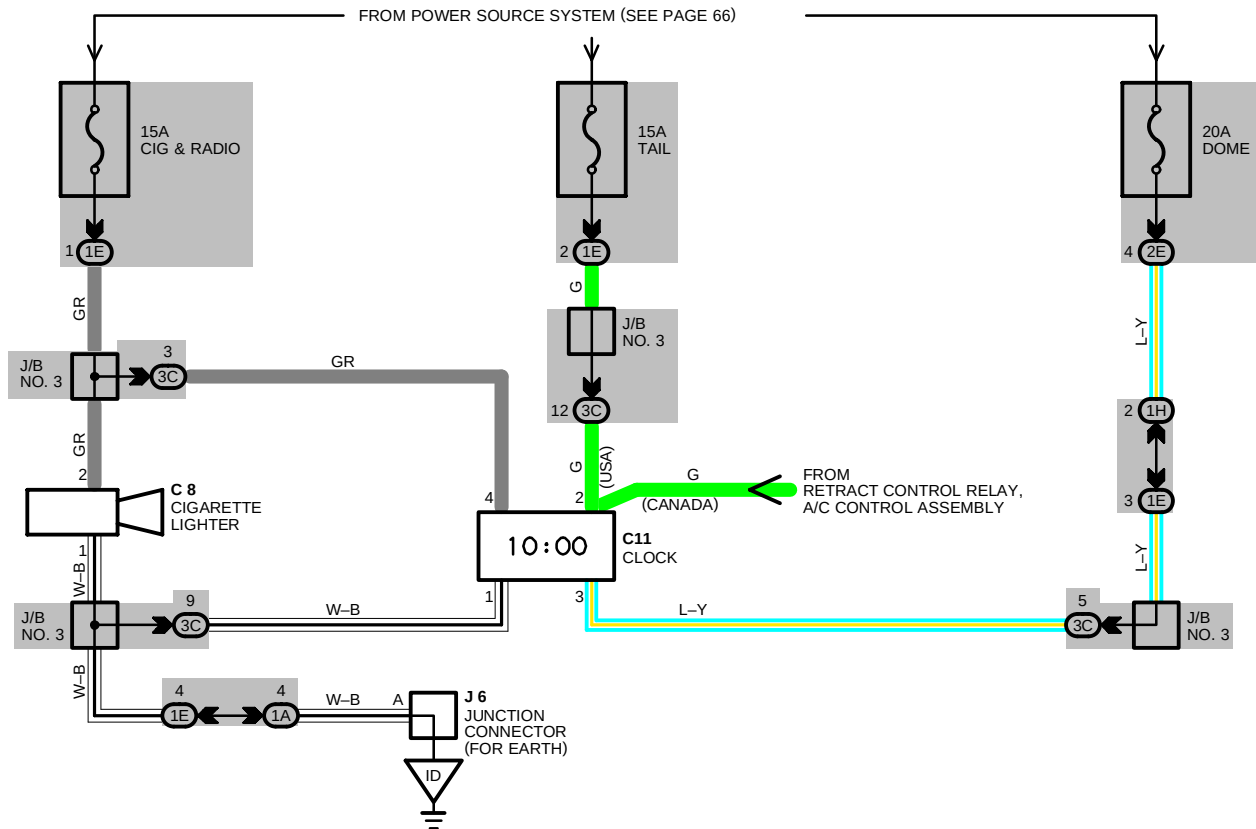
CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I4	46	COWL WIRE			

SRS AIRBAG



•1 :WITH ELECTRICAL CONNECTION CHECK MECHANISM (SEE PAGE 197)

CIGARETTE LIGHTER AND CLOCK



SERVICE HINTS

C 8 CIGARETTE LIGHTER

- 2-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ACC** OR **ON** POSITION
- 1-GROUND : ALWAYS CONTINUITY

C11 CLOCK

- 3-GROUND : ALWAYS APPROX. 12 VOLTS (POWER FOR CLOCK)
- 4-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ACC** OR **ON** POSITION (POWER FOR INDICATION)
- 2-GROUND : APPROX. 12 VOLTS WITH LIGHT CONTROL SW AT **TAIL** OR **HEAD** POSITION
- APPROX. 12 VOLTS WITH ENGINE RUNNING AND PARKING BRAKE LEVER RELEASED (CANADA)
- 1-GROUND : ALWAYS CONTINUITY

○ : PARTS LOCATION

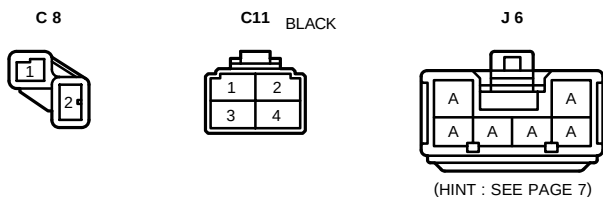
CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C 8	32	C11	32	J 6	33

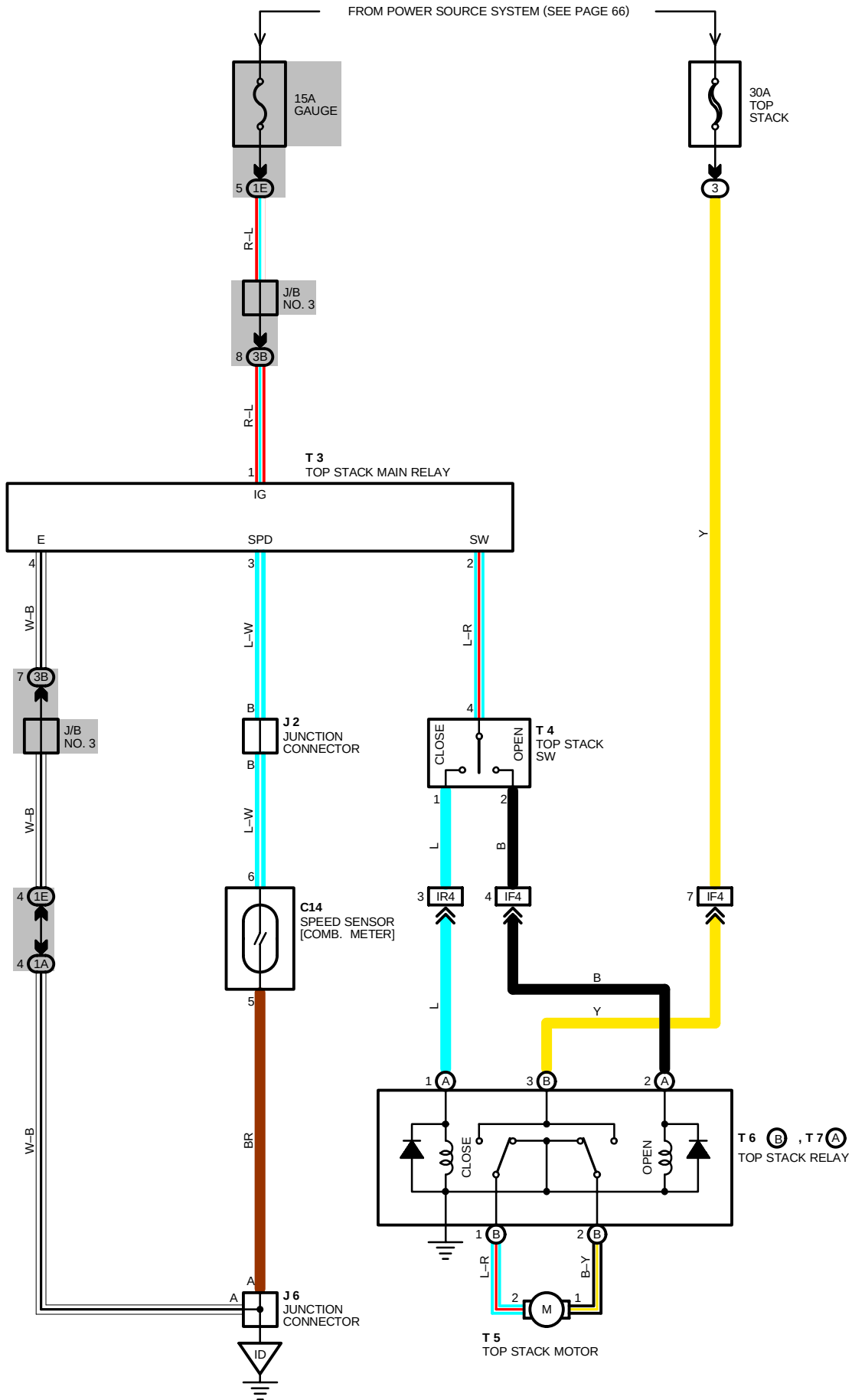
○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2E	22 (2WD) 23 (ALL-TRAC/4WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
3C	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
ID	44	LEFT KICK PANEL





SERVICE HINTS

T 3 TOP STACK MAIN RELAY

- 1-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ACC** OR **ON** POSITION
- 3-GROUND : 4 PULSE WITH 1 ROTATION OF ROTOR SHAFT
- 4-GROUND : ALWAYS CONTINUITY

T 4 TOP STACK SW

- 4-2 : CLOSED WITH TOP STACK SW AT **OPEN** POSITION
- 4-1 : CLOSED WITH TOP STACK SW AT **CLOSE** POSITION

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
C14	32	T 3	33	T 6	B 36
J 2	32	T 4	33	T 7	A 36
J 6	32	T 5	36		

○ : RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
3	26	R/B NO. 3 (RIGHT KICK PANEL)

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

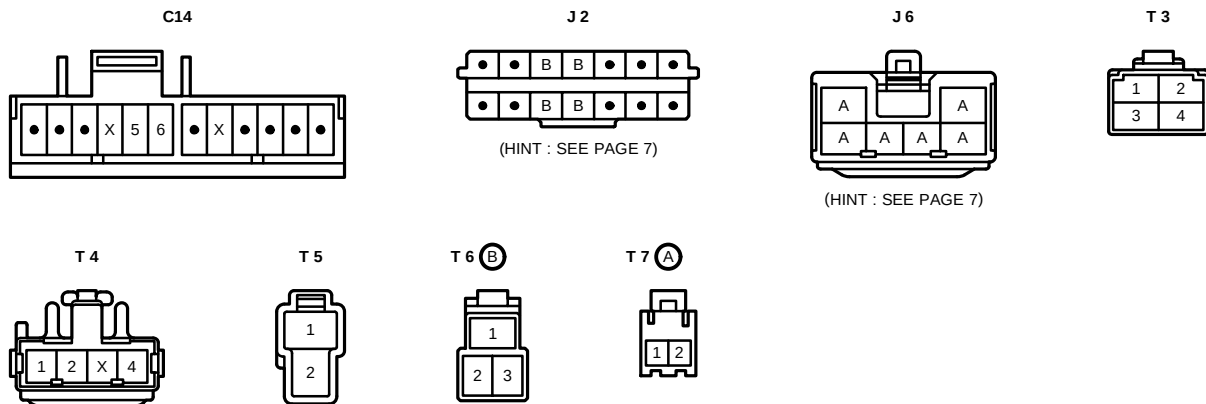
CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
3B	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

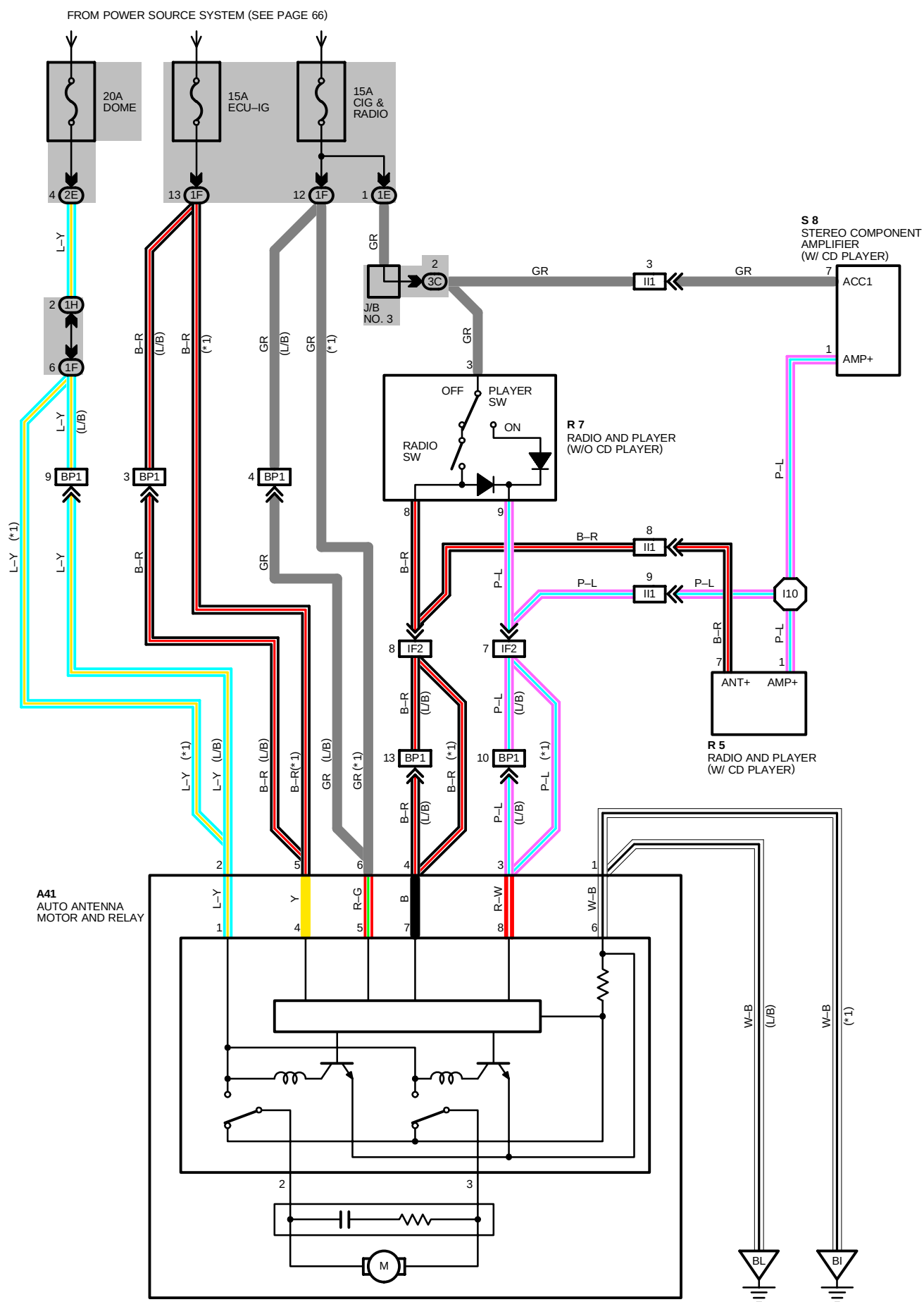
□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IF4	46	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)

▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
ID	46	LEFT KICK PANEL





SERVICE HINTS

A41(B) AUTO ANTENNA CONTROL RELAY

- 9-GROUND : APPROX. **12** VOLTS WITH IGNITION SW AT **ON** POSITION
- 5-GROUND : APPROX. **12** VOLTS WITH IGNITION SW AT **ACC** OR **ON** POSITION
- 7-GROUND : ALWAYS APPROX. **12** VOLTS
- 8-GROUND : APPROX. **12** VOLTS WITH IGNITION SW AT **ACC** OR **ON** POSITION AND RADIO SW ON
- 3-GROUND : CONTINUITY (UPPER LIMIT SW ON) UNLESS ANTENNA AT **UP** STOP
- 2-GROUND : CONTINUITY (DOWN LIMIT SW ON) UNLESS ANTENNA AT **DOWN** STOP
- 4-3 : CLOSED WITH IGNITION SW AT **ACC** OR **ON** POSITION AND RADIO SW ON AND PLAYER SW OFF UNTIL ANTENNA AT **UPPERMOST** POSITION
- 1-2 : CLOSED WITH IGNITION SW AT **ACC** OR **ON** POSITION AND RADIO SW OFF AND PLAYER SW OFF UNTIL ANTENNA AT **LOWERMOST** POSITION
- 1-2 : CLOSED WITH IGNITION SW OFF UNTIL ANTENNA AT **LOWERMOST** POSITION

○ : PARTS LOCATION

CODE		SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
A41	A	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)	R 5	33	S 8	33
	B	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)	R 7	33		

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1E	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1F	20	FLOOR WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2E	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
	23(ALL-TRAC/4WD)	
3C	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

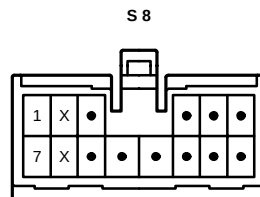
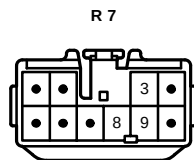
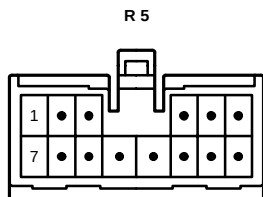
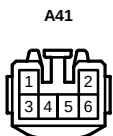
CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IF2	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
II1	46	COWL WIRE AND CONSOLE BOX WIRE (INSTRUMENT PANEL CENTER)
BP1	48 (L/B)	FLOOR WIRE AND LUGGAGE ROOM WIRE (LEFT QUARTER PANEL CENTER)

▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
BI	50 (C/P)	UNDER THE LEFT CENTER PILLAR
	52 (CONVERTIBLE)	
BL	48 (L/B)	BACK PANEL CENTER

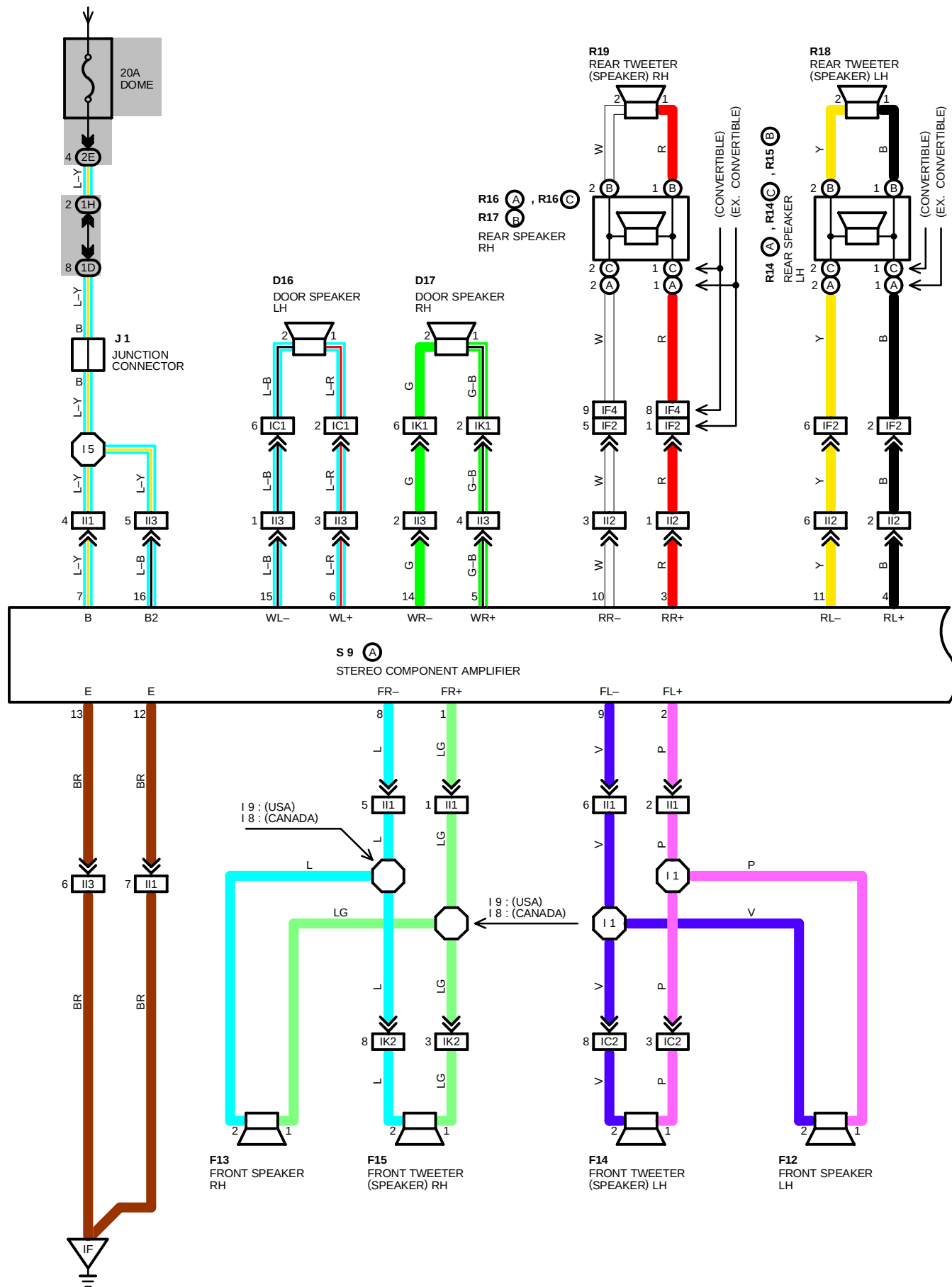
○ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I10	46	CONSOLE BOX WIRE (2WD)			

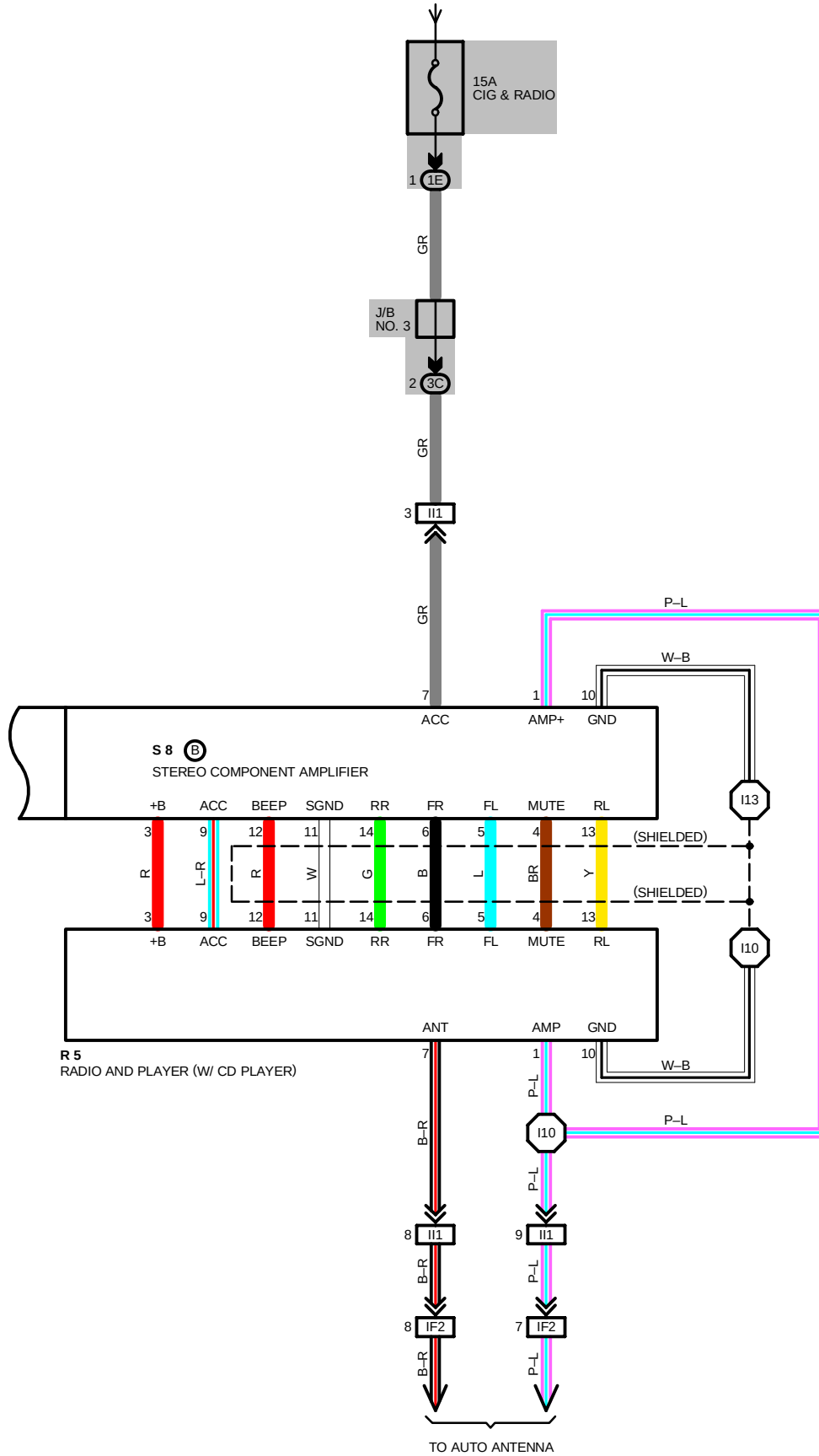


RADIO AND PLAYER (w/ CD PLAYER)

FROM POWER SOURCE SYSTEM (SEE PAGE 66)



FROM POWER SOURCE SYSTEM (SEE PAGE 66)



RADIO AND PLAYER (w/ CD PLAYER)

SERVICE HINTS

S 8 (B), S 9 (A) STEREO COMPONENT AMPLIFIER
(A) 7, (A)16-GROUND : ALWAYS APPROX. 12 VOLTS
(B) 7-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT ON OR ACC POSITION
(A)12, (A)13-GROUND : ALWAYS CONTINUITY

⊖ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
D16	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)	J 1	33	R16	C 36 (CONVERTIBLE)
D17	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)	R 5	33	R17	B 34 (L/B), 35 (C/P)
F12	33	R14	A 34 (L/B), 35 (C/P)	R18	34 (L/B), 35 (C/P)
F13	33	R14	C 36 (CONVERTIBLE)	R19	34 (L/B), 35 (C/P)
F14	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)	R15	B 34 (L/B), 35 (C/P)	S 8	B 33
F15	34 (L/B), 35 (C/P), 36 (CONVERTIBLE)	R16	A 34 (L/B), 35 (C/P)	S 9	A 33

⊖ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1D	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO.1 (LEFT KICK PANEL)
2E	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO.2 (NEAR THE BATTERY)
	23 (ALL-TRAC/4WD)	
3C	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

⊖ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

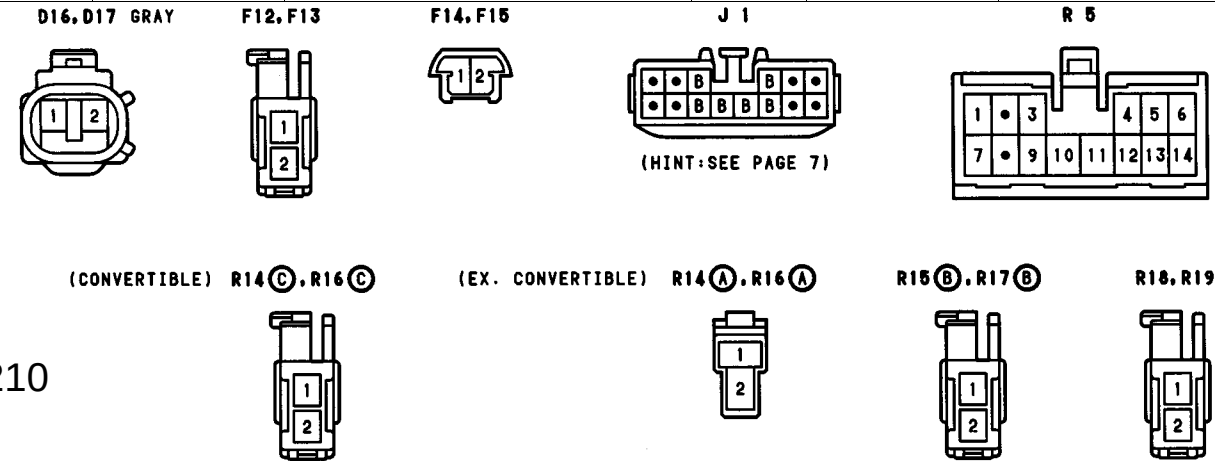
CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IC1	44	FRONT DOOR LH WIRE AND COWL WIRE (LEFT KICK PANEL)
IC2		
IF2	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
IF4	46	
II1	46	COWL WIRE AND CONSOLE BOX WIRE (INSTRUMENT PANEL CENTER)
II2		
II3	46	COWL WIRE AND CONSOLE BOX WIRE (RIGHT KICK PANEL OF ALL-TRAC/4WD)
	46	COWL WIRE AND CONSOLE BOX WIRE (RIGHT KICK PANEL OF 2WD)
IK1	46	FRONT DOOR RH WIRE AND COWL WIRE (RIGHT KICK PANEL)
IK2		

▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
IF	44	INSTRUMENT PANEL BRACE RH

⊖ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I 1	46	COWL WIRE	I 9	46	COWL WIRE
I 5			I10	46	CONSOLE BOX WIRE (2WD)
I 8			I13	46	CONSOLE BOX WIRE



S 8 ⑧

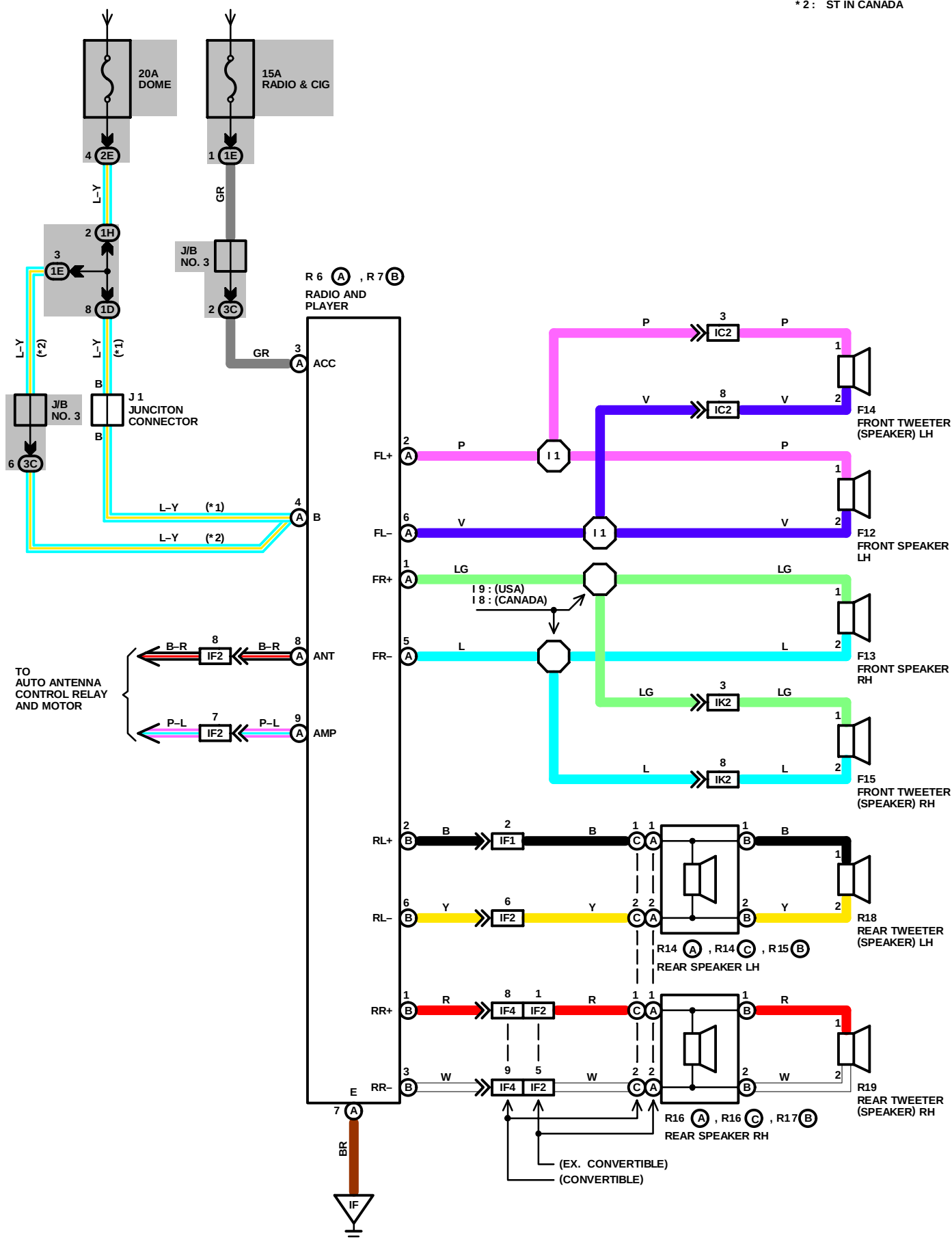
1	X	3			4	5	6
7	X	9	10	11	12	13	14

S 9 ⑨

1	2	3	4	5	6	7
8	9	10	11	12	13	14
				15		16

212

* 1: EX. ST IN CANADA
* 2: ST IN CANADA



SERVICE HINTS

R 6(A), R 7(B) RADIO AND PLAYER

(A) 4-GROUND : ALWAYS APPROX. 12 VOLTS

(A) 3-GROUND : APPROX. **12** VOLTS WITH IGNITION SW AT **ON** OR **ACC** POSITION

(A) 7-GROUND : ALWAYS CONTINUITY

○ : PARTS LOCATION

CODE	SEE PAGE	CODE		SEE PAGE	CODE		SEE PAGE
F12	33	R 6	A	33	R16	A	34 (L/B), 35 (C/P)
F13	33	R 7	B	33	R16	C	36 (CONVERTIBLE)
F14	34 (L/B), 35 (C/P), 36 (CONVERT.)	R14	A	34 (L/B), 35 (C/P)	R17	B	34 (L/B), 35 (C/P)
F15	34 (L/B), 35 (C/P), 36 (CONVERT.)	R14	C	36 (CONVERTIBLE)	R18		34 (L/B), 35 (C/P)
J 1	33	R15	B	34 (L/B), 35 (C/P)	R19		34 (L/B), 35 (C/P)

 : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1D	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2E	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
	23 (ALL-TRAC/4WD)	
3C	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

 : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

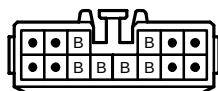
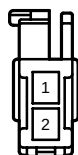
CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IC2	44	FRONT DOOR LH WIRE AND COWL WIRE (LEFT KICK PANEL)
IF2	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
IF4	46	
IK2	46	FRONT DOOR RH WIRE AND COWL WIRE (RIGHT KICK PANEL)

 : GROUND POINTS

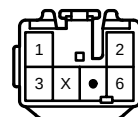
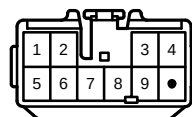
CODE	SEE PAGE	GROUND POINTS LOCATION
IF	44	INSTRUMENT PANEL BRACE RH

 : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I 1	46	COWL WIRE	I 9	46	COWL WIRE
I 8					



(HINT : SEE PAGE 7)



(EX. CONVERTIBLE) R14 , R16



(CONVERTIBLE) R14 ©, R16 ©



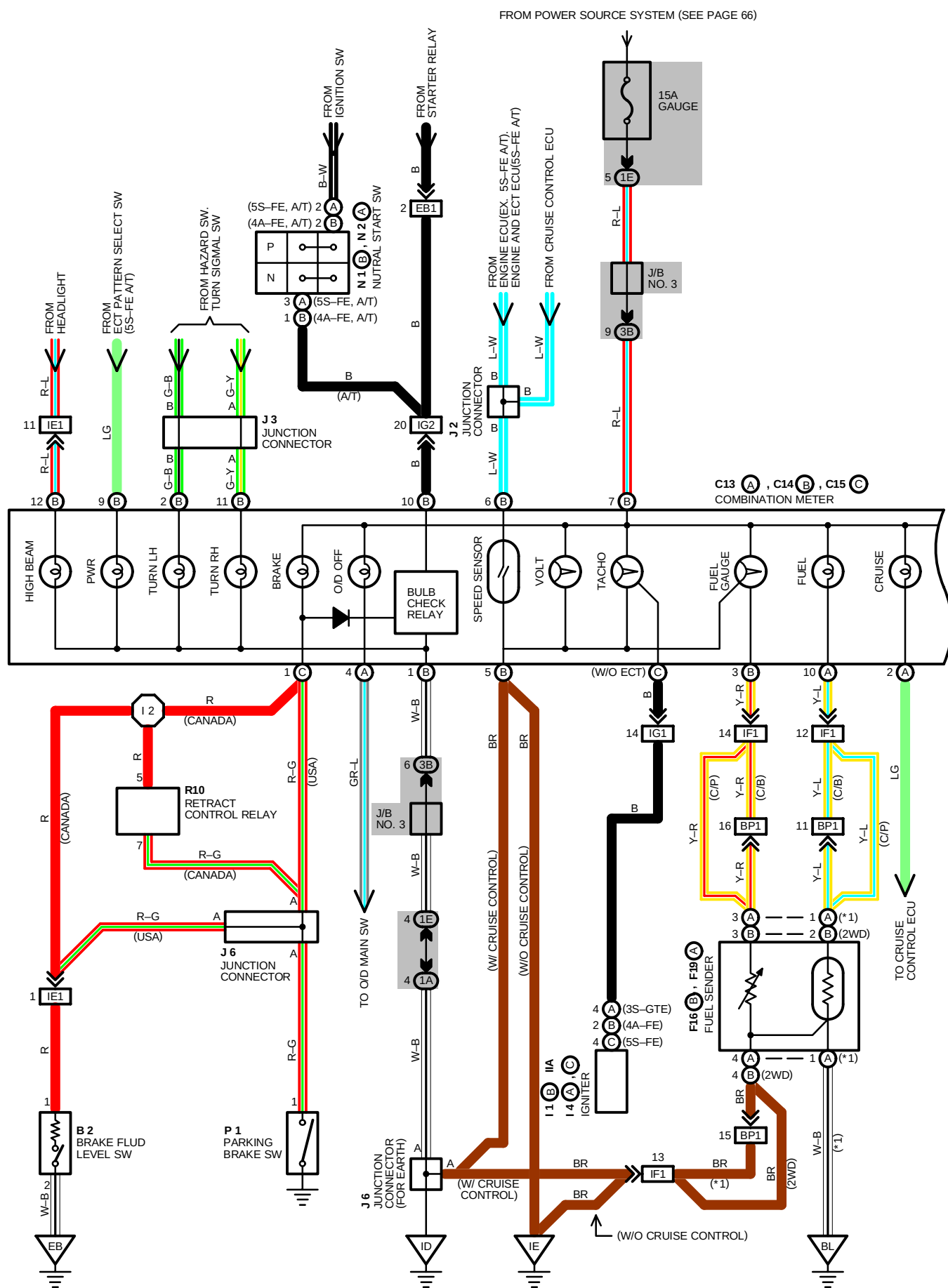
R15 (B) , R17 (B)



R18, R19



COMBINATION METER





COMBINATION METER

SERVICE HINTS

B 2 BRAKE FLUID LEVEL SW

1-2 : CLOSED WITH FLOAT DOWN

C13(A), C14(B), C15(C) COMBINATION METER

(A) 7, (C) 3-GROUND : ALWAYS APPROX. 12 VOLTS

(B) 7, (C) 9-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT ON POSITION

(A) 5, (B) 1, (B) 5-GROUND : ALWAYS CONTINUITY

F16, F19 FUEL SENDER

3-4 : APPROX. 3 (WITH FUEL FULL

APPROX. 110.0 (WITH FUEL EMPTY

O 2 OIL PRESSURE SW

1-GROUND : CLOSED WITH OIL PRESSURE BELOW 0.2 KG/CM² (2.84 SPI, 19.61 KPA)

P 1 PARKING BRAKE SW

1-GROUND : CLOSED WITH PARKING BRAKE LEVER PULLED UP

W 3 WATER TEMP. SENDER

1-GROUND : APPROX. 226 (AT 50°C (122°F)

APPROX. 26.4 (AT 115°C (239°F)



: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
B 2	28 (3S-GTE), 30 (5S-FE)	I 4	C	30 (5S-FE)	P 1
	31 (4S-FE)	J 1		33	P 9
C13	A	32	J 2	33	R
C14	B	32	J 3	33	0
C15	C	32	J 6	33	S15
F16	B	34 (L/B), 35 (C/P)	N 1	B	A
F19	A	34 (L/B)	N 2	A	S16
I 1	B	31 (4A-FE)	O 2	29 (3S-GTE), 30 (5S-FE)	T 3
I 4	A	29 (3S-GTE)		31 (4A-FE)	W 3



: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1C		
1D		
1E		
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2E	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
	21 (ALL-TRAC/4WD)	
3B	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)



: CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EB1	38 (3S-GTE)	ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER)
	40 (5S-FE)	
	42 (4A-FE)	
ES2	38 (3S-GTE)	ENGINE WIRE AND ENGINE ROOM NO. 2 WIRE (NEAR THE DISTRIBUTOR)
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)
IF1	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
IG1	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)
IG2		
BP1	48 (L/B)	FLOOR WIRE AND LUGGAGE ROOM WIRE (LEFT QUARTER PANEL CENTER)

 : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
EB	38 (3S-GTE)	FRONT LEFT FENDER
	40 (5S-FE)	
	42 (4A-FE)	
EC	38 (3S-GTE)	INTAKE MANIFOLD
	40 (5S-FE)	
	42 (4A-FE)	
ID	44	LEFT KICK PANEL
IE	44	INSTRUMENT PANEL BRACE LH
BL	48 (L/B)	BACK PANEL CENTER
	50 (C/P)	
	52 (CONVERTIBLE)	

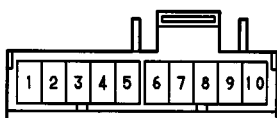
 : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
I 2	46	COWL WIRE	I 6	46	COWL WIRE
I 4					

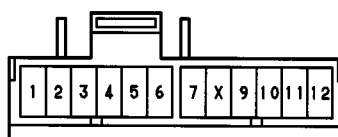
B 2 GRAY



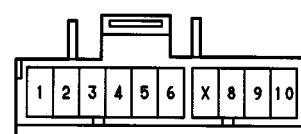
C13 (A) BROWN



C14 (B)



C15 (C) GRAY



F16 DARK GRAY



F19



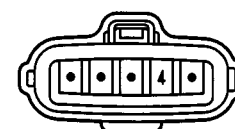
(4A-FE) I 1 (B) DARK GREEN



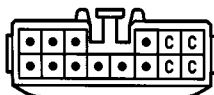
(3S-GT) I 4 (A) BLACK



(5S-FE) I 4 (C) BLACK

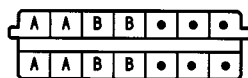


J 1



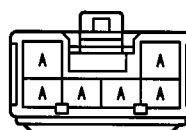
(HINT:SEE PAGE 7)

J 2, J 3



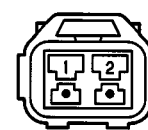
(HINT:SEE PAGE 7)

J 6

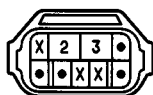


(HINT:SEE PAGE 7)

(2WD W/O ECT) N 1 (B) GRAY



(2WD W/ ECT) N 2 (A) GRAY



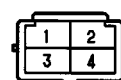
O 2 BLACK



P 1 BLACK



P 9



R10



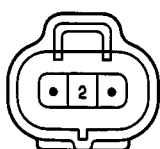
S15 (A) GRAY



S16 (B) GRAY



T 3 BLACK

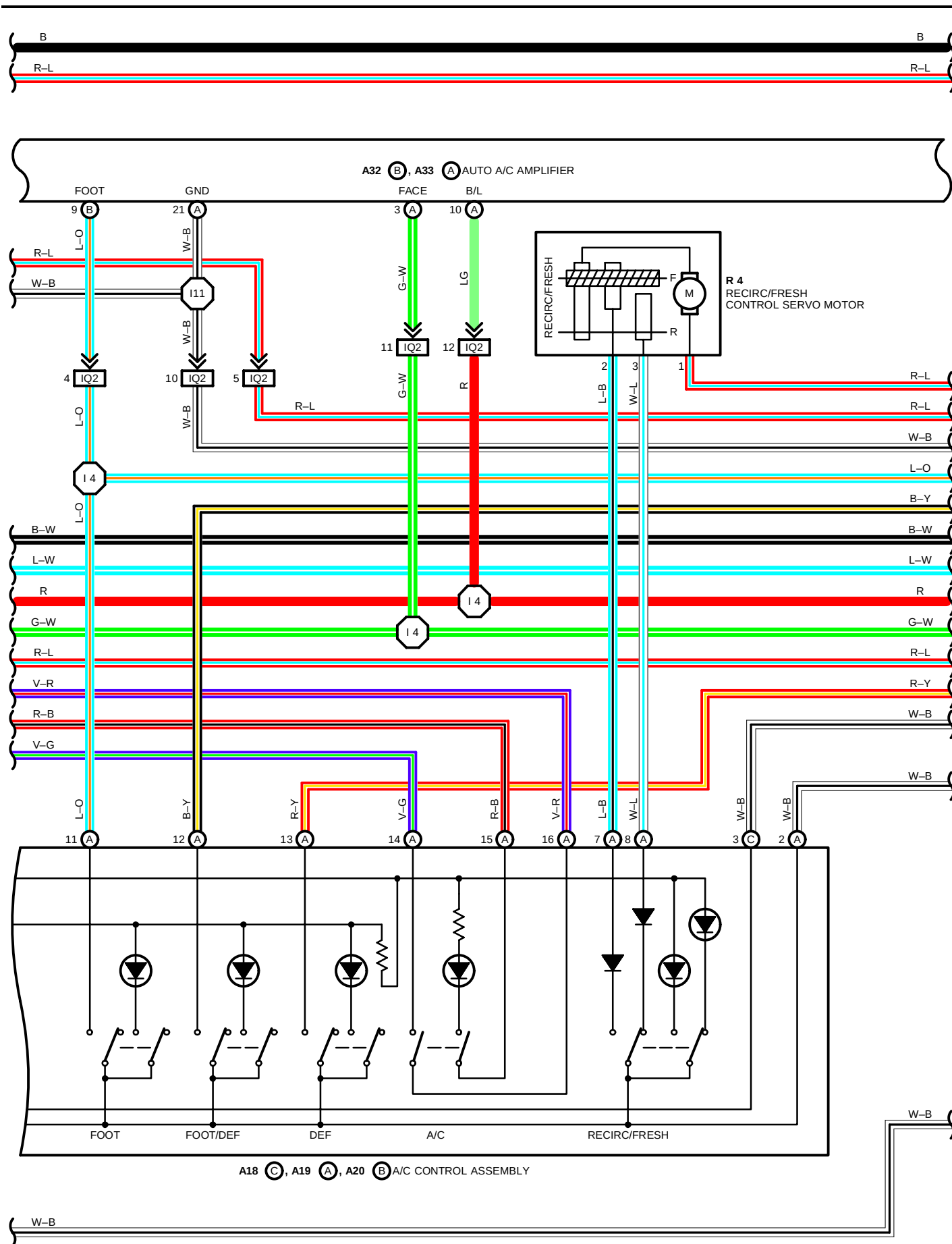


W 3 GRAY

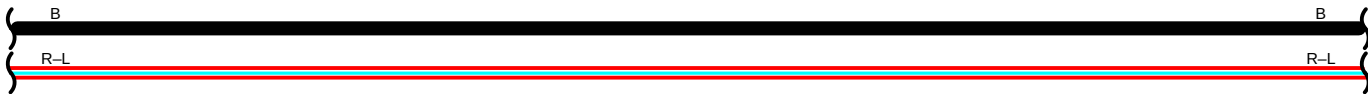




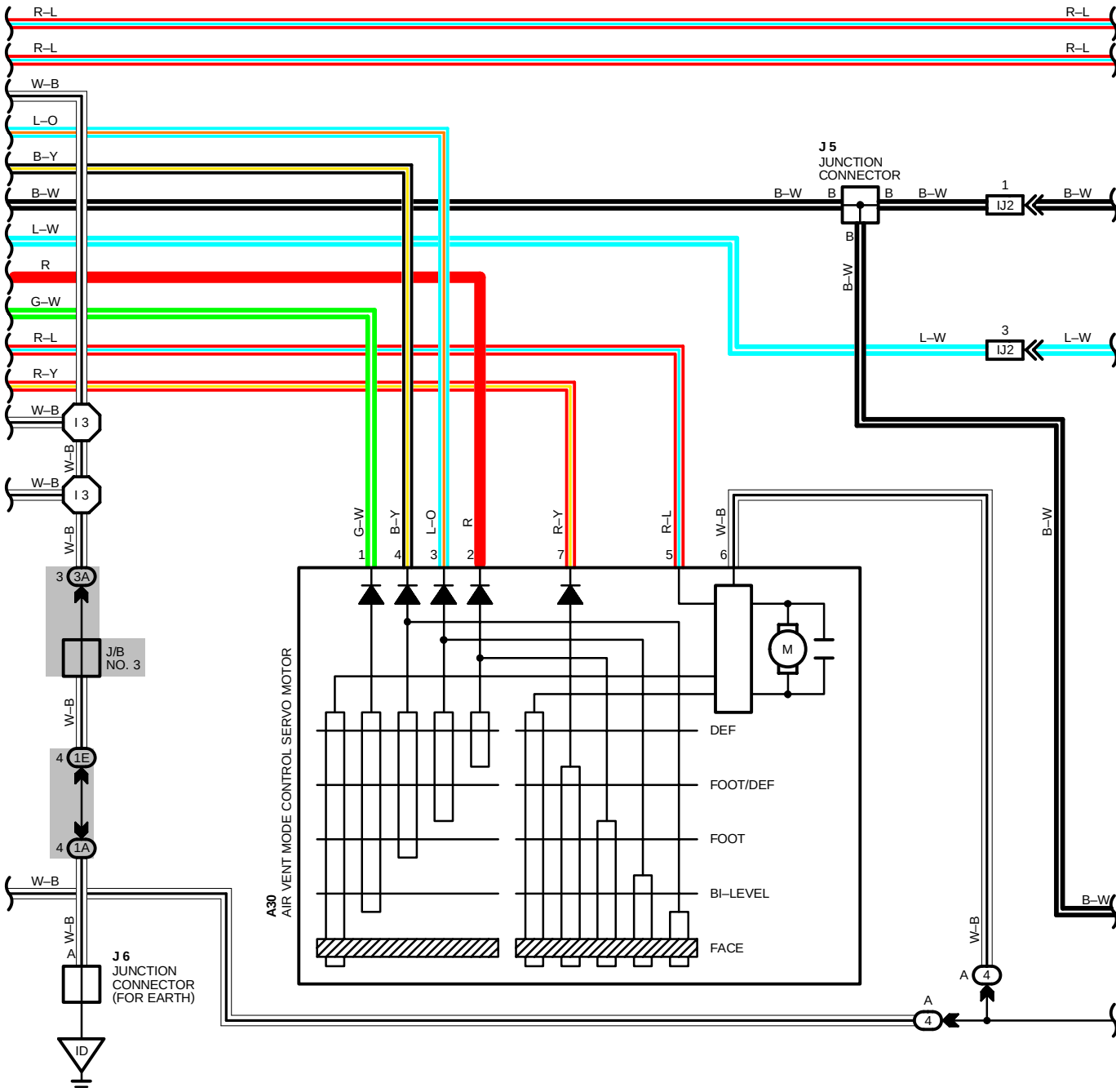


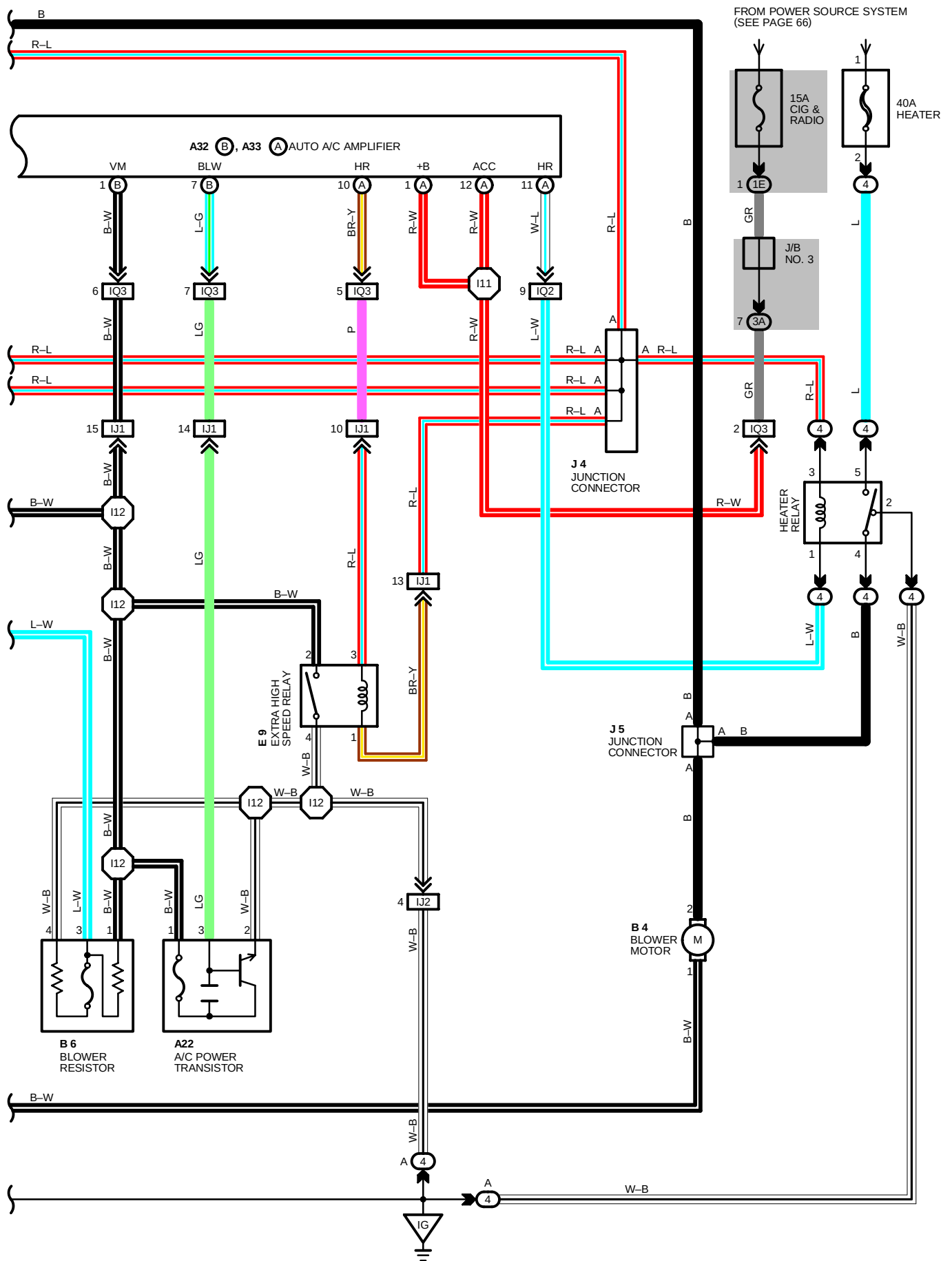


RADIATOR FAN AND AIR CONDITIONER (AUTOMATIC AIR CONDITIONER)



A32 (B), A33 (A) AUTO A/C AMPLIFIER





RADIATOR FAN AND AIR CONDITIONER (AUTOMATIC AIR CONDITIONER)

SYSTEM OUTLINE

1. COOLING FAN OPERATION

WHEN THE IGNITION SW IS TURNED ON, THE CURRENT FROM IGN FUSE FLOWS TO **TERMINAL 1** OF RADIATOR FAN RELAY NO. 1 → **TERMINAL 2** → **TERMINAL 3** OF THE A/C PRESSURE SW → **TERMINAL 2** → **TERMINAL 2** OF WATER TEMP. SW (FOR FANS CONTROL) → **TERMINAL 1** → **GROUND**, FROM **TERMINAL 1** OF A/C FAN RELAY NO. 2 → **TERMINAL 2** → **TERMINAL 3** OF A/C PRESSURE SW → **TERMINAL 2** → **TERMINAL 2** OF WATER TEMP. SW (FOR FANS CONTROL) → **TERMINAL 1** → **GROUND**, CAUSING RELAY NO. 1 AND RELAY NO. 2 OF EACH FAN TO TURN ON.

* OPERATION AT LOW SPEED

WHEN THE A/C SW (A/C CONTROL ASSEMBLY) IS TURNED ON AND THE AIR CONDITIONER OPERATES, THE CURRENT FLOWS FROM GAUGE FUSE FLOWS TO **TERMINAL 1** OF A/C MAGNETIC CLUTCH RELAY → **TERMINAL 2** → **TERMINAL 8** OF A/C AMPLIFIER CAUSING A/C MAGNETIC CLUTCH RELAY TO TURN ON.

AT THAT TIME, THE CURRENT FROM GAUGE FUSE FLOWS TO **TERMINAL 5** OF A/C MAGNETIC CLUTCH RELAY → **TERMINAL 3** → **TERMINAL 4** OF A/C MAGNETIC CLUTCH → **GROUND**, AND FROM **TERMINAL 3** OF A/C MAGNETIC CLUTCH RELAY → **TERMINAL 1** OF A/C FAN RELAY NO. 3 → **TERMINAL 2** → **GROUND**.

AS A RESULT, A/C MAGNETIC CLUTCH AND A/C FAN RELAY NO. 3 TURNS ON AND THE CURRENT FLOWS FROM CDS FAN FUSE → **TERMINAL 2** OF A/C CONDENSER FAN MOTOR → **TERMINAL 1** → **TERMINAL 5** OF A/C FAN RELAY NO. 3 → **TERMINAL 3** → **TERMINAL 3** OF A/C FAN RELAY NO. 2 → **TERMINAL 5** → **TERMINAL 2** OF RADIATOR FAN MOTOR → **TERMINAL 1** → **GROUND**, FLOWING TO EACH FAN MOTOR IN SERIES, CAUSING THE COOLING FAN TO ROTATE AT LOW SPEED.

* OPERATION AT HIGH SPEED

DURING A/C OPERATION, WHEN THE PRESSURE OF A/C COMPRESSOR BECOMES HIGHER THAN NORMAL PRESSURE (MORE THAN 14.3 KG/CM² 1401 KPA, 203 PSI), THE A/C PRESSURE SW TURNS OFF.

AS A RESULT, FAN RELAY NO. 1 AND NO. 2 TURNS OFF AND THE CURRENT FLOWS FROM RDI FAN FUSE → **TERMINAL 4** OF RADIATOR FAN RELAY NO. 1 → **TERMINAL 3** → **TERMINAL 2** OF RADIATOR FAN MOTOR → **TERMINAL 1** → **GROUND**, AND FROM CDS FAN FUSE → **TERMINAL 2** OF A/C CONDENSER FAN MOTOR → **TERMINAL 1** → **TERMINAL 5** OF A/C FAN RELAY NO. 3 → **TERMINAL 3** → **TERMINAL 1** OF A/C FAN RELAY NO. 2 → **TERMINAL 3** → **GROUND**, FLOWING TO EACH FAN MOTOR IN PARALLEL CAUSING THE COOLING FAN TO ROTATE AT HIGH SPEED. WHEN THE ENGINE COOLANT TEMPERATURE BECOMES MORE THAN ABOUT 90° (194°F), THE WATER TEMP. SW TURNS OFF AND THE SAME OPERATION AS ABOVE IS PERFORMED.

2. HEATER BLOWER MOTOR OPERATION (DIAL TYPE BLOWER CONTROL SW (W/ AUTO A/C))

CURRENT IS APPLIED AT ALL TIMES THROUGH THE HEATER FUSE TO **TERMINAL 5** OF HEATER RELAY. WHEN THE IGNITION SW IS TURNED ON, CURRENT FLOWS THROUGH GAUGE FUSE TO **TERMINAL 3** OF HEATER RELAY → **TERMINAL 1** → **TERMINAL HR** OF AUTO A/C AMPLIFIER. AT THE SAME TIME, CURRENT ALSO FLOWS FROM GAUGE FUSE TO **TERMINAL IG** OF AUTO A/C AMPLIFIER, AND **TERMINAL (A) 1** OF A/C CONTROL ASSEMBLY.

* LOW SPEED OPERATION (OPERATION AT MANUAL)

CURRENT TO **TERMINAL 3** OF HEATER RELAY FLOWING TO **TERMINAL 1** OF HEATER RELAY → **TERMINAL HR** → **TERMINAL BSW** → **TERMINAL (C) 2** OF A/C CONTROL ASSEMBLY → **TERMINAL (C) 3** → **GROUND** AND TURNS THE HEATER RELAY ON. THIS CAUSES THE CURRENT TO **TERMINAL 5** OF THE HEATER RELAY TO FLOW TO **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → MOTOR → **TERMINAL 1** → **TERMINAL 1** OF BLOWER RESISTOR → **TERMINAL 4** → **GROUND**, CAUSING THE BLOWER MOTOR TO ROTATE AT LOW SPEED.

* HIGH SPEED OPERATION (OPERATION AT MANUAL)

CURRENT TO **TERMINAL 3** OF HEATER RELAY FLOWS TO **TERMINAL 1** → **TERMINAL HR** OF THE AUTO A/C AMPLIFIER → **TERMINAL BSW** → **TERMINAL (C) 2** OF A/C CONTROL ASSEMBLY → **TERMINAL (C) 3** → **GROUND**, TURNING THE HEATER RELAY ON.

AT THE SAME TIME, CURRENT FLOW TO **TERMINAL 5** OF HEATER RELAY → **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → **TERMINAL 1** → **TERMINAL (C) 5** OF A/C CONTROL ASSEMBLY → **TERMINAL (C) 3** → **GROUND**, CAUSING THE BLOWER MOTOR TO ROTATE AT HIGH SPEED.

* MEDIUM SPEED OPERATION (OPERATION AT MANUAL M1, M2)

THE CURRENT TO **TERMINAL 3** OF HEATER RELAY → **TERMINAL 1** → **TERMINAL HR** OF THE A/C AMPLIFIER → **TERMINAL BSW** → **TERMINAL (C) 2** OF A/C CONTROL ASSEMBLY → **TERMINAL (C) 3** → **GROUND**, AND TURNS THE HEATER RELAY ON.

AS A RESULT, THE CURRENT TO **TERMINAL 5** OF HEATER RELAY → **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → **TERMINAL 1** → **TERMINAL 1** OF BLOWER RESISTOR → **TERMINAL 3** → **TERMINAL (C) 4** OF A/C CONTROL ASSEMBLY → **TERMINAL (C) 3** → **GROUND**, AND CAUSES THE BLOWER MOTOR TO ROTATE AT MEDIUM SPEED.

* AUTO FUNCTION

WHEN THE AUTO SW (A/C CONTROL ASSEMBLY) IS TURNED ON, SIGNALS ARE INPUT TO **TERMINAL BSW** AND **TERMINAL BAUTO** OF AUTO A/C AMPLIFIER CONTROLLING THE CURRENT FLOW FROM **TERMINAL BLW** OF THE AUTO A/C AMPLIFIER TO **TERMINAL 3** OF POWER TRANSISTOR → **TERMINAL 2** → **GROUND**, THE AUTO A/C AMPLIFIER CONTROLS THE BLOWER MOTOR STEPLESSLY.

3. OPERATION OF RECIRC/FRESH CONTROL SERVO MOTOR

(SWITCHING FROM FRESH TO RECIRC)

WITH IGNITION SW TURNED ON, THE CURRENT FLOWS FROM GAUGE FUSE TO **TERMINAL 1** OF AIR INLET CONTROL SERVO MOTOR. WHEN THE RECIRC/FRESH SW IS SWITCHED TO THE RECIRC SIDE, THE CURRENT FLOWS FROM **TERMINAL 1** OF AIR INLET CONTROL SERVO MOTOR → **TERMINAL 2** → **TERMINAL (A) 7** OF A/C CONTROL ASSEMBLY → **TERMINAL (A) 2** → **GROUND**. THE MOTOR ROTATES AND THE DAMPER MOVES TO THE RECIRC SIDE.

WHEN IT IS IN THE **RECIRC** POSITION, THE CURRENT IS CUT INSIDE THE SERVO MOTOR AND THE DAMPER STOPS AT THAT POSITION.

(SWITCHING FROM RECIRC TO FRESH)

WITH IGNITION SW ON, WHEN THE RECIRC/FRESH SW IS SWITCHED TO THE FRESH SIDE, THE CURRENT FLOWS FROM **TERMINAL 1** OF AIR INLET CONTROL SERVO MOTOR → **TERMINAL 3** → **TERMINAL (A) 8** OF A/C CONTROL ASSEMBLY → **TERMINAL (A) 2** → **GROUND**, THE MOTOR ROTATES AND THE DAMPER MOVES TO THE FRESH SIDE. WHEN IT IS IN THE **FRESH** POSITION, THE CURRENT IS CUT INSIDE THE SERVO MOTOR AND THE DAMPER STOPS AT THAT POSITION.

4. OPERATION OF AIR VENT MODE CONTROL SERVO MOTOR

WITH IGNITION SW TURNED ON, THE CURRENT FLOWS FROM GAUGE FUSE TO **TERMINAL 5** OF AIR VENT MODE CONTROL SERVO MOTOR → **TERMINAL 6** → **GROUND**, AND THE DAMPER MOVES TO THE POSITION OF THE MODE SELECTION SW OF THE CONTROL ASSEMBLY SWITCH IS ON.

WHEN THE MODE SELECTION SW OF A/C CONTROL ASSEMBLY IS MOVED TO **DEF** POSITION FROM THE DAMPER IN THE **FACE** POSITION, THE CURRENT FLOWS FROM **TERMINAL 7** OF AIR VENT MODE CONTROL SERVO MOTOR TO **TERMINAL (A) 13** OF A/C CONTROL ASSEMBLY → **TERMINAL (A) 2** → **GROUND**.

AS A RESULT, THE SERVO MOTOR OPERATES UNTIL THE DAMPER REACHES **DEF** POSITION.

WHEN THIS OCCURS THE CURRENT TO THE A/C CONTROL ASSEMBLY IS SHUT OFF AND ROTATION OF THE MOTOR STOPS. SWITCHING TO OTHER MODES IS CONTROLLED BY THE SERVO MOTOR ACCORDING THE FLOWING CURRENT:

1. **FOOT/DEF** POSITION: THE CURRENT FLOWS FROM **TERMINAL 4** OF SERVO MOTOR TO **TERMINAL (A) 12** OF A/C CONTROL ASSEMBLY.
2. **FOOT** POSITION: THE CURRENT FLOWS FROM **TERMINAL 3** OF SERVO MOTOR TO **TERMINAL (A) 11** OF A/C CONTROL ASSEMBLY.
3. **BI-LEVEL** POSITION: THE CURRENT FLOWS FROM **TERMINAL 2** OF SERVO MOTOR TO **TERMINAL (A) 10** OF A/C CONTROL ASSEMBLY.

5. OPERATION OF AIR MIX CONTROL SERVO MOTOR

WHEN THE TEMPERATURE CONTROL VOLUME IS TURNED TO THE COOL SIDE, A SIGNAL IS INPUT TO **TERMINAL TEST** OF AUTO A/C AMPLIFIER FROM **TERMINAL (B) 1** OF A/C CONTROL ASSEMBLY.

AS A RESULT, A SIGNAL IS OUTPUT FROM **TERMINAL AMH** OF AUTO A/C AMPLIFIER TO **TERMINAL 3** OF A/C SYSTEM AMPLIFIER AND THE CURRENT FLOWING TO **TERMINAL 7** OF A/C SYSTEM AMPLIFIER FROM THE GAUGE FUSE FLOWS FROM **TERMINAL 1** OF A/C SYSTEM AMPLIFIER → **TERMINAL 2** OF AIR MIX CONTROL SERVO MOTOR → **TERMINAL 6** → IF /A/C SYSTEM AMPLIFIER → **TERMINAL 9** → **GROUND**. CAUSING THE AIR MIX CONTROL SERVO MOTOR TO ROTATE.

AT THIS TIME IS INPUT THE DAMPER OPENING ANGLE FROM **TERMINAL 4** OF SERVO MOTOR TO **TERMINAL TP** OF AUTO A/C AMPLIFIER THIS IS USED TO DETERMINE THE DAMPER **STOP** POSITION AND MAINTAIN THE SET TEMPERATURE.

WHEN THE TEMPERATURE CONTROL VOLUME IS TURNED TO THE WARM SIDE, IN A/C SYSTEM AMPLIFIER THE CURRENT FLOWS FROM SERVO MOTOR → **TERMINAL 6** OF A/C SYSTEM AMPLIFIER → **TERMINAL 6** OF AIR MIX CONTROL SERVO MOTOR → MOTOR → **TERMINAL 2** → **TERMINAL 1** OF A/C SYSTEM AMPLIFIER → **TERMINAL 9** → **GROUND**, ROTATING THE MOTOR IN REVERSE AND SWITCHING THE DAMPER FROM COOL TO WARM SIDE.

RADIATOR FAN AND AIR CONDITIONER (AUTOMATIC AIR CONDITIONER)

SYSTEM OUTLINE

6. AIR CONDITIONER OPERATION

THE A/C AMPLIFIER RECEIVES VARIOUS SIGNALS, THE ENGINE RPM SIGNAL FROM THE IGNITER, OUTLET TEMPERATURE SIGNAL FROM THE A/C THERMIST, COOLANT TEMPERATURE FROM THE WATER TEMP. SENSOR, AND THE LOCK SIGNAL FROM THE A/C COMPRESSOR, ETC.

WHEN THE ENGINE IS STARTED AND THE A/C SW (A/C CONTROL ASSEMBLY) IS TURNED ON, THE CURRENT FLOWS FROM A/C FUSE TO **TERMINAL 3** OF A/C AMPLIFIER.

AS A RESULT, THE CURRENT FLOWING FROM **TERMINAL 2** OF A/C MAGNETIC CLUTCH RELAY TO **TERMINAL 8** OF A/C AMPLIFIER FLOWS FROM **TERMINAL 15** OF A/C AMPLIFIER TO **GROUND** AND TURNS IN THE MAGNETIC CLUTCH RELAY.

BECAUSE THE MAGNETIC CLUTCH IS ON, THE A/C COMPRESSOR OPERATES, CAUSING THE CURRENT FLOWING FROM A/C IDLE-UP VSV TO **TERMINAL 6** OF A/C AMPLIFIER TO FLOW TO **TERMINAL 15** OF A/C AMPLIFIER → **GROUND**, AND TURNS ON THE VSV TO AVOID LOWERING THE ENGINE RPM DURING AIR CONDITIONER OPERATION.

WHEN ANY OF THE FOLLOWING SIGNALS ARE INPUT TO THE A/C AMPLIFIER, THE AMPLIFIER OPERATES TO TURN OFF THE AIR CONDITIONER:

- * ENGINE HIGH RPM SIGNAL
- * COOLANT TEMP. SIGNAL IS HIGH.
- * A SIGNAL THAT THE TEMPERATURE AT THE AIR OUTLET IS LOW.
- * A SIGNAL THAT THERE IS A LARGE DIFFERENCE BETWEEN ENGINE RPM AND COMPRESSOR RPM
- * A SIGNAL THAT THE REFRIGERANT PRESSURE IS ABNORMALLY HIGH OR LOW.

SERVICE HINTS

A 4 A/C MAGNETIC CLUTCH

4-GROUND : APPROX. **3.7 Ω**

A 5 A/C PRESSURE SW

3-2 : OPEN ABOVE APPROX. **13.5 KG/CM² (192 PSI, 1323 KPA)**

CLOSED BELOW APPROX. **10 KG/CM² (142 PSI, 980 KPA)**

1-4 : OPEN WITH PRESSURE LESS THAN **2.1 KG/CM² (30 PSI, 206 KPA)** OR ABOVE **27 KG/CM² (384 PSI, 2648 KPA)**

A17 A/C AMPLIFIER

8-15 : CONTINUITY WITH A/C SW (A/C CONTROL ASSEMBLY) ON AND IGNITION SW **ON** POSITION

14-15 : ALWAYS CONTINUITY

14-GROUND: ALWAYS CONTINUITY

15-GROUND: ALWAYS CONTINUITY

13-GROUND: APPROX. **12 VOLTS** WITH IGNITION SW ON

A23 A/C ROOM TEMP. SENSOR

1-2 : APPROX. **1.7 KΩ** AT **25°C (77°F)**

A26 A/C THERMISTOR

1-2, 3-4 : APPROX. **4852 Ω** AT **0°C (59°F)**

APPROX. **2341 Ω** AT **15°C (59°F)**

APPROX. **1500 Ω** AT **25°C (77°F)**

A29 AIR MIX CONTROL SERVO MOTOR

2-GROUND : APPROX. **12 VOLTS** WITH TEMPERATURE CONTROL VOLUME AT **WARM** TO **COOL** POSITION

6-GROUND : APPROX. **12 VOLTS** WITH TEMPERATURE CONTROL VOLUME AT **COOL** TO **WARM** POSITION

1-3 : ALWAYS APPROX. **6 KΩ**

A32(B), A34(A) AUTO A/C AMPLIFIER

B-GROUND : APPROX. **12 VOLTS** WITH IGNITION SW AT **ACC** OR **ON** POSITION

IG-GROUND : APPROX. **12 VOLTS** WITH IGNITION SW AT **ON** POSITION

HR-GROUND : CONTINUITY WITH BLOWER SW (A/C CONTROL ASSEMBLY) ON OR AUTO SW (A/C CONTROL ASSEMBLY) ON

S5-GROUND : APPROX. **5 VOLTS** WITH IGNITION SW ON

SG-GROUND : ALWAYS CONTINUITY

REC-GROUND: APPROX. **12 VOLTS** WITH A/C CONTROL ASSEMBLY AT **RECIRC** POSITION

FRS-GROUND : APPROX. **12 VOLTS** WITH A/C CONTROL ASSEMBLY AT **FRESH** POSITION

GND-GROUND: ALWAYS CONTINUITY

B 5 BLOWER RESISTOR

1-3 : APPROX. **0.84 Ω**

3-4 : APPROX. **1.48 Ω**

W 4 WATER TEMP. SW (FOR FANS CONTROL)

1-2 : OPEN ABOVE APPROX. **90°C (194°F)**

CLOSED BELOW APPROX. **83°C (181.4°F)**

 : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
A 1	28 (3S-GTE), 30 (5S-FE)	A24	32	C 2	A 28 (3S-GTE), 30 (5S-FE)
A 2	28 (3S-GTE), 30 (5S-FE)	A25	32	C 3	B 28 (3S-GTE), 30 (5S-FE)
A 4	28 (3S-GTE), 30 (5S-FE)	A26	32	E 9	33
A 5	28 (3S-GTE), 30 (5S-FE)	A27	32	J 4	33
A17	32	A29	32	J 5	33
A18	C 32	A30	32	J 6	33
A19	A 32	A32	B 32	R 1	28 (3S-GTE), 30 (5S-FE)
A20	B 32	A33	A 32	R 4	33
A22	32	B 4	32	W 4	28 (3S-GTE), 30 (5S-FE)
A23	32	B 6	32		

 : RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
4	27	R/B NO. 4 (RIGHT KICK PANEL)
5	27	R/B NO. 5 (ENGINE COMPARTMENT FRONT RIGHT)

 : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2A	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2C	22 (2WD)	ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2D	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2E		
3A	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

 : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EA1	38 (3S-GTE)	COWL WIRE AND ENGINE ROOM MAIN WIRE (FRONT SIDE OF RIGHT FRONT FENDER)
	40 (5S-FE)	
EB2	38 (3S-GTE)	ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER)
	40 (5S-FE)	
IG1	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)
IG2		
IH1	44	ENGINE WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)
IJ1	46	COWL WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)
IJ2		
IQ2	46	COWL WIRE AND A/C NO. 2 WIRE (BESIDE HEATER UNIT)
IQ3		

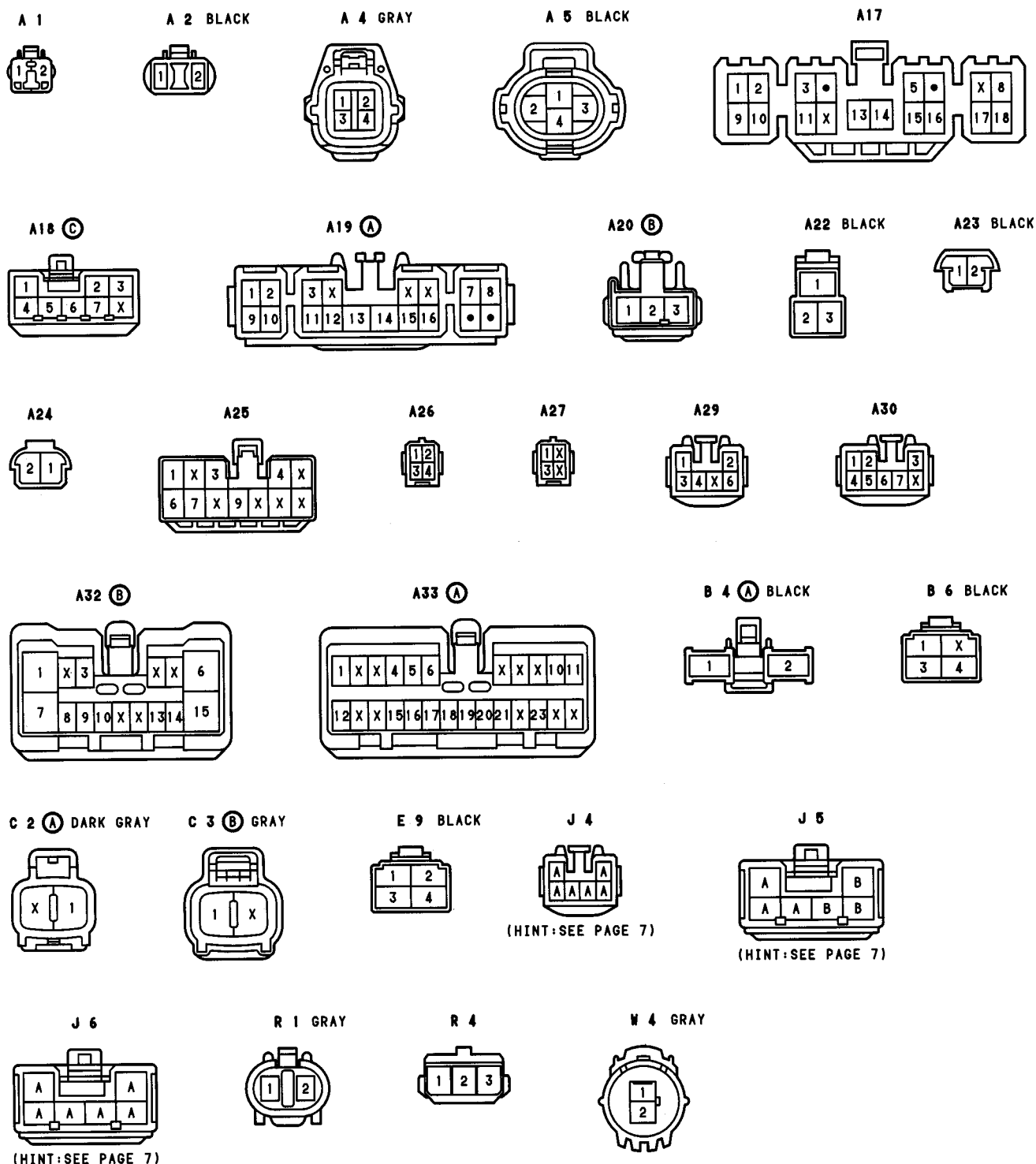
 : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
EA	40 (5S-FE)	FRONT RIGHT FENDER
EB	40 (5S-FE)	FRONT LEFT FENDER
ID	44	LEFT KICK PANEL
IG	44	R/B NO. 4 SET BOLT

 : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 2	38 (3S-GTE)	ENGINE ROOM MAIN WIRE	E13	40	ENGINE ROOM MAIN WIRE
	40 (5S-FE)		I 3	46	COWL WIRE
E 3	38 (3S-GTE)		I 4		
	40 (5S-FE)		I 8		
E 5	38 (3S-GTE)	ENGINE WIRE	I 9		
	40 (5S-FE)		I11	46	A/C NO. 2 WIRE
E10	38 (3S-GTE)		I12	46	A/C NO. 1 WIRE
	40 (5S-FE)				

RADIATOR FAN AND AIR CONDITIONER (AUTOMATIC AIR CONDITIONER)

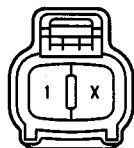


(AUTOMATIC AIR CONDITIONER, FOR PUSH TYPE OF BLOWER CONTROL SW)

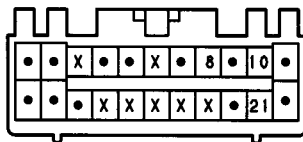
C 2 (A) GRAY



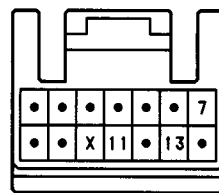
C 3 (B) GRAY



E 7 (A) (B) GRAY



E 7 (C) GRAY

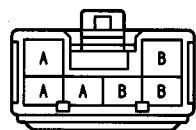


J 4



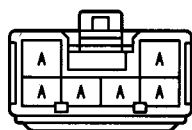
(HINT:SEE PAGE 7)

J 5



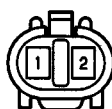
(HINT:SEE PAGE 7)

J 6



(HINT:SEE PAGE 7)

R 1 GRAY



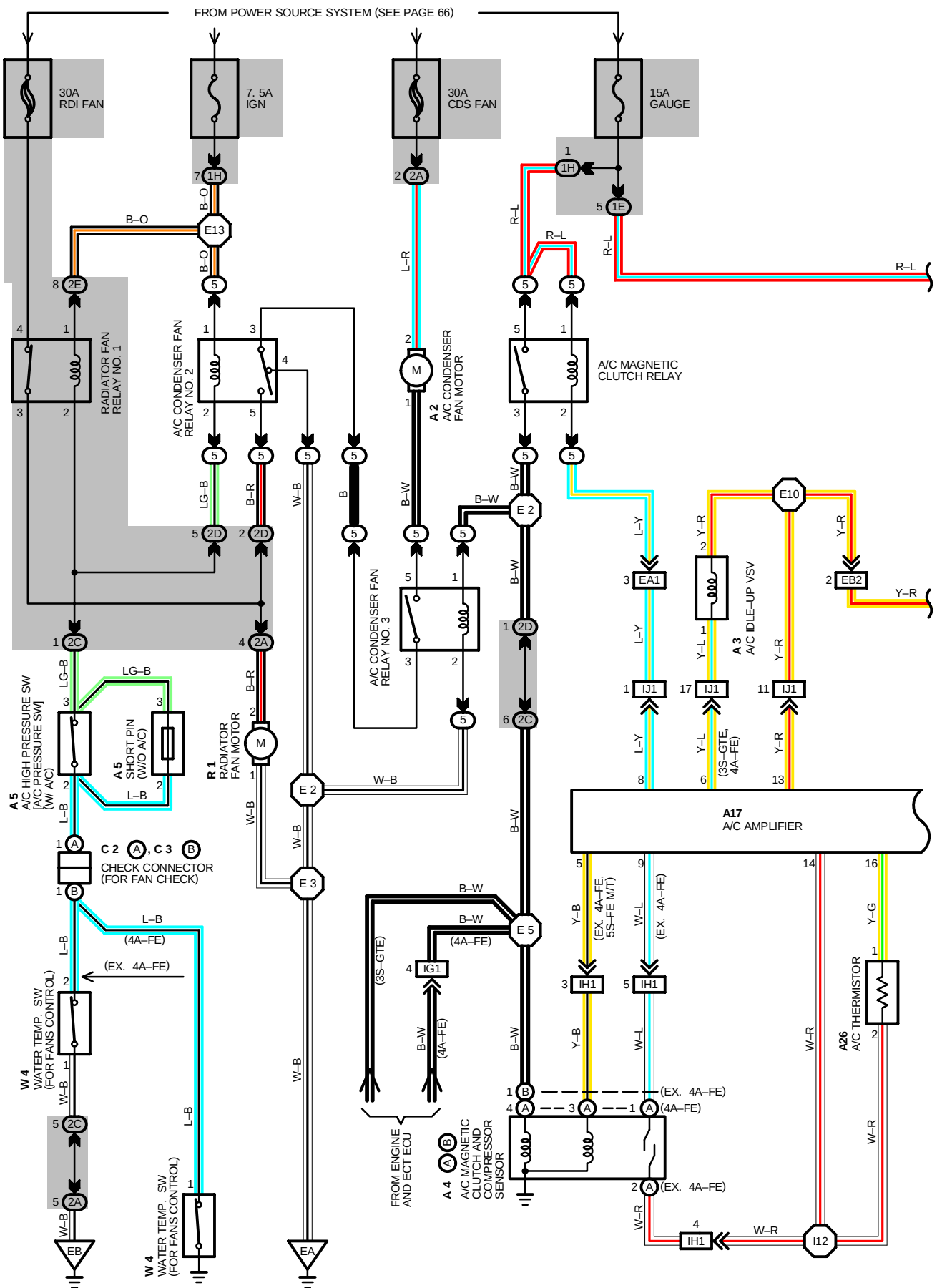
R 4

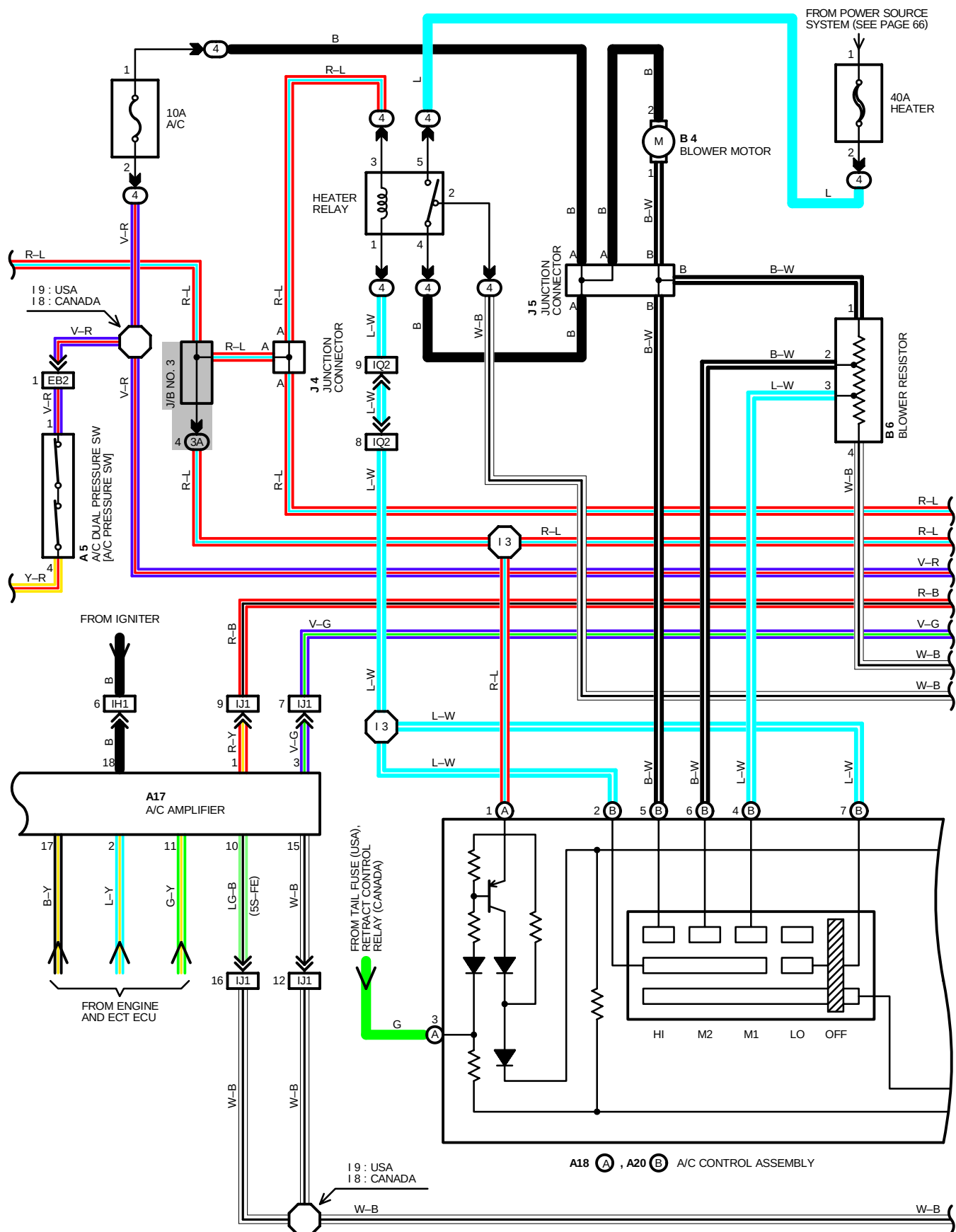


W 4 GRAY

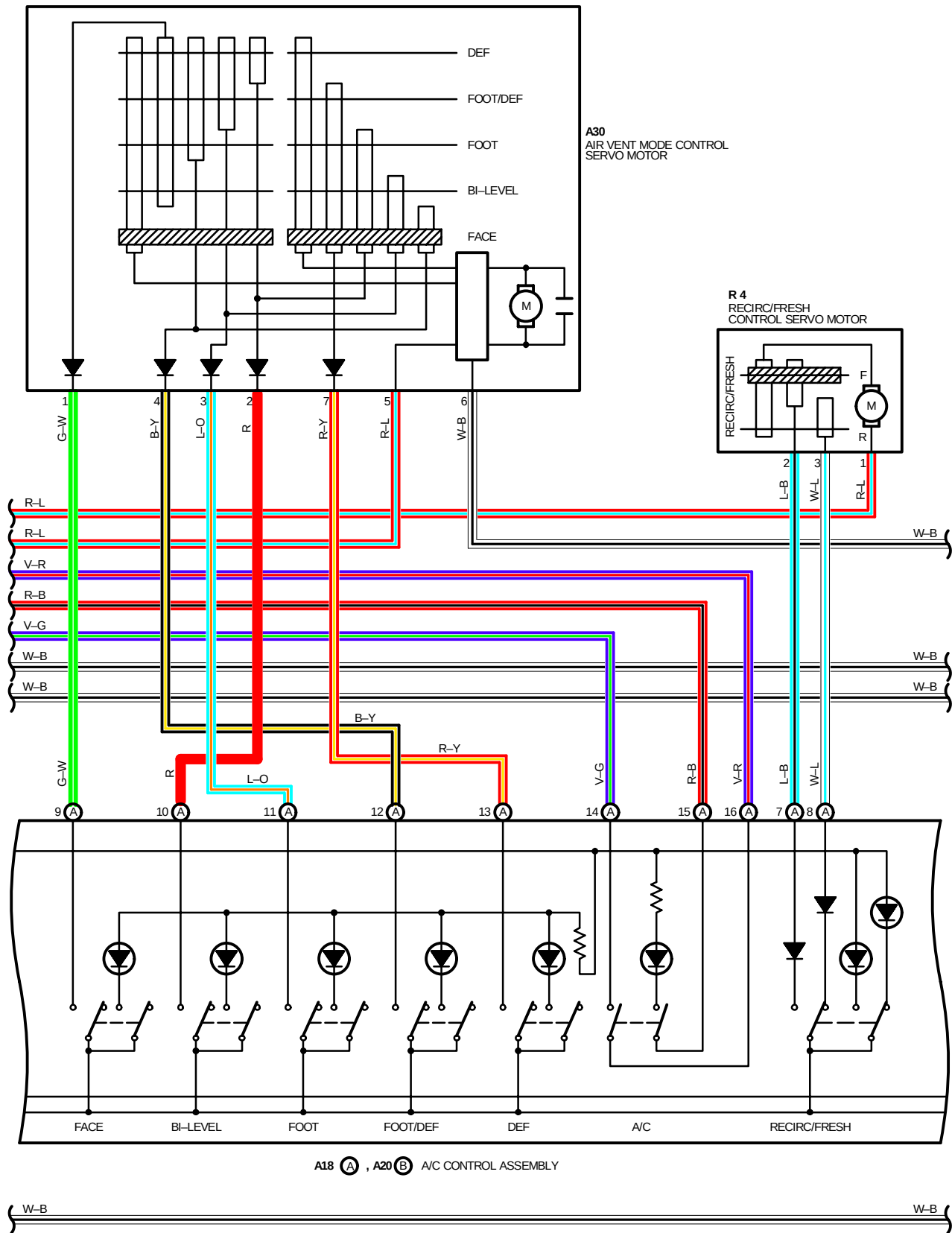


RADIATOR FAN AND AIR CONDITIONER (MANUAL AIR CONDITIONER)





RADIATOR FAN AND AIR CONDITIONER (MANUAL AIR CONDITIONER)



RADIATOR FAN AND AIR CONDITIONER (MANUAL AIR CONDITIONER)

SYSTEM OUTLINE

1. COOLING FAN OPERATION

WHEN THE IGNITION SW IS TURNED ON, THE CURRENT FROM IGN FUSE FLOWS TO **TERMINAL 1** OF RADIATOR FAN RELAY NO. 1 → **TERMINAL 2** → **TERMINAL 3** OF THE A/C PRESSURE SW → **TERMINAL 2** → **TERMINAL 1** (4S-FE) OR **TERMINAL 2** (EX. 4A-FE) OF THE WATER TEMP. SW → **TERMINAL 1** (EX. 4A-FE) → **GROUND**, FROM **TERMINAL 1** OF A/C FAN RELAY NO. 2 → **TERMINAL 2** → **TERMINAL 3** OF A/C PRESSURE SW → **TERMINAL 2** → **TERMINAL 1** (4A-FE) OR **TERMINAL 2** (EX. 4A-FE) OF THE WATER TEMP. SW → **TERMINAL 1** (EX. 4A-FE) → **GROUND**, CAUSING RELAY NO. 1 AND RELAY NO. 2 OF EACH FAN TO TURN ON.

* OPERATION AT LOW SPEED

WHEN THE A/C SW (A/C CONTROL ASSEMBLY) IS TURNED ON AND THE AIR CONDITIONER OPERATES, THE CURRENT FLOWS FROM GAUGE FUSE FLOWS TO **TERMINAL 1** OF A/C MAGNETIC CLUTCH RELAY → **TERMINAL 2** → **TERMINAL 8** OF A/C AMPLIFIER CAUSING A/C MAGNETIC CLUTCH RELAY TO TURN ON.

AT THAT TIME, THE CURRENT FROM GAUGE FUSE FLOWS TO **TERMINAL 5** OF A/C MAGNETIC CLUTCH RELAY → **TERMINAL 3** → **TERMINAL 4** (5S-FE), **1** (4A-FE) OF A/C MAGNETIC CLUTCH → **GROUND**, AND FROM **TERMINAL 3** OF A/C MAGNETIC CLUTCH RELAY → **TERMINAL 2** OF A/C FAN RELAY NO. 3 → **TERMINAL 2** → **GROUND**.

AS A RESULT, A/C MAGNETIC CLUTCH AND A/C FAN RELAY NO. 3 TURNS ON AND THE CURRENT FLOWS FROM CDS FAN FUSE → **TERMINAL 2** OF A/C CONDENSER FAN MOTOR → **TERMINAL 1** → **TERMINAL 5** OF A/C FAN RELAY NO. 3 → **TERMINAL 3** → **TERMINAL 3** OF A/C FAN RELAY NO. 2 → **TERMINAL 5** → **TERMINAL 2** OF A/C CONDENSER FAN MOTOR → **TERMINAL 1** → **GROUND**, FLOWING TO EACH FAN MOTOR IN SERIES, CAUSING THE COOLING FAN TO ROTATE AT LOW SPEED.

* OPERATION AT HIGH SPEED

DURING A/C OPERATION, WHEN THE PRESSURE OF A/C COMPRESSOR BECOMES HIGHER THAN NORMAL PRESSURE (MORE THAN 14.3 KG/CM² 1401 KPA, 203 PSI), THE A/C PRESSURE SW TURNS OFF.

AS A RESULT, FAN RELAY NO. 1 AND NO. 2 TURNS OFF AND THE CURRENT FLOWS FROM RDI FAN FUSE → **TERMINAL 4** OF RADIATOR FAN RELAY NO. 1 → **TERMINAL 3** → **TERMINAL 2** OF RADIATOR FAN MOTOR → **TERMINAL 1** → **GROUND**, AND FROM CDS FAN FUSE → **TERMINAL 2** OF A/C CONDENSER FAN MOTOR → **TERMINAL 1** → **TERMINAL 5** OF A/C FAN RELAY NO. 3 → **TERMINAL 3** → **TERMINAL 3** OF A/C FAN RELAY NO. 2 → **TERMINAL 4** → **GROUND**, FLOWING TO EACH FAN MOTOR IN PARALLEL CAUSING THE COOLING FAN TO ROTATE AT HIGH SPEED. WHEN THE ENGINE COOLANT TEMPERATURE BECOMES MORE THAN ABOUT 90°C (194°F), THE WATER TEMP. SW TURNS OFF AND THE SAME OPERATION AS ABOVE IS PERFORMED.

2. HEATER BLOWER MOTOR OPERATION (DIAL TYPE BLOWER (W/O AUTO A/C))

CURRENT IS APPLIED AT ALL TIMES THROUGH THE HEATER FUSE TO **TERMINAL 5** OF HEATER RELAY. WHEN THE IGNITION SW IS TURNED ON, CURRENT FLOWS THROUGH GAUGE FUSE TO **TERMINAL 3** OF HEATER RELAY → **TERMINAL 1** → **TERMINAL (B) 2** AND **TERMINAL (B) 7** OF A/C CONTROL ASSEMBLY.

* LOW SPEED OPERATION

WHEN THE BLOWER SW (A/C CONTROL ASSEMBLY) IS MOVED TO **LO** POSITION. THE CURRENT FLOWS TO **TERMINAL (B) 7** OF THE A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND** AND TURNS THE HEATER RELAY ON.

AS A RESULT, THE CURRENT TO **TERMINAL 5** OF THE HEATER RELAY → **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → **TERMINAL 1** → **TERMINAL 1** OF BLOWER RESISTOR → **TERMINAL 4** → **GROUND**, CAUSING THE BLOWER MOTOR TO ROTATE AT LOW SPEED.

* HIGH SPEED OPERATION

WHEN THE BLOWER SW (A/C CONTROL ASSEMBLY) IS MOVED TO **HI** POSITION. THE CURRENT TO **TERMINAL (B) 2** OF A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND** AND TURNS THE HEATER RELAY ON.

AS A RESULT, THE CURRENT TO **TERMINAL 5** OF HEATER RELAY → **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → **TERMINAL 1** → **TERMINAL (B) 5** OF A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND**, AND CAUSES THE BLOWER MOTOR TO ROTATE AT HIGH SPEED.

* MEDIUM SPEED OPERATION (OPERATION AT MANUAL M1, M2)

WHEN THE BLOWER SW (A/C CONTROL ASSEMBLY) IS MOVED TO **M1** POSITION. THE CURRENT FLOWING TO **TERMINAL 3** OF HEATER RELAY FLOWS TO **TERMINAL 1** → **TERMINAL (B) 2** OF A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND** AND TURNS THE HEATER RELAY ON.

AS A RESULT, CURRENT FROM THE HEATER FUSE TO **TERMINAL 5** OF HEATER RELAY FLOW TO **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → **TERMINAL 1** OF BLOWER RESISTOR → **TERMINAL 3** → **TERMINAL (B) 4** OF A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND**, CAUSING THE BLOWER MOTOR TO ROTATE AT MEDIUM LOW SPEED.

WHEN THE BLOWER SW (A/C CONTROL ASSEMBLY) IS MOVED TO **M2** POSITION. THE CURRENT TO **TERMINAL 5** OF HEATER RELAY FLOWS TO **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → **TERMINAL 1** → **TERMINAL 1** OF BLOWER RESISTOR → **TERMINAL 2** → **TERMINAL (B) 6** OF A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND**.

AS THE CURRENT FLOW FROM BLOWER MOTOR TO GROUND IS GREATER THAN FOR M1. THE BLOWER MOTOR ROTATES AT HIGH SPEED.

3. OPERATION OF RECIRC/FRESH CONTROL SERVO MOTOR

(SWITCHING FROM FRESH TO RECIRC)

WITH IGNITION SW TURNED ON, THE CURRENT FLOWS FROM GAUGE FUSE TO **TERMINAL 1** OF AIR INLET CONTROL SERVO MOTOR. WHEN THE RECIRC/FRESH SW IS SWITCHED TO THE RECIRC SIDE, THE CURRENT FLOWS FROM **TERMINAL 1** OF AIR INLET CONTROL SERVO MOTOR → **TERMINAL 2** → **TERMINAL (A) 7** OF RECIRC/FRESH CONTROL SERVO MOTOR → **TERMINAL (A) 2** → **GROUND**, THE MOTOR ROTATES AND THE DAMPER MOVES TO THE RECIRC SIDE.

WHEN IT IS IN THE **RECIRC** POSITION, THE CURRENT IS CUT INSIDE THE SERVO MOTOR AND THE DAMPER STOPS AT THAT POSITION.

(SWITCHING FROM RECIRC TO FRESH)

WITH IGNITION SW ON, WHEN THE RECIRC/FRESH SW IS SWITCHED TO THE FRESH SIDE, THE CURRENT FLOWS FROM **TERMINAL 1** OF AIR INLET CONTROL SERVO MOTOR → **TERMINAL 3** → **TERMINAL (A) 8** OF A/C CONTROL ASSEMBLY → **TERMINAL (A) 2** → **GROUND**, THE MOTOR ROTATES AND THE DAMPER MOVES TO THE FRESH SIDE. WHEN IT IS IN THE **FRESH** POSITION, THE CURRENT IS CUT INSIDE THE SERVO MOTOR AND THE DAMPER STOPS AT THAT POSITION.

4. OPERATION OF AIR VENT MODE CONTROL SERVO MOTOR

WITH IGNITION SW TURNED ON, THE CURRENT FLOWS FROM GAUGE FUSE TO **TERMINAL 5** OF AIR VENT MODE CONTROL SERVO MOTOR → **TERMINAL 6** → **GROUND**, AND THE DAMPER MOVES TO THE POSITION OF THE MODE SELECTION SW OF THE CONTROL ASSEMBLY SWITCH IS ON.

WHEN THE MODE SELECTION SW OF A/C CONTROL ASSEMBLY IS MOVED TO **DEF** POSITION FROM THE DAMPER IN THE **FACE** POSITION, THE CURRENT FLOWS FROM **TERMINAL 7** OF AIR VENT MODE CONTROL SERVO MOTOR TO **TERMINAL (A) 13** OF A/C CONTROL ASSEMBLY → **TERMINAL (A) 2** → **GROUND**.

AS A RESULT, THE SERVO MOTOR OPERATES UNTIL THE DAMPER REACHES **DEF** POSITION.

WHEN THIS OCCURS THE CURRENT TO THE A/C CONTROL ASSEMBLY IS SHUT OFF AND ROTATION OF THE MOTOR STOPS. SWITCHING TO OTHER MODES IS CONTROLLED BY THE SERVO MOTOR ACCORDING TO THE FLOWING CURRENT:

1. **FOOT/DEF** POSITION: THE CURRENT FLOWS FROM **TERMINAL 4** OF SERVO MOTOR TO **TERMINAL (A) 12** OF A/C CONTROL ASSEMBLY.
2. **FOOT** POSITION: THE CURRENT FLOWS FROM **TERMINAL 3** OF SERVO MOTOR TO **TERMINAL (A) 11** OF A/C CONTROL ASSEMBLY.
3. **BI-LEVEL** POSITION: THE CURRENT FLOWS FROM **TERMINAL 2** OF SERVO MOTOR TO **TERMINAL (A) 10** OF A/C CONTROL ASSEMBLY.

5. AIR CONDITIONER OPERATION

THE A/C AMPLIFIER RECEIVES VARIOUS SIGNALS, THE ENGINE RPM SIGNAL FROM THE IGNITER, OUTLET TEMPERATURE SIGNAL FROM THE A/C THERMISTOR, COOLANT TEMPERATURE FROM THE WATER TEMP. SENSOR, AND THE LOCK SIGNAL FROM THE A/C COMPRESSOR, ETC.

WHEN THE ENGINE IS STARTED AND THE A/C SW (A/C CONTROL ASSEMBLY) IS TURNED ON, THE CURRENT FLOWS FROM A/C FUSE TO **TERMINAL 13** OF A/C AMPLIFIER.

AS A RESULT, THE CURRENT FLOWING FROM **TERMINAL 2** OF A/C MAGNETIC CLUTCH RELAY TO **TERMINAL 8** OF A/C AMPLIFIER FLOWS FROM **TERMINAL 15** OF A/C AMPLIFIER TO **GROUND** AND TURNS ON THE MAGNETIC CLUTCH RELAY.

BECAUSE THE MAGNETIC CLUTCH IS ON, THE A/C COMPRESSOR OPERATES, CAUSING THE CURRENT FLOWING FROM A/C IDLE-UP VSV TO **TERMINAL 6** OF A/C AMPLIFIER TO FROM TO **TERMINAL 15** OF A/C AMPLIFIER → **GROUND**, AND TURNS ON THE VSV TO AVOID LOWERING THE ENGINE RPM DURING AIR CONDITIONER OPERATION.

WHEN ANY OF THE FOLLOWING SIGNALS ARE INPUT TO THE A/C AMPLIFIER, THE AMPLIFIER OPERATES TO TURN OFF THE AIR CONDITIONER:

- * ENGINE HIGH RPM SIGNAL
- * COOLANT TEMP. SIGNAL IS HIGH
- * A SIGNAL THAT THE TEMPERATURE AT THE AIR OUTLET IS LOW
- * A SIGNAL THAT THERE IS A LARGE DIFFERENCE BETWEEN ENGINE RPM AND COMPRESSOR RPM
- * A SIGNAL THAT THE REFRIGERANT PRESSURE IS ABNORMALLY HIGH OR LOW

RADIATOR FAN AND AIR CONDITIONER (MANUAL AIR CONDITIONER)

SERVICE HINTS

A 4 (A)(B) A/C MAGNETIC CLUTCH

(A)4, (B)1—GROUND : APPROX. 3.7 Ω

A 5 A/C PRESSURE SW

3-2 : OPEN ABOVE APPROX. 13.5 KG/CM² (192 PSI, 1323 KPA)

CLOSED BELOW APPROX. 10 KG/CM² (142 PSI, 980 KPA)

1-4 : OPEN WITH PRESSURE LESS THAN 2.1 KG/CM² (30 PSI, 206 KPA) OR ABOVE 27 KG/CM² (384 PSI, 2648 KPA)

A17 A/C AMPLIFIER

8-15 : CONTINUITY WITH A/C SW (A/C CONTROL ASSEMBLY) ON AND IGNITION SW AT ON POSITION

14-15 : ALWAYS CONTINUITY

14-GROUND: ALWAYS CONTINUITY

15-GROUND: ALWAYS CONTINUITY

13-GROUND: APPROX. 12 VOLTS WITH IGNITION SW ON

A26 A/C THERMISTOR

1-2 : APPROX. 4852 Ω AT 0°C (32°F)

APPROX. 2341 Ω AT 15°C (39°F)

APPROX. 1500 Ω AT 25°C (77°F)

B 6 BLOWER RESISTOR

1-3 : APPROX. 0.48 Ω

3-2 : APPROX. 0.94 Ω

2-4 : APPROX. 0.91 Ω

W 4 WATER TEMP. SW (FOR FANS CONTROL)

1-2 : OPEN ABOVE APPROX. 90°C (194°F)

CLOSED BELOW APPROX. 83°C (181.4°F)

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
A 2	28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)	A20	B 32	J 4	33
A 3	28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)	A26	32	J 5	33
A 4	A 28 (3S-GTE), 31 (4A-FE)	A30	32	J 6	33
	B 28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)	B 4	32	R 1	28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)
A 5	28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)	B 6	32	R 4	33
A17	32	C 2	A 28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)	W 4	28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)
A18	A 32	C 3	B 28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)		

○ : RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
4	27	R/B NO. 4 (RIGHT KICK PANEL)
5	27	R/B NO. 5 (ENGINE COMPARTMENT FRONT RIGHT)

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2A	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2C	22 (2WD)	ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2D	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2E		
3A	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EA1	38 (3S-GTE)	COWL WIRE AND ENGINE ROOM MAIN WIRE (FRONT SIDE OF RIGHT FRONT FENDER)
	40 (5S-FE)	
	42 (4A-FE)	
EB2	38 (3S-GTE)	ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER)
	40 (5S-FE)	
	42 (4A-FE)	
IG1	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)
IH1	44	ENGINE WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)
IJ1	46	COWL WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)
IQ2	46	COWL WIRE AND A/C NO. 2 WIRE (BESIDE HEATER UNIT)

 : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
EA	38 (3S-GTE)	FRONT RIGHT FENDER
	40 (5S-FE)	
	42 (4A-FE)	
EB	38 (3S-GTE)	FRONT LEFT FENDER
	40 (5S-FE)	
ID	44	LEFT KICK PANEL
IG	44	R/B NO. 4 SET BOLT

 : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 2	38 (3S-GTE)	ENGINE ROOM MAIN WIRE	E10	40 (5S-FE)	COWL WIRE
	40 (5S-FE)			42 (4A-FE)	
	42 (4A-FE)		E13	38 (3S-GTE)	ENGINE ROOM MAIN WIRE
E 3	38 (3S-GTE)			40 (5S-FE)	
	40 (5S-FE)			42 (4A-FE)	
42 (4A-FE)	I 3		46	COWL WIRE	
E 5	38 (3S-GTE)	I 8			
	40 (5S-FE)	I 9			
42 (4A-FE)	I12	46	A/C NO. 1 WIRE		
E10	38 (3S-GTE)	COWL WIRE			

A 2 BLACK



A 3 BLACK



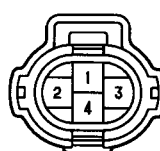
(5S-FE) A 4 (A) GRAY



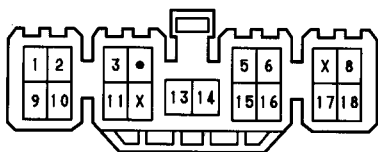
(4A-FE) A 4 (B) GRAY



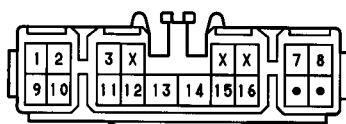
A 5 BLACK



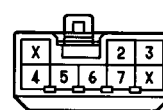
A17



A18 (A)



A20 (B)



A26



A30



B 4 (A) BLACK



B 6 BLACK



C 2 (A) GRAY



C 3 (B) GRAY

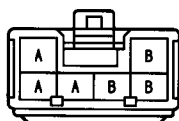


J 4



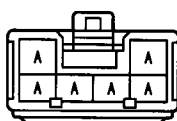
(HINT:SEE PAGE 7)

J 5



(HINT:SEE PAGE 7)

J 6



(HINT:SEE PAGE 7)

R 1



R 4



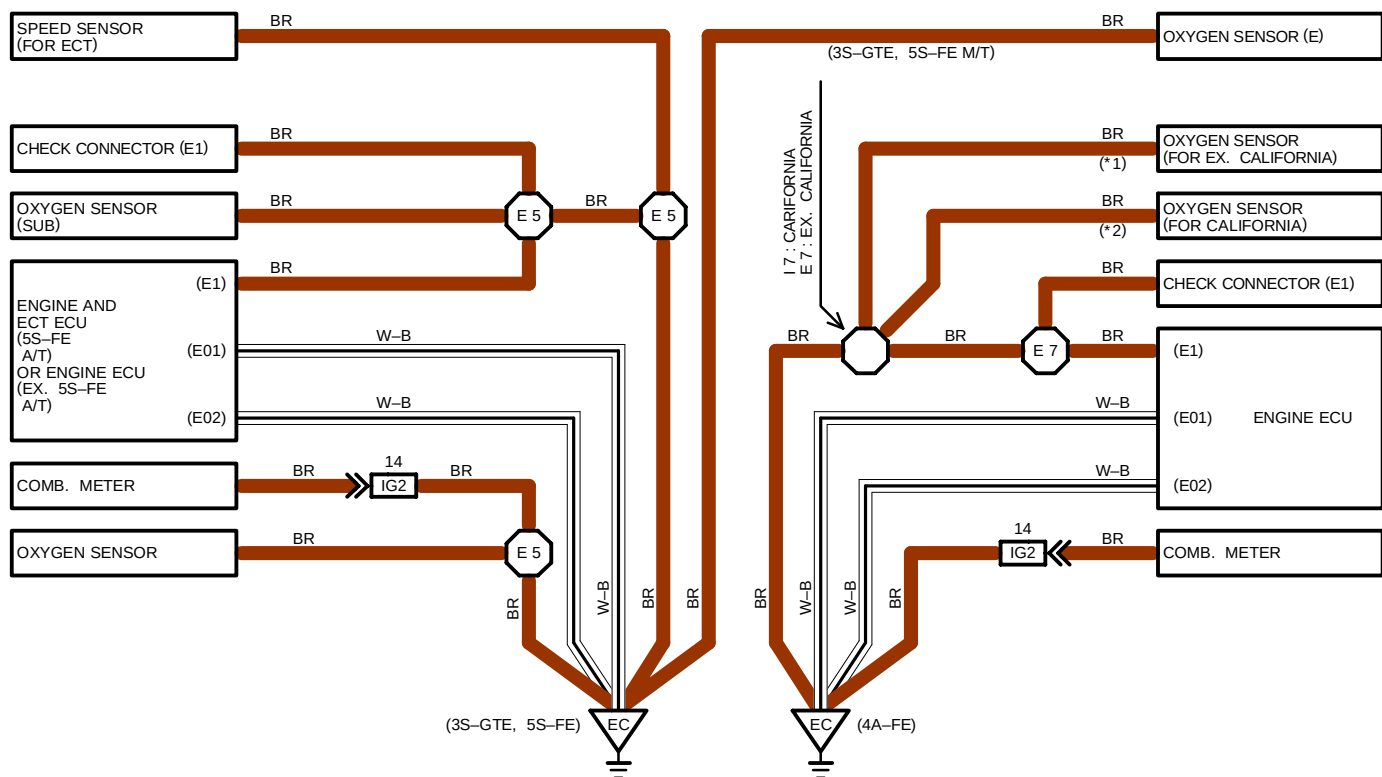
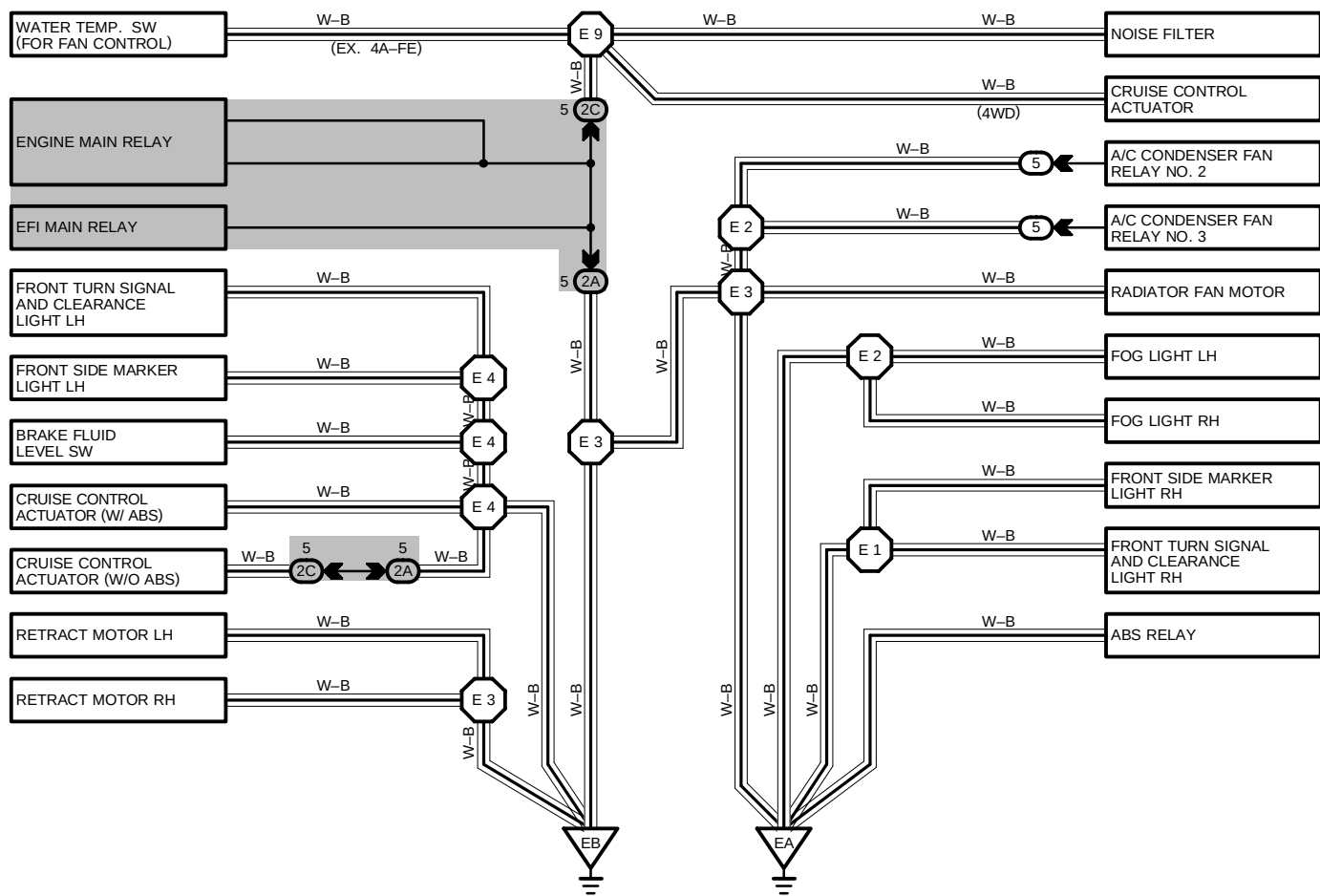
(EX. 4A-FE) W 4 GRAY

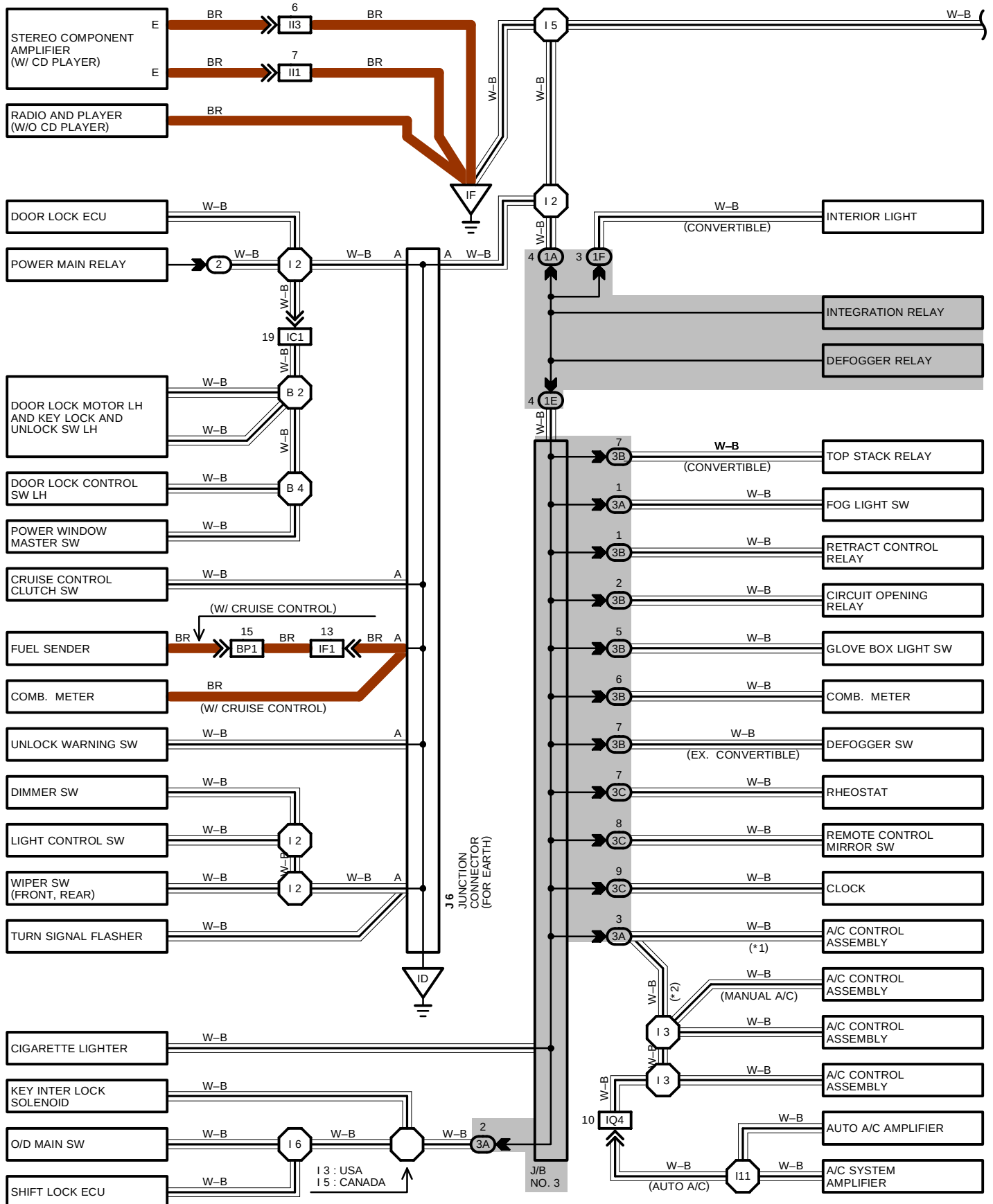


(4A-FE) W 4 GRAY

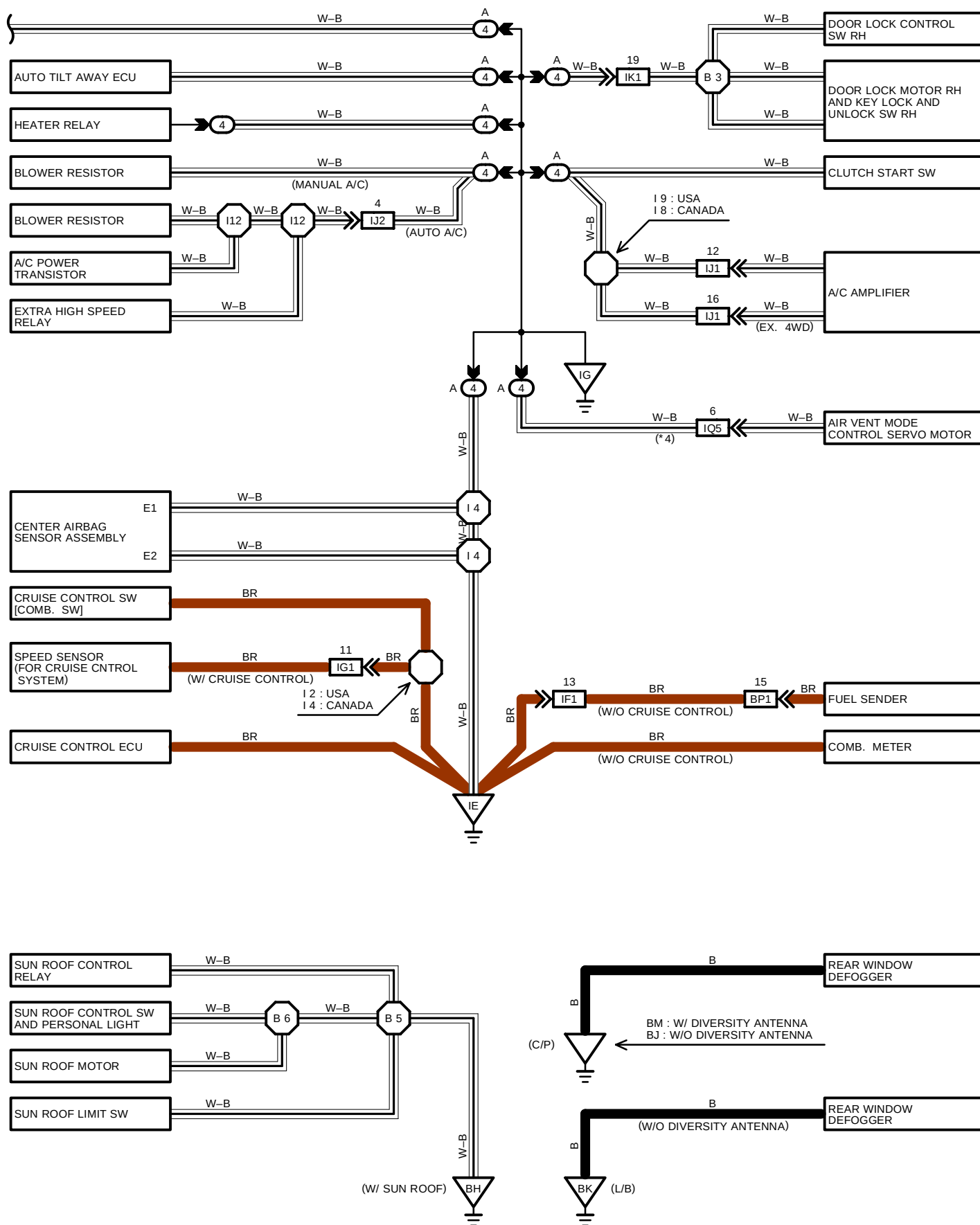


GROUND POINT





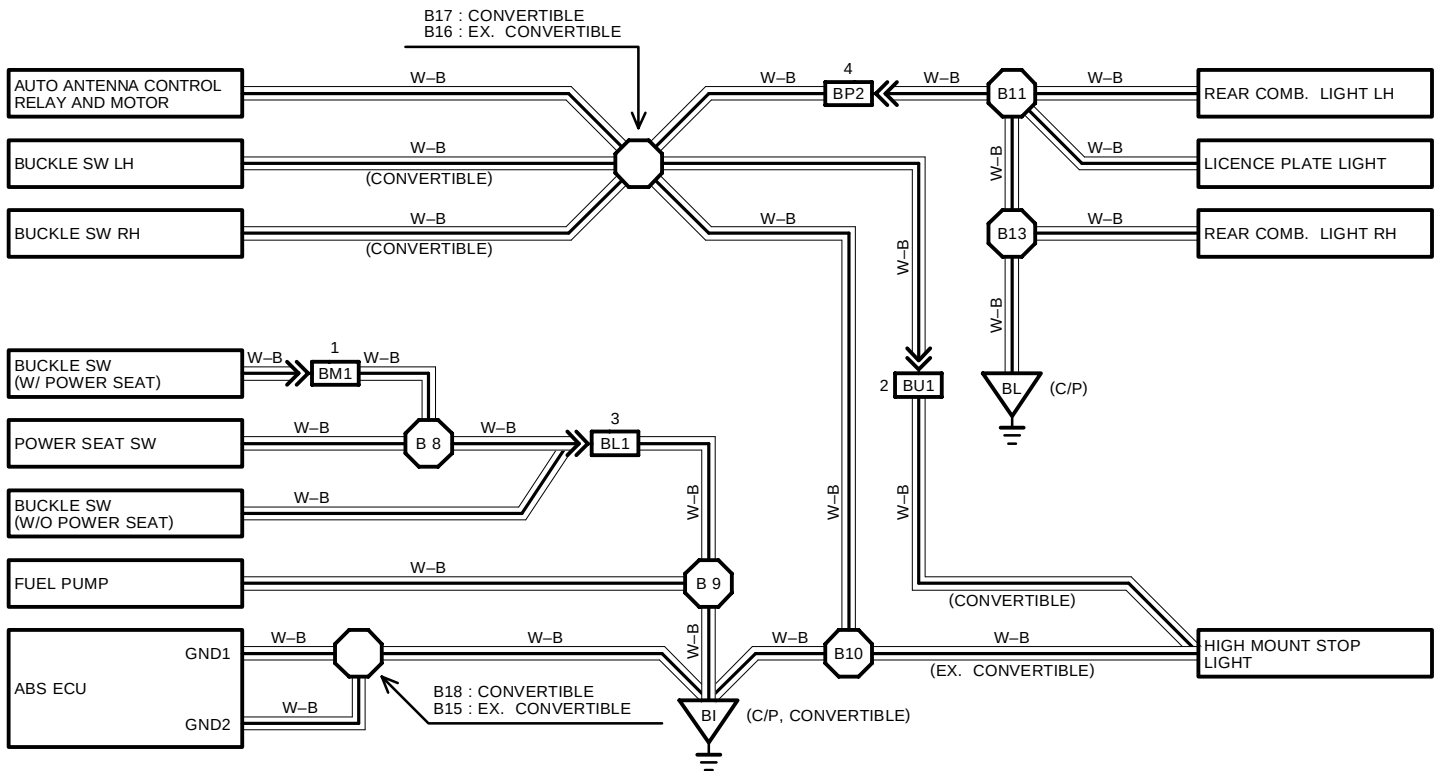
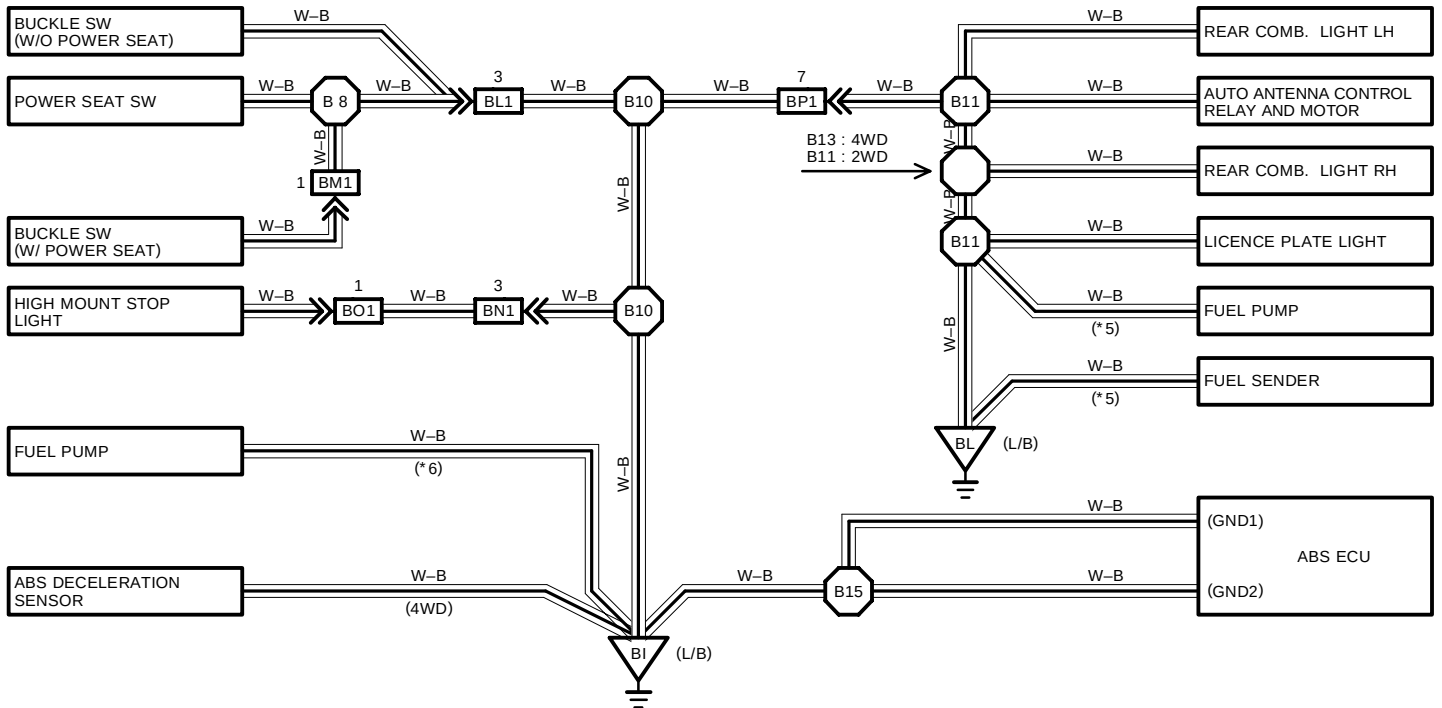
GROUND POINT



* 1 : EX. AUTO IR CONDITIONER
* 2 : AUTO AIR CONDITIONER

* 3 : PUSH TYPE OF BLOWER CONTROL SW
* 4 : DIAL TYPE OF BLOWER CONTROL SW

* 5 : ALL-TRAC/4WD
* 6 : 2WD



GROUND POINT

: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
J 6	33				

: RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
2	26	R/B NO. 2 (LEFT KICK PANEL)
4	27	R/B NO. 4 (RIGHT KICK PANEL)
5	27	R/B NO. 5 (ENGINE COMPARTMENT FRONT RIGHT)

: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
1F	20	FLOOR NO. 1 WIRE AND J/B NO. 1 WIRE (LEFT KICK PANEL)
2A	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2C	22 (2WD)	ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)
3A	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)
3B		
3C		

: CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IC1	44	FRONT DOOR LH WIRE AND COWL WIRE (LEFT KICK PANEL)
IE1	44	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)
IF1	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
IG1	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)
IG2		
II1	46	COWL WIRE AND CONSOLE BOX WIRE (INSTRUMENT PANEL CENTER)
II3	46	COWL WIRE AND CONSOLE BOX WIRE (RIGHT KICK PANEL OF ALL-TRAC/4WD)
	46	COWL WIRE AND CONSOLE BOX WIRE (RIGHT KICK PANEL OF 2WD)
IJ1	46	COWL WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)
IJ2		
IJ3		
IK1	46	FRONT DOOR RH WIRE AND COWL WIRE (RIGHT KICK PANEL)
IQ1	46	COWL WIRE AND A/C NO. 2 WIRE (BESIDE HEATER UNIT)
IQ4		
IQ5		
BL1	48 (L/B)	FLOOR WIRE AND FRAME WIRE (LEFT SIDE OF FRONT FLOOR PANEL)
	50 (C/P)	
BM1	48 (L/B)	FRAME WIRE AND SEAT WIRE (UNDER THE DRIVER'S SEAT)
	50 (C/P)	
BN1	48 (L/B)	BACK DOOR NO. 1 WIRE AND FLOOR WIRE (LEFT SIDE OF PACKAGE TRAY TRIM)
BO1	48 (L/B)	BACK DOOR NO. 1 WIRE AND BACK DOOR NO. 1 SUB WIRE (BACK DOOR UPPER LEFT)
BP1	48 (L/B)	FLOOR WIRE AND LUGGAGE ROOM WIRE (LEFT QUARTER PANEL CENTER)
BU1	48 (L/B)	FLOOR WIRE AND REAR WINDOW NO. 2 WIRE (LEFT REAR PILLAR)



: GROUND POINTS

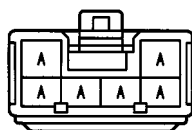
CODE	SEE PAGE	GROUND POINTS LOCATION
EA	38 (3S-GTE)	FRONT RIGHT FENDER
	40 (5S-FE)	
	42 (4A-FE)	
EB	38 (3S-GTE)	FRONT LEFT FENDER
	40 (5S-FE)	
	42 (4A-FE)	
EC	38 (3S-GTE)	INTAKE MANIFOLD
	40 (5S-FE)	
	42 (4A-FE)	
ID	44	LEFT KICK PANEL
IE	44	INSTRUMENT PANEL BRACE LH
IF	44	INSTRUMENT PANEL BRACE RH
IG	44	R/B NO. 4 SET BOLT
BH	48 (L/B)	ROOF LEFT
	50 (C/P)	
BI	48 (L/B)	UNDER THE LEFT CENTER PILLAR
	50 (C/P)	
BJ	50 (C/P)	RIGHT REAR PILLAR (C/P)
BK	48 (L/B)	BACK DOOR RIGHT (L/B)
BL	48 (L/B)	BACK PANEL CENTER
	50 (C/P)	
BM	50 (C/P)	LEFT REAR PILLAR (C/P)



: SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 1	38 (3S–GTE)	ENGINE ROOM MAIN WIRE	I11	46	A/C NO. 2 WIRE
	40 (5S–FE)		I12	46	A/C NO. 1 WIRE
	42 (4A–FE)		B 2	48 (L/B)	FRONT DOOR LH WIRE
E 2	38 (3S–GTE)			50 (C/P)	
	40 (5S–FE)		B 3	48 (L/B)	FRONT DOOR RH WIRE
	42 (4A–FE)			50 (C/P)	
E 3	38 (3S–GTE)		B 5	48 (L/B)	ROOF WIRE
	40 (5S–FE)			50 (C/P)	
	42 (4A–FE)		B 6	48 (L/B)	
E 4	38 (3S–GTE)			50 (C/P)	
	40 (5S–FE)		B 8	48 (L/B)	FRAME WIRE
	42 (4A–FE)			50 (C/P)	
E 5	38 (3S–GTE)	ENGINE WIRE	B10	48 (L/B)	FLOOR WIRE
	40 (5S–FE)			50 (C/P)	
	42 (4A–FE)		B11	48 (L/B)	LUGGAGE ROOM WIRE
E 7	38 (3S–GTE)			50 (C/P)	
	40 (5S–FE)		B13	48 (L/B)	FLOOR WIRE
	42 (4A–FE)			50 (C/P)	
E 8	38 (3S–GTE)		B15	48 (L/B)	
	40 (5S–FE)		B16	48 (L/B)	
I 2	46	COWL WIRE		50 (C/P)	
I 3		B17	48 (L/B)		
I 4			50 (C/P)		
I 5			52 (CONVERTIBLE)		
I 6		B18	48 (LB)		
I 7	46		ENGINE WIRE	50 (C/P)	
I 8	46		COWL WIRE		
I 9					

J 6



(HINT:SEE PAGE 7)